On a metaplectic Jacquet–Langlands transfer for division algebras of prime degree

Draft 2

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Abstract

This book was written by David Joyner together with the AI systems Claude (Anthropic) and ChatGPT (OpenAI), which drafted, revised and refereed the text. The Acknowledgments (page 200) describe each contribution in detail.

Let K be a totally imaginary number field containing the n-th roots of unity, with n odd. Let r be a prime not dividing n, and let D be a central division algebra of degree r over K. The book compares the n-fold cover of $D_{\mathbb{A}}^{\times}$ constructed in [JC26a] with the Kazhdan–Patterson cover of $\mathrm{GL}(r,\mathbb{A}_K)$. It grew out of [Jo88]. Remark 10.3 of the corrected version [JC26a] of that paper outlines a step that was left open there, namely a metaplectic Jacquet–Langlands transfer between these two covers. The book takes the first steps toward that transfer.

The contributions are as follows. Both covers are realized inside one cover of $\mathrm{GL}(r,\mathbb{A}_L)$, where $L \subset D$ is a cyclic maximal subfield, and their covered elliptic tori are identified on good elements at the ramified places (Theorem A). Genuine test functions are matched, first on the regular set (Theorem B) and then in general (Theorems D and E). A global transfer of automorphic representations is proved under a non-cancellation hypothesis (Theorems C and C$'$). On the division-algebra side, a unitary transform L_D of class functions is shown to carry irreducible characters to irreducible genuine characters at the places not dividing n. At the places dividing n this is proved only in part. Toral Fourier positivity is proved for all quadratic extensions. At the wild place of the running example ($K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$) the same statement is established for the trivial character and for every representation trivial on the third congruence subgroup. These last results are proved modulo exhaustive exact computations, whose scripts are in the repository (label (P^{c})).

Every statement carries one of the status labels of §3.1. The full metaplectic Jacquet–Langlands correspondence is not claimed. Chapter 9 lists the results with their status, and §1.4 lists the main open problems.

Preface

This book collects a series of research manuscripts on a metaplectic Jacquet–Langlands transfer between the n-fold cover of the multiplicative group of a central division algebra D of prime degree r over a number field and the Kazhdan–Patterson cover of $\mathrm{GL}(r)$. It is ongoing research: the text distinguishes complete proofs, results established in the literature or in other chapters, conditional statements and open problems by the labels of §3.1, and **the full metaplectic Jacquet–Langlands correspondence is not claimed**.

Chapters 4–8 began as five separate manuscripts, written in November 2026. Those manuscripts, the revision logs and the referee reports are in the project repository (`github.com/wdjoyner/metaplectic-jl`). The revision log of this draft records which manuscript each chapter came from and what was changed when the book was assembled. Independent human review is the next step, and the mathematical claims should be assessed independently.

Chapter 1

Introduction

This chapter describes the setting of the book, summarizes each chapter, fixes the running example and lists the main open problems.

1.1 The setting

Motivation. The classical Jacquet–Langlands correspondence relates automorphic representations of $\mathrm{GL}(r)$ to those of the multiplicative group of a central division algebra D of the same dimension, by comparing the two groups' trace formulas: elements of $D^\times$ correspond to the elliptic (non-split) conjugacy classes of $\mathrm{GL}(r)$, so the correspondence transports the harder, compact part of harmonic analysis on $\mathrm{GL}(r)$ into the smaller, compact-modulo-center group $D^\times$. A *metaplectic*, or covering-group, analog of this correspondence asks the same question one level up: instead of $\mathrm{GL}(r)$ and $D^\times$ themselves, one compares an n-fold topological cover of $\mathrm{GL}(r)$ with a corresponding cover of $D^\times$, and asks whether genuine representations (those that are honestly projective and do not descend to the base group) still match. This book builds and studies such a correspondence for a division algebra of prime degree.

The groups. Let K be a totally imaginary number field (one with no real embeddings, so every archimedean place of K is complex) containing the group μ_n of n-th roots of unity, with n odd; let r be a prime with $(n,r)=1$; and let D be a central division algebra of dimension r^2 over K, ramified at the finite set S of places, containing a cyclic maximal subfield L. "Ramified at S" means that D is a twisted form of the matrix algebra $M_r(K)$ – isomorphic to it almost everywhere – that becomes a genuine division ring, with no zero divisors, at the places of S; the cyclic maximal subfield L, of degree r over K, exhibits D as a cyclic algebra and supplies the auxiliary field used throughout to compare $D^\times$ with $\mathrm{GL}(r)$. (For the general theory of central division algebras, ramification and cyclic algebras behind this paragraph, with worked examples, see §3.3.)

The two covers. The book compares two n-fold covers: the cover of $D^\times_{\mathbb{A}}$ from the author's corrected 1988 paper on the metaplectic analog of Kazhdan's "endoscopic" lifting [JC26a], and the Kazhdan–Patterson cover of $\mathrm{GL}(r,\mathbb{A}_K)$ with twisting parameter c_v, where $n \mid r-1+2rc_v$ (this numerical condition is exactly what makes the local cocycle of [KP84] well defined for the chosen n and r). Both are subgroups of one cover of $\mathrm{GL}(r,\mathbb{A}_L)$ (Proposition 4.2.4). Comparing two

covers built by different methods, for two different groups, raises an identification problem before any harmonic analysis can begin: an isomorphism of central extensions is canonical only up to a character valued in μ_n, so an unexamined choice of isomorphism could leave the comparison of genuine functions, and so of genuine representations, ambiguous. Chapter 4 resolves this by realizing both covers as subgroups of the single ambient cover of $\mathrm{GL}(r, \mathbb{A}_L)$ just mentioned; conjugation inside that one group then supplies the comparison, and its restriction to the centers of the relevant tori is canonical (Theorem A, §1.2 below). (Covers, the cocycle σ_F and the twisting parameter c_v are defined in general in §10.1.)

The method. The approach follows the trace-formula proof of the classical Jacquet–Langlands correspondence and uses the metaplectic correspondence of Flicker and Kazhdan [FK86] with the corrections of Zou [Zo23]. Concretely: test functions on the two groups are matched by matching their orbital integrals; the geometric sides of the resulting trace formulas are then compared term by term; and the automorphic spectra are compared by reading off what survives. Two further points are special to the covered setting. First, orbital integrals on a cover are well behaved only on the *good* elements – regular elements whose centralizing torus the cover splits over canonically – so matching and transfer are proved first in that generality, with the fully general case reached afterward through germ expansions near the center. Second, the metaplectic correspondence of [FK86], as corrected by [Zo23], is used as the already-available harmonic analysis for the Kazhdan–Patterson side; the corresponding harmonic analysis for the division-algebra side is what this book supplies. (Good elements are defined in §10.2; orbital integrals, matching and germs in §10.3.)

The global data and conventions – the measures, the test functions, and the cocycle normalization used in every later chapter – are fixed once, in Chapters 3 and 10 (see §3.2, §10.4 and §10.1), rather than being restated each time they are needed.

1.2 The chapters

Chapters 4–7 and 8 began as five separate papers and are now assembled into one book; each proves one stage of the correspondence and states explicitly which earlier stages it relies on.

- Chapter 2 reviews the literature on the classical Jacquet–Langlands correspondence, on covering groups, and on metaplectic correspondences, in order to place this book's results among them, and Chapter 3 collects the background on division algebras and representation theory, the notation, and the status labels (P), (E), (O), (C) used throughout to flag how completely each result is established.

- Chapter 4 constructs the two covers inside one ambient cover, solving the identification problem of §1.1 above; it identifies their covered elliptic tori on good elements (Theorem A), so that a genuine character of such a torus and a genuine test function both have an unambiguous meaning on either side. It proves matching of test functions supported on the regular set (Theorem B), and uses this matching, inside a comparison of the two groups' trace formulas, to prove a global transfer of automorphic representations under a non-cancellation hypothesis (Hyp_D) (Theorem C), a hypothesis that rules out an otherwise-possible vanishing of a packet-counting coefficient for reasons unconnected to the transfer itself.

- Chapter 5 removes the regular-set restriction of Theorem B: by computing the germ expansion of orbital integrals near the center, it proves matching of arbitrary genuine test functions, including those not vanishing near the identity (Theorems D and E). This stronger matching improves the global transfer of Theorem C to a version needing a weaker hypothesis at only one place of S (Theorem C$'$).

- Chapter 6 turns from matching test functions to the finer question of matching individual representations. It studies the division-algebra side through a unitary transform L_D of class functions, modeled on the classical character transform between $\mathrm{GL}(r)$ and $D^\times$; it proves that L_D carries irreducible characters to irreducible genuine characters at places not dividing n (the *tame* places), obtains the character relation there in the range of the corrected Flicker–Kazhdan correspondence, and gives the obstruction – a common factor forced on every genuine irreducible dimension – at places dividing n (the *wild* places), where the same argument breaks down.

- Chapter 7 reconciles the differing cocycle conventions of [FK86], [Zo23], and the present book, so that a result proved under one convention can be quoted under another; it studies power maps and good elements for every maximal torus, not only the one singled out in Chapter 4; it proves a toral positivity criterion – a sign condition on a discriminant ratio – at places dividing n (Toral positivity, §10.6); and it proves a conditional global comparison of characters, conditional on that positivity holding everywhere it is needed.

- Chapter 8 treats the wild place of the running example in degree two, resolving the Chapter 6 obstruction completely in this one case: the transform of the trivial character, and of every representation trivial on the third congruence subgroup, is a genuine irreducible character, exhibited explicitly as induced from a canonical genuine character of a congruence subgroup.

- Chapter 9 lists the results of Chapters 4–8 with their labels and status, as a single reference table.

- Chapter 10 is a glossary of the technical terms used in every chapter. It includes the cocycle normalization.

1.3 The running example

Every chapter tests its constructions on one case, chosen to be as small as possible while still showing every phenomenon the theory needs to see.

Example 1.3.1 (The running example)**.** Take $K=\mathbb{Q}(\zeta_3)$, $n=3$, $r=2$ (the smallest available prime, giving the quaternion case), $L=K(i)=\mathbb{Q}(\zeta_{12})$, and the quaternion algebra $D=(-1,4-\zeta_3)_K$. Then D is ramified exactly at $S=\{\mathfrak{p}_3,\mathfrak{p}_7\}$, with $\mathfrak{p}_3=(1-\zeta_3)$ the prime above 3 (a wild place, where $b_D=3$ and $c_D=27$ are, respectively, the forced common dimension of genuine representations and the discriminant of the relevant congruence filtration) and $\mathfrak{p}_7=(3+\zeta_3)$ a prime above 7 (a tame place). In Assumption 4.1.1 we take $r'=2$, $c=1$ and $c_v=r'c=2$.

Having one place of each kind in a single fixed example lets every chapter illustrate its tame and wild behavior side by side, in a setting small enough to compute with directly; it is also exactly the case in which the wild obstruction of Chapter 6 is resolved completely, in Chapter 8. Each chapter

has a short subsection on it – the full technical instantiation of Assumption 4.1.1, with the explicit factorization of $4-\zeta_3$ into $\mathfrak{p}_3\mathfrak{p}_7$, is repeated where it is first used, in §4.1.6 – and [JC26b] works the whole case out in detail, with an independent machine check of the ramification set, across Chapters 4–7.

Remark 1.3.2 (The example meets the hypotheses in play)**.** Three outside results are drawn on in Chapters 4–6, and Example 1.3.1 meets the standing hypotheses of each.

[JC26a]. Assumption 4.1.1, which fixes the data of [JC26a, §1], holds termwise: K is totally imaginary and contains $\mu_n = \mu_3$; $L = K(i)$ is cyclic of degree $r = 2$ over K – since -1 is not a square in $K = \mathbb{Q}(\sqrt{-3})$, the subalgebra $K[i] \subset D$ generated by the first Hilbert-symbol entry of $D = (-1, 4-\zeta_3)_K$ is a field, and it is exactly L – with $(n,r) = (3,2) = 1$; and $r' = 2$, $c = 1$, $c_v = 2$ satisfy $rr' \equiv 1 \pmod{n}$, $n \mid r-1+2c$ and $n \mid r-1+2rc_v$ (all three reduce to $3 \mid 3$, $3 \mid 3$ and $3 \mid 9$).

[FK86]. Two of its hypotheses are in play, at different strengths. The local unitary transform (Theorem 6.4.4) needs only n odd and $(n,r) = 1$ (Remark 6.12.1), both immediate here. The stronger tame character relation (D2), which draws on the corrected Flicker–Kazhdan–Zou correspondence, has the published sufficient hypothesis $(n, N_r) = 1$ (Remark 4.8.1), where N_r is the least common multiple of the composite integers at most r. For $r = 2$ there are no composite integers at all, so $N_2 = 1$ and $(n, N_2) = 1$ automatically – for every odd n, not just $n = 3$, simply because the running example is a quaternion algebra.

[FK88]. It is cited only for the method of the simple trace formula; the instance the book actually uses (Proposition 4.6.2) is proved by following [FK86, §18] instead. Its own standing requirement – a test function with a single supercuspidal component at a place outside S – is a condition on which automorphic representations are considered in Theorem C, already built into that theorem's hypotheses, not a numerical condition on (K, n, r, D) for this example to satisfy.

1.4 Main open problems

Three problems are left open by the book, roughly in order of how much new machinery each is likely to need.

- *Integrality of L_D at wild places in general.* Chapter 8 proves integrality – the property that L_D sends irreducible characters to irreducible, rather than merely virtual, genuine characters – only for the running example's own wild place, by an explicit combinatorial argument special to that case. Whether the same conclusion holds at every wild place, for every division algebra of prime degree and every admissible odd n, is open.

- *The full global comparison of the cover of $D^\times$ with the cover of* $\mathrm{GL}(r)$. Chapters 4–7 compare the two covers' automorphic representations at the ramified places of S and, conditionally on toral positivity, at the wild places. A complete global comparison would also need matching at the places where D splits – where $D^\times$ and $\mathrm{GL}(r)$ coincide as groups, but the two covers can still differ – and at the complex archimedean places; this book treats neither.

- *Removing the auxiliary support and packet hypotheses of Chapters 4–5.* The global transfer theorems (Theorems C and C′) are proved by fixing one auxiliary place w_0 carrying a supercuspidal component and by assuming a non-cancellation hypothesis on the relevant packet. Removing both

devices, so that the transfer holds unconditionally and without reference to an auxiliary place, would likely need either the covering-group trace-formula techniques of W.-W. Li [Li10, Li12, Li14] or a comparable alternative to the packet-counting argument used here.

Chapter 8, §8.15, discusses in more detail what a fully general wild result would need.

Chapter 2

Literature

This chapter reviews the literature on which the book draws. Section 2.1 reviews the literature of the whole series; the later sections add what is specific to Chapters 5, 6 and 8. Chapter 7 has no separate related-work section; its sources are discussed in its own text.

2.1 The literature of the series

This section reviews the literature on which the whole series draws, and places the series within it. Chapters 5–7 refer to this review and add only what is specific to them. It proves nothing and introduces no notation.

The Jacquet–Langlands correspondence. Jacquet and Langlands [JL70] related automorphic representations of the multiplicative group of a quaternion algebra to those of $\mathrm{GL}(2)$ by comparing trace formulas. For a central division algebra of arbitrary degree the local correspondence is due to Deligne–Kazhdan–Vignéras [DKV84] and Rogawski [Rog83], and Badulescu [Bad08] proved the global correspondence for all inner forms of $\mathrm{GL}(n)$, with multiplicity one and strong multiplicity one. Flicker and Kazhdan [FK88] gave a simple trace formula for test functions with a single supercuspidal component and used it to extend both the simple-algebras correspondence and the metaplectic correspondence to automorphic forms with one supercuspidal component; Proposition 4.6.2 is a statement of the same kind for the cover $\overline{G}(\mathbb{A}_K)$, proved here by following [FK86, §18], and we cite [FK88] for its approach only. The transfer of orbital integrals between $\mathrm{GL}(n)$ and its inner forms, with explicit matching functions, is studied by Cohen [Coh18]; harmonic-analytic background is in Kottwitz [Kot05] and Harish-Chandra [HC70]. Our definition of matching and the use of a simple trace formula follow the pattern of these works, with two new features: only good elements carry orbital integrals, and the two covers must first be identified on tori (§4.3). The sign convention is compared with Badulescu's in §4.4.2.

Covering groups. The n-fold covers of $\mathrm{SL}(2)$ and $\mathrm{GL}(2)$ over local and global fields go back to Kubota [Kub67], [Kub69]. Moore [Moo68] and Matsumoto [Mat69] classified the topological central extensions of the split simply connected groups by Steinberg symbols (see also Milnor [Mil71]); this classification is the main input of Proposition 4.2.4. Kazhdan and Patterson [KP84] constructed the n-fold covers of $\mathrm{GL}(r)$ with a determinantal twisting parameter, developed their

representation theory, including the exceptional representations, and formulated the generalized Shimura correspondence [KP86]. Explicit block-compatible cocycles are in Banks–Levy–Sepanski [BLS99]. The modern framework is that of Brylinski and Deligne [BD01], for which Weissman [Wei18] and Gan–Gao [GG18] formulated an L-group and a Langlands program; the historical introduction [GGW18] surveys the subject. This series uses only the Kazhdan–Patterson covers and their restrictions to subgroups.

Metaplectic correspondences. For $n = 2$ the correspondence goes back to Shimura's work on modular forms of half-integral weight [Shi73]; its representation-theoretic form is due to Gelbart [Gel76] and Waldspurger [Wal80], [Wal91], through the Weil representation. Flicker [Fli80] treated the n-fold cover of $\mathrm{GL}(2)$ by the trace formula, and Flicker and Kazhdan [FK86] established the metaplectic correspondence between genuine representations of the n-fold cover of $\mathrm{GL}(r)$ and representations of $\mathrm{GL}(r)$, using a simple trace formula [FK86, §18]; our GL-side trace formula is theirs. Mezo generalized the local correspondence of [FK86] and prepared a comparison of the invariant trace formulas of $\mathrm{GL}(r)$ and its covers [Mez02], [Mez01], and constructed metaplectic tensor products for irreducible representations [Mez04]. Zou [Zo23] revisited the local metaplectic correspondence, restored and generalized parts of the local theory of [FK86], and pointed out an error in [FK86, §26.2]; Chapter 6 uses his corrected form. Savin [Sav88] proved a local Shimura correspondence at the level of Iwahori–Hecke algebras. W.-W. Li developed the Arthur–Selberg trace formula for covers of connected reductive groups [Li10], [Li12], [Li14]; it is one of the two routes to removing the auxiliary supercuspidal place (goal (iii) of §4.1.3). Kazhdan's lifting from tori [Kaz84] has a metaplectic analog [Jo88], corrected in [JC26a], which is the source of the cover of $D^{\times}_{\mathbb{A}}$ used here. This chapter does not compose with the Flicker–Kazhdan correspondence; that would be the step after (D2).

Central extensions of division algebras. The topological central extensions of $\mathrm{SL}_1(D)$, and the metaplectic kernel for anisotropic groups, were determined by Prasad–Raghunathan [PR88] and Prasad–Rapinchuk [PR96]; see also Rapinchuk [Rap86], [Rap88] and Sury [Sur05]. These works concern the extensions themselves rather than automorphic forms on them.

The work closest to ours on the side of the division algebra is that of Klose, in his Göttingen dissertation (1983, written under S. J. Patterson) and the two papers drawn from it. He studied central extensions by μ_n of the multiplicative group of a quaternion division algebra. The local paper [Klo86a] concerns a p-adic quaternion division algebra; according to its indexing data, its subject is the commutator structure of $D^{\times}$, the structure of its group of one-units, and the classes in the second cohomology that describe its central extensions, including the metaplectic ones. The global paper [Klo86b] concerns a quaternion division algebra D over a number field. It constructs explicit cocycles, and in the description of Prasad and Rapinchuk [PR96, Introduction] it obtains partial results on the topological central extensions of the adele group of $\mathrm{GL}_{1,D}$; their own computation of the metaplectic kernel is more precise. Hill [Hil08, §§2.3, 3.8] attributes to Klose the facts on commutators and symmetric cocycles of abelian groups that were used there to determine the metaplectic extensions of $D^{\times}$; among them is a canonical isomorphism $H^2_{\mathrm{sym}}(\mu_m, \mu_m) \cong \mathbb{Z}/m$. In [Jo88] it is noted that a result of Klose allows one to choose the quaternion algebra D so that the cover of $G_D(\mathbb{A}_K)$ exists without assuming $(n, r) = 1$, that is, also for even n. In this series n is odd, so for $r = 2$ that hypothesis holds automatically, and the cover of $D^{\times}$ is defined by restriction from

the Kazhdan–Patterson cover of $\mathrm{GL}(r)$ over the splitting field L ([JC26a], §4.2). We have not been able to consult [Klo86a], [Klo86b] or the dissertation. The summary above rests on their indexing data (keywords and subject classification), on the descriptions in [PR96], [Hil08] and [Jo88], and on the titles; the theorem statements of Klose's papers are not reproduced here for that reason. We have not compared Klose's extensions with the cover of [JC26a] for $r = 2$. Patel [Pat14] shows that the 2-fold metaplectic cover of $\mathrm{GL}(2,E)$, for a quadratic extension E/F, splits over $\mathrm{GL}(2,F)$ and over the multiplicative group of the quaternion division algebra over F; this is the same mechanism of restricting a cover over an extension field to a subgroup, and it is the mechanism by which our ambient cover realizes both $\overline{G}_D$ and $\overline{G}$.

The present series. This chapter supplies the first steps of [JC26a, Remark 10.3]. We found no existing metaplectic Jacquet–Langlands correspondence for covers of inner forms of $\mathrm{GL}(r)$. In particular, among the papers that zbMATH lists as citing [FK86] (Zbl 0616.10024), only [Jo88] concerns a cover of $D^{\times}$; those mentioning division or simple algebras treat linear groups. Chapter 5 proves the matching of arbitrary functions, using germ expansions [FK86, §8] and results on the linear groups (Shalika germs, Steinberg pseudo-coefficients), and improves Theorem C. Chapter 6 studies the division-algebra side through a unitary transform of class functions, using Adams operations at tame places and Kottwitz's framework for rational conjugacy. Chapter 7 reconciles the cocycle conventions with [FK86] and [Zo23], studies power maps and toral positivity at the places dividing n, and proves a conditional global comparison. The companion chapter (Chapter 8) studies the remaining wild local problem in degree two, using the Heisenberg–Weil formalism and character theory of compact p-adic groups.

2.2 Work related to Chapter 5

This section discusses the work related to Chapter 5. The literature of the whole book (the Jacquet–Langlands correspondence, covering groups, metaplectic correspondences, central extensions of division algebras) is reviewed in §2.1; we add only what is specific to Chapter 5. The section proves nothing and introduces no notation.

The local results of Chapter 5 concern harmonic analysis near the center, and their inputs are of two kinds. For covers of $\mathrm{GL}(r)$, Kazhdan–Patterson [KP86, §1] and Flicker–Kazhdan [FK86, §8] proved the germ expansion and the characterization of orbital integrals of genuine functions (after Vignéras), and [FK86] constructed the local metaplectic correspondence for square-integrable representations; Zou [Zo23] corrected and extended parts of that theory. W.-W. Li developed the local harmonic analysis needed for the trace formula on covers [Li12], including semisimple descent, the Plancherel formula and the density of tempered characters, and proved the invariant trace formula for covers [Li14]. On the linear side we use the Shalika germ expansion [Sha72], Howe's work on the Fourier transform and germs of characters for GL_n [How74], Harish-Chandra's theory of admissible invariant distributions [HC70], [HC78], Rogawski's computation of the trivial germ on elliptic tori via the building [Rog81], van Dijk's formula for induced characters [vD72], and, for parabolic descent of germs, Kivinen–Tsai [KT26]. Pseudo-coefficients of square-integrable representations and the density theorem are taken in the form stated in [FK86, §§25.1, 27.3], which goes back to Kazhdan [Kaz86]; the linear Jacquet–Langlands inputs are from [DKV84], [Rog83], [Bad02] and [Bad08, Theorem 2.2]. The Plancherel formula and Harish-Chandra's finiteness

theorem are used in the form of Waldspurger [Wal03], with explicit formulas for $\mathrm{GL}(n,F)$ and $\mathrm{GL}(m,D)$ in Aubert–Plymen [AP05]; and Kottwitz's normalization of measures on inner forms is in [Kot88]. For the classification of tempered representations of $\mathrm{GL}(r,F)$ we use Zelevinsky [Zel80]; general background on smooth representations is in Bernstein–Zelevinsky [BZ77] and Bernstein [Ber84].

Our matching theorem uses none of the deeper parts of the theory of covers. Its main tool is the section σ of §5.3, which reduces genuine harmonic analysis near the center of either cover to linear harmonic analysis, compatibly with the identification of elliptic tori of Chapter 4. That covers split over small open subgroups is standard [FK86, §2]; what is used here is the stronger statement that the splitting given by n-th powers of n-th roots is equivariant and consistent with $\jmath$ on a full conjugation-invariant neighborhood of the unipotent set, at every finite place including those dividing n.

2.3 Work related to Chapter 6

This section discusses the work related to Chapter 6. The literature of the whole book is reviewed in §2.1; we add only what is specific to Chapter 6. The section proves nothing and introduces no notation.

The division-algebra side. The local Jacquet–Langlands correspondence for $D^\times$ is due to Deligne, Kazhdan and Vignéras [DKV84], and Badulescu [Bad08] proved the global correspondence for inner forms of GL_n; Cohen [Coh18] studies the transfer of representations and of orbital integrals between GL_n and its inner forms. The norm-one group $H^1 = \mathrm{SL}_1(D)$ of a p-adic division algebra was studied by Riehm [Rie70], and its topological central extensions by Prasad and Raghunathan [PR88]; Klose [Klo86a] studied commutators and the second cohomology of p-adic quaternion division algebras, including their metaplectic extensions (see §2.1). Proposition 6.7.2 (unique splitting over H^1 when $p \nmid n$) and Proposition 6.8.1 (non-splitting when $p \mid n$) are statements of this kind for the particular cover of Chapter 4; they are proved here directly, and we cite these works for their subject only.

Power operations. The Adams operations come from topological K-theory [Ada62]; Atiyah's account of power operations [Ati66] and Segal's equivariant K-theory [Seg68] give the representation-ring form used in §6.6 and in the tame proof (Lemma 6.7.5).

Metaplectic correspondences. The square-integrable metaplectic correspondence for GL_r is due to Flicker and Kazhdan [FK86]; Zou [Zo23] restored and generalized it and gave the root-sum relation used in Corollary 6.7.8, whose constant is compared with b_D in Remark 6.4.5. The transfer L_D is defined through the n-th power map on good elements and the central character, after the model of Kazhdan's lifting and its metaplectic analog (§2.1).

Arithmetic. Section 6.10 follows Kottwitz [Kot84] on rational and stable conjugacy, transfer of tori to inner forms and Tamagawa factors; the Galois cohomology of Sha^1 is in Milne [Mil06]. Proposition 6.11.3 is a statement modulo scalars of the kind given, for n-th powers themselves,

by the classical local–global principle whose sharp form is the Grunwald–Wang theorem [AT68, Ch. X]; for odd n the exceptional case of that theorem does not arise. The p-adic change-of-variables formula used in §6.7 is quoted from [Loe08].

2.4 Work related to Chapter 8

This section places Chapter 8 among the works it draws on or runs parallel to. It is selective; §2.1 has the fuller discussion of the metaplectic correspondence itself.

Metaplectic covers. The n-fold covers of SL_2 and GL_2 over a local field containing μ_n go back to Kubota [Kub67]; [Kub69], who gave the explicit cocycle used in Section 8.11; Moore [Moo68] and Matsumoto classified the topological central extensions of simply connected groups by Steinberg symbols, which is why the Hilbert symbols of F and E_0 could be pinned down in Lemma 8.11.1 as the unique Steinberg symbols. Kazhdan and Patterson [KP84] constructed the n-fold covers of GL_r with the determinantal twist parameter c, developed their representation theory (the metaplectic tensor product, the exceptional representations), and formulated the generalized Shimura correspondence; the parameter $c_v = -1/4 \bmod p$ of Section 8.14, forced by scalar centrality, is a point on that family. The modern framework is that of Brylinski–Deligne central extensions [BD01], for which Weissman [Wei18] and Gan–Gao [GG18] formulated an L-group and a Langlands program; a Satake isomorphism for covers is in [McN12]; [Gao18]. The cover of $D^\times$ used here is defined by restriction from $\mathrm{GL}_2(E_0)$ (Chapter 4); we are not aware of an independent literature on n-fold covers of division algebras beyond [JC26a] and this series.

Metaplectic correspondences. For $n = 2$ the Shimura correspondence was realized by Waldspurger [Wal80]; [Wal91] through the theta correspondence, as recalled in Section 8.2.3; Flicker [Fli80] treated the n-fold cover of GL_2 by the trace formula, and Flicker–Kazhdan [FK86] established the metaplectic correspondence between the n-fold cover of GL_r and GL_r in general, at the cost of a simple trace formula and hypotheses at the places dividing n. Kazhdan's lifting [Kaz84] and its metaplectic analogue [JC26a] are the local models for the division-side transfer L_D of Chapter 6: a transfer of characters defined on good elements through the n-th power map and the central character. At the level of the Iwahori–Hecke algebra, Savin [Sav88] proved a local Shimura correspondence for the tame covers; Zou [Zo23] revisits the local metaplectic correspondence and its applications. None of these works addresses integrality at a place dividing n for a division algebra, which is the problem of Targets A–C; Chapter 6's tame argument is an Adams-operation argument in the sense of Section 8.2.3, and the obstruction at $p \mid n$ is the one described in Remark 8.12.2 and Section 8.15.

Representations of division algebras and Jacquet–Langlands. The irreducible representations of $D^\times$ were constructed by Corwin [Cor74] and Howe [How77b] in the tame case and by Koch–Zink [KZ80] and Corwin in general; the local Jacquet–Langlands correspondence is due to Jacquet–Langlands [JL70] for $r = 2$ and to Deligne–Kazhdan–Vignéras [DKV84] and Rogawski [Rog83] in general, with the global correspondence completed by Badulescu [Bad08]. Bushnell and Henniart made the correspondence explicit in the essentially tame case [BH11] and treat GL_2 and the

quaternion algebra in detail in [BH06], including the conductor and depth bookkeeping behind the notion of level in Section 8.2.1. The induced models of Sections 8.12 and 8.13 are of the shape of Howe's construction, with a character of a covered congruence subgroup in place of a character of a torus.

Heisenberg groups, Weil representations and the Howe correspondence. Weil [Wei64] introduced the metaplectic representation; Howe [How79] formulated the duality conjecture, proved by Waldspurger [Wal90] for $p \neq 2$ and by Gan–Takeda [GT16] in general; [MVW87] is the standard reference for the p-adic theory. The finite Weil representations and their characters are in Gérardin [Ger77]. We use this circle of ideas only as a guide (Section 8.2.3 and Remark 8.16.1); the Stone–von Neumann theorem for covered tori is used literally (Proposition 4.3.5).

Characters of compact p-adic groups. Howe's Kirillov theory [How77a] describes the characters of congruence subgroups by coadjoint orbits when the group is large relative to p; the orbit method for uniform pro-p groups with p odd was developed further by Jaikin-Zapirain [Jai06] and Boyarchenko–Sabitova [BS08], and the cohomology of uniform groups by Lazard [Laz65] (see [DDMS; SW]), which enters the alternative route to (H1) in Remark 8.11.5. The finite-group tools are Brauer's characterization of characters and Clifford theory [Isa76]. The Hilbert-symbol facts of Section 8.11 are in Fesenko–Vostokov [FV02] and Serre [Ser79].

Computation. The exact computations behind the **($\mathbf{P}^{\mathrm{c}}$)** labels are small-scale: each is a short, purpose-written Python script – for instance an orbit enumeration on a specific group of order 3^9, or a character table obtained numerically and then verified exactly against the class-sum eigenvector criterion – rather than a call to a general-purpose computer-algebra library (Appendix A.2). A general-purpose system such as GAP or Magma could recompute the character tables of Section 8.13 directly, and the design search of Section 8.10 is an enumeration that any system can reproduce. The methodological choice to separate hand proofs, exhaustive exact computations and sampled corroboration, and to label them, follows the practice of the series.

Chapter 3

Background and notation

This chapter collects the background and the notation of the book. Sections 3.1–3.6 hold for every chapter. They give the status labels (§3.1), the global and local data (§§3.2, 3.4 and 3.5), and two background sections with examples, on division algebras (§3.3) and on representation theory (§3.6). The remaining sections list the notation specific to individual chapters, with pointers to the definitions. The technical terms for covers, genuine objects, matching, measures and named results are in the glossary, Chapter 10, which includes the cocycle normalization (§10.1). The chapter proves nothing and can be consulted at any point.

3.1 Status labels

Every statement in the book has one of the following statuses. A status is never upgraded without a proof, and every change is recorded in the revision log of the draft in which it is made.

Proved here is the default and carries no visible tag. A Lemma, Proposition, Theorem or Corollary with no label after its name has a complete argument in the chapter in which it appears, possibly using results labeled **(E)** below; "proved here, using (using **(E)**)" marks a proof that rests on such inputs. A statement proved only under an explicitly stated hypothesis says so in words, as in "proved here, under (Hyp_D)", and the hypothesis itself is listed separately with its own status.

Every other status is written explicitly after the statement's name, and a remark immediately follows the statement giving the reason: where it is proved, if the status is **(E)**; what is missing from the outline, if **(O)**; or the mathematical motivation for believing it, if **(C)**. The statuses are:

(E) *Established elsewhere*: a precisely cited theorem from the literature or from an earlier chapter, used as a black box. "**(E)**, partly proved here" marks a statement that is partly cited and partly proved; the following remark says which part is which.

(O) *Proof outlined*: the steps are identified, but at least one of them remains to be written out or checked; the following remark says which.

(C) *Conjectural*: new mathematical input is required. This includes the goals deferred to later chapters and the hypotheses under which conditional statements are proved; the following remark gives the evidence or motivation for the conjecture.

(P^{c}) *Proved modulo an exhaustive exact computation*: the proof consists of hand arguments together with a computation that is exhaustive over an explicitly described finite set and uses exact arithmetic only (no sampling, and no floating point without exact re-verification). Each such script is committed to the repository, cited by commit hash, and described in an appendix and in the scripts' README. At present this label is used only in Chapter 8.

A status table near the front of each chapter lists every noteworthy statement with its status, for quick reference; such a table may still show **(P)** explicitly as a row label, even though the statement itself carries no visible tag in the body.

3.2 Global data

This section fixes the global objects that every chapter uses: the number field, the covering degree, the division algebra and its ramification set. They are fixed once here so that later chapters can refer to them without restating them. The last item instantiates them in the running example.

K, n, μ_n K is a totally imaginary number field containing the group μ_n of n-th roots of unity, and $n \geq 1$ is odd. All archimedean places of K are complex.

r, D, L, S r is a prime with $(n,r) = 1$; D is a central division algebra over K of dimension r^2; $L \subset D$ is a cyclic maximal subfield, so $[L : K] = r$; S is the (finite) set of places at which D ramifies. Since r is prime, every $v \in S$ is finite and D_v is a division algebra; $|S| \geq 2$, and $|S|$ is even if $r = 2$ (§4.1.1). For the general theory behind this definition, why r is taken prime, and worked local and global examples, see §3.3.

r', c, c_v Integers with $rr' \equiv 1 \pmod{n}$ and $n \mid r - 1 + 2c$; then $c_v := r'c$ satisfies $n \mid r - 1 + 2rc_v$ (Assumption 4.1.1). Despite the subscript, c_v does not depend on the place v. The parameter c occurs *before* the relabeling of μ_n described in §10.1, and c_v *after* it.

$\mathbb{A}_K$, L_v $\mathbb{A}_K$ is the adele ring of K (written $\mathbb{A}$ when no confusion is possible), and $L_v := L \otimes_K K_v = \prod_{w|v} L_w$.

G_D The algebraic group $D^\times$ over K, so that $G_D(K_v) = D_v^\times$.

Running example $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$ with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$; $L = K(i) = \mathbb{Q}(\zeta_{12})$, $c = 1$, $r' = 2$, $c_v = 2$. The place $\mathfrak{p}_3$ divides n and $\mathfrak{p}_7$ does not. See Example 1.3.1 for the choice of this example and [JC26b] for the case worked out across the series.

3.3 Division algebras

This section expands on the single line in §3.2 introducing D: it collects, with examples, the facts about central division algebras actually used in the book, and explains why the series restricts to *prime* degree r. Nothing here is new; the point of writing it out is to make the single running example of §3.2 serve as a worked illustration of each fact as it is introduced, rather than to prove anything (see, e.g., [Rei75, Pie82]).

Central simple algebra, central division algebra A finite-dimensional algebra D over a field F is *central simple* if its center is exactly F and it has no two-sided ideals other than 0 and D; every central simple F-algebra has F-dimension m^2 for some integer $m \geq 1$, called its *degree*. By Wedderburn's theorem, $D \cong M_s(\Delta)$ for a division algebra Δ with center F and some $s \geq 1$; D itself is a *division algebra* (every nonzero element invertible) exactly when $s = 1$. The matrix algebra $M_m(F)$ is the *split* central simple algebra of degree m: it is central simple but, for $m \geq 2$, not a division algebra, so it is the standing non-example against which "division algebra" is a real restriction. The running example's $D = (-1, 4-\zeta_3)_K$ (Example 1.3.1) is such an algebra, of degree $r = 2$ over K.

Brauer group and local invariant Two central simple F-algebras are *Brauer equivalent* if $M_a(D_1) \cong M_b(D_2)$ for some a, b; equivalence classes form a group $\mathrm{Br}(F)$ under tensor product, the *Brauer group*, in which every class has a unique division-algebra representative. When F is a non-archimedean local field, $\mathrm{Br}(F) \cong \mathbb{Q}/\mathbb{Z}$ canonically, and the image $\mathrm{inv}(D) \in \mathbb{Q}/\mathbb{Z}$ of a class is its *local invariant*; D is split (Brauer-trivial) exactly when $\mathrm{inv}(D) = 0$. For a field extension E/F of degree e, restriction $\mathrm{Br}(F) \to \mathrm{Br}(E)$ corresponds under this isomorphism to multiplication by e on $\mathbb{Q}/\mathbb{Z}$: $\mathrm{inv}(D \otimes_F E) = e \cdot \mathrm{inv}(D)$. This is the single computation behind Example 5.5.13 below and behind Remark 6.3.2. Since the running example's D is ramified, not split, at both places of $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$ (Example 1.3.1), and a quaternion algebra has only one nonzero invariant to be, $\mathrm{inv}(D_{\mathfrak{p}_3}) = \mathrm{inv}(D_{\mathfrak{p}_7}) = \frac{1}{2}$: the same value at both places, which is exactly why $|S| = 2$ is even (see Global existence below).

Local classification; the standard construction Over a non-archimedean local field F, for every $m \geq 1$ there is, up to isomorphism, exactly one central division algebra of degree m: the one with $\mathrm{inv}(D) = 1/m$ (equivalently $-1/m$, $\gcd(k,m)/m$ for any k coprime to m, ... – any generator of the cyclic group $\frac{1}{m}\mathbb{Z}/\mathbb{Z} \subset \mathbb{Q}/\mathbb{Z}$ gives the same algebra up to the labeling of the isomorphism $\mathrm{Br}(F) \cong \mathbb{Q}/\mathbb{Z}$). It can be built explicitly: let F_m/F be the unramified extension of degree m, with Frobenius-twisted conjugation σ, and let ϖ be a uniformizer of F; then

$$D = F_m\langle\Pi\rangle, \qquad \Pi^m = \varpi, \qquad \Pi x \Pi^{-1} = \sigma(x) \ (x \in F_m),$$

is a central division algebra of degree m over F with $\mathrm{inv}(D) = 1/m$. This is the *standard construction*; the degree-2 case is the usual quaternion division algebra over F, and §6.4 reuses exactly this construction (with $m = r$) to build D_v at a wild place. At the running example's own wild place, $F = K_{\mathfrak{p}_3} = \mathbb{Q}_3(\zeta_3)$ ($m = r = 2$); the construction and its uniformizer Π are written out explicitly in Example 8.7.1.

Congruence filtration and level The standard construction's valuation ring $\mathscr{O}_D = \mathscr{O}_{F_m}[\Pi]$ has a unique maximal two-sided ideal $\mathfrak{P} = \Pi\mathscr{O}_D$, with residue field $k_D = \mathscr{O}_D/\mathfrak{P}$ the degree-m extension of the residue field of F; this gives $D^\times$ its *congruence filtration* by the open normal pro-p subgroups $K^j := 1 + \mathfrak{P}^j$ ($j \geq 1$), with $K^j/K^{j+1} \cong k_D$ (additively) and $\mathscr{O}_D^\times/K^1 \cong k_D^\times$. Since $D^\times/F^\times$ is compact, every smooth irreducible representation ρ of $D^\times$ is finite-dimensional and trivial on some K^j; its *level* is the least j with $\rho|_{K^{j+1}}$ trivial (so level 0 means ρ is trivial on K^1). Level is the integer form of the Moy–Prasad depth – a representation of level j has depth j/m – and depth is preserved by the local Jacquet–Langlands correspondence. The same notion, for the groups $\overline{G}_D(K_v)$ of this book rather than for $D_v^\times$ itself, is what is meant by

"level" in Proposition 5.6.11 and Corollary 5.6.13 below. At the running example's wild place ($F = \mathbb{Q}_3(\zeta_3)$, $m = r = 2$, residue field $\mathbb{F}_3$), $D^\times/F^\times K^1 \cong D_8$ (order 8), and the representations of level ≤ 2 with trivial central character are exactly the irreducible representations of a group of order 216: five of level 0 (the representations of D_8), four of level 1, and four of level 2 – worked out in full in Example 8.7.1 and §8.8. Levels are unbounded on $\mathrm{Irr}(D^\times)$, which is exactly why Chapter 8's Theorems 8.13.5 and 8.13.8, which settle Target B at levels ≤ 2 (§8.13), do not by themselves settle it for every representation, and why §8.15 asks for an inductive mechanism in the level.

Embedding criterion For D a central division algebra of degree m over a local field F, a degree-m field extension E/F embeds as a maximal subfield of D if and only if E *splits* D, i.e. $\mathrm{inv}(D \otimes_F E) = 0$; by the restriction formula above this says exactly $m \cdot \mathrm{inv}(D) = 0$ in $\mathbb{Q}/\mathbb{Z}$, which always holds, so *every* degree-m extension of F embeds in the degree-m division algebra. (For D of degree m and E/F of smaller degree $e < m$ properly dividing m, the same formula need not vanish; this is exactly the $r = 2$ instance worked out in Example 5.5.13 and the $r \mapsto n$ failure in Remark 6.3.2.)

Quaternion algebras, Hilbert symbols The $r = 2$ case of the standard construction is classically written $(a,b)_K := K\langle i,j\rangle$ with $i^2 = a$, $j^2 = b$, $ij = -ji$, for $a,b \in K^\times$; it is a division algebra unless b is a norm from $K(\sqrt{a})$, in which case $(a,b)_K \cong M_2(K)$. (This is the *classical, quadratic* Hilbert symbol – valued in $\{\pm 1\} = \mu_2$ – and is unrelated to the book's own n-th Hilbert symbol $(a,b)_F \in \mu_n$ of §3.5: the two never coincide, since n is odd throughout.) The running example's global algebra $D = (-1, 4-\zeta_3)_K$ (§3.2) and its wild-place localization $D_{\mathfrak{p}_3}$ are both of this form, as is the tame-place localization $D_{\mathfrak{p}_7} \cong (-1,7)_{\mathbb{Q}_7}$ used in Example 5.5.13.

Cyclic algebras More generally, for L/F cyclic of degree m with $\mathrm{Gal}(L/F) = \langle\sigma\rangle$ and $a \in F^\times$, the *cyclic algebra* $(L/F,\sigma,a) := L\langle\Pi\rangle$ with $\Pi^m = a$ and $\Pi x \Pi^{-1} = \sigma(x)$ is central simple of degree m over F; the standard construction above is the local case $L = F_m$ unramified, $a = \varpi$. Every central division algebra over a local field is Brauer-equivalent to a cyclic algebra, and (Albert) every central division algebra of *prime* degree over any field of characteristic 0 containing a cyclic degree-r subfield of K is itself a cyclic algebra – which is one reason, alongside the one below, that cyclic maximal subfields $L \subset D$ (§3.2) are available by construction rather than as an extra hypothesis. The running example's quaternion presentation $D = (-1, 4-\zeta_3)_K$ (Example 1.3.1) is exactly this cyclic algebra $(L/K, \sigma, 4-\zeta_3)$ for $L = K(\sqrt{-1}) = K(i) = \mathbb{Q}(\zeta_{12})$ – the cyclic maximal subfield of §3.2 – with σ complex conjugation on L/K and $\Pi = j$.

Why prime degree The *index* of a central simple algebra is the degree of its division-algebra representative in $\mathrm{Br}(F)$; it always divides the degree m of the algebra one started with. When $m = r$ is prime, the only divisors of r are 1 and r, so a central simple algebra built by the cyclic-algebra construction at prime degree r is *automatically* either fully split ($M_r(F)$, index 1) or a genuine division algebra (index r) – there is no intermediate case to rule out. This dichotomy is the structural reason the series takes r prime throughout: it is what makes "ramified at v" (D_v a division algebra) and "split at v" ($D_v \cong M_r(K_v)$) the only two possibilities at every place, with no third case for Chapters 4–7 to treat separately. The running example

exhibits exactly this dichotomy and nothing else: at the two places of $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, D_v is the genuine quaternion division algebra (index 2); at every other place, including $\mathfrak{p}_{13}$ of Example 4.2.8, $D_v \cong M_2(K_v)$ is fully split (index 1) – there is no place where D_v is only "partly" nonsplit.

Global existence and the reciprocity obstruction Over a global field K, the Albert–Brauer–Hasse–Noether theorem identifies $\mathrm{Br}(K)$ with the classes $(\mathrm{inv}_v(D))_v \in \bigoplus_v \mathbb{Q}/\mathbb{Z}$ summing to 0 (the reciprocity law), subject only to: finitely many local invariants nonzero, and each archimedean one of order dividing 2. Since K is totally imaginary (§3.2), the archimedean constraint is vacuous, so a central division algebra of prime degree r over K exists with *any* prescribed finite ramification set S of size ≥ 2, the local invariants at S being any nonzero elements of $\frac{1}{r}\mathbb{Z}/\mathbb{Z}$ summing to 0 – this is exactly why $|S|$ is even when $r = 2$ (every nonzero element of $\frac{1}{2}\mathbb{Z}/\mathbb{Z}$ is the same element, so an even number of them is forced to sum to 0), and why no such constraint appears for $r \geq 3$. The running example's own $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, $r = 2$ (§3.2) is the smallest possible instance of this existence theorem, and Remark 6.3.2 constructs, for contrast, a local degree-3 division algebra over the running example's own wild completion $K_{\mathfrak{p}_3} = \mathbb{Q}_3(\zeta_3)$.

3.4 Local fields and absolute values

This section fixes the notation for a nonarchimedean local field and its absolute value. It also fixes the words *tame* and *wild* for places, because the two kinds of place behave differently in every later chapter.

F, $\mathscr{O}_F$, ϖ, q_F, p F is a nonarchimedean local field of characteristic zero containing μ_n (usually $F = K_v$), with ring of integers $\mathscr{O}_F$, uniformizer ϖ, residue field of order q_F, and residual characteristic p.

$|\cdot|_F$, $|\cdot|_p$ $|\cdot|_F$ is the normalized absolute value, $|\varpi|_F = q_F^{-1}$; on $\mathbb{Q}_p$ it is written $|\cdot|_p$, with $|p|_p = p^{-1}$. In particular $|n|_F = 1$ if $p \nmid n$, and $|F^\times/F^{\times n}| = n^2|n|_F^{-1}$ (proof of Proposition 4.3.5).

Places dividing n; tame and wild places A place v (or the field $F = K_v$) *divides* n if $p \mid n$. The places $v \in S$ with $p \nmid n$ are called *tame*, and those with $p \mid n$ *wild*. In the running example $\mathfrak{p}_7$ is tame and $\mathfrak{p}_3$ is wild ($|3|_{\mathfrak{p}_3} = 3^{-2}$).

3.5 Hilbert symbols

This section records the n-th Hilbert symbol and the two properties of it that the book uses. The symbol enters every cocycle and every commutator formula for the covers, so its normalization has to be fixed before the covers are defined.

$(a,b)_F$ The n-th Hilbert symbol of F, with values in μ_n; Chapter 7 and the companion chapter also write $(a,b)_{n,F}$ or $(a,b)_n$. It is bimultiplicative, nondegenerate and antisymmetric, $(a,b)_F = (b,a)_F^{-1}$. At complex places it is trivial. The classical $n = 2$ analog of this symbol builds

quaternion division algebras; see the Quaternion algebras entry of §3.3 for that construction and why it is a different object from the one here.

Norm compatibility For a finite extension F'/F, $a \in F^\times$ and $b \in F'^\times$, $(a,b)_{F'} = (a, N_{F'/F} b)_F$; in particular $(a,b)_{F'} = (a,b)_F^{[F':F]}$ for $a,b \in F^\times$ [Serre, *Local Fields*, Ch. XIV].

$(a,b)_{\mathbb{A}}$ For a number field $M \supseteq K$, $(a,b)_{\mathbb{A}_M} = \prod_w (a_w, b_w)_{M_w}$. Hilbert reciprocity: $(a,b)_{\mathbb{A}_M} = 1$ for $a,b \in M^\times$.

3.6 Representation-theoretic background

This section collects, in one place, the representation-theoretic facts used throughout the series but not specific to any one chapter: the structure of smooth representations of the groups at hand, the behavior of characters on compact-modulo-center groups, and the germ expansion of orbital integrals near singular elements. It is meant as a single point of reference for §5.3 (Chapter 5), which transports genuine orbital integrals near the center to a germ expansion for the linear group, and for §6.5 (Chapter 6), which turns a unitary transform of class functions into a question about honest (integral) correspondence of irreducible characters; both sections assume the background collected here.

Smooth, admissible representation A representation of a p-adic group G (or of a cover $\overline{G}$) on a complex vector space is *smooth* if every vector is fixed by some open subgroup, and *admissible* if in addition the space of vectors fixed by any given open compact subgroup is finite-dimensional. Every representation occurring in this book, genuine or not, is smooth and admissible; for a genuine representation, "fixed by U" means fixed by $U \cap \ker(\mu_n\text{-action})$, since μ_n itself never fixes a nonzero genuine vector (§10.2).

Compact-modulo-center groups and finite dimensionality If G/Z is compact – as for $D_v^\times/F^\times$ (since r is prime and D_v a division algebra), or for a covered torus modulo its center – every smooth irreducible representation with a fixed central character is automatically finite-dimensional: its matrix coefficients are then literally the matrix coefficients of a unitary representation of a compact group (twisted by the central character), which decomposes discretely with finite multiplicities by the Peter–Weyl theorem. For covered tori this is the Stone–von Neumann-type finiteness of [KP84, §0.3], used throughout §4.3 (Proposition 4.3.5); for $H = D_v^\times$ and its cover $\widetilde{H}$ (§10.5) it is the starting point of §6.5: "since H/Z is compact, every smooth irreducible representation with a fixed central character is finite dimensional."

Character of a representation For a finite-dimensional smooth representation ρ, its character $\Theta_\rho(g) := \operatorname{tr}\rho(g)$ is a class function (constant on conjugacy classes), locally constant on G; a genuine character transforms under the center by the central character, exactly as a genuine test function does (§10.2). Two finite-dimensional irreducible representations with the same central character are isomorphic if and only if they have the same character, by the usual linear independence of characters.

Schur orthogonality, compact-modulo-center case For G/Z compact and a fixed unitary central character, the characters $\{\Theta_\rho\}$ of the irreducible representations with that central character form an orthonormal basis of the space of class functions on G transforming by that central character, for the invariant inner product normalized by $\mathrm{vol}(G/Z) = 1$. This is the fact underlying Corollary 6.5.1: it is what makes the matrix of a unitary transform of class functions, expanded in the two bases of irreducible characters, a genuinely unitary matrix in the ℓ^2 sense, rather than merely an operator preserving some other norm.

Unitary versus integral A unitary matrix in an orthonormal basis need not have integer, or even rational, entries; unitarity only forces each row and column to have squared norm 1. An *integral* unitary matrix is forced to be a signed permutation matrix (Corollary 6.5.2), which is the precise sense in which "the transform matches characters one-to-one." Distinguishing these two notions is the point of §6.5 and of Remark 6.5.6: unitarity of the normalized transform L_D of §10.5 is proved uniformly, but integrality, genuine irreducible characters going to a single genuine irreducible character rather than to a signed sum of several, is a separate arithmetic question, open at wild places.

Orbital integral For $\gamma \in G$ regular semisimple with centralizer T_γ and f a suitable function on G, the orbital integral $\Phi(\gamma, f) := \int_{T_\gamma \backslash G} f(x^{-1}\gamma x)\, d\dot{x}$ is the basic local invariant distribution of harmonic analysis on p-adic groups: by the Weyl integration formula it reconstructs the invariant integral of f from its values on each conjugacy class, and, normalized by a discriminant factor Δ, it is the local factor of the geometric side of a trace formula. The genuine, normalized version $F = \Delta\Phi$ used throughout the series is defined in §10.3 below.

Unipotent classes of $\mathrm{GL}(r, F)$ The unipotent conjugacy classes of $\mathrm{GL}(r, F)$ are classified, exactly as over $\mathbb{C}$, by the partitions of r (via Jordan type); this is the index set $\mathscr{O}$ of Corollary 5.3.8. On the division-algebra side, since $D_v^\times / F^\times$ is compact, the only unipotent class is the trivial one $\{1\}$, which is why the germ expansion on that side (Proposition 4.4.8) has a single term.

Shalika germ, germ expansion An orbital integral $\Phi(\gamma, f)$ need not extend continuously as γ approaches a singular (non-regular) semisimple element δ; however, Shalika's germ expansion theorem [Sha72] (see also [HC70]) says that near δ it agrees, up to bounded error, with a *finite* linear combination

$$\Phi(\gamma, f) = \sum_{\mathscr{O}} \Gamma_{\mathscr{O}}(\gamma)\, \Phi(u_{\mathscr{O}}, f)$$

of the orbital integrals of f at representatives $u_{\mathscr{O}}$ of the unipotent classes of the centralizer of δ, with universal coefficients $\Gamma_{\mathscr{O}}$ (the *germs*) depending only on γ and δ, not on f. This is the classical fact that §5.3 transports to genuine functions on a cover: Proposition 5.3.7 shows that, sufficiently near a central good element, a genuine orbital integral on the cover literally *equals* an ordinary orbital integral on the linear group, not just up to germs, so the Shalika germ expansion applies verbatim and gives Corollary 5.3.8.

Harish-Chandra's local character expansion The representation-theoretic counterpart of the germ expansion: the character Θ_ρ of an admissible representation ρ, near the identity, is a linear combination of the Fourier transforms of nilpotent orbital integrals, with coefficients (rather than the germs) carrying the representation-theoretic information. This book does

not use the local character expansion itself, but it is the reason germ expansions of orbital integrals (§5.3) are the geometric mirror of questions about characters of representations (§6.5): the same local, purely geometric data that describes test functions near a singular element also governs, dually, how characters of representations behave there.

Genuine representations as a parallel representation theory A genuine representation of a cover $\overline{G}$ is, in effect, an ordinary representation of $\overline{G}$ on which the central μ_n acts tautologically; equivalently, after fixing a section (never a homomorphism, in general), it is a projective representation of G with a prescribed cocycle. Every general fact recalled above, admissibility, finite-dimensionality modulo a compact center, characters as class functions, Schur orthogonality, and germ expansions, applies to genuine representations and genuine functions exactly as stated, with "class function" and "conjugacy-invariant" always understood modulo the extra care needed at non-good elements (§10.2): a genuine class function is automatically forced to vanish at regular elements that are not good, which is why every statement above, applied to a genuine representation or function, is really a statement about good elements only.

3.7 Notation specific to Chapter 4

This section lists the symbols and terms introduced in Chapter 4, with a pointer to the place where each is defined. It is a lookup table and introduces nothing new.

ι_D, ι_{GL}; $\overline{G}_D(K_v)$, $\overline{G}_v$, $\overline{Z}_v$
The embeddings of $D_v^\times$ and $\mathrm{GL}(r,K_v)$ in $\mathscr{G}_v = \mathrm{GL}(r,L_v)$, and the inverse images of their images and of $K_v^\times$ in the ambient cover $\overline{\mathscr{G}}_v$ (§4.2.2, Definition 4.2.1).

α_v, ψ_v
For $v \notin S$: an isomorphism $D_v \cong M_r(K_v)$, and the induced isomorphism of covers $\overline{G}_D(K_v) \to \overline{G}_v$ given by conjugation in $\overline{\mathscr{G}}_v$ (Proposition 4.2.7). $\psi^V = \otimes_{v\notin V}\psi_v$ (§4.7.2).

E, j, N; $\overline{E}_D$, $\overline{E}_{\mathrm{GL}}$
At $v \in S$: a maximal subfield $E \subset D_v$, an embedding $j : E \hookrightarrow M_r(F)$, the norm $N = N_{E/F}$, and the covered tori $p^{-1}(\iota_D(E^\times))$ and $p^{-1}(\iota_{\mathrm{GL}} j(E^\times))$ (§4.3).

$\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$
Isomorphisms of covered tori over $\iota_{\mathrm{GL}} j \iota_D^{-1}$ that are the identity on μ_n and on $\overline{Z}_v$; a torsor under the characters $E^\times \to \mu_n$ trivial on $F^\times$ (Theorem 4.3.10).

$\mathscr{C}^g(\overline{G}_D(K_v))$, $\mathscr{C}^g_{\mathrm{ell}}(\overline{G}_v)$
The good regular, respectively good regular elliptic, conjugacy classes (§4.3.5).

E_i, j_i, $\jmath_i$; dg_D, dg, dt_i
Representatives of the degree-r extensions of F with fixed embeddings, the corresponding identifications, and the local Haar measures (§4.4.1).

Admissible function; $W_{i,\bullet}$
A locally constant, $\tilde{\theta}_v^{-1}$-equivariant, $N_{G_\bullet}(T_{i,\bullet})$-invariant function on good regular elements of a covered torus with support compact modulo $\overline{Z}_v$; $W_{i,\bullet} = N_{G_\bullet}(T_{i,\bullet})/T_{i,\bullet}$ (§4.4.3).

$\overline{\mathscr{G}}(\mathbb{A}_L)$, $\overline{G}_D(\mathbb{A}_K)$, $\overline{G}(\mathbb{A}_K)$, $\overline{Z}(\mathbb{A}_K)$
The global ambient cover and its subgroups (§4.5.1).

ι, κ_v^*, κ_w, τ_w, s_N
The rational splitting, the compact splittings, the Flicker–Kazhdan function κ_w, the normalized cocycle (Lemma 4.5.1), and the unipotent splittings (Proposition 4.5.6(e)).

S_D
A finite set of places containing S, the archimedean places, the places above n, and the places where L or $\tilde{\theta}$ ramifies or the integrality conditions of Proposition 4.2.7(c) fail (§4.5.3).

W_γ, $c(\gamma)$, $\mathscr{O}_H$
For a regular elliptic rational γ: the automorphisms of $K(\gamma)/K$ fixing $K^\times\gamma$, the class-counting constant, and the set of good regular elliptic rational classes (§4.6).

w_0, u; admissible pair
A finite place $w_0 \notin S$ carrying the supercusp form, a place $u \in S$ with elliptic support, and the pairs of test functions of Definition 4.6.5.

(A1), (A2); (R1)–(R5)
The analytic inputs of the proof of Proposition 4.6.2, and the properties of $\overline{G}(\mathbb{A}_K)$ verified in §4.5.4.

σ, $\tilde{f}_\sigma$, $d(\sigma)$
A genuine supercuspidal representation of $\overline{G}_{w_0}$, its normalized matrix coefficient, and its formal degree (Lemma 4.7.1).

V, ρ
$V = S \cup \{w_0\}$, and ρ an admissible genuine irreducible representation of $\overline{G}(\mathbb{A}^V)$ (§4.7.2).

Corner Hecke algebra; e_v, e_{t}, A_v, A_∞, A, $e\tau$
For an idempotent (or idempotent multiplier) e, the algebra $e\mathscr{H}e$; the central-character idempotents, the archimedean K_∞-type projector, the corner algebras and the module $e\tau$ of Lemma 4.7.5. Irreducible τ with $e\tau \neq 0$ correspond to simple $e\mathscr{H}e$-modules.

Near-equivalence packets $P_D(\rho)$, $P(\rho)$
The automorphic (resp. cuspidal) representations with components ρ away from V and σ at w_0, counted with multiplicity; they need not be finite (Proposition 4.7.6). Ω_D, Ω are the finite sets of representations that can contribute at a fixed level (proof of Proposition 4.7.6).

$\mathscr{T}_v$, a, Λ_a
The genuine irreducibles of $\overline{G}_D(K_v)$ with fixed vectors at a given level, a coefficient function on $\prod_{v\in S}\mathscr{T}_v$, and the corresponding trace functional (Lemma 4.7.8).

(Hyp_D)**, (Hyp); elliptic detectability**
The condition that the character at S of the D-side (resp. GL-side) packet is not identically zero on matching regular-supported test functions (Definition 4.7.9, Theorem 4.7.13).

(D1)–(D4)
The deferred steps of §4.8.

Ψ**,** $\tilde{\Psi}$**,** $\overline{G}_T$**,** T^{reg}**,** $\overline{T}^{g,\mathrm{reg}}$
The conjugation maps, the open set of good elements conjugate into T^{reg}, the regular elements of an elliptic torus T, and the good regular elements of $\overline{T}$ (§4.4.6).

3.8 Notation specific to Chapter 5

This section lists the symbols and terms introduced in Chapter 5, with a pointer to the place where each is defined. It is a lookup table and introduces nothing new.

G**,** $\overline{G}$**,** Z**,** $\overline{Z}$**;** P_g**,** A_g**; eigenvalues**
At a finite place v with $F = K_v$: G is $\mathrm{GL}(r,F)$ or $D_v^\times$, $\overline{G} = p^{-1}(G)$, $Z = F^\times$, $\overline{Z} = p^{-1}(Z)$; P_g is the (reduced) characteristic polynomial of g, $A_g = F[g]$, and the eigenvalues of g are the roots of P_g in $\overline{F}$ (§5.2.1).

Φ^z
The orbital integral over $Z\backslash G$, $\Phi^z(\gamma,f) = \mathrm{vol}(T_\gamma/Z)\,\Phi(\gamma,f)$ for regular elliptic γ [FK86, §6] (§5.2.1).

$d(\mathrm{St})$**, compatible measures,** λ_v
The formal degree of the Steinberg representation of $\mathrm{GL}(r,F)$; the normalization $\mathrm{vol}(D_v^\times/F^\times) = d(\mathrm{St})^{-1}$ at $v \in S$ (Definition 5.2.1); and $\lambda_v = d(\mathrm{St})\,\mathrm{vol}(D_v^\times/F^\times)$, which equals 1 for these measures (Remark 5.2.3).

(I1)–(I11)
The results of Chapter 4 used here, restated in the form used (§5.2.3).

Ω_M**,** V_M**,** $V_{D,M}$
The monic degree-r polynomials congruent to $(X-1)^r$ modulo $\mathfrak{p}^M$, and the conjugation-invariant open and closed set of elements of G (resp. $D_v^\times$) whose characteristic polynomial lies in Ω_M (§5.3.1).

δ_n**,** B**,** M_0**,** M_1
The constants of Lemmas 5.3.1 and 5.3.6: $\delta_n = 1$ if $p \nmid n$ and $q^{-e/(p-1)}$ if $p \mid n$, $B = |n|_F^{-1} q^{e/(p-1)}$, M_0 the level from which the binomial series converges, and $M_1 \geq M_0$ the level from which $\tilde{\theta}\circ\sigma$ is trivial on the relevant scalars.

$g^{1/n}$
The canonical n-th root $\sum_k \binom{1/n}{k}(g-1)^k \in A_g$ for $g \in V_M$, $M \geq M_0$ (Lemma 5.3.1).

σ**,** σ_D**,** σ_{GL}
The section $\sigma(g) = \tilde{h}^n$ with $h = g^{1/n}$ (§5.3.2, Proposition 5.3.4).

$\mathscr{T}$, $\mathscr{T}_D$; $C_c^\infty(\overline{G},\tilde{\theta})_{\mathrm{gen},M}$, $C_c^\infty(G/Z)_M$
The transport $(\mathscr{T}\tilde{f})(p(z)g) = \tilde{\theta}(z)\tilde{f}(z\sigma(g))$, a bijection between genuine functions supported on $p^{-1}(ZV_M)$ and functions on G/Z supported on ZV_M (Proposition 5.3.7).

$\Gamma_{\mathscr{O}}$, $u_{\mathscr{O}}$
Shalika germs of $\mathrm{GL}(r,F)$ for the fixed measures, indexed by the unipotent classes $\mathscr{O}$ (partitions of r), with representatives $u_{\mathscr{O}}$; $\Gamma_{\{1\}}$ is the trivial germ (Corollaries 5.3.8 and 5.5.11).

(L1), (L2); f_{St}
The two linear inputs: the Steinberg pseudo-coefficient f_{St} and its orbital integrals, and linear matching of functions with vanishing non-elliptic orbital integrals (§5.5.1).

e_D, $\tilde{f}^{\mathrm{St}}$
The genuine function on $\overline{G}_D(K_v)$ equal to $\tilde{\theta}_v^{-1}$ on $\overline{Z}_v\sigma_D(V_{D,M})$, and the transport of $d(\mathrm{St})^{-1}f_{\mathrm{St}}$ cut off to ZV_M; they match (Proposition 5.5.5).

$\tilde{f}_{\mathrm{reg}}$
The part of the matching function of Theorem E supported on the good regular elliptic set (Theorem 5.5.8).

(Ell), (Non); (G1), (G2)
The orbital-integral form and the unipotent (germ) form of the condition that a function match $\tilde{f}_D$ near $\overline{Z}_v$ (§5.5.4, Theorem 5.5.14).

***u*-admissible pair**
A pair of global test functions with elliptic support at one place $u \in S$ only and arbitrary matching functions at the other places of S (Definition 5.6.1).

(Hyp'_D), (Hyp')
The weakened non-cancellation hypotheses (Definition 5.6.5).

3.9 Notation and terminology specific to Chapter 6

This section lists the symbols and terms introduced in Chapter 6, with a pointer to the place where each is defined. It is a lookup table and introduces nothing new.

Adams operation ψ^n
The λ-ring operation on the representation ring satisfying $\chi_{\psi^n V}(g) = \chi_V(g^n)$; it can produce a virtual character rather than an actual representation.

Equivariant *K*-theory
At a point, $K_Q^0(\mathrm{pt}) = R(Q)$, the representation ring of a finite group Q. Exterior powers give its λ-operations.

Virtual character
An integral linear combination of irreducible characters.

Hall π-subgroup
A subgroup whose order involves only primes in π and whose index is prime to every prime in π; in the tame proof it separates the central n-part of a finite quotient.

Normalized orbital integral
Here $J(t,f)=\Delta(t)O(t,f)$, where Δ is the square root of the Weyl discriminant and quotient Haar measures are specified.

Transforms $L_D, \mathscr{O}_D, \mathscr{B}$
Respectively the unitary invariant class-function transform, the forward test-function transfer, and its inverse modulo orbital-integral kernels; see Definitions 6.4.1 and Theorem 6.9.1.

Stable/rational conjugacy
Conjugacy over an algebraic closure (with descent) versus conjugacy by the rational points $G(K)$; one stable class may contain several rational classes for a general reductive group.

Inner form A twist of a reductive group by inner automorphisms; $\mathrm{GL}_1(D)$ is an inner form of GL_r.

Induced torus
In Chapter 6 $T=\mathrm{Res}_{E/K}\mathbb{G}_m$, with $T(K)=E^\times$ and $H^1(K,T)=1$ by Hilbert 90.

Kottwitz torus obstruction
A global cohomological obstruction to transferring an admissible torus to an inner form despite local embeddings, discussed in [Kot84, §9.5].

Relative Tamagawa number
The ratio $\tau_1(G)=\tau(G)/\tau(G_{\mathrm{sc}})$; it is not a local probability Haar measure.

Brauer class The class $[D]\in \mathrm{Br}(K)$ of a central simple algebra; E embeds maximally in D exactly when $D\otimes_K E$ splits.

Scalar power quotient
$\mathscr{Q}(E/K,n)=E^\times/(K^\times E^{\times n})$; it measures global scalar-modulo-nth-power classes.

Kummer and Sha^1
The exact sequence $1\to T[n]\to T\xrightarrow{n} T\to 1$ relates powers to Galois cohomology. $\mathrm{Sha}^1(K,T[n])$ is the kernel of localization of $H^1(K,T[n])$ at every place.

$H, Z, \widetilde{H}, \widetilde{Z}, \widetilde{T}$ $H=D^\times$ for a local division algebra D of prime degree r, $Z=F^\times$, and their inverse images in the cover of Chapter 4; $\widetilde{T}=p^{-1}(T)$ for a maximal torus $T=E^\times$ (§6.2).

H^1, U_D^1 The reduced-norm-one group $\ker(\mathrm{Nrd})$ and the principal units $1+\mathfrak{P}_D$ (§6.7).

Tame, wild In Chapter 6 *tame* means $p\nmid n$; it imposes no condition on the ramification of maximal subfields (§6.7).

$\mathscr{H}_\theta(H)$, $\mathscr{H}_{\widetilde{\theta}}(\widetilde{H})$
: The Hilbert spaces of θ-equivariant, respectively genuine $\tilde{\theta}$-equivariant, invariant L^2 functions, with the quotient probability measures (§6.4).

b_D, c_D; **character matrix**
: $b_D = |n|_F^{-(r-1)/2}$ (Theorem 6.4.4) and $c_D = |n|_F^{-(r^2-1)/2}$ (Corollary 6.5.2); the matrix of L_D in the bases of irreducible characters (Corollary 6.5.1).

$\mathscr{C}_{\eta^{-1}}(H)$, $L_D^{(\widetilde{\eta})}$
: Locally constant functions transforming by η^{-1} under Z, and the transform of Definition 6.4.1 for the genuine central character $\widetilde{\eta}$ (§6.9).

Ψ_D, **(6.38)**
: The conjectural map from genuine irreducible representations to $\mathrm{Irr}(D_v^\times)$ and the target spectral identity of §6.12.

3.10 Terminology specific to Chapter 7

This section lists the symbols and terms introduced in Chapter 7, with a pointer to the place where each is defined. It is a lookup table and introduces nothing new.

Good element
: A regular lift commuting with the full preimage of its ordinary centralizer; genuine invariant characters vanish on the nongood regular locus.

Scalar-central cover
: A cover whose entire inverse image of $F^\times I_r$ is central; here forced by $n \mid r-1+2rc_v$.

Power-map kernel
: $\ker(P_n)$ on $T(F)/F^\times$, distinct from the image index. The kernel can be nontrivial on product tori even when the corresponding field-torus map is injective.

Star map
: $h^* = \widetilde{h}^n$, independent of lift; multiplication on tori uses n odd.

Weyl discriminant
: The factor $\Delta = |\det(1-\mathrm{Ad}(t)|\mathfrak{g}/\mathfrak{t})|^{1/2}$; it is not the unnormalized characteristic-polynomial discriminant.

Toral Fourier test
: Expand the discriminant ratio R_E in characters of the compact abelian group $E^\times/F^\times$. Integral positive coefficients ensure torus-restriction positivity, not full-group integrality.

Nonabelian gluing
: The additional requirement that compatible genuine covered-torus characters arise from an integral character of the full group $\widetilde{D^\times}$.

Twisted finite quotient
: A finite quotient of H/Z with the central character retained as a multiplier, used to compute projective/genuine irreducible characters.

Packet identity An identity of traces over collections of automorphic representations; it does not assert a bijection between individual local irreducibles.

Formal degree The Plancherel normalization of a square-integrable representation relative to a specified quotient Haar measure.

Root-sum relation
Zou's equation [Zo23, (3.2)], which sums over all nth roots satisfying the stated regularity condition rather than choosing one arbitrarily.

Chapter 4

Local matching and a global transfer under elliptic-support conditions

Overview. Let D be a central division algebra of prime degree r over a totally imaginary number field K containing μ_n, with n odd and $(n,r)=1$. We compare the n-fold cover of $D^\times_{\mathbb{A}}$ constructed in [JC26a] with the Kazhdan–Patterson cover of $\mathrm{GL}(r,\mathbb{A}_K)$, realizing both as subgroups of one cover of $\mathrm{GL}(r,\mathbb{A}_L)$, where $L\subset D$ is a cyclic maximal subfield. *Theorem A:* at each ramified place of D, the covered elliptic tori on the two sides have the same commutator pairing and the same good elements, and they are identified on good elements by a canonical isomorphism. *Theorem B:* genuine test functions supported on the regular set have matching functions on the other side, in both directions. *Theorem C:* using a variant of the simple trace formula of Flicker–Kazhdan, we prove a locally finite identity of packets at the ramified places. From it we obtain a cuspidal transfer, with the same components away from the ramified places, for every genuine automorphic representation of the cover of $D^\times_{\mathbb{A}}$ that has a supercuspidal component at some split place, under a non-cancellation hypothesis that holds whenever the D-side packet is finite. Chapter 5 proves the matching of arbitrary functions and replaces this hypothesis by a weaker one; Chapter 6 studies the local character relations from the side of the division algebra.

4.1 Introduction

This section states the main results, fixes the standing assumptions and conventions, explains how results are labeled, and describes what the later parts of the series do. It depends on nothing else in the chapter. The global data K,n,r,L,D,S,r',c,c_v are fixed in Assumption 4.1.1; all notation, including the status labels and the cocycle normalization, is collected in §3, and the results are summarized, with labels and pointers, in §9.1 at the end of the chapter. Throughout, [JC26a] refers to the corrected version (`metaplectic_kazhdan_revised12.tex`) of [Jo88], and a reference such as [JC26a, Lemma 6.1] is to the numbering there.

4.1.1 Setting and aim

We fix the global data, which are those of [JC26a, §1], and explain where this chapter fits.

Standing assumptions 4.1.1. K is a totally imaginary number field containing μ_n, with $n \geq 1$ odd; L/K is cyclic of prime degree r with $(n,r) = 1$; D is a central division algebra over K of dimension r^2 containing L; and S is the set of places where D ramifies. Fix $r' \in \mathbb{Z}$ with $rr' \equiv 1 \pmod{n}$ and $c \in \mathbb{Z}$ with $n \mid r-1+2c$, and put $c_v := r'c$, so that $n \mid r-1+2rc_v$.

The paper [JC26a] lifts genuine automorphic representations of an n-fold cover of the torus $L_{\mathbb{A}}^{\times}$ to ε-invariant genuine automorphic representations of an n-fold cover $\overline{G}_D(\mathbb{A}_K)$ of $D_{\mathbb{A}}^{\times}$. Its Remark 10.3 outlines the missing step needed to reach the Kazhdan–Patterson cover $\overline{G}(\mathbb{A}_K)$ of $\mathrm{GL}(r,\mathbb{A}_K)$: a metaplectic Jacquet–Langlands transfer from $\overline{G}_D$ to $\overline{G}$. Chapter 4 establishes the local matching of test functions that the transfer requires, and a global transfer for representations with suitable local components. The later parts of the series (§4.1.3) remove some of those conditions and study the local character relations.

Since r is prime, every $v \in S$ is finite (the archimedean places are complex), D_v is a division algebra, and $L_v := L \otimes_K K_v$ is a field. The local invariants of D are nonzero elements of $\frac{1}{r}\mathbb{Z}/\mathbb{Z}$ with sum 0, so $|S| \geq 2$, and $|S|$ is even if $r = 2$.

4.1.2 Main results

We state the three main results informally; precise statements are referenced.

Covers. Both $\overline{G}_D(\mathbb{A}_K)$ and $\overline{G}(\mathbb{A}_K)$ are realized as subgroups of one cover of $\mathrm{GL}(r,\mathbb{A}_L)$ (§4.2, §4.5), so that their centers, rational splittings and central characters are literally shared.

Theorem A (covered tori; Theorem 4.3.10). Let $v \in S$, let $E \subset D_v$ be a maximal subfield, and embed E in $M_r(K_v)$. The covered tori $\overline{E}_D \subset \overline{G}_D(K_v)$ and $\overline{E}_{\mathrm{GL}} \subset \overline{G}(K_v)$ have the same commutator pairing $(x,y) \mapsto ((Nx)^{1+2c_v}x^{-1}, y)_E$. Their good elements are the inverse image of $K_v^{\times}E^{\times n}$, of index $|n|_{K_v}^{-(r-1)}$. There is an isomorphism $\overline{E}_D \cong \overline{E}_{\mathrm{GL}}$ induced by conjugation in the ambient cover, and its restriction to the good elements is canonical: it is the same for every isomorphism that is the identity on the common center.

Theorem B (local matching; Theorem 4.4.7). With matching defined by $F(\tilde{\gamma}, \tilde{f}_D) = (-1)^{r-1}F(\jmath(\tilde{\gamma}), \tilde{f})$ at good regular elements, and vanishing of the GL-side orbital integrals at non-elliptic elements, every genuine $\tilde{f}_D$ on $\overline{G}_D(K_v)$ vanishing near the center has a matching $\tilde{f}$ supported on the good regular elliptic set of $\overline{G}(K_v)$. Conversely, every regular-supported $\tilde{f}$ with vanishing non-elliptic orbital integrals has a matching $\tilde{f}_D$. For general $\tilde{f}_D$ the method necessarily fails (Proposition 4.4.8), and germ expansions are needed.

Theorem C (global transfer; Proposition 4.7.6, Theorems 4.7.12 and 4.7.13). Let $\overline{\pi}_D$ be a genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ whose component at some finite place $w_0 \notin S$ is supercuspidal. Let $P_D \ni \overline{\pi}_D$ and P be the packets of D-side and cuspidal GL-side representations with the same components outside S. These packets satisfy an identity of traces at the places of S against all matching regular-supported test functions; the sums are locally finite. If the D-side packet is *not cancelled* on regular-supported test functions (hypothesis (Hyp_D), which holds whenever P_D is finite), then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with $\overline{\pi}_v \cong \overline{\pi}_{D,v}$ for all $v \notin S$. A converse holds under the analogous hypothesis (Hyp) on the GL-side packet.

Lemma 4.7.8 shows that (Hyp_D) can fail only if, at every level, the coefficient function of the packet at S lies in the sum of the subspaces having a Plancherel factor at one place of S.

Theorem C rests on the simple trace formula of Flicker and Kazhdan [FK86, §18]. We need it in a slightly different form, with the supercusp form at a place outside S; this variant is proved in Proposition 4.6.2. Since the multiplicity-one and local character theory needed to isolate single representations is not available here, Theorem C is a statement about packets.

4.1.3 Goals of the later parts

The goals below are needed to complete the metaplectic Jacquet–Langlands correspondence by removing the restrictions of Theorems B and C. Each item lists the inputs it needs, followed by its current status in the later parts of the series.

(i) *Matching of arbitrary functions.* Extend Theorem B to all $\tilde{f}_D$, and to all $\tilde{f}$ with vanishing non-elliptic orbital integrals. This needs the germ expansions of [FK86, §8] at central good elements, their restriction to the elliptic set, and a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center, and an analog of [DKV84, A.3] (§4.4.5, item (D1)). *Status:* proved in Chapter 5 (Theorem E).

(ii) *Local character relations.* Prove a metaplectic local Jacquet–Langlands correspondence: a bijection $\overline{\pi}_v \leftrightarrow \overline{\pi}_{D,v}$ between genuine square-integrable representations of $\overline{G}(K_v)$ and genuine irreducible representations of $\overline{G}_D(K_v)$, with $\Theta_{\overline{\pi}_{D,v}} = (-1)^{r-1}\Theta_{\overline{\pi}_v} \circ \jmath$ on good regular elements (item (D2)). This needs (i), the elliptic orthogonality relations of [FK86, §§24–25], and multiplicity one and rigidity for $\overline{G}(\mathbb{A}_K)$ ([FK86, §§27–28], as corrected in [Zo23]). *Status:* Chapter 6 constructs a unitary transform L_D between ordinary and genuine invariant class functions on $D_v^\times$ and its cover, and proves that it bijects irreducible characters at the places $v \in S$ of residue characteristic prime to n. When the square-integrable Flicker–Kazhdan correspondence is available in Zou's corrected form [Zo23, Proposition 3.6, Definition 3.7], Corollary 6.7.8, proves (D2) at such tame places; the published sufficient range $(n, N_r) = 1$ and the precise compatibility checks are recorded in Remark 4.8.1. Outside that range an appropriate metaplectic correspondence theorem is an additional input. At places dividing n the irreducible-character integrality needed for this composition remains open.

(iii) *An unconditional global correspondence.* Remove the auxiliary supercuspidal place in Theorem C, either via W.-W. Li's trace formula for covers [Li14] or via (i), and refine the packet identity of Theorem C to a correspondence of individual representations using (ii). *Status:* open; see §5.7(iii).

(iv) *Removing* (Hyp_D) *and (Hyp).* Prove finiteness of near-equivalence packets for $\overline{G}_D(\mathbb{A}_K)$ (for example, by bounding the level at S in terms of the components outside S), and linear independence of elliptic characters in GL-side packets, so that Theorem C becomes unconditional (Remark 4.7.14, item (D3)). *Status:* Chapter 5 replaces (Hyp_D) by the weaker (Hyp'_D) and reduces its removal to a bound on levels at S (Corollary 5.6.13); open in general.

(v) *Completing [JC26a, Remark 10.3].* Compose the lifting of [JC26a] with the transfer, to obtain genuine automorphic representations of the Kazhdan–Patterson cover of $\mathrm{GL}(r, \mathbb{A}_K)$

attached to genuine characters of the covered torus (item (D4)). Then compare the result with the Flicker–Kazhdan correspondence [FK86] to $\mathrm{GL}(r,\mathbb{A}_K)$ and with Kazhdan's endoscopic lifting. *Status:* open.

4.1.4 Conventions

The conventions below are repeated, with the rest of the series' notation, in §3 (Hilbert symbols in §3.5, covers and the relabeling of μ_n in §10.1, genuine objects and good elements in §10.2).

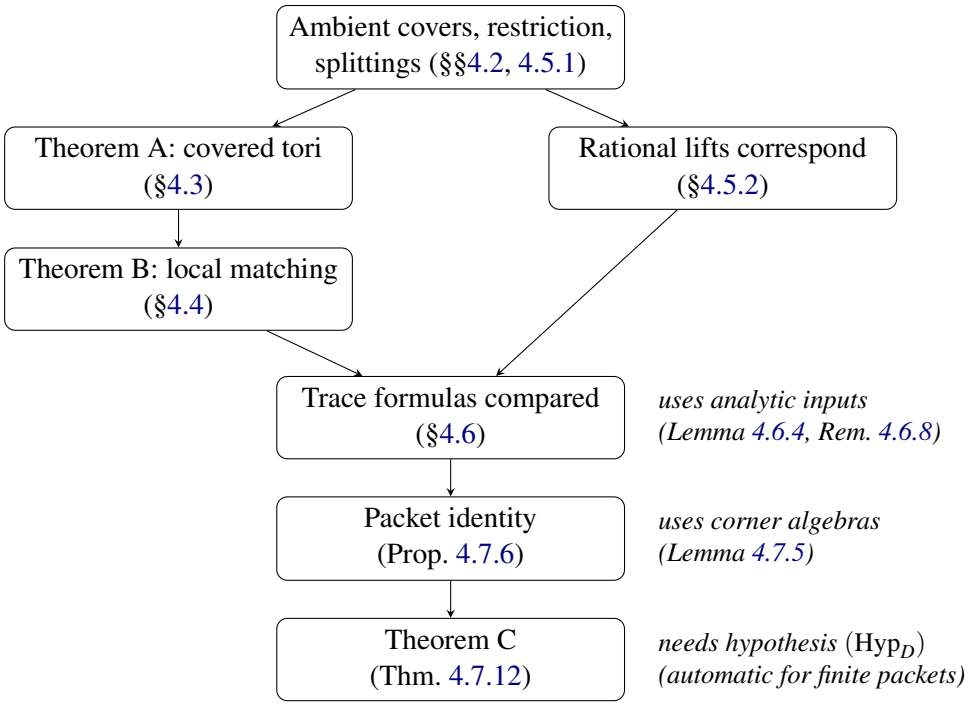
All Hilbert symbols are n-th Hilbert symbols. For a finite extension F'/F of local fields containing μ_n, $a \in F^\times$ and $b \in F'^\times$, we use the norm compatibility $(a,b)_{F'} = (a, N_{F'/F} b)_F$ [Ser79, Ch. XIV]; in particular $(a,b)_{F'} = (a,b)_F^{[F':F]}$ for $a,b \in F^\times$. For lifts $\tilde{x},\tilde{y}$ of commuting elements, $[x,y] = \tilde{x}\tilde{y}\tilde{x}^{-1}\tilde{y}^{-1}$. For a cover $p : \overline{H} \to H$ and $H' \subset H$ we write $\overline{H'} = p^{-1}(H')$. Genuine functions and representations, good elements, and $C_c^\infty(\overline{H},\tilde{\theta})_{\mathrm{gen}}$ are as in [JC26a, §§1–2]. As in [JC26a, §6.1], we relabel μ_n once, by $\zeta \mapsto \zeta^{r'}$, at every place and globally, and we say explicitly when a formula is stated before or after relabeling.

4.1.5 How results are labeled

Statements proved in this chapter carry no visible tag; the labels **(E)**, **(O)**, **(C)** mark everything else and are defined in §3.1 (which also defines **($\mathbf{P}^{\mathbf{c}}$)**, used only in the companion chapter and not in this one), with a remark immediately after each tagged statement explaining its status. The table below applies the labels to the results of this chapter.

Statement	Where	Status
Both covers are subgroups of one ambient cover; the GL side is the Kazhdan–Patterson cover with parameter c_v	Prop. 4.2.4	**(P)**
Section and splitting conventions	Lemma 4.5.1	**(P)**
Canonical identification of the covers at $v \notin S$	Prop. 4.2.7	**(P)**
Covered elliptic tori at $v \in S$: pairings, good elements, centers; canonical identification (Theorem A)	§4.3	**(P)**
Matching for regular-supported functions (Theorem B)	Thm. 4.4.7, §4.4.6	**(P)**
Global covers; rational lifts correspond	§4.5	**(P)**
Geometric expansions, including class-counting constants	Props. 4.6.1, 4.6.2	**(P)**
Variant of the simple trace formula of [FK86, §18]	Prop. 4.6.2	**(P)**
Comparison of the trace formulas	Thm. 4.6.7	**(P)**
Packet identity at S (locally finite)	Prop. 4.7.6	**(P)**
Characterization of cancellation	Lemma 4.7.8	**(P)**
Global transfer (Theorem C)	Thm. 4.7.12	**(P)** under (Hyp_D); unconditional for finite packets
Converse transfer	Thm. 4.7.13	**(P)** under (Hyp)
Finiteness of packets; removal of (Hyp_D), (Hyp)	§4.8	**(C)** (weakened in Chapter 5)
Matching of arbitrary functions; local character relations	§4.8	**(C)** (matching: Chapter 5; tame character relations with FK–Zou input: Chapter 6)

The logical dependencies are as follows. Arrows mean “is used in”, and the notes on the right list the additional inputs of each step.



Everything except the last box is unconditional, given the cited **(E)** inputs. The **(E)** inputs used in the global sections are listed in Remark 4.6.8.

4.1.6 A running example

Continuing Example 1.3.1: $K=\mathbb{Q}(\zeta_3)$, $n=3$, $r=2$, $D=(-1,4-\zeta_3)_K$, which is ramified exactly at $S=\{\mathfrak{p}_3,\mathfrak{p}_7\}$, with $\mathfrak{p}_3=(1-\zeta_3)$ and $\mathfrak{p}_7=(3+\zeta_3)$; explicitly, $4-\zeta_3=(1-\zeta_3)(3+\zeta_3)=\mathfrak{p}_3\mathfrak{p}_7$. We take $L=K(i)=\mathbb{Q}(\zeta_{12})$, and $c=1$, $r'=2$, $c_v=2$ in Assumption 4.1.1. At $\mathfrak{p}_3$, where $K_v=\mathbb{Q}_3(\zeta_3)$, the covered elliptic tori are non-abelian: the good elements have index 9, and genuine irreducibles of a covered torus have dimension 3 (Example 4.3.7). At $\mathfrak{p}_7$, where $K_v=\mathbb{Q}_7\supset\mu_3$, they are abelian. The sign in Theorem 4.4.7 is -1. [JC26b] works this case out in detail across Chapters 4–7, including an independent machine check ([JC26b, §1]) that D is ramified exactly at $\mathfrak{p}_3$ and $\mathfrak{p}_7$.

4.1.7 Organization

Section 2.1 reviews the literature for the whole series (Chapters 5–7 refer to it), and Section 3 collects the notation and glossary of the series (shared by all four Parts) together with the notation specific to Chapter 4. Sections 4.2–4.4 contain the local theory; the facts about orbital integrals on elliptic tori that it needs are proved in §4.4.6. Sections 4.5–4.7 contain the global theory, and §4.5.4 supplies its analytic input for cusp forms on the cover. Section 4.8 lists what is left after this chapter, Section 4.9 summarizes what the local theory gives on its own, and Section 9.1 lists the results with their labels.

4.2 Local covers

This section realizes the cover of $D_v^\times$ from [JC26a] and the Kazhdan–Patterson cover of $\mathrm{GL}(r,K_v)$ as subgroups of one cover of the larger group $\mathrm{GL}(r,L\otimes_K K_v)$ (§4.2.1–4.2.2). It then shows that the GL-side subgroup is the Kazhdan–Patterson cover with parameter c_v, as an extension (§4.2.3), and identifies the two covers canonically at the places where D splits (§4.2.4). It depends only on the Kazhdan–Patterson cocycle [KP84, §0.1], [FK86, §2] and the Matsumoto–Moore classification [Mat69], [Moo68], and uses from [JC26a] only the definition of the cover of $D_v^\times$. It introduces the ambient cover $\overline{\mathscr{G}}_v$, the embeddings ι_D and ι_{GL}, the local covers $\overline{G}_D(K_v)$ and $\overline{G}_v$, their common central subgroup $\overline{Z}_v$, and the identifications ψ_v at $v\notin S$.

4.2.1 Why an ambient cover

We explain the idea before the computations.

Equality of commutator pairings does not, by itself, give a preferred isomorphism between two central extensions. Any two isomorphisms differ by a character with values in μ_n, and such characters can be nontrivial on the center. A careless choice would therefore make the comparison of central characters, and hence of genuine functions, ambiguous. Our approach is to realize both covers as *subgroups* of a single cover of $\mathrm{GL}(r,L\otimes_K K_v)$. Conjugation in this larger group then produces isomorphisms between the relevant subgroups. These isomorphisms are in general not unique, but their restrictions to the centers of covered tori are canonical (Theorem 4.3.10). Because both covers sit in one group, the common central subgroup $p^{-1}(K_v^\times)$ is literally the same subgroup on the two sides. This uses no choice of cocycle, so the question of whether two cocycles agree exactly or only up to a coboundary does not arise for it.

4.2.2 The ambient cover

We define the ambient cover at each place and the two subgroups; the notation $\overline{\mathscr{G}}_v$, ι_D, ι_{GL}, $\overline{G}_D(K_v)$, $\overline{G}_v$, $\overline{Z}_v$ introduced here is used throughout.

Fix once and for all an isomorphism $D\otimes_K L\cong M_r(L)$. For a place v of K put $L_v=L\otimes_K K_v=\prod_{w|v}L_w$ and $\mathscr{G}_v:=\mathrm{GL}(r,L_v)=\prod_{w|v}\mathrm{GL}(r,L_w)$. Let $\overline{\mathscr{G}}_v$ be the cover of $\mathscr{G}_v$ obtained from the product of the Kazhdan–Patterson covers of the $\mathrm{GL}(r,L_w)$ with twisting parameter c_v, by identifying their copies of μ_n (i.e. the quotient of the product by $\{(\zeta_w):\prod_w\zeta_w=1\}$). A cocycle for $\overline{\mathscr{G}}_v$ is

$$\prod_{w|v}\sigma_{L_w}(g_w,h_w)\,(\det g_w,\det h_w)_{L_w}^{c_v},\tag{4.1}$$

where σ_{L_w} is the Kazhdan–Patterson cocycle with parameter 0 [KP84, §0.1], [FK86, §2]. Let

$$\iota_D:D_v^\times\hookrightarrow\mathscr{G}_v,\qquad \iota_{\mathrm{GL}}:\mathrm{GL}(r,K_v)\hookrightarrow\mathscr{G}_v$$

be the embedding induced by $D\otimes_K L\cong M_r(L)$, and the diagonal inclusion. Note that $\det\circ\iota_D=\mathrm{Nrd}$, and ι_D and ι_{GL} agree on the scalars $K_v^\times$.

Definition 4.2.1 (Local covers $\overline{G}_D(K_v)$, $\overline{G}_v$)**.** $\overline{G}_D(K_v):=p^{-1}(\iota_D(D_v^\times))\subset\overline{\mathscr{G}}_v$ and $\overline{G}_v:=p^{-1}(\iota_{\mathrm{GL}}(\mathrm{GL}(r,K_v)))\subset\overline{\mathscr{G}}_v$. Let $\overline{Z}_v:=p^{-1}(K_v^\times)$; it is a subgroup of both.

Example 4.2.2 (Continuing Example 1.3.1: the local covers at $\mathfrak{p}_7$)**.** Take $v = \mathfrak{p}_7$. Then $K_v = \mathbb{Q}_7$, because 7 splits in $K = \mathbb{Q}(\zeta_3)$. Since -1 is not a square in $\mathbb{Q}_7$, the algebra $L_v = \mathbb{Q}_7(i)$ is a field, namely the unramified quadratic extension of $\mathbb{Q}_7$. So $\mathscr{G}_v = \mathrm{GL}(2, \mathbb{Q}_7(i))$, and $\overline{\mathscr{G}}_v$ is its cover with cocycle $\sigma_{L_v}(g,h)\,(\det g, \det h)_{L_v}^2$, because $c_v = 2$. Write $D_v = L_v \oplus L_v j$ with $j^2 = 4 - \zeta_3$ and $jx = \sigma(x)j$ for $x \in L_v$, where σ is the nontrivial automorphism of L_v/K_v. One choice of the isomorphism $D \otimes_K L \cong M_2(L)$ gives

$$\iota_D(x+yj) = \begin{pmatrix} x & (4-\zeta_3)y \\ \sigma(y) & \sigma(x) \end{pmatrix}, \qquad \det \iota_D(x+yj) = x\sigma(x) - (4-\zeta_3)\,y\sigma(y) = \mathrm{Nrd}(x+yj).$$

Then $\overline{G}_D(K_v)$ is the preimage in $\overline{\mathscr{G}}_v$ of the matrices of this form with nonzero determinant, and $\overline{G}_v$ is the preimage of $\mathrm{GL}(2, \mathbb{Q}_7)$. The group $\overline{Z}_v$ is the preimage of the scalar matrices with entry in $\mathbb{Q}_7^\times$.

$$\begin{array}{ccccc} \overline{G}_D(K_v) & \hookrightarrow & \overline{\mathscr{G}}_v & \hookleftarrow & \overline{G}_v \\ \downarrow & & \downarrow & & \downarrow \\ D_v^\times & \xrightarrow{\iota_D} & \mathrm{GL}(r, L_v) & \xleftarrow{\iota_{\mathrm{GL}}} & \mathrm{GL}(r, K_v) \end{array} \qquad \overline{Z}_v = p^{-1}(K_v^\times) \subset \overline{G}_D(K_v) \cap \overline{G}_v. \tag{4.2}$$

Lemma 4.2.3. *The inverse image in $\overline{\mathscr{G}}_v$ of the center $L_v^\times$ of $\mathscr{G}_v$ is central in $\overline{\mathscr{G}}_v$. In particular $\overline{Z}_v$ is central in $\overline{\mathscr{G}}_v$, in $\overline{G}_D(K_v)$ and in $\overline{G}_v$.*

Proof. By [FK86, §2] (see [JC26a, Lemma 2.1]), the center of the parameter-c_v cover of $\mathrm{GL}(r, L_w)$ is the inverse image of $\{z : z^{r-1+2rc_v} \in L_w^{\times n}\}$, which is all of $L_w^\times$ since $n \mid r - 1 + 2rc_v$. This is an **(E)** input. Centrality in the product with identified μ_n's follows factor by factor. □

4.2.3 The restriction lemma

We now identify the two subgroups of Definition 4.2.1. The D side is the cover of [JC26a] by construction. The GL side is isomorphic, as a central extension, to the Kazhdan–Patterson cover with parameter c_v. This replaces the cocycle-level claim in the proof of [JC26a, Lemma 6.1(c)] by a statement about extensions, with a proof that does not depend on the internal form of the Kazhdan–Patterson cocycle.

Proposition 4.2.4. *(using **(E)** inputs) Let v be a finite place.*

(a) The pull-back of the cocycle (4.1) along ι_D is the local component at v of the cocycle $\beta = \sigma_L \cdot (\mathrm{Nrd}, \mathrm{Nrd})^c$ of [JC26a, equation (4)]. Hence $\overline{G}_D(K_v)$ is the local cover of [JC26a, §6.1].

(b) There is an isomorphism of central extensions of $\mathrm{GL}(r, K_v)$ by μ_n (the identity on μ_n and over $\mathrm{GL}(r, K_v)$)

$$\overline{G}_v \;\cong\; \text{the extension with cocycle } \sigma_{K_v}^r\,(\det, \det)_{K_v}^c.$$

After relabeling by $\zeta \mapsto \zeta^{r'}$, the right side is the Kazhdan–Patterson cover $\overline{G}^{(c_v)}$ of $\mathrm{GL}(r, K_v)$, normalized as in [JC26a, Lemma 2.1].

(c) Any two isomorphisms as in (b) differ by an automorphism $\tilde{g} \mapsto \chi(\det p(\tilde{g}))\tilde{g}$ with $\chi : K_v^\times \to \mu_n$ a continuous character. Every result about $\overline{G}^{(c_v)}$ that we import from [FK86] or [KP84] is stated in terms of the group $\overline{G}^{(c_v)}$, the projection p and μ_n alone (commutators, good elements, orbital integrals, characters, supercusp forms, germ expansions). It therefore holds for $\overline{G}_v$ via any such isomorphism. The only exception is the use of the splittings κ_v^* over maximal compact subgroups; these are defined directly on $\overline{G}_v$ in Proposition 4.5.6(d), and the local results of [JC26a] hold for any splitting [JC26a, Remarks after Lemma 6.1].

Proof. (a) For $g,h \in D_v^\times$ we have $\det \iota_D(g)_w = \operatorname{Nrd} g \in K_v^\times$ for each $w \mid v$. By norm compatibility, $\prod_{w|v}(\operatorname{Nrd} g, \operatorname{Nrd} h)_{L_w}^{c_v} = (\operatorname{Nrd} g, \operatorname{Nrd} h)_{K_v}^{rc_v} = (\operatorname{Nrd} g, \operatorname{Nrd} h)_{K_v}^{c}$, since $\sum_w [L_w : K_v] = r$, $rc_v = rr'c \equiv c \pmod n$, and the values lie in μ_n. The remaining factor is the restriction of the parameter-0 cocycle of $\mathrm{GL}(r, L_v)$, which is the local component of σ_L in [JC26a, equation (4)]. (Globally one uses the normalized cocycle τ_L of [FK86, §2], which differs from the product of the σ_{L_w} by the coboundary of κ. This changes the cocycle model of $\overline{\mathscr{G}}_v$ but not the group $\overline{\mathscr{G}}_v$, nor the subgroups of Definition 4.2.1.)

(b) The twisting factor restricts to $(\det, \det)_{K_v}^c$ exactly as in (a). So it suffices to show that, for each $w \mid v$ with $d_w := [L_w : K_v]$, the pull-back X_1 of the parameter-0 cover of $\mathrm{GL}(r, L_w)$ to $\mathrm{GL}(r, K_v)$ is isomorphic to the extension X_2 with cocycle $\sigma_{K_v}^{d_w}$, i.e. the push-out of the parameter-0 cover of $\mathrm{GL}(r, K_v)$ along $\zeta \mapsto \zeta^{d_w}$. Taking the Baer sum over $w \mid v$ then gives the cocycle $\sigma_{K_v}^r$. Write $F = K_v$ and $F' = L_w$.

Step 1: the cocycles agree on the diagonal torus. By [FK86, §2], the Kazhdan–Patterson cocycle restricted to the diagonal torus is $\sigma_{F'}(a,a') = \prod_{i<j}(a_i, a'_j)_{F'}$. For $a, a' \in A(F)$, norm compatibility gives $\sigma_{F'}(a,a') = \prod_{i<j}(a_i, a'_j)_F^{d_w} = \sigma_F(a,a')^{d_w}$. So the cocycles of X_1 and X_2 (in these models) agree *literally* on $A(F) \times A(F)$.

Step 2: the restrictions to $\mathrm{SL}(r,F)$ *are isomorphic.* By Matsumoto and Moore [Mat69], [Moo68], topological central extensions of $\mathrm{SL}(r,F)$ by μ_n are classified by continuous Steinberg symbols $c : F^\times \times F^\times \to \mu_n$. Every such symbol is $(\cdot,\cdot)_F^k$ for some $k \in \mathbb{Z}/n$ (Moore), and $\mathrm{SL}(r,F)$ is perfect, so an isomorphism between two extensions, if it exists, is unique **(E)**. In the extension with symbol c, the commutator of lifts of $a, a' \in A(F) \cap \mathrm{SL}(r,F)$ is $\prod_i c(a_i, a'_i)^\varepsilon$ for a universal sign ε [Mat69, §5], [Mil71, §§8–9] **(E)**. Let $X_1|_{\mathrm{SL}}$ and $X_2|_{\mathrm{SL}}$ have symbols $(\cdot,\cdot)^{k_1}$ and $(\cdot,\cdot)^{k_2}$. Their torus commutators agree by Step 1. Take $a = \mathrm{diag}(x, x^{-1}, 1, \dots, 1)$ and $a' = \mathrm{diag}(y, y^{-1}, 1, \dots, 1)$; this gives $(x,y)^{2\varepsilon k_1} = (x,y)^{2\varepsilon k_2}$ for all x, y. As the Hilbert symbol is surjective onto μ_n and n is odd, $k_1 = k_2$. Hence the Baer difference $X := X_1 - X_2$ splits over $\mathrm{SL}(r,F)$, by a unique homomorphism $s : \mathrm{SL}(r,F) \to X$.

Step 3: X splits over $\mathrm{GL}(r,F)$. Let $A_1 = \{\mathrm{diag}(a, 1, \dots, 1)\} \cong F^\times$, so that $\mathrm{GL}(r,F) = \mathrm{SL}(r,F) \rtimes A_1$. By Step 1 the cocycle of X (the quotient of the two cocycles) is identically 1 on $A(F) \times A(F)$, so $t(a) = (a, 1)$ is a homomorphism $A_1 \to X$. For $a \in A_1$, $h \mapsto t(a)s(a^{-1}ha)t(a)^{-1}$ is again a splitting of X over $\mathrm{SL}(r,F)$, so it equals s by uniqueness. Hence $S(ha) := s(h)t(a)$ ($h \in \mathrm{SL}(r,F)$, $a \in A_1$) is a homomorphism: $S(ha\,h'a') = s(h)\,s(ah'a^{-1})\,t(aa') = s(h)t(a)s(h')t(a)^{-1}t(a)t(a') = S(ha)S(h'a')$. So X is split, i.e. $X_1 \cong X_2$.

For the relabeling, raising $\sigma_{K_v}^r(\det, \det)_{K_v}^c$ to the power r' gives $\sigma_{K_v}(\det, \det)_{K_v}^{c_v}$, the Kazhdan–Patterson cocycle with parameter c_v.

(c) Two isomorphisms of extensions over the identity differ by a continuous homomorphism $\mathrm{GL}(r, K_v) \to \mu_n$. Since $\mathrm{SL}(r, K_v)$ is perfect, such a homomorphism factors through det. The rest is a

remark on how the cited results are used. □

Remark 4.2.5. (1) Proposition 4.2.4(b) is the form in which [JC26a, Lemma 6.1(c)] should be stated: an isomorphism of extensions, not an equality of cocycles. The proof there, which appeals to the Bruhat decomposition, is replaced by Steps 1–3 above. An equality at the level of explicit cocycles would follow from the formula of [BLS99], where σ is a product of Hilbert symbols of ratios of minors (which lie in K_v when $g,h \in \mathrm{GL}(r,K_v)$); we do not need it.

(2) For v split in L, (b) is literal: the diagonal restriction of a product of r copies of a cocycle is its r-th power.

(3) At archimedean v all Hilbert symbols are trivial, and all covers in sight are split.

Remark 4.2.6 (Central characters and the choice of isomorphism)**.** An isomorphism $\phi : \overline{G}_v \to \overline{G}^{(c_v)}$ as in Proposition 4.2.4(b) is determined up to twisting by $\chi \circ \det$. On the inverse image of a scalar z this twist multiplies by $\chi(z^r)$. So ϕ transports a genuine character $\tilde{\theta}_v$ of $\overline{Z}_v$ to a central character of $\overline{G}^{(c_v)}$ that depends on ϕ. We avoid this ambiguity as follows.

(1) Central characters live on $\overline{Z}_v \subset \overline{\mathscr{G}}_v$ and are *never* transported through ϕ. The same $\tilde{\theta}_v$ serves $\overline{G}_D(K_v)$ and $\overline{G}_v$ (Lemma 4.2.3).

(2) The results imported through ϕ are statements about all genuine functions or representations, for all central characters at once, or about objects that do not involve central characters: commutators and good elements [FK86, §3], convergence of orbital integrals [FK86, §6], supercusp forms and Selberg's principle [FK86, §13], Harish-Chandra's character theory, and the analytic steps of [FK86, §18]. Such a statement for $\overline{G}^{(c_v)}$ with central character $\tilde{\theta}_v \circ \phi^{-1}$ is equivalent, via ϕ, to the same statement for $\overline{G}_v$ with central character $\tilde{\theta}_v$. So these results hold for $\overline{G}_v$ independently of the choice of ϕ.

(3) Results tied to a normalized identification are *not* used in Chapter 4. These include the unramified theory relative to the splitting of [KP84], the Flicker–Kazhdan correspondence relating specific central characters of $\overline{G}^{(c_v)}$ and $\mathrm{GL}(r,K_v)$, and exceptional representations. For the later parts we note the following. For almost all v, two splittings over $\mathrm{GL}(r,\mathscr{O}_v)$ differ by $\chi_0 \circ \det$ with χ_0 a character of $\mathscr{O}_v^\times$. Extending χ_0 to $K_v^\times$, one can choose ϕ to carry κ_v^* of Proposition 4.5.6(d) to the Kazhdan–Patterson splitting. Such ϕ are then unique up to $\chi \circ \det$ with χ unramified, and the remaining ambiguity on $\overline{Z}_v$ is $\chi(\varpi^r) \in \mu_n$, which must be fixed separately when the Flicker–Kazhdan correspondence is applied.

4.2.4 Places where D splits

At $v \notin S$ the groups $D_v^\times$ and $\mathrm{GL}(r,K_v)$ are isomorphic. We show that the corresponding covers are identified by conjugation in $\overline{\mathscr{G}}_v$, canonically once the isomorphism of the base groups is fixed, and compatibly with the splittings over maximal compact subgroups at almost all places. This is step (2) of [JC26a, Remark 10.3].

Proposition 4.2.7. *Let $v \notin S$ and fix a K_v-algebra isomorphism $\alpha_v : D_v \xrightarrow{\sim} M_r(K_v)$.*

(a) There is $g_v \in \mathscr{G}_v$ with $\mathrm{Ad}(g_v) \circ \iota_D = \iota_{\mathrm{GL}} \circ \alpha_v$ on $D_v^\times$. It is unique up to the center $L_v^\times$ of $\mathscr{G}_v$.

(b) $\psi_v := \mathrm{Ad}(\tilde{g}_v)|_{\overline{G}_D(K_v)} : \overline{G}_D(K_v) \to \overline{G}_v$ *(any lift* $\tilde{g}_v$*) is an isomorphism of covers over* α_v*. It is the identity on* μ_n *and on* $\overline{Z}_v$*, and depends only on* α_v*. Replacing* α_v *by* $\mathrm{Ad}(h)\circ\alpha_v$ *with* $h \in \mathrm{GL}(r,K_v)$ *replaces* ψ_v *by* $\mathrm{Ad}(\tilde{h})\circ\psi_v$*.*

(c) Suppose v *is finite,* $v \nmid n$*,* $\alpha_v(\mathscr{O}_{D,v}) = M_r(\mathscr{O}_v)$ *for the global order* $\mathscr{O}_D$ *used in [JC26a, Lemma 6.1(d)], and* $\iota_D(\mathscr{O}_{D,v}) \otimes \mathscr{O}_{L_v} = M_r(\mathscr{O}_{L_v})$ *(true for almost all* v*). Then* g_v *may be taken in* $\mathrm{GL}(r,\mathscr{O}_{L_v})$*, and* ψ_v *carries the splitting* $\kappa_v^*(G_D(\mathscr{O}_v))$ *of [JC26a, Lemma 6.1(d)] to the splitting* $\kappa_v^*(\mathrm{GL}(r,\mathscr{O}_v))$ *of Proposition 4.5.6(d).*

Proof. (a) Both ι_D and $\iota_{\mathrm{GL}}\circ\alpha_v$ extend L_v-linearly to L_v-algebra isomorphisms $D_v \otimes_{K_v} L_v \to M_r(L_v)$, so they differ by an L_v-algebra automorphism of $M_r(L_v) = \prod_w M_r(L_w)$. Such an automorphism preserves the idempotents of L_v, hence each factor, and on each factor it is inner (Skolem–Noether). Uniqueness up to the centralizer of $\iota_{\mathrm{GL}}(M_r(K_v)) \otimes L_v = M_r(L_v)$, which is $L_v^\times$, is clear.

(b) $\mathrm{Ad}(\tilde{g}_v)$ is an automorphism of $\overline{\mathscr{G}}_v$, trivial on μ_n, over $\mathrm{Ad}(g_v)$, so it maps $p^{-1}\iota_D(D_v^\times)$ onto $p^{-1}\iota_{\mathrm{GL}}(\mathrm{GL}(r,K_v))$. Changing $\tilde{g}_v$ within $p^{-1}(g_v L_v^\times)$ does not change $\mathrm{Ad}(\tilde{g}_v)$, since $p^{-1}(L_v^\times)$ is central (Lemma 4.2.3). The same lemma shows ψ_v is the identity on $\overline{Z}_v$. The last assertion is clear.

(c) Under the hypotheses, $\mathrm{Ad}(g_v)$ restricts to an automorphism of the Azumaya algebra $M_r(\mathscr{O}_{L_v})$ over $\mathscr{O}_{L_v} = \prod_w \mathscr{O}_{L_w}$, a product of local rings. Such an automorphism is inner by an element of $\mathrm{GL}(r,\mathscr{O}_{L_v})$ (Skolem–Noether for Azumaya algebras over local rings, whose Picard groups are trivial). So $g_v \in L_v^\times \mathrm{GL}(r,\mathscr{O}_{L_v})$, and we may take $g_v \in \mathrm{GL}(r,\mathscr{O}_{L_v})$. The splittings in question are the restrictions of the splitting $\kappa^*_{\mathscr{G}_v}$ of $\overline{\mathscr{G}}_v$ over $\mathrm{GL}(r,\mathscr{O}_{L_v})$ (Proposition 4.5.6(d) and [JC26a, Lemma 6.1(d)]). Since $\kappa^*_{\mathscr{G}_v}$ is a homomorphism, $\mathrm{Ad}(\kappa^*_{\mathscr{G}_v}(g_v))$ carries $\kappa^*_{\mathscr{G}_v}(\iota_D(k))$ to $\kappa^*_{\mathscr{G}_v}(g_v\iota_D(k)g_v^{-1}) = \kappa^*_{\mathscr{G}_v}(\iota_{\mathrm{GL}}\alpha_v(k))$. □

Example 4.2.8 (Continuing Example 1.3.1: a split place, and matching costs nothing there)**.** Take v the place of $K = \mathbb{Q}(\zeta_3)$ above 13. Since $13 \equiv 1 \pmod 3$, 13 splits completely in K (consistent with the Sage check behind Example 1.3.1), so $K_v = \mathbb{Q}_{13}$; and $13 \notin \{3,7\} = S$, $13 \nmid n = 3$. Since D is ramified exactly at S (Example 1.3.1), $v \notin S$ means $D_v \cong M_2(\mathbb{Q}_{13})$ is already split, so v is exactly the kind of place Proposition 4.2.7 covers, and α_v is any $\mathbb{Q}_{13}$-algebra isomorphism $D_v \xrightarrow{\sim} M_2(\mathbb{Q}_{13})$.

Concretely: let $\gamma = \iota_D(t)$ for t a generator of a maximal subfield $E \subset D$ good at v, and let $\tilde{f}_{D,v}$ be any genuine test function on $\overline{G}_D(K_v)$. Set $\tilde{f}_v := \tilde{f}_{D,v}\circ\psi_v^{-1}$ as in part (b). Because ψ_v is a group isomorphism carrying $\iota_D(\gamma)$ to $\iota_{\mathrm{GL}}(\alpha_v(\gamma))$ and the Haar measures are transported by $dg_{D,v} = \psi_v^* dg_v$ (the convention fixed in §4.6.1), the two orbital integrals are literally the same integral written in two sets of coordinates:

$$\Phi\big(\iota_D(\gamma), \tilde{f}_{D,v}\big) = \Phi\big(\iota_{\mathrm{GL}}(\alpha_v(\gamma)), \tilde{f}_v\big),$$

with no normalizing factor on either side. This is what "matching is the identity at $v \notin S$" means in practice: at the thirteen-adic place there is nothing to solve, no germ to match, and no constant to track, because the two sides are not merely isomorphic but presented, after ψ_v, as the same group acting on the same test function. All of the content of Chapters 4–6 is about the two places $\mathfrak{p}_3, \mathfrak{p}_7 \in S$, precisely because every other place looks like this one.

4.3 Comparison of covered elliptic tori at the ramified places

This section shows that, at a place $v \in S$, the covered torus $\overline{E}_D \subset \overline{G}_D(K_v)$ attached to a maximal subfield $E \subset D_v$ and the covered torus $\overline{E}_{\mathrm{GL}} \subset \overline{G}_v$ attached to an embedding $E \hookrightarrow M_r(K_v)$ have the

same commutator pairing, the same good elements and the same centers. They are identified on good elements by a canonical isomorphism. We compute the pairing (§4.3.1) and the good elements (§4.3.2), give an example (§4.3.3), construct the identification (§4.3.4), and record its consequences (§4.3.5). It depends on §4.2 (Lemma 4.2.3 and Proposition 4.2.4) and on the commutator formula of [FK86, §3], [KP84, Prop. 0.1.5]. It introduces the covered tori $\overline{E}_D$ and $\overline{E}_{\mathrm{GL}}$, the set $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$, the canonical identification $\jmath_{E,j}$ and the induced bijection $\jmath$ of good regular elliptic classes.

Throughout this section, $v \in S$, $F := K_v$, w is the unique place of L over v, so that $\mathscr{G}_v = \mathrm{GL}(r, L_w)$, and L_w/F is cyclic of degree r. Let $E \subset D_v$ be a maximal subfield, so $[E:F] = r$, and let $j : E \hookrightarrow M_r(F)$ be an F-algebra embedding. (One exists, e.g. the regular representation; any two are $\mathrm{GL}(r,F)$-conjugate.) We write $N = N_{E/F}$, $\overline{E}_D := p^{-1}(\imath_D(E^\times)) \subset \overline{G}_D(K_v)$ and $\overline{E}_{\mathrm{GL}} := p^{-1}(\imath_{\mathrm{GL}} j(E^\times)) \subset \overline{G}_v$.

4.3.1 The commutator pairings

We compute the commutator pairing of the covered torus on each side, inside $\overline{\mathscr{G}}_v$, and find the same answer.

For $x, y \in E^\times$ write $[x,y]_D$ and $[x,y]_{\mathrm{GL}}$ for the commutators of lifts of $\imath_D(x), \imath_D(y)$ and of $\imath_{\mathrm{GL}} j(x), \imath_{\mathrm{GL}} j(y)$. They are well defined and bimultiplicative [FK86, §3].

Proposition 4.3.1. *For all $x, y \in E^\times$,*

$$[x,y]_D = [x,y]_{\mathrm{GL}} = (Nx, Ny)_F^{r+2c}\,(x,y)_E^{-r} \qquad \textit{(before relabeling)}. \tag{4.3}$$

After relabeling by $\zeta \mapsto \zeta^{r'}$ this becomes

$$[x,y] = (Nx, Ny)_F^{1+2c_v}\,(x,y)_E^{-1} = \big((Nx)^{1+2c_v} x^{-1}, y\big)_E, \tag{4.4}$$

which is [JC26a, equation (2)] for the elliptic torus $j(E^\times)$ of $\overline{G}^{(c_v)}$.

Proof. Commutators are invariants of the extension, so we may compute in any model. We use the formula of [FK86, §3] and [KP84, Prop. 0.1.5] **(E)**: for a maximal torus $T' = \prod_i F_i'^\times$ of $\mathrm{GL}(r, F')$ and the parameter-0 cover, $[x,y] = (\det x, \det y)_{F'} \prod_i (x_i, y_i)_{F_i'}^{-1}$. Twisting by $(\det, \det)^m$ multiplies commutators by $(\det x, \det y)^{2m}$.

The D side. $\imath_D(x), \imath_D(y)$ lie in the maximal torus $\imath_D(E \otimes_F L_w)^\times$ of $\mathscr{G}_v$. Write $E \otimes_F L_w = \prod_i E_i$ with fields $E_i \supseteq E$; then $\sum_i [E_i : L_w] = r$, and $[E_i : E] = [E_i : L_w]$ since $[E:F] = [L_w : F]$. We have $\det \imath_D(x) = Nx \in F^\times$, so

$$(\det x, \det y)_{L_w} = (Nx, Ny)_F^r, \qquad \prod_i (x,y)_{E_i} = \prod_i (x, N_{E_i/E}\, y)_E = (x,y)_E^r.$$

The twist by $(\det, \det)_{L_w}^{c_v}$ contributes $(Nx, Ny)_F^{2rc_v} = (Nx, Ny)_F^{2c}$.

The GL *side.* By Proposition 4.2.4(b) the extension $\overline{G}_v$ has cocycle $\sigma_F^r (\det, \det)_F^c$ up to isomorphism. For the parameter-0 cover and the elliptic torus $j(E^\times)$, [FK86, §3] gives $(Nx, Ny)_F (x,y)_E^{-1}$. Raising to the r-th power and multiplying by $(Nx, Ny)_F^{2c}$ gives the same expression.

Relabeling. Raise (4.3) to the power r'. For the last equality in (4.4), $(Nx, Ny)_F = (Nx, y)_E$ by norm compatibility, since $Nx \in F^\times$. □

Remark 4.3.2. The equality $[x,y]_D = [x,y]_{\mathrm{GL}}$ uses the formula of [FK86, §3] only through its multiplicative shape. If the formula were $(\det x, \det y)^a \prod_i (x_i, y_i)^b$, both sides of (4.3) would become $(Nx,Ny)_F^{ra+2c}(x,y)_E^{rb}$.

Remark 4.3.3 (The twisting parameter is forced)**.** If the GL-side cover had parameter c' instead of c_v, its pairing on $j(E^\times)$ would be $(Nx,Ny)_F^{1+2c'}(x,y)_E^{-1}$. This agrees with (4.4) if and only if $(Nx,Ny)_F^{2(c'-c_v)} = 1$ for all $x,y \in E^\times$. Since $N(E^\times) \supseteq F^{\times r}$ and $(F^{\times r}, F^{\times r})_F = \mu_n$, and n is odd, this holds if and only if $c' \equiv c_v \pmod n$. So matching requires exactly $\overline{G}^{(c_v)}$, compared with the D side through the same relabeling of μ_n.

4.3.2 Good elements and centers

From the explicit pairing we determine the good elements on both sides, the index of the center, and hence the dimension of genuine irreducible representations of the covered torus.

Lemma 4.3.4. *$F^\times \cap E^{\times n} = F^{\times n}$. Hence $F^\times/F^{\times n} \to E^\times/E^{\times n}$ is injective.*

Proof. If $z = y^n$ with $y \in E^\times$, then $z^r = N(z) = N(y)^n \in F^{\times n}$. Also $z^n \in F^{\times n}$, and $(r,n) = 1$, so $z \in F^{\times n}$. □

Proposition 4.3.5. *Let $\bullet \in \{D, \mathrm{GL}\}$.*

(a) An element $\tilde{x} \in \overline{E}_\bullet$ is good (central in $\overline{E}_\bullet$) if and only if $p(\tilde{x}) \in F^\times E^{\times n}$. No regularity assumption is needed.

(b) $[\overline{E}_\bullet : Z(\overline{E}_\bullet)] = [E^\times : F^\times E^{\times n}] = |n|_F^{-(r-1)}$. In particular $\overline{E}_\bullet$ is abelian if and only if $v \nmid n$.

(c) Every genuine irreducible smooth representation of $\overline{E}_\bullet$ has dimension $|n|_F^{-(r-1)/2}$ (an integer: if $p \mid n$ and $p^k \| n$, then $\mu_{p^k} \subset F$ forces $p-1 \mid [F:\mathbb{Q}_p]$, so $|n|_F^{-1} = p^{k[F:\mathbb{Q}_p]}$ is a square), and is determined by its central character on $Z(\overline{E}_\bullet)$ [JC26a, Prop. 2.4].

(d) If $x \in E^\times \smallsetminus F^\times$, then $F(x^n) = E$. In particular, if x is regular, so is x^n.

Proof. (a) By (4.4), $\tilde{x}$ is central if and only if $((Nx)^{1+2c_v}x^{-1}, y)_E = 1$ for all $y \in E^\times$. By nondegeneracy of the Hilbert symbol, this holds if and only if $x \equiv (Nx)^{1+2c_v} \pmod{E^{\times n}}$. If so, $x \in F^\times E^{\times n}$. Conversely, if $x = zy^n$ with $z \in F^\times$, then $(Nx)^{1+2c_v} \equiv z^{r(1+2c_v)} = z \cdot z^{r-1+2rc_v} \equiv z \equiv x \pmod{E^{\times n}}$.

(b) Since $\mu_n \subset F$, we have $|F^\times/F^{\times n}| = n^2|n|_F^{-1}$ and $|E^\times/E^{\times n}| = n^2|n|_E^{-1} = n^2|n|_F^{-r}$ [Neu99, Ch. II, §5] **(E)**. Now use Lemma 4.3.4.

(c) Stone–von Neumann: the dimension is the square root of the index.

(d) Put $F' = F(x^n) \subseteq E$. Since $\mu_n \subset F'$, the degree of the Kummer extension $F'(x)/F'$ divides n. It also divides r, so it is 1. As r is prime and $x \notin F$, $F(x) = E$, so $F' = E$. □

Remark 4.3.6. Part (b) generalizes [JC26a, Lemma 2.5] from unramified E to all maximal subfields when $v \nmid n$. Part (d) shows that for regular elliptic elements the condition "x^* regular" in [FK86, §18] is automatic.

4.3.3 An example

We illustrate the index computation with the smallest case in which a place above n makes the covered torus non-abelian.

Example 4.3.7. Take $r = 2$, $n = 3$, and $F = K_v = \mathbb{Q}_3(\zeta_3)$, so D_v is the quaternion division algebra over F. We may take $c = 1$ (so $3 \mid r-1+2c = 3$), $r' = 2$ and $c_v = 2$, and indeed $3 \mid r-1+2rc_v = 9$. For any quadratic extension E/F, (4.4) reads $[x,y] = (Nx,Ny)_F^5(x,y)_E^{-1} = (Nx,Ny)_F^2(x,y)_E^{-1}$. The field F is totally ramified of degree 2 over $\mathbb{Q}_3$, so $|3|_F = 3^{-2}$ and

$$[\overline{E} : Z(\overline{E})] = |3|_F^{-1} = 9.$$

So every genuine irreducible representation of $\overline{E}_D$ or $\overline{E}_{\mathrm{GL}}$ has dimension 3, and the good elements form a subgroup of index 9 in $E^\times$. (As a check, $|F^\times/F^{\times 3}| = 3 \cdot 3 \cdot 9 = 81$.) By contrast, at a place $v \in S$ of residual characteristic $\neq 3$, every covered elliptic torus is abelian.

4.3.4 A canonical identification on good elements

Equal commutator pairings leave open whether the two covered tori are isomorphic, and if so how canonically. We show that they are isomorphic, by conjugation inside $\overline{\mathscr{G}}_v$, and that every isomorphism that is the identity on the common central subgroup $\overline{Z}_v$ has the same restriction to good elements. This defines the identification $\jmath_{E,j}$ used in the rest of the paper.

Definition 4.3.8 (Isomorphisms of covered tori)**.** Let $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ be the set of isomorphisms of topological groups $\varphi : \overline{E}_D \to \overline{E}_{\mathrm{GL}}$ with $p \circ \varphi = (\iota_{\mathrm{GL}} j \iota_D^{-1}) \circ p$ that are the identity on μ_n and on $\overline{Z}_v$.

Example 4.3.9 (Continuing Example 1.3.1: the set $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ at the two ramified places)**.** By Theorem 4.3.10(a) and (c) below, the set $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ is nonempty and is a torsor under the group X of continuous characters $E^\times \to \mu_n$ that are trivial on $F^\times$. Such a character is trivial on $F^\times E^{\times n}$. So X is the group of characters of the finite abelian group $E^\times/F^\times E^{\times n}$ with values in μ_n. That group has exponent dividing n, so $|X| = [E^\times : F^\times E^{\times n}]$, and this index is $|n|_F^{-(r-1)}$ by Proposition 4.3.5(b).

At $v = \mathfrak{p}_7$ we have $|3|_F = 1$. So X is trivial and there is exactly one isomorphism. Here the covered tori are abelian, every element is good, and the single isomorphism is $\jmath_{E,j}$ itself. At $v = \mathfrak{p}_3$ we have $|3|_F^{-1} = 9$ (Example 4.3.7), so there are nine isomorphisms. They differ by the nine characters in X. By Theorem 4.3.10(b) they all agree on the good elements, which form a subgroup of index 9.

Theorem 4.3.10. *(a) There is $g \in \mathscr{G}_v$ with $g\,\iota_D(e)\,g^{-1} = \iota_{\mathrm{GL}}(j(e))$ for all $e \in E$. For any lift $\tilde{g}$,* $\mathrm{Ad}(\tilde{g})$ *restricts to an element $\varphi_g \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$.*

(b) Any two elements of $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ *agree on $Z(\overline{E}_D)$. The common restriction*

$$\jmath_{E,j} : Z(\overline{E}_D) \xrightarrow{\sim} Z(\overline{E}_{\mathrm{GL}})$$

is an isomorphism and a homeomorphism. It is the identity on $\overline{Z}_v$, satisfies $p \circ \jmath_{E,j} = (\iota_{\mathrm{GL}} j \iota_D^{-1}) \circ p$, commutes with n-th powers, and preserves regularity.

(c) $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ *is a torsor under the group of continuous characters $E^\times \to \mu_n$ trivial on $F^\times$; all of these are trivial on $F^\times E^{\times n}$.*

Proof. (a) Via ι_D and $\iota_{\mathrm{GL}} \circ j$, the space L_w^r is a module over the étale L_w-algebra $A := E \otimes_F L_w = \prod_i E_i$, faithful in both cases. A faithful A-module of L_w-dimension $r = \dim_{L_w} A$ is free of rank one, since each E_i must act nontrivially and dimensions add up. So the two module structures are conjugate by some $g \in \mathscr{G}_v$. Then $\mathrm{Ad}(\tilde{g})$ is an automorphism of $\overline{\mathscr{G}}_v$ over $\mathrm{Ad}(g)$, trivial on μ_n, mapping $\overline{E}_D$ onto $\overline{E}_{\mathrm{GL}}$. It is the identity on $\overline{Z}_v$ by Lemma 4.2.3, and it is continuous with continuous inverse.

(b) If $\varphi, \varphi' \in \mathrm{Iso}_{\overline{Z}}$, then $\varphi'\varphi^{-1}$ is an automorphism of $\overline{E}_{\mathrm{GL}}$ over the identity, trivial on μ_n, so it is $\tilde{x} \mapsto \chi(x)\tilde{x}$ for a continuous character $\chi : E^\times \to \mu_n$. Then $\chi|_{F^\times} = 1$, and χ is trivial on $E^{\times n}$ since it has values in μ_n. So $\varphi = \varphi'$ on $p^{-1}(F^\times E^{\times n}) = Z(\overline{E}_D)$ (Proposition 4.3.5(a)). The other properties hold for any φ, and φ_g is a homeomorphism by (a).

(c) This is the computation in (b), together with the fact that $\tilde{x} \mapsto \chi(x)\tilde{x}$ is an automorphism for every such χ. □

$$\begin{array}{ccc} Z(\overline{E}_D) & \xrightarrow[\sim]{\jmath_{E,j}} & Z(\overline{E}_{\mathrm{GL}}) \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ F^\times E^{\times n} & \xrightarrow{=} & F^\times E^{\times n} \end{array} \qquad \text{(identity on } \overline{Z}_v\text{, on the bottom identified via } \iota_D,\ \iota_{\mathrm{GL}} \circ j\text{).} \tag{4.5}$$

Remark 4.3.11. Off the good elements, φ depends on choices through a character of $E^\times/F^\times E^{\times n}$, a group that is nontrivial exactly when $v \mid n$. This does no harm, since genuine orbital integrals vanish at non-good elements (§4.4) and genuine representations of covered tori are determined by their central characters.

4.3.5 Consequences

We record that the identification is natural for conjugation, independent of j, and induces bijections of good regular elliptic classes and of genuine representations of covered tori. The class sets $\mathscr{C}^g(\overline{G}_D(K_v))$, $\mathscr{C}^g_{\mathrm{ell}}(\overline{G}_v)$ and the bijection $\jmath$ defined here enter the definition of matching in §4.4.

Proposition 4.3.12 (Naturality)**.** *Let $y \in D_v^\times$ and $h \in \mathrm{GL}(r,F)$, and put $E' = yEy^{-1}$ and $j' = \mathrm{Ad}(h) \circ j \circ \mathrm{Ad}(y)^{-1} : E' \hookrightarrow M_r(F)$. Then on $Z(\overline{E}_D)$*

$$\jmath_{E',j'} \circ \mathrm{Ad}(\tilde{y}) = \mathrm{Ad}(\tilde{h}) \circ \jmath_{E,j}.$$

In particular $\jmath_{E,j}$ depends on j only through its $\mathrm{GL}(r,F)$-conjugacy class, which is unique. If y normalizes E and h normalizes $j(E)$, inducing the same automorphism of E, then $\jmath_{E,j} \circ \mathrm{Ad}(\tilde{y}) = \mathrm{Ad}(\tilde{h}) \circ \jmath_{E,j}$.

Proof. Put $j_1 := \mathrm{Ad}(h) \circ j = j' \circ \mathrm{Ad}(y)$. Let $\varphi \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ and $\varphi' \in \mathrm{Iso}_{\overline{Z}}(\overline{E}'_D, \overline{E}'_{\mathrm{GL}})$, relative to j and j'. Then $\mathrm{Ad}(\tilde{h}) \circ \varphi$ and $\varphi' \circ \mathrm{Ad}(\tilde{y})$ are isomorphisms $\overline{E}_D \to \overline{E}'_{\mathrm{GL}}$ over j_1. Both are the identity on μ_n and on the central subgroup $\overline{Z}_v$ (Lemma 4.2.3). By Theorem 4.3.10(b) for j_1, they agree on $Z(\overline{E}_D)$. □

A regular element $\tilde{\gamma}$ of $\overline{G}_D(K_v)$ or $\overline{G}_v$ is *good* if it is central in $\overline{T}_\gamma$, where T_γ is the centralizer of $\gamma = p(\tilde{\gamma})$. Let $\mathscr{C}^g(\overline{G}_D(K_v))$ be the set of conjugacy classes of good regular elements of $\overline{G}_D(K_v)$ (all regular elements of $D_v^\times$ are elliptic), and $\mathscr{C}^g_{\mathrm{ell}}(\overline{G}_v)$ the set of conjugacy classes of good regular elliptic elements of $\overline{G}_v$.

Corollary 4.3.13. *The maps $\jmath_{E,j}$ induce a well-defined bijection*

$$\jmath : \mathscr{C}^g(\overline{G}_D(K_v)) \xrightarrow{\sim} \mathscr{C}^g_{\mathrm{ell}}(\overline{G}_v), \qquad [\tilde{\gamma}] \mapsto [\jmath_{F(\gamma),j}(\tilde{\gamma})].$$

It preserves (reduced) characteristic polynomials and commutes with multiplication by $\overline{Z}_v$.

Proof. Well defined: by Proposition 4.3.12. *Surjective:* if $\tilde{\gamma}'$ is good regular elliptic, $E' = F(p(\tilde{\gamma}'))$ is a field of degree r, which embeds in D_v [Wei73, Ch. XIII]. Choose $E \subset D_v$ and j with $j(E) = E'$; then $\tilde{\gamma}' \in Z(\overline{E}_{\mathrm{GL}})$ is $\jmath_{E,j}$ of a unique element. *Injective:* if $\jmath(\tilde{\gamma}_1)$ and $\jmath(\tilde{\gamma}_2)$ are conjugate, then $p(\tilde{\gamma}_1)$ and $p(\tilde{\gamma}_2)$ have the same characteristic polynomial, hence are conjugate in $D_v^\times$. By naturality we may assume $p(\tilde{\gamma}_1) = p(\tilde{\gamma}_2)$, so $\tilde{\gamma}_2 = \zeta\tilde{\gamma}_1$ with $\zeta \in \mu_n$ and $\zeta\jmath(\tilde{\gamma}_1)$ is conjugate to $\jmath(\tilde{\gamma}_1)$. If $x^{-1}\tilde{\delta}x = \zeta\tilde{\delta}$ with $\tilde{\delta}$ regular and good, then $p(x)$ centralizes $p(\tilde{\delta})$, so $x \in \overline{T_\delta}$ commutes with $\tilde{\delta}$, and $\zeta = 1$. □

Corollary 4.3.14. *For $\varphi \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$, $\chi \mapsto \chi \circ \varphi^{-1}$ induces a bijection between isomorphism classes of genuine irreducible smooth representations of $\overline{E}_D$ and $\overline{E}_{\mathrm{GL}}$, independent of φ. On good elements $\operatorname{tr}\chi(\tilde{t}) = \operatorname{tr}(\chi \circ \varphi^{-1})(\jmath_{E,j}(\tilde{t}))$, and both characters vanish off the good elements.*

Proof. By [JC26a, Prop. 2.4] the isomorphism class is determined by the central character ζ_χ on $Z(\overline{E}_\bullet)$, and the central character of $\chi \circ \varphi^{-1}$ is $\zeta_\chi \circ \jmath_{E,j}^{-1}$, independent of φ by Theorem 4.3.10(b). □

4.4 Orbital integrals and matching

This section defines normalized orbital integrals of genuine functions and the notion of matching (§§4.4.1–4.4.2). It then proves that matching functions exist, in both directions, for functions supported on the regular set (§§4.4.3–4.4.4). Finally it explains exactly what fails for general functions (§4.4.5). The last subsection, §4.4.6, proves the facts about orbital integrals on elliptic tori that the section uses. It depends on §4.3. It introduces the normalized orbital integrals Φ and $F = \Delta\Phi$, the matching conditions (M1)–(M2), the representatives E_i, j_i, $\jmath_i$ of elliptic tori, and admissible functions on covered tori. Throughout, $v \in S$ and $F = K_v$, and $\tilde{\theta}_v$ is a genuine character of $\overline{Z}_v$; by Lemma 4.2.3 the same character serves both sides.

4.4.1 Orbital integrals and measures

We fix measures, define the normalized orbital integrals, and record their basic properties. The representatives E_i, j_i, $\jmath_i$ and the measures dg_D, dg, dt_i fixed here are used in the rest of the paper.

Representatives of tori. The field extensions of F of degree r fall into finitely many isomorphism classes; fix representatives $E_1, \dots, E_m$. For each i fix an embedding $E_i \subset D_v$ (it exists [Wei73, Ch. XIII]) and an embedding $j_i : E_i \hookrightarrow M_r(F)$. Every maximal subfield of D_v is $D_v^\times$-conjugate to exactly one E_i, and every elliptic maximal torus of $\mathrm{GL}(r,F)$ is conjugate to exactly one $j_i(E_i^\times)$ (Skolem–Noether). Write $\jmath_i := \jmath_{E_i,j_i}$.

Measures. Fix Haar measures dg_D on $D_v^\times/F^\times$ and dg on $\mathrm{GL}(r,F)/F^\times$, and for each i a Haar measure dt_i on $E_i^\times/F^\times$. Use dt_i on both $\iota_D(E_i^\times)/F^\times$ and $j_i(E_i^\times)/F^\times$, and transport it by conjugation to every torus conjugate to these. Quotient measures on $T\backslash G$ are dg/dt. The matching relation below does not depend on the choice of the dt_i: rescaling dt_i by λ multiplies the orbital integrals on both sides at elements of tori of type i by λ^{-1}. It does depend on dg_D and dg, which are fixed.

Definition 4.4.1 (Orbital integral)**.** For $\bullet \in \{D, \mathrm{GL}\}$, $G_\bullet \in \{D_v^\times, \mathrm{GL}(r,F)\}$, a genuine $\tilde{f} \in C_c^\infty(\overline{G_\bullet}, \tilde{\theta}_v)_{\mathrm{gen}}$ and a good regular semisimple $\tilde{\gamma}$ with centralizer T_γ, put

$$\Phi(\tilde{\gamma}, \tilde{f}) := \int_{T_\gamma \backslash G_\bullet} \tilde{f}(\tilde{x}^{-1}\tilde{\gamma}\tilde{x})\, d\dot{x}, \qquad F(\tilde{\gamma}, \tilde{f}) := \Delta(\gamma)\, \Phi(\tilde{\gamma}, \tilde{f}),$$

with $\Delta(\gamma) = |D(\gamma)|_F^{1/2} |\det \gamma|_F^{-(r-1)/2}$ as in [FK86, §6], where $D(\gamma)$ is the discriminant of the (reduced) characteristic polynomial and det = Nrd on the D side. For non-good regular $\tilde{\gamma}$ put $F(\tilde{\gamma}, \tilde{f}) := 0$.

Example 4.4.2 (Continuing Example 1.3.1: the factor Δ and an orbital integral on the D side)**.** Take $v = \mathfrak{p}_7$, so $F = \mathbb{Q}_7$ and $r = 2$. Let $\gamma = a + bi \in L_v^\times \subset D_v^\times$ with $a, b \in F$ and $b \neq 0$. Its reduced characteristic polynomial is $X^2 - 2aX + (a^2 + b^2)$, with discriminant $D(\gamma) = -4b^2$. So γ is regular and

$$\Delta(\gamma) = |-4b^2|_F^{1/2} \, |a^2 + b^2|_F^{-1/2} = |b|_F \, |a^2 + b^2|_F^{-1/2}.$$

For example, $\Delta(1+i) = 1$ and $\Delta(1+7i) = 7^{-1}$. Since $v \nmid n$, every lift $\tilde{\gamma}$ of γ is good (Proposition 4.3.5(b)). Now suppose that $\tilde{f}$ is invariant under conjugation. Then the integrand of Φ is constant, and

$$\Phi(\tilde{\gamma}, \tilde{f}) = \mathrm{vol}(T_\gamma \backslash D_v^\times)\, \tilde{f}(\tilde{\gamma}), \qquad F(\tilde{\gamma}, \tilde{f}) = \Delta(\gamma)\, \mathrm{vol}(T_\gamma \backslash D_v^\times)\, \tilde{f}(\tilde{\gamma}).$$

The volume is finite because $D_v^\times / F^\times$ is compact.

Lemma 4.4.3. *(convergence: **(E)**)*

(a) The integrand of Φ is independent of the lift $\tilde{x}$ and left T_γ-invariant (because $\tilde{\gamma}$ is good). The integral converges absolutely. On the D side $T_\gamma \backslash D_v^\times$ is compact, since $D_v^\times / F^\times$ is. On the GL *side the orbit of a regular semisimple element is closed, and convergence is [FK86, §6] (following Rao) **(E)**.*

(b) $F(\cdot, \tilde{f})$ is constant on conjugacy classes, and $F(z\tilde{\gamma}, \tilde{f}) = \tilde{\theta}_v(z)^{-1} F(\tilde{\gamma}, \tilde{f})$ for $z \in \overline{Z}_v$; in particular it is genuine.

(c) If $\tilde{\gamma}$ is regular elliptic and not good, then the formal integral $\int_{T_\gamma \backslash G_\bullet}$ (over $\overline{T_\gamma} \backslash \overline{G_\bullet}$) vanishes; this is [JC26a, Prop. 2.7] with $\varepsilon = 1$. Thus the convention $F = 0$ at non-good elements loses nothing.

(d) For $\gamma \in E_i^\times$ regular, $\Delta(\gamma)$ is the same on the two sides. If E_i/F is cyclic, then Δ equals $|t_0^2|_F^{1/2}$ times the factor Δ of [JC26a, §4.1] with $\varepsilon = 1$, for any t_0 as there, so that $F(\cdot, \tilde{f}) = |t_0^2|_F^{1/2}\, \tilde{f}^T$ in the notation of [JC26a, §4.1].

Proof. (a) and (b) are immediate from the definitions, Lemma 4.2.3 and conjugation invariance of the measures. (c) is [JC26a, Prop. 2.7]. (d) The characteristic polynomials agree. For [JC26a]'s factor (defined only for cyclic E_i): $\tilde{\Delta}(t)^2 = \pm D(t)$ and $t_0^2 \in F^\times$, so $|\tilde{\Delta}(t) t_0^{-1}|_F = |D(t)|_F^{1/2} |t_0^2|_F^{-1/2}$. □

4.4.2 Definition of matching

We define matching with the sign $(-1)^{r-1}$ and with vanishing at good regular non-elliptic elements, using the bijection of Corollary 4.3.13.

Definition 4.4.4 (Matching)**.** Functions $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ and $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ *match* if

(M1) $F(\tilde{\gamma}, \tilde{f}_D) = (-1)^{r-1} F(\jmath(\tilde{\gamma}), \tilde{f})$ for every good regular $\tilde{\gamma} \in \overline{G}_D(K_v)$, and

(M2) $F(\tilde{\delta}, \tilde{f}) = 0$ for every good regular non-elliptic $\tilde{\delta} \in \overline{G}_v$.

Example 4.4.5 (Continuing Example 1.3.1: matching when $r = 2$)**.** In the running example $r = 2$, so the sign in (M1) is -1, and (M1) reads $F(\tilde{\gamma}, \tilde{f}_D) = -F(\jmath(\tilde{\gamma}), \tilde{f})$ for good regular $\tilde{\gamma}$. A regular semisimple element of $\mathrm{GL}(2,F)$ is either elliptic, when its eigenvalues generate a quadratic field over F, or conjugate to a diagonal matrix with distinct entries in F. So (M2) says that the orbital integrals of $\tilde{f}$ vanish at the good lifts of the elements conjugate to $\mathrm{diag}(a,b)$ with $a \neq b$. Such elements have no counterpart in $D_v^\times$, where every noncentral element generates a quadratic field. The simplest matching pair is $\tilde{f}_D = 0$ and $\tilde{f} = 0$. A larger family comes from Theorem 4.4.7(a) below. Every $\tilde{f}_D$ that vanishes near $\overline{Z}_v$ has a matching $\tilde{f}$ supported on the good regular elliptic set. For such $\tilde{f}$ condition (M2) holds because no conjugate of a non-elliptic element lies in its support.

By Corollary 4.3.13 and Lemma 4.4.3(b), (M1) is a condition on conjugacy classes, and it is compatible with the common central character $\tilde{\theta}_v$. With the sign placed here, the expected character relation $\Theta_{\overline{\pi}_D} = (-1)^{r-1} \Theta_{\overline{\pi}} \circ \jmath$ on good regular elements would give $\mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D) = \mathrm{tr}\,\overline{\pi}(\tilde{f})$ for matching functions. Badulescu [Bad08, §4.6] puts no sign in the matching of functions and keeps the sign $(-1)^{nd-n}$ in the character relation; the two conventions are equivalent.

4.4.3 Transported orbital-integral functions

We show that the orbital integrals of a regular-supported function on one side, transported to the other side by the identifications $\jmath_i$, have exactly the properties that characterize orbital integrals of regular-supported functions there. The proofs, for arbitrary (not necessarily cyclic) maximal subfields, are in §4.4.6. The notion of an admissible function defined here is used again in §4.4.6.

For $\bullet \in \{D, \mathrm{GL}\}$ and each i, let $\overline{E}_{i,\bullet}$ be the covered torus of type i (Definition in §4.3, with $E = E_i$, $j = j_i$), $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ its good regular elements, and $W_{i,\bullet} := N_{G_\bullet}(T_{i,\bullet})/T_{i,\bullet}$, which acts on $Z(\overline{E}_{i,\bullet})$ by conjugation and injects into $\mathrm{Aut}(E_i/F)$. Call a function h on $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ *admissible* if

(i) h is locally constant;

(ii) $h(z\tilde{t}) = \tilde{\theta}_v(z)^{-1} h(\tilde{t})$ for $z \in \overline{Z}_v$ (in particular h is genuine);

(iii) the support of h is compact modulo $\overline{Z}_v$ and contained in $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$, i.e. its closure in $Z(\overline{E}_{i,\bullet})$ avoids the non-regular elements;

(iv) $h(\tilde{y}\tilde{t}\tilde{y}^{-1}) = h(\tilde{t})$ for $y \in N_{G_\bullet}(T_{i,\bullet})$.

Lemma 4.4.6. *(a) If $\tilde{f} \in C_c^\infty(\overline{G_\bullet}, \tilde{\theta}_v)_{\mathrm{gen}}$ is supported in the regular set, then for each i the restriction of $F(\cdot, \tilde{f})$ to $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ is admissible.*

(b) $h \mapsto h \circ \jmath_i^{-1}$ *is a bijection from admissible functions on* $Z(\overline{E}_{i,D})^{\mathrm{reg}}$ *to admissible functions on* $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$.

(c) Conversely, every admissible h *on* $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ *is of the form* $F(\cdot,\tilde{f}_i)|_{Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}}$ *for some* $\tilde{f}_i \in C_c^\infty(\overline{G}_\bullet,\tilde{\theta}_v)_{\mathrm{gen}}$ *supported on the set of* $G_\bullet$*-conjugates of* $\operatorname{supp} h$. *In particular* $\tilde{f}_i$ *is supported in the good regular set of type* i, *and* $F(\tilde{\delta},\tilde{f}_i)=0$ *for every regular* $\tilde{\delta}$ *not conjugate into* $Z(\overline{E}_{i,\bullet})$.

Proof. (a) This is Lemma 4.4.12, applied to $G_\bullet$ and the elliptic torus $T_{i,\bullet}$. For cyclic E_i it is also [JC26a, Lemma 4.5(a)] with $\varepsilon = 1$, up to the constant of Lemma 4.4.3(d).

(b) $\jmath_i$ is a homeomorphism (Theorem 4.3.10(b)), so (i) is preserved. It is the identity on $\overline{Z}_v$ (so (ii) and compactness modulo $\overline{Z}_v$ in (iii) are preserved) and preserves regularity in both directions (so the rest of (iii) is preserved). For (iv): let $y \in N_{D_v^\times}(E_i)$ induce $\sigma \in \mathrm{Aut}(E_i/F)$. By Skolem–Noether there is $h \in N_{\mathrm{GL}(r,F)}(j_i(E_i))$ inducing $j_i \sigma j_i^{-1}$, and conversely. By Proposition 4.3.12, $\jmath_i \circ \mathrm{Ad}(\tilde{y}) = \mathrm{Ad}(\tilde{h}) \circ \jmath_i$, so invariance under $\mathrm{Ad}(\tilde{y})$ on one side is equivalent to invariance under $\mathrm{Ad}(\tilde{h})$ on the other. (Elements of $T_{i,\bullet}$ itself act trivially on good elements.) The same argument with $\jmath_i^{-1}$ gives the inverse.

(c) This is Lemma 4.4.13, applied to $G_\bullet$ and $T_{i,\bullet}$. A regular $\tilde{\delta}$ is conjugate into $Z(\overline{E}_{i,\bullet})$ only if its centralizer is conjugate to $T_{i,\bullet}$. □

4.4.4 The easy direction, in both directions

We combine the previous lemmas into the existence theorem for matching functions supported on the regular set, and check the hypotheses of the simple trace formula of [FK86, §18].

Theorem 4.4.7. *(a) Let* $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v),\tilde{\theta}_v)_{\mathrm{gen}}$ *vanish on a neighborhood of* $\overline{Z}_v$. *Then there is* $\tilde{f} \in C_c^\infty(\overline{G}_v,\tilde{\theta}_v)_{\mathrm{gen}}$, *supported on the good regular elliptic set, matching* $\tilde{f}_D$.

(b) Let $\tilde{f} \in C_c^\infty(\overline{G}_v,\tilde{\theta}_v)_{\mathrm{gen}}$ *be supported in the regular set, with* $F(\tilde{\delta},\tilde{f})=0$ *for all good regular non-elliptic* $\tilde{\delta}$. *Then there is* $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v),\tilde{\theta}_v)_{\mathrm{gen}}$ *vanishing on a neighborhood of* $\overline{Z}_v$ *and matching* $\tilde{f}$.

(c) Every $\tilde{f}$ *as in (a) satisfies, at the place* v, *both local hypotheses of [FK86, §18, Theorem]:* $F(x,\tilde{f})=0$ *for all regular non-elliptic* x, *and* $\tilde{f}$ *vanishes on* $\{x : x^* \text{ singular}\}$, *where* $x^* = \tilde{x}^n$.

Proof. (a) Since r is prime, the non-regular elements of $D_v^\times$ are the central ones. So $\tilde{f}_D$, which vanishes near $\overline{Z}_v$ and has support compact modulo $\overline{Z}_v$, is supported in the regular set. For each i, Lemma 4.4.6(a) shows that $h_i^D := F(\cdot,\tilde{f}_D)|_{Z(\overline{E}_{i,D})^{\mathrm{reg}}}$ is admissible. By (b) of that lemma, $h_i := (-1)^{r-1} h_i^D \circ \jmath_i^{-1}$ is admissible on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$, and by (c) there is $\tilde{f}_i$ on $\overline{G}_v$ with $F(\cdot,\tilde{f}_i) = h_i$ on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$ and $F(\tilde{\delta},\tilde{f}_i)=0$ for regular $\tilde{\delta}$ not conjugate into $Z(\overline{E}_{i,\mathrm{GL}})$. Put $\tilde{f} := \sum_i \tilde{f}_i$, which is supported on the good regular elliptic set.

(M2): a good regular non-elliptic $\tilde{\delta}$ is not conjugate into any elliptic torus, so $F(\tilde{\delta},\tilde{f})=0$.

(M1): let $\tilde{\gamma} \in \overline{G}_D(K_v)$ be good regular. It is conjugate to some $\tilde{t} \in Z(\overline{E}_{i,D})^{\mathrm{reg}}$ for a unique i, and $\jmath(\tilde{\gamma})$ is conjugate to $\jmath_i(\tilde{t})$ (Corollary 4.3.13). The tori of types $k \neq i$ do not contribute at $\jmath_i(\tilde{t})$. Hence

$$F(\jmath(\tilde{\gamma}),\tilde{f}) = F(\jmath_i(\tilde{t}),\tilde{f}_i) = h_i(\jmath_i(\tilde{t})) = (-1)^{r-1} F(\tilde{t},\tilde{f}_D) = (-1)^{r-1} F(\tilde{\gamma},\tilde{f}_D).$$

(b) For each i, the restriction of $F(\cdot,\tilde f)$ to $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$ is admissible by Lemma 4.4.6(a). Transport it by $\jmath_i^{-1}$, multiply by $(-1)^{r-1}$, realize it on $\overline{G}_D(K_v)$ by Lemma 4.4.6(c), and sum over i. (M1) follows as in (a), and (M2) is the hypothesis. The support of $\tilde f_D$ is closed, compact modulo $\overline{Z}_v$, and contained in the regular set, so it avoids a neighborhood of $\overline{Z}_v$.

(c) The first hypothesis is (M2) together with the vanishing at non-good elements. For the second, every $x\in\operatorname{supp}\tilde f$ is regular elliptic, so $p(x)^n$ is regular by Proposition 4.3.5(d) (applied in the torus $F(p(x))^\times$), and $p(x^*)=p(x)^n$. □

4.4.5 What fails for general functions

We explain why Theorem 4.4.7(a) cannot be extended to all $\tilde f_D$ by the same method, by computing the orbital integrals of $\tilde f_D$ near the center exactly.

Proposition 4.4.8. *Let $\tilde f_D\in C_c^\infty(\overline{G}_D(K_v),\tilde\theta_v)_{\mathrm{gen}}$. There are an open compact subgroup $U\subset D_v^\times$, normal in $D_v^\times$, and a splitting $\kappa_U:U\to\overline{G}_D(K_v)$ with the following properties.*

(a) $\mathrm{Ad}(\tilde x)\circ\kappa_U=\kappa_U\circ\mathrm{Ad}(x)$ on U for every $x\in D_v^\times$, and $\kappa_U(t)$ is good for every regular $t\in U$.

(b) For every $z\in\overline{Z}_v$, every maximal subfield E and every regular $t\in E^\times\cap U$,

$$F(z\,\kappa_U(t),\tilde f_D)=\Delta(t)\,\mathrm{vol}(T\backslash D_v^\times)\,\tilde f_D(z),\qquad T=E^\times.$$

Proof. Let $U_k=1+\mathfrak{P}_D^k$, where $\mathfrak{P}_D$ is the maximal ideal of the maximal order of D_v. These are normal subgroups of $D_v^\times$ and form a basis of neighborhoods of 1. The cover splits over $U_{k'}$ for k' large, by restriction from $\overline{\mathscr{G}}_v$ [FK86, §2]; fix such k' and a splitting κ over $U_{k'}$. For $k\ge k'$ large enough, every $t\in U_k$ is the n-th power of an element $s\in U_{k'}$ that is a limit of polynomials in t. If $p\nmid n$, take s in the pro-p group $\overline{\langle t\rangle}$, on which n-th powering is bijective. If $p\mid n$, take $s=\sum_i\binom{1/n}{i}(t-1)^i$, which converges for k large. Put $U=U_k$ and $\kappa_U=\kappa|_U$.

(a) For $x\in D_v^\times$, $\mathrm{Ad}(\tilde x)\circ\kappa\circ\mathrm{Ad}(x)^{-1}$ is another splitting over $U_{k'}$, so it differs from κ by a character $U_{k'}\to\mu_n$. Such a character is trivial on n-th powers, hence on U. If $t\in U\cap E^\times$ is regular, then $s\in F[t]\subseteq E$ (as E is closed), so $t\in E^{\times n}$ and $\kappa_U(t)$ is good (Proposition 4.3.5(a)).

(b) Since $\tilde f_D$ is smooth with compact support modulo $\overline{Z}_v$, after enlarging k we have $\tilde f_D(z\,\kappa_U(u))=\tilde f_D(z)$ for all $z\in\overline{Z}_v$ and $u\in U$. Then, by (a) and centrality of z, $\tilde f_D(\tilde x^{-1}z\kappa_U(t)\tilde x)=\tilde f_D(z\,\kappa_U(x^{-1}tx))=\tilde f_D(z)$ for all x. Integrate over the compact quotient $T\backslash D_v^\times$. □

Remark 4.4.9. Proposition 4.4.8 shows that $F(\cdot,\tilde f_D)$ is *not* compactly supported in the regular set unless $\tilde f_D$ vanishes on $\overline{Z}_v$: near each $z\in\overline{Z}_v$ it is a constant times $\Delta(t)$, which tends to 0 but is nonzero. A matching $\tilde f$ on $\overline{G}_v$ must then have orbital integrals with this behavior near z on every elliptic torus, and vanishing orbital integrals on all non-elliptic tori. Such an $\tilde f$ cannot be supported in the regular set. Producing it requires the germ expansion of [FK86, §8] at the central good element z, the restriction of the germs to the elliptic set, and a function realizing exactly the "identity" germ on the elliptic set (a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center; Remark 5.5.6, explains why it is not a pseudo-coefficient of a single genuine representation). This is item (D1) of §4.8.

4.4.6 Orbital integrals on an elliptic torus

This subsection proves, directly and for an arbitrary elliptic maximal torus (cyclic or not), the facts about orbital integrals of regular-supported functions used in §4.4: their properties (Lemma 4.4.12), their realization (Lemma 4.4.13), and the fact that traces of regular-supported functions depend only on orbital integrals (Lemma 4.4.14). It makes the local matching independent of [JC26a, Lemma 4.5], which is stated there only for cyclic extensions. The only input is Harish-Chandra's submersion principle and a properness statement (Lemma 4.4.10). The subsection introduces the maps Ψ and $\tilde{\Psi}$ and the open set $\overline{G}_T$, which are used only here.

Setting. F is a p-adic field containing μ_n, and G is $D_F^\times$ or $\mathrm{GL}(r,F)$, with D_F a central division algebra of degree r. The group $\overline{G}$ is an n-fold cover in which $\overline{Z} = p^{-1}(Z)$, $Z = F^\times$, is central, and $\tilde{\theta}$ is a genuine character of $\overline{Z}$. $T = E^\times \subset G$ is an elliptic maximal torus, with E/F separable of degree r, and $W = N_G(T)/T$ (a finite group). T^{reg} is the set of regular elements of T (here $T^{\mathrm{reg}} = E^\times \smallsetminus F^\times$ when r is prime), and $\overline{T}^{g,\mathrm{reg}}$ is the set of good regular elements of $\overline{T}$. Measures dg, dt on G/Z, T/Z and $d\dot{x}$ on $T\backslash G$ are as in §4.4.1. Note that $Z(\overline{T})$ is an open and closed subgroup of $\overline{T}$, since its image $F^\times E^{\times n}$ (Proposition 4.3.5) contains the open subgroup $E^{\times n}$.

Lemma 4.4.10. ***(E)***

(a) The map $\Psi : T\backslash G \times T^{\mathrm{reg}} \to G$, $(\dot{x},t) \mapsto x^{-1}tx$, is a local analytic isomorphism onto the open set G_T of elements conjugate into T^{reg}. W acts freely on the source by $w\cdot(\dot{x},t) = (T\tilde{w}x, wtw^{-1})$, and the fibers of Ψ are exactly the W-orbits.

(b) If $C \subset T^{\mathrm{reg}}$ is compact modulo Z and $\Omega \subset G$ is compact modulo Z, then $\{\dot{x} \in T\backslash G : x^{-1}Cx \cap \Omega \neq \emptyset\}$ is compact.

Proof. (a) For regular t, the differential of Ψ is surjective, because $\mathrm{Ad}(t^{-1}) - 1$ is invertible on a complement of $\mathrm{Lie}\,T$. Dimensions agree, so the inverse function theorem for analytic maps over F applies (Harish-Chandra's submersion principle [HC70], [Kot05, §§7, 13]). If $x^{-1}tx = y^{-1}t'y$, then $w := yx^{-1}$ conjugates t to t'; since t is regular, w normalizes T, which gives the fibers. Freeness holds because $w \in T$ if $Twx = Tx$. (b) This is [HC70, Chapter 4], [Kot05, §13]. On the D side it is trivial, since $T\backslash G$ is compact. □

Lemma 4.4.11. *The map $\tilde{\Psi} : T\backslash G \times \overline{T}^{g,\mathrm{reg}} \to \overline{G}$, $(\dot{x},\tilde{t}) \mapsto \tilde{x}^{-1}\tilde{t}\tilde{x}$, is well defined. It commutes with multiplication by $\overline{Z}$ (in particular by μ_n) on the second factor, and it is a covering map of degree $|W|$ onto the open set $\overline{G}_T := p^{-1}(G_T) \cap \{\text{good elements}\}$. Its fibers are the W-orbits for the action $w\cdot(\dot{x},\tilde{t}) = (T\tilde{w}x, \tilde{w}\tilde{t}\tilde{w}^{-1})$.*

Proof. Conjugation by $\tilde{x}$ does not depend on the lift. Replacing x by sx with $s \in T$ does not change $\tilde{x}^{-1}\tilde{t}\tilde{x}$, because $\tilde{t}$ is good. Similarly $\tilde{w}\tilde{t}\tilde{w}^{-1}$ does not depend on the lift $\tilde{w}$ of $w \in N_G(T)$ or on the representative of w modulo T. The fiber statement follows from Lemma 4.4.10(a) by lifting: if $\tilde{x}^{-1}\tilde{t}\tilde{x} = \tilde{y}^{-1}\tilde{t}'\tilde{y}$, then $w = yx^{-1}$ normalizes T and $\tilde{t}' = \tilde{w}\tilde{t}\tilde{w}^{-1}$. The induced map from the quotient by W is therefore injective. It is a local homeomorphism, because Ψ is and because $p : \overline{G} \to G$ and $p : \overline{T} \to T$ are finite coverings that are trivial over small open sets (the cover splits over small open compact subgroups). So it is a homeomorphism onto its open image. Since W is finite and acts freely, $\tilde{\Psi}$ is a covering of degree $|W|$ onto its image. The image consists of good elements, because conjugates of good elements are good, and it is all of $\overline{G}_T$, because every good regular element conjugate into T is conjugate to an element of $\overline{T}^{g,\mathrm{reg}}$. □

Lemma 4.4.12. *Let $\tilde f \in C_c^\infty(\overline{G},\tilde\theta)_{\mathrm{gen}}$ be supported in the regular set. Then $h(\tilde t) := F(\tilde t,\tilde f)$, for $\tilde t \in \overline{T}^{g,\mathrm{reg}}$, is admissible in the sense of §4.4.3: (i) locally constant; (ii) $h(z\tilde t) = \tilde\theta(z)^{-1}h(\tilde t)$; (iii) with support compact modulo $\overline{Z}$ inside $\overline{T}^{g,\mathrm{reg}}$; (iv) invariant under $N_G(T)$.*

Proof. (ii) holds because $\tilde f$ transforms by $\tilde\theta^{-1}$ and $\Delta(zt)=\Delta(t)$.

(iv) For $m \in N_G(T)$, substitute $x \mapsto mx$. The map $Tx \mapsto Tmx$ is a well-defined, measure-preserving bijection of $T\backslash G$, since $\mathrm{Ad}(m)$ preserves dt and W is finite. Also $\Delta(mtm^{-1})=\Delta(t)$.

(iii) The function $\delta(g) := |D(g)|_F\,|\det g|_F^{-(r-1)}$ is continuous, conjugation invariant and Z-invariant, and it vanishes exactly at the non-regular elements. Since $p(\operatorname{supp}\tilde f)$ is compact modulo Z and regular, $\delta \ge c > 0$ on it. If $h(\tilde t)\neq 0$, some conjugate of t lies in $p(\operatorname{supp}\tilde f)$, so $\delta(t)\ge c$. The set $B=\{t\in T : \delta(t)\ge c\}$ is closed, Z-stable and regular, and T/Z is compact, so B is compact modulo Z. Hence $\operatorname{supp} h \subseteq p^{-1}(B)\cap Z(\overline{T})$, which is compact modulo $\overline{Z}$ and contained in $\overline{T}^{g,\mathrm{reg}}$.

(i) Fix $\tilde t_0 \in \overline{T}^{g,\mathrm{reg}}$. Let U_T be an open compact subgroup of T with $t_0U_T \subset T^{\mathrm{reg}}$, over which the cover splits by some κ. Every element of $\overline{T}$ near $\tilde t_0$ is of the form $\tilde t_0\kappa(u)$ with $u\in U_T$ (these are good, $Z(\overline{T})$ being open). By Lemma 4.4.10(b) applied to $C = Zt_0U_T$ and $\Omega = p(\operatorname{supp}\tilde f)$, the set X of $\dot x$ with $x^{-1}Cx\cap\Omega\neq\emptyset$ is compact. The function $(\dot x,u)\mapsto \tilde f(\tilde\Psi(\dot x,\tilde t_0\kappa(u)))$ is locally constant on $X\times U_T$ and vanishes for $\dot x\notin X$. By compactness of X there is an open subgroup $U'\subset U_T$ such that its value at $(\dot x,u)$ equals its value at $(\dot x,1)$ for all $\dot x$ and all $u\in U'$. Integrating over $T\backslash G$ gives $\Phi(\tilde t_0\kappa(u),\tilde f)=\Phi(\tilde t_0,\tilde f)$ for $u\in U'$, and Δ is locally constant on T^{reg}. □

Lemma 4.4.13. *Let h be admissible on $\overline{T}^{g,\mathrm{reg}}$. Choose $\beta\in C_c^\infty(T\backslash G)$ with $\int\beta\,d\dot x = 1$, and define $\tilde f$ on $\overline{G}$ by*

$$\tilde f(\tilde y) := \frac{1}{|W|}\sum_{(\dot x,\tilde t)\in\tilde\Psi^{-1}(\tilde y)} \beta(\dot x)\,h(\tilde t)\,\Delta(t)^{-1} \quad (\tilde y\in\overline{G}_T), \qquad \tilde f(\tilde y):=0 \quad (\tilde y\notin\overline{G}_T).$$

Then $\tilde f\in C_c^\infty(\overline{G},\tilde\theta)_{\mathrm{gen}}$, its support is contained in $\tilde\Psi(\operatorname{supp}\beta\times\operatorname{supp}h)$ (good regular elements conjugate into $\overline{T}$), $F(\cdot,\tilde f)=h$ on $\overline{T}^{g,\mathrm{reg}}$, and $F(\tilde\delta,\tilde f)=0$ for every regular $\tilde\delta$ not conjugate into $\overline{T}$.

Proof. Genuine, with central character $\tilde\theta^{-1}$. We have $\tilde\Psi^{-1}(z\tilde y)=\{(\dot x,z\tilde t):(\dot x,\tilde t)\in\tilde\Psi^{-1}(\tilde y)\}$, $\Delta(zt)=\Delta(t)$, and (ii).

Smooth, with compact support modulo $\overline{Z}$. The set $K_f := \tilde\Psi(\operatorname{supp}\beta\times\operatorname{supp}h)$ is compact modulo $\overline{Z}$ (by (iii) and continuity), hence closed, and it is contained in the open set $\overline{G}_T$. On $\overline{G}_T$, $\tilde f$ is locally constant, because near any point $\tilde\Psi$ has $|W|$ continuous local inverses (Lemma 4.4.11) and β, h, Δ are locally constant. Outside the closed set K_f, $\tilde f$ vanishes. The two open sets $\overline{G}_T$ and $\overline{G}\smallsetminus K_f$ cover $\overline{G}$, since $K_f\subseteq\overline{G}_T$, and $\tilde f$ is locally constant on each. So $\tilde f$ is locally constant on $\overline{G}$; there is no boundary issue, because the support K_f stays inside the open set $\overline{G}_T$ and away from its boundary.

Orbital integrals. For $\tilde t_0\in\overline{T}^{g,\mathrm{reg}}$ and $x\in G$, the fiber of $\tilde\Psi$ over $\tilde x^{-1}\tilde t_0\tilde x$ is $\{(T\tilde wx,\tilde w\tilde t_0\tilde w^{-1}) : w\in W\}$. By (iv) and W-invariance of Δ,

$$\tilde f(\tilde x^{-1}\tilde t_0\tilde x) = \frac{h(\tilde t_0)}{|W|\Delta(t_0)}\sum_{w\in W}\beta(Twx).$$

Integrating over $T\backslash G$, and using that $Tx\mapsto Twx$ preserves $d\dot x$, gives $\Phi(\tilde t_0,\tilde f)=h(\tilde t_0)\Delta(t_0)^{-1}$, i.e. $F(\tilde t_0,\tilde f)=h(\tilde t_0)$. If $\tilde\delta$ is regular and not conjugate into $\overline{T}$, its conjugacy class does not meet $\overline{G}_T\supseteq\operatorname{supp}\tilde f$, so $\Phi(\tilde\delta,\tilde f)=0$. □

Lemma 4.4.14. *(using (E)) Let $\overline{\pi}$ be an irreducible admissible genuine representation of $\overline{G}$ with central character $\tilde{\theta}$, and let $\tilde{f}_1, \tilde{f}_2 \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ be supported in the regular set. If $F(\tilde{\gamma}, \tilde{f}_1) = F(\tilde{\gamma}, \tilde{f}_2)$ for every good regular $\tilde{\gamma}$, then $\operatorname{tr}\overline{\pi}(\tilde{f}_1) = \operatorname{tr}\overline{\pi}(\tilde{f}_2)$.*

Proof. The character $\Theta_{\overline{\pi}}$ is a locally constant function on the regular set with $\operatorname{tr}\overline{\pi}(\tilde{f}) = \int_{\overline{G}/\overline{Z}} \tilde{f}\,\Theta_{\overline{\pi}}$ for regular-supported $\tilde{f}$ ([FK86, §13], after Harish-Chandra) **(E)**. The product $\tilde{f}\Theta_{\overline{\pi}}$ is invariant under $\overline{Z}$ (hence under μ_n), so it is a function on G/Z. The Weyl integration formula on G/Z [HC70] **(E)** gives

$$\operatorname{tr}\overline{\pi}(\tilde{f}) = \sum_T \frac{1}{|W_T|} \int_{T/Z} \Delta(t)^2\, \Theta_{\overline{\pi}}(\tilde{t})\, \Phi(\tilde{t}, \tilde{f})\, dt,$$

summed over representatives T of the (finitely many) conjugacy classes of maximal tori, with $W_T = N_G(T)/T$. The measures are dg on G/Z, a Haar measure dt on T/Z for each T, and $d\dot{x} = dg/dt$ on $T\backslash G$ in Φ; the formula holds with these choices for any dt, since rescaling dt rescales $d\dot{x}$ inversely [HC70], [Kot05, §7]. Here $\Delta(t)^2 = |D(t)|_F |\det t|_F^{-(r-1)}$ is the Weyl discriminant $|\det(1 - \mathrm{Ad}(t^{-1}))|_{\mathfrak{g}/\mathfrak{t}}|_F$. We used $\Theta_{\overline{\pi}}(\tilde{x}^{-1}\tilde{t}\tilde{x}) = \Theta_{\overline{\pi}}(\tilde{t})$. At a non-good regular $\tilde{t}$ the integrand vanishes, because $\Theta_{\overline{\pi}}(\tilde{t}) = 0$ (conjugating by $\overline{T}$ multiplies $\tilde{t}$ by some $\zeta \neq 1$). So the trace depends only on the values of $F(\cdot, \tilde{f}) = \Delta\Phi$ at good regular elements. □

4.5 Global covers and rational lifts

This section constructs the global GL-side cover inside the same ambient adelic cover as the D-side cover of [JC26a] (§4.5.1). It then proves that, under the local identifications of §§4.2–4.4, the rational lift of an elliptic element on the D side corresponds to the rational lift of the corresponding element on the GL side, so that the elliptic terms of the two trace formulas agree term by term (§4.5.2). The measure conventions are fixed in §4.5.3, and §4.5.4 supplies the sup-norm bound for cusp forms on the cover that the trace formula of §4.6 uses. The section depends on §§4.2–4.4 and on the global normalization of the Kazhdan–Patterson cocycle [FK86, §2], [KP84, §0.1]. It introduces the global covers $\overline{\mathscr{G}}(\mathbb{A}_L)$, $\overline{G}_D(\mathbb{A}_K)$, $\overline{G}(\mathbb{A}_K)$ and $\overline{Z}(\mathbb{A}_K)$, the splittings ι and κ_v^*, the global central character $\tilde{\theta}$ (unitary from here on), and the finite set of places S_D.

4.5.1 The global covers

We define the global ambient cover and the two global covers as subgroups, and list the properties needed for the trace formulas. The splitting conventions of Lemma 4.5.1 and the unitarity of $\tilde{\theta}$ are used in all later sections.

Let $\overline{\mathscr{G}}(\mathbb{A}_L)$ be the cover of $\mathrm{GL}(r, \mathbb{A}_L)$ with cocycle $\tau_L \cdot (\det, \det)_{\mathbb{A}_L}^{c_v}$, where $\tau_L = \prod_w \tau_{L,w}$ is the normalized Kazhdan–Patterson cocycle of [FK86, §2] (parameter 0) and $(a,b)_{\mathbb{A}_L} = \prod_w (a_w, b_w)_{L_w}$; here $c_v = r'c$ is independent of v. It is the restricted product of the local covers $\overline{\mathscr{G}}_v$ with respect to the splittings $\kappa^*_{\overline{\mathscr{G}}_v}$ over $\mathrm{GL}(r, \mathscr{O}_{L_v})$ for $v \nmid n\infty$, modulo $\{(\zeta_v) : \prod \zeta_v = 1\}$. Let $\iota : \mathrm{GL}(r, L) \to \overline{\mathscr{G}}(\mathbb{A}_L)$ be the rational splitting described in Lemma 4.5.1.

We first make the section and splitting conventions explicit, since the normalized and unnormalized cocycle models describe the same splittings by different formulas.

Lemma 4.5.1 (Splitting conventions)**.** *(using **(E)**) For a place w of L let $\sigma_w = \sigma_{L_w}$ be the Kazhdan–Patterson cocycle (parameter 0) of $\mathrm{GL}(r,L_w)$, and $s_\sigma(g) = (g,1)_\sigma$ the corresponding section. For $w \nmid n\infty$ let $\kappa_w : \mathrm{GL}(r,L_w) \to \mu_n$ be the function of [FK86, §2]: on $K_w = \mathrm{GL}(r,\mathscr{O}_{L_w})$ it satisfies*

$$\sigma_w(k,k') = \kappa_w(kk')\,\kappa_w(k)^{-1}\kappa_w(k')^{-1} \qquad (k,k' \in K_w), \tag{4.6}$$

and it is extended to $\mathrm{GL}(r,L_w)$ as in [FK86, §2] (with $\kappa_w = 1$ for $w \mid n\infty$). Put

$$\tau_w(g,h) = \sigma_w(g,h)\,\kappa_w(g)\,\kappa_w(h)\,\kappa_w(gh)^{-1}, \tag{4.7}$$

let $s_\tau(g) = (g,1)_\tau$ be the section of the τ-model, and let $\phi_w : (g,\zeta)_\sigma \mapsto (g,\zeta\,\kappa_w(g)^{-1})_\tau$. Then:

(a) ϕ_w is an isomorphism of extensions over the identity, trivial on μ_n.

(b) Compact splitting. *For $w \nmid n\infty$, $\tau_w \equiv 1$ on $K_w \times K_w$, so $\kappa_w^* := s_\tau|_{K_w}$, i.e. $\kappa_w^*(k) = (k,1)_\tau$, is a homomorphism. In the σ-model it is $k \mapsto (k,\kappa_w(k))_\sigma$, i.e. $\phi_w^{-1} \circ \kappa_w^* = s_\sigma \cdot \kappa_w$.*

(c) Rational splitting. *For $x \in \mathrm{GL}(r,L)$, $\kappa_w(x) = 1$ for almost all w [KP84, Prop. 0.1.3], so $\kappa_L(x) := \prod_w \kappa_w(x)$ is defined. The map*

$$\iota(x) = s_\tau(x)\,\kappa_L(x)^{-1} = \big((x,\kappa_w(x)^{-1})_\tau\big)_w$$

is a homomorphism $\mathrm{GL}(r,L) \to \overline{\mathscr{G}}(\mathbb{A}_L)$ splitting p. Componentwise, $\iota(x)_w = \phi_w(s_\sigma(x))$, i.e. in the σ-models the rational splitting is $x \mapsto (x,1)_\sigma$ at every place.

(d) For $x \in \mathrm{GL}(r,L)$, $\iota(x)_w = \kappa_w^*(x)$ if and only if $x \in K_w$ and $\kappa_w(x) = 1$, which holds for almost all w.

(e) The twisting factor $(\det,\det)_{\mathbb{A}_L}^{c_v}$ does not affect (a)–(d): it is a common factor of both models, it is trivial on $\mathrm{GL}(r,L) \times \mathrm{GL}(r,L)$ by Hilbert reciprocity, and it is trivial on $K_w \times K_w$ for $w \nmid n$.

Proof. (a) Using (4.7),

$$\begin{aligned}\phi_w(g,\zeta)\,\phi_w(h,\eta) &= (gh,\ \zeta\eta\,\kappa_w(g)^{-1}\kappa_w(h)^{-1}\tau_w(g,h))\\ &= (gh,\ \zeta\eta\,\sigma_w(g,h)\,\kappa_w(gh)^{-1}) = \phi_w\big((g,\zeta)(h,\eta)\big).\end{aligned}$$

(b) By (4.6) and (4.7), $\tau_w(k,k') = 1$, so $s_\tau(k)s_\tau(k') = \tau_w(k,k')s_\tau(kk') = s_\tau(kk')$. Also $\phi_w((k,\kappa_w(k))_\sigma) = (k,1)_\tau$. (c) The global τ-cocycle is $\tau = \prod_w \tau_w$, a finite product for each pair [FK86, §2]. For $x,x' \in \mathrm{GL}(r,L)$, the values $\sigma_w(x,x')$ are products of Hilbert symbols of elements of $L^\times$ [KP84, §0.1], [BLS99], so they equal 1 for almost all w and $\prod_w \sigma_w(x,x') = 1$ by Hilbert reciprocity. Hence

$$\iota(x)\iota(x') = \tau(x,x')\kappa_L(x)^{-1}\kappa_L(x')^{-1}s_\tau(xx') = \Big(\prod_w \sigma_w(x,x')\Big)\kappa_L(xx')^{-1}s_\tau(xx') = \iota(xx').$$

This is the statement of [FK86, §2] that $x \mapsto s(x)/\kappa(x)$ is a homomorphism on $G(F)$, with $s = s_\tau$. The componentwise formula is $\phi_w((x,1)_\sigma) = (x,\kappa_w(x)^{-1})_\tau$. (d) Compare (b) and (c). (e) is clear. □

Remark 4.5.2 (Conventions)**.** Throughout, ι and κ_v^* are the splittings of Lemma 4.5.1: $\iota(x) = s(x)/\kappa_L(x)$ and $\kappa_w^*(k) = s(k)$ in the normalized model. These are the conventions of [JC26a, Lemma 6.1(a),(d)].

Remark 4.5.3 (No circularity with [JC26a])**.** [JC26a] cites Proposition 4.2.4 of the present chapter for the local identification of covers. That proposition is purely local. Its proof uses only the explicit restriction of the Kazhdan–Patterson cocycle to the diagonal torus [FK86, §2], the Matsumoto–Moore classification [Mat69], [Moo68], and the commutator formula [Mil71]. It uses no result of [JC26a]. Conversely, the present chapter uses from [JC26a] only the definition of the global cover of $D_{\mathbb{A}}^{\times}$ and the local lemmas cited by number (all from [JC26a, §§2–4]). It uses none of [JC26a]'s global theorems.

Unitarity. From now on $\tilde{\theta} : \overline{Z}(\mathbb{A}_K) \to \mathbb{C}^{\times}$ is a genuine *unitary* character trivial on $\iota(K^{\times})$, and its local components $\tilde{\theta}_v$ are unitary. This is needed for the L^2 spaces, adjoints, formal degrees and Schur orthogonality used in §§4.6–4.7.

Definition 4.5.4 (Global covers $\overline{G}_D(\mathbb{A}_K)$, $\overline{G}(\mathbb{A}_K)$)**.** $\overline{G}_D(\mathbb{A}_K) := p^{-1}(\iota_D(G_D(\mathbb{A}_K)))$ and $\overline{G}(\mathbb{A}_K) := p^{-1}(\iota_{\mathrm{GL}}(\mathrm{GL}(r,\mathbb{A}_K)))$, subgroups of $\overline{\mathscr{G}}(\mathbb{A}_L)$, and $\overline{Z}(\mathbb{A}_K) := p^{-1}(\mathbb{A}_K^{\times})$.

Example 4.5.5 (Continuing Example 1.3.1: the global covers)**.** In the running example $L = \mathbb{Q}(\zeta_{12})$, $r = 2$ and $n = 3$, so $\overline{\mathscr{G}}(\mathbb{A}_L)$ is a 3-fold cover of $\mathrm{GL}(2,\mathbb{A}_L)$. The group $\overline{G}_D(\mathbb{A}_K)$ is the preimage of $D_{\mathbb{A}}^{\times}$, embedded by ι_D, and $\overline{G}(\mathbb{A}_K)$ is the preimage of $\mathrm{GL}(2,\mathbb{A}_K)$, embedded diagonally. At $v = \mathfrak{p}_3$ and $v = \mathfrak{p}_7$ the group $D_v^{\times}$ is the unit group of a quaternion division algebra. It is compact modulo its center, while $\mathrm{GL}(2,K_v)$ is not. At every other place $D_v^{\times} \cong \mathrm{GL}(2,K_v)$, and Proposition 4.2.7 identifies the local covers $\overline{G}_D(K_v)$ and $\overline{G}_v$. So the two global covers are built from the same local data away from the two places of S.

Proposition 4.5.6. *(a) The cocycle of $\overline{\mathscr{G}}(\mathbb{A}_L)$ restricts on $G_D(\mathbb{A}_K)$ to $\beta_\tau := \tau_L \cdot (\mathrm{Nrd},\mathrm{Nrd})^c$. This is the cocycle β of [JC26a, equation (4)] when σ_L there is read as the normalized cocycle τ_L, which is the reading under which [JC26a]'s global cover is defined and split over $\mathrm{GL}(r,L)$. The rational splitting $\iota|_{G_D(K)}$ is that of [JC26a, Lemma 6.1(a),(d)], and the compact splittings κ_v^* are those of [JC26a, Lemma 6.1(d)] (Remark 4.5.2).*

(b) ι splits $\overline{G}(\mathbb{A}_K)$ over $\mathrm{GL}(r,K)$, and $\overline{G}_D(\mathbb{A}_K)$ over $G_D(K)$.

(c) $\overline{Z}(\mathbb{A}_K)$ is a common subgroup of $\overline{G}_D(\mathbb{A}_K)$ and $\overline{G}(\mathbb{A}_K)$, central in both, and $\iota(K^{\times})$ is a common subgroup. Hence a genuine unitary character $\tilde{\theta}$ of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^{\times})$ serves both sides.

(d) For almost all v, $\kappa_{\mathscr{G}_v}^*$ restricts to splittings κ_v^* of $\overline{G}_D(K_v)$ over $G_D(\mathscr{O}_v)$ and of $\overline{G}_v$ over $\mathrm{GL}(r,\mathscr{O}_v)$; these agree with [JC26a, Lemma 6.1(d)] on the D side and are carried to each other by ψ_v (Proposition 4.2.7(c)). For γ rational and integral outside a finite set, $\iota(\gamma)_v = \kappa_v^*(\gamma)$ for almost all v. The covers are the restricted products of their local components with respect to these splittings.

(e) Let N be the unipotent radical of a parabolic subgroup of $\mathrm{GL}(r)$ defined over K. There is a unique splitting s_N of $\overline{G}(\mathbb{A}_K)$ over $N(\mathbb{A}_K)$, and $s_N|_{N(K)} = \iota|_{N(K)}$.

(f) The local components of $\overline{G}(\mathbb{A}_K)$ are the covers $\overline{G}_v$ of Definition 4.2.1, isomorphic to $\overline{G}^{(c_v)}$ after relabeling (Proposition 4.2.4(b)).

Proof. (a) As in Proposition 4.2.4(a), $(\det,\det)^{c_v}_{\mathbb{A}_L}$ restricts on $G_D(\mathbb{A}_K)$ to $\prod_v(\mathrm{Nrd},\mathrm{Nrd})^{rc_v}_{K_v} = (\mathrm{Nrd},\mathrm{Nrd})^{c}$, so the restricted cocycle is β_τ. In [JC26a], σ_L is described as the Kazhdan–Patterson cocycle of $\mathrm{GL}(r,\mathbb{A}_L)$ "which splits over $\mathrm{GL}(r,L)$", with rational splitting $x \mapsto s(x)/\kappa_L(x)$. A global product of local cocycles is defined only after the normalization (4.7) [FK86, §2]. So σ_L in [JC26a] is the normalized cocycle τ_L, and its rational splitting is ι of Lemma 4.5.1(c). The compact splittings are those of Lemma 4.5.1(b) (Remark 4.5.2). Central characters are defined on our $\overline{Z}(\mathbb{A}_K)$ and are not transported.

(b) τ_L is split over $\mathrm{GL}(r,L)$ by ι (Lemma 4.5.1(c)), and $(\det g,\det h)_{\mathbb{A}_L} = 1$ for $g,h \in \mathrm{GL}(r,L)$ by Hilbert reciprocity for L. So ι splits $\overline{\mathscr{G}}(\mathbb{A}_L)$ over $\mathrm{GL}(r,L)$, and hence over its subgroups $\iota_{\mathrm{GL}}(\mathrm{GL}(r,K))$ and $\iota_D(G_D(K))$.

(c) ι_D and ι_{GL} agree on $\mathbb{A}_K^\times$, and on $K^\times$ the splitting is ι in both cases. Centrality is Lemma 4.2.3 at each place.

(d) For almost all v (namely $v \nmid n\infty$, v unramified in L, and the integrality conditions of Proposition 4.2.7(c)), $\iota_D(G_D(\mathscr{O}_v))$ and $\iota_{\mathrm{GL}}(\mathrm{GL}(r,\mathscr{O}_v))$ lie in $\mathrm{GL}(r,\mathscr{O}_{L_v})$. There $\kappa^*_{\mathscr{G}_v} := \prod_{w|v}\kappa^*_w$, with $\kappa^*_w(k) = (k,1)_\tau$, is a homomorphism by Lemma 4.5.1(b),(e). This agrees with [JC26a, Lemma 6.1(d)] (Remark 4.5.2). Compatibility with ψ_v is Proposition 4.2.7(c). The statement about $\iota(\gamma)$ is Lemma 4.5.1(d) in $\overline{\mathscr{G}}(\mathbb{A}_L)$, hence in the subgroups.

(e) Locally, the section s of [FK86, §2] splits the cover over $N(K_v)$. A splitting is unique, because $N(K_v)$ is divisible (it is unipotent in characteristic 0), so it has no nontrivial homomorphisms to μ_n. At almost all v the local splitting agrees with κ^*_v on $N(\mathscr{O}_v)$, again by uniqueness: $N(\mathscr{O}_v)$ is pro-p with $p \nmid n$. Hence the local splittings define a splitting of the restricted product over $N(\mathbb{A}_K)$. It is unique, since $N(\mathbb{A}_K)$ is divisible (for X in the Lie algebra with integral components almost everywhere, X/n is again such). The restriction to $N(K)$ is a splitting of $\overline{G}(\mathbb{A}_K)$ over the divisible group $N(K)$, as is $\iota|_{N(K)}$, so they agree.

(f) is the definition of the local components together with Proposition 4.2.4. □

4.5.2 Rational lifts correspond

We prove that the local identifications of §§4.2–4.4 carry the rational lift of a rational elliptic element of $G_D(K)$ to the rational lift of the corresponding element of $\mathrm{GL}(r,K)$, so that the elliptic terms of the two trace formulas agree.

For $\gamma \in G_D(K)$ and genuine factorizable $\tilde{f}_D = \otimes_v \tilde{f}_{D,v}$, write $F(\iota(\gamma),\tilde{f}_D) := \prod_v F(x_v,\tilde{f}_{D,v})$, where (x_v) is any representative of $\iota(\gamma)$ in the restricted product. This is well defined because the product of the local phases is 1 [JC26a, Lemma 6.1, Remark (3)]. Similarly on the GL side.

Theorem 4.5.7. *Let $E \subset D$ be a maximal subfield, $j : E \hookrightarrow M_r(K)$ a K-embedding, and $\gamma \in E^\times$ regular and good at every place on the D side. Fix α_v and ψ_v for $v \notin S$ as in Proposition 4.2.7. Let $\tilde{f}_D = \otimes \tilde{f}_{D,v}$ and $\tilde{f} = \otimes \tilde{f}_v$ be genuine and factorizable with $\tilde{f}_{D,v} = \tilde{f}_v \circ \psi_v$ for $v \notin S$ (with $dg_{D,v} = \psi_v^* dg_v$) and $\tilde{f}_{D,v}$, $\tilde{f}_v$ matching for $v \in S$. Then $j(\gamma)$ is good at every place on the* GL *side, and*

$$F(\iota(\gamma),\tilde{f}_D) = F(\iota(j(\gamma)),\tilde{f}).$$

Proof. A rational conjugating element. As in Theorem 4.3.10(a), L^r is a free module of rank one over the étale L-algebra $E \otimes_K L$ via both ι_D and $\iota_{\mathrm{GL}} \circ j$, so there is $g \in \mathrm{GL}(r,L)$ with $\mathrm{Ad}(g) \circ \iota_D = \iota_{\mathrm{GL}} \circ j$ on E. Since ι is a homomorphism on $\mathrm{GL}(r,L)$, $\mathrm{Ad}(\iota(g))$ maps $\iota(\iota_D\gamma)$ to $\iota(\iota_{\mathrm{GL}} j\gamma)$. Conjugation in

the restricted product is componentwise, so if (x_v) represents $\iota(\gamma)$ on the D side, then $(\mathrm{Ad}(\tilde{g}_v)x_v)$ represents $\iota(j\gamma)$ on the GL side, where $\tilde{g}_v$ are any lifts of the components of g. Hence

$$F(\iota(j\gamma),\tilde{f}) = \prod_v F(\mathrm{Ad}(\tilde{g}_v)x_v,\tilde{f}_v).$$

Places $v \in S$. Here $E_v = E \otimes_K K_v$ is a maximal subfield of D_v, and g conjugates $\iota_D|_{E_v}$ to $\iota_{\mathrm{GL}} \circ j$. So $\mathrm{Ad}(\tilde{g}_v)|_{\overline{E}_{D,v}} \in \mathrm{Iso}_{\overline{Z}}$, and by Theorem 4.3.10(b) it equals $\jmath_{E_v,j}$ on the good element x_v. Hence $\mathrm{Ad}(\tilde{g}_v)x_v$ is good, and by (M1) $F(\mathrm{Ad}(\tilde{g}_v)x_v,\tilde{f}_v) = (-1)^{r-1}F(x_v,\tilde{f}_{D,v})$.

Places $v \notin S$. The two embeddings $\alpha_v|_{E_v}$ and j of the étale K_v-algebra E_v into $M_r(K_v)$ are conjugate by some $h_v \in \mathrm{GL}(r,K_v)$ (again by the free-module argument). Then $\mathrm{Ad}(\tilde{h}_v)\circ\psi_v$ and $\mathrm{Ad}(\tilde{g}_v)$, restricted to $\overline{E}_{D,v} = p^{-1}(\iota_D(E_v^\times))$, are isomorphisms onto $p^{-1}(\iota_{\mathrm{GL}}j(E_v^\times))$ lying over j. Both are the identity on μ_n and on the central $\overline{Z}_v$. As in Theorem 4.3.10(b), they differ by a character of $E_v^\times$ with values in μ_n, trivial on $K_v^\times$. Such a character is trivial on $K_v^\times E_v^{\times n}$, which contains the good regular elements of $\overline{E}_{D,v}$ [FK86, §3, Proposition] **(E)**. So they agree at x_v. Hence $\mathrm{Ad}(\tilde{g}_v)x_v$ is good and

$$F(\mathrm{Ad}(\tilde{g}_v)x_v,\tilde{f}_v) = F(\mathrm{Ad}(\tilde{h}_v)\psi_v(x_v),\tilde{f}_v) = F(\psi_v(x_v),\tilde{f}_v) = F(x_v,\tilde{f}_{D,v}),$$

using conjugation invariance and the fact that ψ_v is an isomorphism of groups carrying $dg_{D,v}$ to dg_v and torus measures to torus measures.

Conclusion. Multiplying over v gives $F(\iota(j\gamma),\tilde{f}) = (-1)^{(r-1)|S|}F(\iota(\gamma),\tilde{f}_D)$, and $(-1)^{(r-1)|S|} = 1$, since $r-1$ is even for odd r and $|S|$ is even for $r=2$. □

Corollary 4.5.8. *The map $\gamma \mapsto j(\gamma)$ induces a bijection between $G_D(K)$-conjugacy classes (modulo $Z(K)$) of regular γ that are good at every place, and $\mathrm{GL}(r,K)$-conjugacy classes (modulo $Z(K)$) of regular elliptic γ' that are good at every place and whose field $K(\gamma')$ embeds in D. For γ' regular elliptic whose field does not embed in D, $F(\iota(\gamma'),\tilde{f}) = 0$ whenever $\tilde{f}_v$ satisfies (M2) at all $v \in S$.*

Proof. Rational regular classes are determined by characteristic polynomials (Skolem–Noether on both sides), and a degree-r field embeds in D if and only if it is a field locally at every $v \in S$ (Albert–Brauer–Hasse–Noether). Goodness corresponds place by place by Theorem 4.5.7. If $K(\gamma')$ does not embed in D, then γ' is non-elliptic at some $v \in S$, and the factor at v vanishes by (M2) (or because γ' is not good there). □

4.5.3 Measures

We fix the global measures and explain why no further compatibility between the measures on the two groups is needed. The finite set S_D of places defined here is used in §§4.6–4.7.

Fix a finite set $S_D \supseteq S$ of places containing the archimedean places, the places above n, the places ramified in L or where $\tilde{\theta}$ is ramified, and the places where the integrality conditions of Proposition 4.2.7(c) fail. For $v \notin S_D$ normalize dg_v on $\mathrm{GL}(r,K_v)/K_v^\times$ by $\mathrm{vol}(\mathrm{GL}(r,\mathcal{O}_v)K_v^\times/K_v^\times) = 1$, and put $dg_{D,v} := \psi_v^* dg_v$; it gives $G_D(\mathcal{O}_v)K_v^\times/K_v^\times$ volume 1 by Proposition 4.2.7(c). For $v \in S_D \smallsetminus S$ take any dg_v and $dg_{D,v} := \psi_v^* dg_v$. For $v \in S$ take any dg_v and $dg_{D,v}$; matching at v is defined with respect to these. Let $dg = \prod dg_v$ and $dg_D = \prod dg_{D,v}$. For a rational maximal subfield E use one product measure $dt = \prod dt_v$ on $\mathbb{A}_E^\times/\mathbb{A}_K^\times$, normalized at almost all places by $\mathrm{vol}(\mathcal{O}_{E_v}^\times K_v^\times/K_v^\times) = 1$, on *both* sides (via ι_D and $\iota_{\mathrm{GL}}\circ j$).

With these choices, the elliptic term of γ on either side is

$$|W_\gamma|^{-1}\,\mathrm{vol}(\mathbb{A}_K^\times E^\times\backslash\mathbb{A}_E^\times;dt)\,F(\iota(\gamma),\cdot),$$

the local orbital integrals being taken with respect to dg_v/dt_v or $dg_{D,v}/dt_v$. The volume factor is literally the same number on the two sides, since it is the same group with the same measure. The spectral sides are $\sum m(\overline{\pi})\,\mathrm{tr}\,\overline{\pi}(\tilde{f};dg)$ and $\sum m(\overline{\pi}_D)\,\mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D;dg_D)$, with the same dg and dg_D used in the orbital integrals. So the comparison requires only that matching at $v\in S$ be defined with respect to the same dg_v and $dg_{D,v}$ that are used in the trace formulas. A canonical normalization (for example, transfer of measures between inner forms, or Euler–Poincaré measures) is needed only to state local character relations with the correct constants (§4.8). That the elliptic term has the stated form, with the class-counting constant $1/|W_\gamma|$, is proved in Propositions 4.6.1 and 4.6.2.

4.5.4 Cusp forms on finite central covers

This subsection supplies the analytic input of Lemma 4.6.4(a): the uniform sup-norm bound for $r(\phi)$ on cusp forms, for the cover $\overline{G}(\mathbb{A}_K)$. For $\mathrm{GL}(r,\mathbb{A})$ the bound is classical [GGPS69, Ch. 3], [God66], [MW95, I.2]. Moeglin and Waldspurger formulate their theory so as to include finite central coverings, and their theory of Eisenstein series is used for covering groups in this generality, for example by Gao [Gao12] for Mp_4. Rather than rely on that alone, we list the properties (R1)–(R5) of the group that the classical proof uses, and we verify each of them for $\overline{G}(\mathbb{A}_K)$. The subsection depends on §4.5 (Proposition 4.5.6(e)) and on Lemma 4.6.4.

Lemma 4.5.9. *(using **(E)**) Let ϕ be a genuine test function on $\overline{G}(\mathbb{A}_K)$ transforming by $\tilde{\theta}^{-1}$ under $\overline{Z}(\mathbb{A}_K)$ (K_∞-finite at the archimedean places). There is C_ϕ with $\sup_x|(r(\phi)\psi)(x)|\le C_\phi\|\psi\|_2$ for all $\psi\in L^2_{\mathrm{cusp}}$.*

Structure of the classical proof. Fix a Siegel set $\mathfrak{S}$ for $\mathrm{GL}(r,\mathbb{A})$ relative to the upper triangular Borel subgroup, so that $\mathrm{GL}(r,\mathbb{A})=\mathrm{GL}(r,K)Z(\mathbb{A})\mathfrak{S}$. For $x\in\mathfrak{S}$, write $(r(\phi)\psi)(x)=\int_X K_\phi(x,y)\psi(y)\,dy$. Since ψ is cuspidal, the constant-term kernels $K_{\phi,P}(x,y)=\int_{N_P(K)\backslash N_P(\mathbb{A})}K_\phi(s(n)x,y)\,dn$ (P standard, proper) integrate to 0 against ψ. One may therefore replace K_ϕ by the modified kernel $K_\phi^\natural:=\sum_P(-1)^{\dim A_P/A_G}K_{\phi,P}$, the sum over all standard P including G. One proves that $\sup_{x\in\mathfrak{S}}\|K_\phi^\natural(x,\cdot)\|_{L^2(X)}<\infty$, whence the bound by Cauchy–Schwarz. The estimate of $K_\phi^\natural$ uses Poisson summation along the abelian quotients of the unipotent radicals, with bounds on the non-constant Fourier coefficients obtained by integration by parts at the archimedean places and from the level at the finite places, together with the contraction of N_P by the torus part of $\mathfrak{S}$ [GGPS69, Ch. 3, §2], [MW95, I.2]. The proof uses exactly the following properties (R1)–(R5) of the group and the test function.

(R1) *Reduction theory and finite volume of X.* For the cover, the quotient

$$X=\iota(\mathrm{GL}(r,K))\overline{Z}(\mathbb{A}_K)\backslash\overline{G}(\mathbb{A}_K)$$

is homeomorphic to $Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$, with matching measures (Lemma 4.6.4, proof). For genuine $\tilde{\theta}$-equivariant ψ, the function $|\psi|$ is a function on X. So Siegel sets, finiteness of volume and the inner product are those of the linear group.

(R2) *Unipotent splittings.* For every standard unipotent radical N there is a unique splitting s_N of $N(\mathbb{A})$, which agrees with ι on $N(K)$ (Proposition 4.5.6(e)). Hence constant terms and Fourier coefficients along $N(K)\backslash N(\mathbb{A})$ of genuine automorphic functions are defined, and cuspidality has its usual meaning.

(R3) *Equivariance of the splittings.* For $m \in \mathrm{GL}(r,\mathbb{A})$ normalizing N and any lift $\tilde{m}$, $\mathrm{Ad}(\tilde{m}) \circ s_N = s_N \circ \mathrm{Ad}(m)$, since both sides are splittings of $N(\mathbb{A})$ and the splitting is unique. This is what is used when the torus part of $\mathfrak{S}$ is conjugated past N.

(R4) *Archimedean places.* K is totally imaginary, and at complex places all Hilbert symbols are trivial, so the cover is canonically split over $\mathrm{GL}(r,K_v)$. Hence derivatives and integration by parts along $N(K_\infty)$ are those of the linear group.

(R5) *Test functions.* For $\tilde{a},\tilde{b} \in \overline{G}(\mathbb{A})$, the function $n \mapsto \phi(\tilde{a}\, s_N(n)\, \tilde{b})$ on $N(\mathbb{A})$ is a Schwartz–Bruhat function. Its seminorms (support, level, and finitely many archimedean derivatives) are bounded, uniformly for $\tilde{a},\tilde{b}$ in compact sets, by finitely many seminorms of ϕ. Indeed, s_N is a continuous homomorphism and the cover is split over small open subgroups and at the archimedean places. The classical proof uses no other property of ϕ; in particular, it never multiplies values of ϕ at different points.

Proof of Lemma 4.5.9. By (R1)–(R5), each step of the classical proof applies verbatim to $\overline{G}(\mathbb{A}_K)$. We list where each property enters.

- The reduction to $x \in \mathfrak{S}$ and the final Cauchy–Schwarz step use (R1).
- The vanishing of $\int K_{\phi,P}(x,y)\psi(y)\,dy$ for cuspidal ψ, and Poisson summation, use (R2).
- The Fourier-coefficient bounds use (R3)–(R5). These bounds depend only on the seminorms in (R5) and on the position of x in $\mathfrak{S}$.

The resulting bound $\sup_{x\in\mathfrak{S}} \|K_\phi^\natural(x,\cdot)\|_{L^2(X)} \le C_\phi$ gives the lemma. □

4.6 The two trace formulas and their comparison

This section computes the geometric side of the trace formula on the D side, including the class-counting constants (§4.6.1). It then proves the variant of the simple trace formula of [FK86, §18] that we need on the GL side, with the supercusp form at a place outside S (§4.6.2), and shows that the two trace formulas agree for matching test functions (§4.6.3). Measures are as in §4.5.3, and $\tilde{\theta}$ is a genuine unitary character of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^\times)$ (Proposition 4.5.6(c)). The section depends on §4.5, on Theorem 4.4.7, and on the analytic inputs listed in Remark 4.6.8 (with §4.5.4). It introduces W_γ, $c(\gamma)$ and the class sets $\mathscr{O}_H$ below, the auxiliary places w_0 and u, and admissible pairs of test functions (Definition 4.6.5).

For $H \in \{G_D, \mathrm{GL}(r)\}$ and a regular elliptic $\gamma \in H(K)$, put $E = K(\gamma)$ (a field of degree r), so that $H_\gamma = E^\times$, and

$$W_\gamma := \{\sigma \in \mathrm{Aut}(E/K) : \sigma(\gamma) \in K^\times \gamma\}, \qquad c(\gamma) := \frac{\mathrm{vol}(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt)}{|W_\gamma|}.$$

Write $\mathscr{O}_H$ for the set of $H(K)$-conjugacy classes in $Z(K)\backslash H(K)$ of regular elliptic elements that are good at every place. (On the D side every regular element is elliptic.)

4.6.1 The D side

We compute the trace of the right regular representation on the compact quotient. This is the untwisted case of [JC26a, Prop. 6.6], except that all maximal subfields of D now contribute, not only L.

Proposition 4.6.1. *Let $\tilde{f}_D = \otimes_v \tilde{f}_{D,v}$ be smooth, genuine, compactly supported modulo $\overline{Z}(\mathbb{A}_K)$, with $\tilde{f}_D(z\tilde{g}) = \tilde{\theta}(z)^{-1}\tilde{f}_D(\tilde{g})$, and let $\tilde{R}_D$ be the right regular representation on $L^2(\iota(G_D(K))\backslash\overline{G}_D(\mathbb{A}_K),\tilde{\theta})$. Then $\tilde{R}_D(\tilde{f}_D)$ is of trace class, and*

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D)\operatorname{tr}\overline{\pi}_D(\tilde{f}_D) = \operatorname{vol}(Z(\mathbb{A})G_D(K)\backslash G_D(\mathbb{A}))\,\tilde{f}_D(1) + \sum_{[\gamma]\in\mathscr{O}_{G_D}} c(\gamma)F(\iota(\gamma),\tilde{f}_D).$$

The left side converges absolutely, and the sum on the right is finite. If $\tilde{f}_{D,u}$ vanishes near $\overline{Z}_u$ for some $u \in S$, the first term on the right vanishes.

Proof. The quotient $Z(\mathbb{A})G_D(K)\backslash G_D(\mathbb{A})$ is compact, and the cover is finite, so $\tilde{R}_D(\tilde{f}_D)$ is an integral operator with smooth kernel on a compact space. It is therefore of trace class, and the spectral expansion converges absolutely **(E)** (Gelfand–Graev–Piatetski-Shapiro). Its kernel is $K(x,y) = \sum_{\gamma\in Z(K)\backslash G_D(K)} \tilde{f}_D(x^{-1}\iota(\gamma)y)$. This is well defined because $\tilde{f}_D(\iota(z)\cdot) = \tilde{\theta}(\iota(z))^{-1}\tilde{f}_D = \tilde{f}_D$ for $z \in K^\times$. The trace is $\int K(x,x)\,dx$ over $Z(\mathbb{A})G_D(K)\backslash G_D(\mathbb{A})$. As in [FK86, §18], only finitely many classes contribute, since characteristic polynomials of rational elements form a discrete set.

Group the γ into classes $[\gamma]$ for the conjugation action of $G_D(K)$ on $Z(K)\backslash G_D(K)$. The stabilizer of $Z(K)\gamma$ is $H^Z_\gamma(K) := \{\delta : \delta\gamma\delta^{-1} \in K^\times\gamma\}$. Since ι is a homomorphism on $G_D(K)$, the contribution of $[\gamma]$ is

$$\int_{Z(\mathbb{A})H^Z_\gamma(K)\backslash G_D(\mathbb{A})} \tilde{f}_D(x^{-1}\iota(\gamma)x)\,dx = \frac{1}{[H^Z_\gamma(K) : Z(K)G_\gamma(K)]}\int_{Z(\mathbb{A})G_\gamma(K)\backslash G_D(\mathbb{A})} \tilde{f}_D(x^{-1}\iota(\gamma)x)\,dx.$$

If γ is central, then $\gamma \in Z(K)$, the class is that of 1, and the contribution is $\operatorname{vol}\cdot\tilde{f}_D(1)$. Otherwise γ is regular, since r is prime, so $G_\gamma = E^\times$ with $E = K(\gamma)$ a field. Any $\delta \in H^Z_\gamma(K)$ normalizes E, and $H^Z_\gamma(K)/G_\gamma(K) \cong W_\gamma$ by Skolem–Noether. The index is therefore $|W_\gamma|$, because $Z(K) \subseteq G_\gamma(K)$. Factoring the integral through $G_\gamma(\mathbb{A})\backslash G_D(\mathbb{A})$ gives

$$\int_{G_\gamma(\mathbb{A})\backslash G_D(\mathbb{A})} \tilde{f}_D(x^{-1}\iota(\gamma)x)\int_{Z(\mathbb{A})G_\gamma(K)\backslash G_\gamma(\mathbb{A})} c_\gamma(s)^{-1}\,ds\,dx,$$

where $c_\gamma(s) \in \mu_n$ is defined by $\tilde{s}^{-1}\iota(\gamma)\tilde{s} = c_\gamma(s)\iota(\gamma)$. This c_γ is a character of $G_\gamma(\mathbb{A})$, trivial on $Z(\mathbb{A})$ (Lemma 4.2.3) and on $G_\gamma(K)$ (as ι is a homomorphism). The inner integral is therefore $\operatorname{vol}(\mathbb{A}_K^\times E^\times\backslash\mathbb{A}_E^\times)$ if $c_\gamma \equiv 1$, and 0 otherwise. As $G_\gamma(\mathbb{A})$ is the restricted product of the $G_\gamma(K_v)$, $c_\gamma \equiv 1$ if and only if $\iota(\gamma)$ is good at every place. In that case the outer integral is the product of the local orbital integrals, i.e. $\Phi(\iota(\gamma),\tilde{f}_D)$, which equals $F(\iota(\gamma),\tilde{f}_D)$ by the product formula $\prod_v \Delta_v(\gamma) = 1$. Interchanging the finite sum with the integral is legitimate, by compactness. The last assertion holds because $\tilde{f}_D(1) = \prod_v \tilde{f}_{D,v}(1)$ and $\tilde{f}_{D,u}(1) = 0$. □

4.6.2 The GL side: a variant of [FK86, §18]

The simple trace formula [FK86, §18, Theorem] requires the supercusp form at one of the two places where non-elliptic orbital integrals vanish. We want it at a place $w_0 \notin S$, where matching is the identity. We state the variant we need and prove it by following the proof in [FK86, §18], noting which hypothesis each step uses. In fact we need only *one* place with elliptic support, which then plays the roles of all three of the places u, u', u'' in [FK86].

Proposition 4.6.2. *(using **(E)**) Let $\tilde f = \otimes_v \tilde f_v$ be smooth, genuine, compactly supported modulo $\overline{Z}(\mathbb{A}_K)$, transforming by $\tilde\theta^{-1}$ under $\overline{Z}(\mathbb{A}_K)$, with $\tilde f_v$ the unit element of the genuine spherical Hecke algebra (relative to κ_v^*) for almost all v. Suppose:*

(i) at some place w_0, $\tilde f_{w_0}$ is a supercusp form [FK86, §13], i.e. $\int_{N(K_{w_0})} \tilde f_{w_0}(\tilde x\, s_N(n)\, \tilde y)\, dn = 0$ for every unipotent radical N of a proper parabolic subgroup and all $\tilde x, \tilde y$;

(ii) at some finite place u, $\tilde f_u$ is supported on the regular elliptic set of $\overline{G}_u$.

Then $r(\tilde f)$ maps $L^2(\iota(\mathrm{GL}(r,K))\backslash\overline{G}(\mathbb{A}_K), \tilde\theta)$ into the cuspidal subspace, is of trace class, and

$$\sum_{\overline{\pi}\ \text{cuspidal}} m(\overline{\pi})\, \mathrm{tr}\, \overline{\pi}(\tilde f) = \sum_{[\gamma]\in\mathscr{O}_{\mathrm{GL}}} c(\gamma)\, F(\iota(\gamma), \tilde f),$$

with absolute convergence on the left and finitely many nonzero terms on the right. The places w_0 and u may be arbitrary (in particular, they may be distinct, and w_0 need not be one of the places of (ii)).

Proof. We follow [FK86, §18] step by step.

Step 1 (spectral side; uses only (i)). For φ in the L^2 space and a unipotent radical N, the constant term of $r(\tilde f)\varphi$ along N is

$$\int_{N(K)\backslash N(\mathbb{A})} (r(\tilde f)\varphi)(s_N(n)\tilde x)\, dn = \int_{N(K)\backslash\overline{G}(\mathbb{A})} \varphi(\tilde y) \int_{N(\mathbb{A})} \tilde f(\tilde x^{-1} s_N(n) \tilde y)\, dn\, d\tilde y.$$

This uses that φ is left invariant under $\iota(N(K)) = s_N(N(K))$ (Proposition 4.5.6(e)) and that s_N splits $N(\mathbb{A})$. The inner integral factors over places, and its factor at w_0 vanishes by (i). So $r(\tilde f)$ maps L^2 into the cuspidal subspace L^2_{cusp}, and so does $r(\tilde f^*)$, since $\tilde f^*(\tilde g) = \overline{\tilde f(\tilde g^{-1})}$ again satisfies (i). Let P be the orthogonal projection onto L^2_{cusp}; it commutes with $r(\phi)$ for every test function ϕ. Then $r(\tilde f) = Pr(\tilde f)$, and taking adjoints in $r(\tilde f^*) = Pr(\tilde f^*)$ gives $r(\tilde f) = r(\tilde f)P$.

We now justify the analytic conclusions explicitly. The inputs **(E)** are:

(A1) for every test function ϕ, $r(\phi)P$ is Hilbert–Schmidt, and for $\phi = \phi_1 * \phi_2$ its kernel $K_\phi^{\mathrm{cusp}}(x,y) = \sum_i (r(\phi_1)\varphi_i)(x)\, \overline{(r(\phi_2^*)\varphi_i)(y)}$ (where $\{\varphi_i\}$ is an orthonormal basis of L^2_{cusp}) converges absolutely and locally uniformly to a continuous function. This is Lemma 4.6.4 below;

(A2) the Dixmier–Malliavin theorem [DM78]: every test function is a finite sum $\sum_j \phi_j * \phi_j'$ of convolutions of test functions.

Write $\tilde f = \sum_j \phi_j * \phi_j'$ by (A2). Then $r(\tilde f) = r(\tilde f)P = \sum_j (r(\phi_j)P)(r(\phi_j')P)$ is a finite sum of products of two Hilbert–Schmidt operators (Lemma 4.6.4(b)), hence of trace class. Its trace is

$$\operatorname{tr} r(\tilde f) = \sum_j \int\!\!\int K^{\mathrm{cusp}}_{\phi_j}(x,y) K^{\mathrm{cusp}}_{\phi_j'}(y,x)\,dy\,dx = \int K^{\mathrm{cusp}}_{\tilde f}(x,x)\,dx,$$

with absolute convergence by Cauchy–Schwarz (both kernels are square integrable), the integrals being over $Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$. Finally, the operator $r(\tilde f)(1-P) = 0$ has kernel $K_{\tilde f} - K^{\mathrm{cusp}}_{\tilde f}$, where $K_{\tilde f}(x,y) = \sum_{\gamma\in Z(K)\backslash\mathrm{GL}(r,K)} \tilde f(x^{-1}\iota(\gamma)y)$ is the full kernel (a locally finite sum of smooth functions). Both kernels are continuous, so $K_{\tilde f} = K^{\mathrm{cusp}}_{\tilde f}$ everywhere, and in particular on the diagonal. Hence $\operatorname{tr} r(\tilde f) = \int K_{\tilde f}(x,x)\,dx$, and the spectral side is $\sum m(\overline\pi)\operatorname{tr}\overline\pi(\tilde f)$ over cuspidal $\overline\pi$, converging absolutely. This is the content of the first sentences of the proof in [FK86, §18]. It is the only step of [FK86]'s proof that uses the supercusp form, and the argument does not use the place at which the supercusp form sits. (The absolute convergence of $\int K_{\tilde f}(x,x)\,dx$ also follows directly from Steps 2–4, since only finitely many elliptic classes contribute.)

Step 2 (finiteness). As in [FK86, §18], only finitely many classes $[\gamma]$ have $\tilde f(x^{-1}\iota(\gamma)x) \neq 0$ for some x, by discreteness of characteristic polynomials of rational elements. This uses no local hypothesis.

Step 3 (only regular elliptic classes contribute; uses only (ii)). If $\tilde f(x^{-1}\iota(\gamma)x) \neq 0$, then $x_u^{-1}\gamma x_u$ lies in the support of $\tilde f_u$, so γ is regular and elliptic in $\mathrm{GL}(r,K_u)$. Its characteristic polynomial is then separable and irreducible over K_u, hence over K, so γ is regular and elliptic over K. In [FK86] this step needs two places: u'' for regularity, and u' together with a modification of $\tilde f_{u'}$ to remove the non-elliptic regular classes. Under (ii) the non-elliptic and singular classes contribute nothing *pointwise*, and no modification is needed.

Step 4 (elliptic terms). For regular elliptic γ the computation in the proof of Proposition 4.6.1 applies verbatim with G_D replaced by $\mathrm{GL}(r)$. The quotient $Z(\mathbb{A})G_\gamma(K)\backslash G_\gamma(\mathbb{A})$ is compact; $H^Z_\gamma(K)/G_\gamma(K) \cong W_\gamma$ by Skolem–Noether in $M_r(K)$; non-good classes contribute 0 by the character c_γ; and the global orbital integral converges absolutely and factors into local ones [FK86, §§6, 18] **(E)**. The interchange of the finite sum and the integral is the one carried out at the end of [FK86, §18]. □

Remark 4.6.3. In the notation of [FK86, §18, Theorem], condition (ii) at u implies the conditions at u' and u'' (Theorem 4.4.7(c)), and the place u of [FK86] (which carries the supercusp form there) is replaced by w_0. In particular u'' need not be distinct from u', and a single place of S suffices for the geometric side. We use $|S| \geq 2$ only to know that $S \neq \emptyset$.

Lemma 4.6.4 (Cuspidal kernels)**.** *(using **(E)**) Let $X := \iota(\mathrm{GL}(r,K))\overline{Z}(\mathbb{A}_K)\backslash\overline{G}(\mathbb{A}_K)$, which is homeomorphic to $Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$ and has finite volume. Let $L^2_{\mathrm{cusp}} \subset L^2(\iota(\mathrm{GL}(r,K))\backslash\overline{G}(\mathbb{A}_K),\tilde\theta)$ be the cuspidal subspace, P the projection onto it, and ϕ a genuine test function transforming by $\tilde\theta^{-1}$.*

*(a) (using **(E)**; §4.5.4) There is C_ϕ such that*

$$\sup_x |(r(\phi)\psi)(x)| \leq C_\phi \|\psi\|_2 \qquad (\psi \in L^2_{\mathrm{cusp}})$$

(Gelfand–Graev–Piatetski-Shapiro, Godement).

(b) $r(\phi)P$ *is Hilbert–Schmidt, with* $\|r(\phi)P\|_{HS}^2 \le C_\phi^2 \operatorname{vol}(X)$.

(c) If $\phi = \phi_1 * \phi_2$, *the series* $K_\phi^{\mathrm{cusp}}(x,y) = \sum_i (r(\phi_1)\varphi_i)(x)\overline{(r(\phi_2^*)\varphi_i)(y)}$ *converges absolutely and uniformly on compact sets to a continuous function, and it is the kernel of* $r(\phi)P$.

Proof. The map $X \to Z(\mathbb{A})\mathrm{GL}(r,K)\backslash \mathrm{GL}(r,\mathbb{A})$. It is a continuous bijection, since $\mu_n \subset \overline{Z}(\mathbb{A}_K)$. It is open, because p is a local homeomorphism [FK86, §2]. Hence it is a homeomorphism, and we give X the transported measure. For genuine $\tilde\theta$-equivariant ψ, ψ', the function $\psi\overline{\psi'}$ is invariant under $\overline{Z}(\mathbb{A}_K)$, since $\tilde\theta$ is unitary, and under $\iota(\mathrm{GL}(r,K))$, so it is a function on X. Thus the inner product is an integral over X.

(a) This is Lemma 4.5.9 of §4.5.4, where the passage from $\mathrm{GL}(r,\mathbb{A})$ to the finite central cover is carried out.

(b) Let $\{\varphi_i\}$ be an orthonormal basis of L^2_{cusp}. For each x and each finite set F of indices, $\sum_{i\in F} |(r(\phi)\varphi_i)(x)|^2 = \sup |(r(\phi)\psi)(x)|^2$, where the supremum is over $\psi = \sum_{i\in F} c_i\varphi_i$ with $\sum |c_i|^2 \le 1$. By (a) this is at most C_ϕ^2. Integrating over X gives $\sum_i \|r(\phi)\varphi_i\|_2^2 \le C_\phi^2 \operatorname{vol}(X)$.

(c) By (a) and (b) applied to ϕ_1 and ϕ_2^*, both $\sum_i |(r(\phi_1)\varphi_i)(x)|^2$ and $\sum_i |(r(\phi_2^*)\varphi_i)(y)|^2$ are bounded by constants. So the series converges absolutely, by Cauchy–Schwarz. Moreover $(r(\phi_1)\psi)(x) = \langle \psi, k_x\rangle$ with $k_x := \overline{K_{\phi_1}(x,\cdot)}$, where

$$K_{\phi_1}(x,y) = \sum_{\gamma \in Z(K)\backslash \mathrm{GL}(r,K)} \phi_1(x^{-1}\iota(\gamma)y)$$

is the full kernel of $r(\phi_1)$. For fixed x this is a locally finite sum, compactly supported in y modulo $\iota(\mathrm{GL}(r,K))\overline{Z}(\mathbb{A}_K)$, so $k_x \in L^2$. Hence $\sum_i |(r(\phi_1)\varphi_i)(x)|^2 = \|Pk_x\|_2^2$. The map $x \mapsto k_x \in L^2$ is continuous, so this sum of continuous nonnegative functions converges to a continuous function. By Dini's theorem the convergence is uniform on compact sets, and likewise in y. Hence the partial sums of K_ϕ^{cusp} converge locally uniformly, and the limit is continuous. It is the kernel of $r(\phi_1)Pr(\phi_2) = r(\phi)P$, by expanding $r(\phi_2)\psi$ in the basis. □

4.6.3 Comparison

We now show that the two trace formulas agree for matching test functions. The places $w_0 \notin S$ and $u \in S$ and the admissible pairs fixed in Definition 4.6.5 are used throughout §4.7.

Definition 4.6.5 (Admissible pairs)**.** Fix a finite place $w_0 \notin S$, a place $u \in S$, and a finite set $S_D \supseteq S \cup \{w_0\}$ as in §4.5.3. A pair $(\tilde f_D, \tilde f)$ of factorizable genuine test functions with central characters $\tilde\theta^{-1}$ is *admissible* if:

(a) for $v \in S$, $\tilde f_{D,v}$ vanishes near $\overline{Z}_v$, $\tilde f_v$ is supported in the regular set, and they match (in particular $F(\tilde\delta, \tilde f_v) = 0$ at good regular non-elliptic $\tilde\delta$); moreover $\tilde f_u$ is supported on the regular elliptic set. (By Theorem 4.4.7, every $\tilde f_{D,v}$ vanishing near $\overline{Z}_v$ has such a partner, supported on the good regular elliptic set; conversely every such $\tilde f_v$ has a partner $\tilde f_{D,v}$.)

(b) for $v \notin S$, $\tilde f_{D,v} = \tilde f_v \circ \psi_v$; for $v \notin S_D$ both are unit elements (Proposition 4.2.7(c)); at the archimedean places they are K_v-finite;

(c) $\tilde f_{w_0}$ is a supercusp form.

Example 4.6.6 (Continuing Example 1.3.1: an admissible pair)**.** In the running example $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$. Take w_0 to be a place of K above 13, which lies outside S and does not divide n (Example 4.2.8), and take $u = \mathfrak{p}_7$. An admissible pair is assembled place by place. At $\mathfrak{p}_3$ and at $\mathfrak{p}_7$ choose any $\tilde{f}_{D,v}$ vanishing near $\overline{Z}_v$, and let $\tilde{f}_v$ be a matching function supported on the good regular elliptic set. It exists by Theorem 4.4.7(a). This gives condition (a), including the extra condition at u. At w_0 let $\tilde{f}_{w_0}$ be a supercusp form, and put $\tilde{f}_{D,w_0} = \tilde{f}_{w_0} \circ \psi_{w_0}$. At the remaining places of S_D choose $\tilde{f}_v$ freely, K_v-finite at the complex places, and put $\tilde{f}_{D,v} = \tilde{f}_v \circ \psi_v$. Outside S_D take the unit elements.

Theorem 4.6.7. *For every admissible pair,*

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D) \operatorname{tr} \overline{\pi}_D(\tilde{f}_D) = \sum_{\overline{\pi} \text{ cuspidal}} m(\overline{\pi}) \operatorname{tr} \overline{\pi}(\tilde{f}),$$

both sides converging absolutely.

Proof. Apply Proposition 4.6.1 to $\tilde{f}_D$; the identity term vanishes by (a). Apply Proposition 4.6.2 to $\tilde{f}$, with $u \in S$ and w_0 as given. We compare the geometric sides class by class.

By Corollary 4.5.8, $\gamma \mapsto j(\gamma)$ is a bijection from $\mathscr{O}_{G_D}$ onto the set of classes in $\mathscr{O}_{\mathrm{GL}}$ whose field embeds in D. It commutes with multiplication by $K^\times$, so it is well defined on classes modulo $Z(K)$. Classes in $\mathscr{O}_{\mathrm{GL}}$ whose field does not embed in D are non-elliptic (and regular) at some $v \in S$, where $F(\cdot, \tilde{f}_v)$ vanishes by (M2), so they contribute 0. For corresponding classes, $F(\iota(\gamma), \tilde{f}_D) = F(\iota(j\gamma), \tilde{f})$ by Theorem 4.5.7. Finally $c(\gamma) = c(j\gamma)$: the field $E = K(\gamma) \cong K(j\gamma)$ and the measure dt are the same (§4.5.3), and $W_\gamma = W_{j\gamma}$ because j is a K-algebra isomorphism of E onto $K(j\gamma)$. □

Remark 4.6.8. The **(E)** inputs of the global sections are: the sup-norm bound for cusp forms on $\mathrm{GL}(r, \mathbb{A})$ [GGPS69], [God66], [MW95, I.2], transferred to the cover in §4.5.4, and the Dixmier–Malliavin theorem (A2) [DM78], used for the trace formula on the cuspidal spectrum; the trace class property on the compact quotient; absolute convergence and factorization of elliptic orbital integrals [FK86, §6]; finiteness of cusp forms of fixed level, type and infinitesimal character [HC68], [MW95, I.2]; Arthur's multipliers [Art83] (as in [JC26a, Lemma 6.3]); and the splitting-off of supercuspidal representations (Lemma 4.7.1).

4.7 Spectral separation and the global transfer

This section deduces Theorem C from Theorem 4.6.7. We first use matrix coefficients at w_0 to fix the component there (§4.7.1). We then separate by archimedean infinitesimal characters and by the local components outside S (§4.7.2). This uses corner Hecke algebras and linear independence of traces of finitely many simple modules, but no multiplicity-one theorem, and it yields the packet identity of Proposition 4.7.6 unconditionally. The section depends on Theorem 4.6.7 and on the local representation theory cited in Lemmas 4.7.1 and 4.7.5. It introduces the set $V = S \cup \{w_0\}$, the packets $P_D(\rho)$ and $P(\rho)$, the corner algebras $e\mathscr{H}e$, and the hypotheses (Hyp_D) and (Hyp).

Dependency structure. The passage from the packet identity to the existence of a transfer (§4.7.3) needs the D-side packet character to be nonzero on matching regular-supported test functions. This is hypothesis (Hyp_D) (Definition 4.7.9). We prove it when the packet is finite (Proposition 4.7.11), and we show exactly what a failure would require (Lemma 4.7.8), but we cannot prove it in general. So Theorem C is unconditional for finite packets and conditional on (Hyp_D) in general. Similarly, the converse is conditional on (Hyp).

4.7.1 The place w_0

We fix the component at w_0 by a choice of test function, so that no separation is needed there. The supercuspidal σ and the test function $\tilde{f}_\sigma$ fixed here are kept for the rest of the section.

Lemma 4.7.1. *(using (E)) Let σ be a genuine irreducible supercuspidal representation of $\overline{G}_{w_0}$ with (unitary) central character $\tilde{\theta}_{w_0}$. Then σ is unitarizable, since its matrix coefficients are compactly supported modulo $\overline{Z}_{w_0}$ and hence square integrable modulo the center; we fix an invariant inner product. Let $v \in \sigma$ and $v^\vee \in \sigma^\vee$ with $\langle v, v^\vee \rangle = 1$, and let $d(\sigma)$ be the formal degree (for dg_{w_0}). Put $\tilde{f}_\sigma(\tilde{g}) := d(\sigma)\langle \sigma(\tilde{g}^{-1})v, v^\vee \rangle$. Then $\tilde{f}_\sigma$ is genuine, transforms by $\tilde{\theta}_{w_0}^{-1}$ under $\overline{Z}_{w_0}$, and is a supercusp form. Moreover $\operatorname{tr}\sigma(\tilde{f}_\sigma) = 1$, and $\pi(\tilde{f}_\sigma) = 0$ for every irreducible smooth $\pi \not\cong \sigma$ with central character $\tilde{\theta}_{w_0}$.*

Proof. Genuineness and the central character follow from $\sigma(\zeta\tilde{g}) = \zeta\sigma(\tilde{g})$ and $\sigma(z) = \tilde{\theta}_{w_0}(z)$. Matrix coefficients of supercuspidal representations are compactly supported modulo the center and are supercusp forms [FK86, §13] **(E)**. The trace formula $\operatorname{tr}\sigma(\tilde{f}_\sigma) = \langle v, v^\vee \rangle = 1$ is Schur orthogonality for representations that are square-integrable modulo the center. The vanishing for $\pi \not\cong \sigma$ is the fact that a supercuspidal representation splits off the category of smooth representations [Ber84, BZ] **(E)**. Its proof uses only that $\overline{G}_{w_0}$ is a totally disconnected group with $\overline{Z}_{w_0}$ central of finite index in the center modulo μ_n and cocompact in the support of matrix coefficients, so it applies to covers. □

Fix a genuine supercuspidal σ of $\overline{G}_{w_0}$ and take $\tilde{f}_{w_0} = \tilde{f}_\sigma$ and $\tilde{f}_{D,w_0} = \tilde{f}_\sigma \circ \psi_{w_0}$ in every admissible pair. Then Theorem 4.6.7 becomes

$$\sum_{\overline{\pi}_D:\ \overline{\pi}_{D,w_0} \cong \sigma\circ\psi_{w_0}} m(\overline{\pi}_D)\operatorname{tr}\overline{\pi}_D^{w_0}(\tilde{f}_D^{w_0}) = \sum_{\overline{\pi}:\ \overline{\pi}_{w_0}\cong\sigma} m(\overline{\pi})\operatorname{tr}\overline{\pi}^{w_0}(\tilde{f}^{w_0}), \tag{4.8}$$

where the superscript w_0 denotes the components away from w_0. All $\overline{\pi}$ on the right are cuspidal.

4.7.2 Separation

We separate (4.8) first by archimedean infinitesimal characters, to reduce to finite sums, and then by the components outside $S \cup \{w_0\}$. For the second step we use corner Hecke algebras $e\mathscr{H}e$ and linear independence of traces of finitely many simple modules. Packets are *not* assumed finite. Instead, the packet identity is proved as an identity of locally finite sums: for each test function, only finitely many terms are nonzero.

Lemma 4.7.2. *(using (E)) For each $W_{\infty,\mathbb{C}}$-orbit ν, (4.8) holds with both sums restricted to representations of archimedean infinitesimal character ν. If all finite components of the test functions are bi-invariant under a fixed open compact subgroup U of the finite adeles (split by κ^*), and the archimedean components lie in $e_{\mathfrak{t}}C_c^\infty e_{\mathfrak{t}}$ for a fixed finite set $\mathfrak{t}$ of K_∞-types, then only finitely many terms on either side are nonzero. Specifically, the nonzero terms come from representations $\overline{\pi}$ with $e_U e_{\mathfrak{t}}\overline{\pi} \neq 0$ and infinitesimal character ν.*

Proof. This is the proof of [JC26a, Lemma 6.12], with the torus side replaced by the GL side. *Finiteness:* a term is nonzero only if $\overline{\pi}(\tilde{f}) \neq 0$, which forces $e_U e_{\mathfrak{t}}\overline{\pi} \neq 0$. On the D side, the automorphic forms of level U, of K_∞-types in $\mathfrak{t}$ and with given Casimir eigenvalue form a finite-dimensional space, by ellipticity of the Casimir operator on the compact quotient, as in [JC26a,

Lemma 6.12]. On the GL side the contributing $\overline{\pi}$ are cuspidal, and the corresponding spaces of cusp forms are finite-dimensional [HC68], [MW95, I.2] **(E)**. *Multipliers:* at the complex places both covers split, $\tilde{f}_{D,v} = \tilde{f}_v \circ \psi_v$, and Arthur's multipliers [JC26a, Lemma 6.3] act by the same scalar $\hat{\mu}(\nu_\pi)$ on both sides, preserving $e_{\mathfrak{t}} C_c^\infty e_{\mathfrak{t}}$ and admissibility. The Paley–Wiener argument of [JC26a, Lemma 6.12] then separates the orbits ν, using the absolute convergence of both sides (Theorem 4.6.7). □

Lemma 4.7.3. ***(E)****Let A be a $\mathbb{C}$-algebra and $M_1, \dots, M_k$ pairwise non-isomorphic simple A-modules of finite dimension. Then the functions $a \mapsto \operatorname{tr}(a|M_i)$ are linearly independent. (By the density theorem, $A \to \prod_i \operatorname{End}(M_i)$ is surjective [Lan02, XVII].)*

Let $V := S \cup \{w_0\}$. For a genuine representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ let $\overline{\pi}^V$ be its component away from V, and use the same notation on the D side. Via $\psi^V := \otimes_{v \notin V} \psi_v$, genuine representations of $\overline{G}_D(\mathbb{A}^V)$ and of $\overline{G}(\mathbb{A}^V)$ correspond; we write $\rho \circ \psi^V$ for the transport of ρ.

Lemma 4.7.4 (Tensor products of simple modules)**.** *Let $B_1, \dots, B_m$ be $\mathbb{C}$-algebras (not necessarily unital), and let M_i, M_i' be finite-dimensional simple B_i-modules on which B_i acts nondegenerately (i.e. $B_i M_i = M_i$), so that $B_i \to \operatorname{End}(M_i)$ is surjective by density. Then $M := \bigotimes_i M_i$ is a simple $B := \bigotimes_i B_i$-module, and if $M \cong M' := \bigotimes_i M_i'$ as B-modules then $M_i \cong M_i'$ for every i.*

Proof. $B \to \bigotimes_i \operatorname{End}(M_i) = \operatorname{End}(M)$ is surjective, so M is simple. For the second assertion, compare traces: $\operatorname{tr}(b_1 \otimes \cdots \otimes b_m \,|\, M) = \prod_i \operatorname{tr}(b_i|M_i)$. Fix i. For $k \neq i$ choose $b_k \in B_k$ mapping to a rank-one idempotent of $\operatorname{End}(M_k)$, so $\operatorname{tr}(b_k|M_k) = 1$. Then $\operatorname{tr}(b_i|M_i) = c \operatorname{tr}(b_i|M_i')$ for all b_i, with $c = \prod_{k \neq i} \operatorname{tr}(b_k|M_k')^{-1}$. Here c is finite and nonzero, since both sides are nonzero functionals of b_i, by surjectivity. Two nonzero proportional trace functionals of simple finite-dimensional modules come from isomorphic modules (Lemma 4.7.3), so $M_i \cong M_i'$. □

Why fixed-central-character Hecke algebras. All test functions in this chapter transform under the full central subgroup $\overline{Z}$ by $\tilde{\theta}^{-1}$, as the L^2 spaces do, and convolution is over G/Z. The resulting Hecke algebras $C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ are not subalgebras of the ordinary Hecke algebra $C_c^\infty(\overline{G}_v)$. In particular, the idempotent of an open compact subgroup $\kappa_v^*(U_v)$, a function supported on $\mu_n \kappa_v^*(U_v)$, does not belong to them when $\overline{Z}_v$ is non-compact. It acts on them as a convolution multiplier, but it is not their unit. The unit of the corner algebra must instead be the *central-character idempotent* supported on $\overline{Z}_v \kappa_v^*(U_v)$, defined below. The archimedean projector $e_{\mathfrak{t}}$ is likewise a multiplier and not a function.

Lemma 4.7.5 (Corner algebras)**.** *(using **(E)**) Let $U^V = \prod_{v \notin V,\, v < \infty} U_v$ be an open compact subgroup, with $U_v = \mathrm{GL}(r, \mathscr{O}_v)$ split by κ_v^* for $v \notin S_D$, and let $\mathfrak{t}$ be a finite set of K_∞-types.*

- *At a finite place $v \notin V$, assume that $\tilde{\theta}_v$ is trivial on $\kappa_v^*(U_v \cap K_v^\times)$; this holds for $v \notin S_D$, and after shrinking U_v in general. Let $\mathscr{H}_v := C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ be the algebra of smooth genuine functions, compactly supported modulo $\overline{Z}_v$, with $f(z\tilde{g}) = \tilde{\theta}_v(z)^{-1} f(\tilde{g})$, under the convolution $(f_1 * f_2)(\tilde{g}) = \int_{\overline{G}_v/\overline{Z}_v} f_1(\tilde{h}) f_2(\tilde{h}^{-1}\tilde{g})\, d\dot{h}$. Let $e_v \in \mathscr{H}_v$ be the central-character idempotent*

$$e_v(z\,\kappa_v^*(u)) = \operatorname{vol}\left(\overline{Z}_v \kappa_v^*(U_v)/\overline{Z}_v\right)^{-1} \tilde{\theta}_v(z)^{-1} \quad (z \in \overline{Z}_v,\ u \in U_v),$$

  and $e_v = 0$ off $\overline{Z}_v \kappa_v^*(U_v)$. It is well defined by the assumption on $\tilde{\theta}_v$, and $e_v * e_v = e_v$. For τ_v with central character $\tilde{\theta}_v$, $\tau_v(e_v)$ is the projection onto $\tau_v^{\kappa_v^*(U_v)}$. Put $A_v := e_v * \mathscr{H}_v * e_v$, a unital algebra with unit e_v.

- *At the archimedean places (where the covers split, and $Z_\infty \cap K_\infty$ acts by $\tilde\theta_\infty$ on every $\delta \in \mathfrak{t}$), let $e_\mathfrak{t}$ be the measure $\sum_{\delta\in\mathfrak{t}} \dim(\delta)\overline{\chi_\delta(k)}\,dk$ on K_∞. This is a distribution, not a function on G_∞, but convolution by it maps $C_c^\infty(G_\infty)_{K\text{-}fin}$ into itself. Let $A_\infty := e_\mathfrak{t} * C_c^\infty(G_\infty, \tilde\theta_\infty)_{K\text{-}fin} * e_\mathfrak{t}$. This is a subalgebra of $C_c^\infty(G_\infty)$ that in general has no unit; $e_\mathfrak{t}$ acts on it as a two-sided multiplier.*

- *Let A be the span of the functions $\phi_\infty \otimes \bigotimes_{v<\infty} \phi_v$ with $\phi_\infty \in A_\infty$, $\phi_v \in A_v$, and $\phi_v = e_v$ for almost all v.*

For an admissible genuine irreducible (unitary) representation τ of $\overline{G}(\mathbb{A}^V)$ with central character $\tilde\theta^V$, put $e\tau := \tau(e_\mathfrak{t}) \prod_v \tau(e_v)\,\tau$. Then:

(a) $e\tau$ is a finite-dimensional, nondegenerate A-module, and $\operatorname{tr}\tau(\phi) = \operatorname{tr}(\phi|e\tau)$ for $\phi \in A$;

(b) if $e\tau \neq 0$, it is a simple A-module;

(c) if $e\tau \cong e\tau' \neq 0$ as A-modules, then $\tau \cong \tau'$.

Proof. (a) Every $\phi \in A$ satisfies $\phi = e_\mathfrak{t} * \phi * e_\mathfrak{t}$ and $\phi = e_v * \phi * e_v$, so $\tau(\phi)$ maps τ into $e\tau$ and kills its complement $\ker\big(\tau(e_\mathfrak{t})\prod_v \tau(e_v)\big)$. So the trace of $\tau(\phi)$ is its trace on $e\tau$, which is finite-dimensional by admissibility. Nondegeneracy follows from (b) and its proof.

(b) Write $\tau = \tau_\infty \otimes \bigotimes_v' \tau_v$. By admissibility there is a finite set T of finite places such that $\dim e_v\tau_v = 1$ for $v \notin T$, and then $e\tau = e_\mathfrak{t}\tau_\infty \otimes \bigotimes_{v\in T} e_v\tau_v$, tensored with the distinguished lines at $v \notin T$. The single-place facts are **(E)**:

- at a finite place, $\tau_v \mapsto e_v\tau_v$ is a bijection from irreducible representations with $e_v\tau_v \neq 0$ to simple A_v-modules [BZ76, §2], [Ber84];

- at the archimedean places, by Harish-Chandra, $e_\mathfrak{t}\tau_\infty$ is a simple, nondegenerate, finite-dimensional A_∞-module whenever it is nonzero, and it determines the irreducible admissible $(\mathfrak{g}, K_\infty)$-module of τ_∞, hence the unitary representation τ_∞ up to isomorphism [HC54], [Wal88, §3.4–3.5].

Thus each factor is a simple, nondegenerate, finite-dimensional module over the corresponding algebra. The subalgebra $A_\infty \otimes \bigotimes_{v\in T} A_v \otimes \bigotimes_{v\notin T} \mathbb{C}e_v$ of A acts on $e\tau$ through the tensor product. By Lemma 4.7.4, $e\tau$ is simple, and nondegenerate, over this subalgebra, hence over A.

(c) Choose a finite set T that works for both τ and τ' as in (b), and enlarge it by any finite set of places. Applying Lemma 4.7.4 to the subalgebra of (b) gives $e_\mathfrak{t}\tau_\infty \cong e_\mathfrak{t}\tau'_\infty$ and $e_v\tau_v \cong e_v\tau'_v$ for $v \in T$, hence $\tau_\infty \cong \tau'_\infty$ and $\tau_v \cong \tau'_v$ for $v \in T$. Since T can contain any given finite place, $\tau_v \cong \tau'_v$ for every v, and so $\tau \cong \tau'$. □

Proposition 4.7.6 (Packet identity)**.** *Let ρ be an admissible genuine irreducible representation of $\overline{G}(\mathbb{A}^V)$. Put*

$$P_D(\rho) := \{\overline{\pi}_D :\ \overline{\pi}_D^V \cong \rho \circ \psi^V,\ \overline{\pi}_{D,w_0} \cong \sigma \circ \psi_{w_0}\},$$
$$P(\rho) := \{\overline{\pi}\ \text{cuspidal} :\ \overline{\pi}^V \cong \rho,\ \overline{\pi}_{w_0} \cong \sigma\},$$

counted with multiplicities $m(\cdot)$. These sets need not be finite. Let $(\tilde f_{D,v}, \tilde f_v)_{v\in S}$ be matching, with $\tilde f_{D,v}$ vanishing near $\overline{Z}_v$ and $\tilde f_v$ supported in the regular set. Then only finitely many terms of each side of

$$\sum_{\overline{\pi}_D \in P_D(\rho)} m(\overline{\pi}_D) \prod_{v\in S} \operatorname{tr}\overline{\pi}_{D,v}(\tilde f_{D,v}) = \sum_{\overline{\pi}\in P(\rho)} m(\overline{\pi}) \prod_{v\in S} \operatorname{tr}\overline{\pi}_v(\tilde f_v) \tag{4.9}$$

are nonzero, and the identity holds.

Proof. Reduction to elliptically supported $\tilde{f}_u$. Fix $u \in S$, and let $\tilde{f}'_u$ be the function of Theorem 4.4.7(a) attached to $\tilde{f}_{D,u}$; it is supported on the regular elliptic set. Both $\tilde{f}_u$ and $\tilde{f}'_u$ match $\tilde{f}_{D,u}$, so their normalized orbital integrals agree at all good regular elements: at elliptic ones by (M1), and at non-elliptic ones both vanish by (M2). By Lemma 4.4.14, $\operatorname{tr}\overline{\pi}_u(\tilde{f}_u) = \operatorname{tr}\overline{\pi}_u(\tilde{f}'_u)$ for every $\overline{\pi}_u$. So we may assume that $\tilde{f}_u$ is supported on the regular elliptic set, i.e. that the family satisfies Definition 4.6.5(a).

Uniform finiteness. Choose an open compact $U_S \subset \prod_{v\in S} D_v^\times$ (split) such that each $\tilde{f}_{D,v}$ is bi-$U_{S,v}$-invariant, and a level U'_S for the $\tilde{f}_v$. Choose a level U_{w_0} under which $\tilde{f}_\sigma$ is bi-invariant. Choose U^V and $\mathfrak{t}$ as in Lemma 4.7.5 with $e\rho \neq 0$, and let ν be the infinitesimal character of ρ_∞. Let Ω_D (resp. Ω) be the set of automorphic $\overline{\pi}_D$ (resp. cuspidal $\overline{\pi}$) with $\overline{\pi}_{w_0} \cong \sigma$ (transported), infinitesimal character ν, nonzero U_S- (resp. U'_S-) fixed vectors at S, and $e\overline{\pi}^V \neq 0$. These conditions do not involve the test functions away from V. By the finiteness part of Lemma 4.7.2, applied at the level $U_S \times U_{w_0} \times U^V$ (resp. $U'_S \times U_{w_0} \times U^V$) with types $\mathfrak{t}$, both Ω_D and Ω are *finite*. For *every* $\phi \in A$, a term in the identity below (with components ϕ away from V) is nonzero only if the representation lies in Ω_D (resp. Ω). Indeed, $\overline{\pi}(\tilde{f}) \neq 0$ forces each local operator to be nonzero, hence nonzero fixed vectors at every place for the level of the corresponding test function; at w_0 it forces $\overline{\pi}_{w_0} \cong \sigma$ (Lemma 4.7.1); and at infinity it forces the type condition. So all the identities below are identities of sums over the fixed finite sets Ω_D and Ω, uniformly in $\phi \in A$. This uniformity is what allows us to apply Lemma 4.7.3, which concerns a fixed finite family of modules.

Separation. Keep the components at S and at w_0 fixed, and let the components away from V run through $A = e\mathscr{H}e$ (transported to the D side by ψ^V). By Lemma 4.7.2 the ν-part of (4.8) becomes the identity, for $\phi \in A$,

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D)\Big(\prod_{v\in S}\operatorname{tr}\overline{\pi}_{D,v}(\tilde{f}_{D,v})\Big)\operatorname{tr}(\phi\,|\,e(\overline{\pi}_D^V\circ(\psi^V)^{-1})) = \sum_{\overline{\pi}} m(\overline{\pi})\Big(\prod_{v\in S}\operatorname{tr}\overline{\pi}_v(\tilde{f}_v)\Big)\operatorname{tr}(\phi\,|\,e\overline{\pi}^V),$$

in which only representations from the finite sets Ω_D, Ω occur. Automorphic representations in L^2 are unitary and admissible, so Lemma 4.7.5 applies. Group the terms by the isomorphism class of the simple A-module $e\overline{\pi}^V$ (or $e(\overline{\pi}_D^V\circ(\psi^V)^{-1})$). By Lemma 4.7.3 the total coefficients of the class of $e\rho$ on the two sides agree. By Lemma 4.7.5(c), the terms in that class are exactly those with $\overline{\pi}^V \cong \rho$ (resp. $\overline{\pi}_D^V \cong \rho\circ\psi^V$). This is (4.9). □

4.7.3 Non-cancellation and the transfer

Proposition 4.7.6 produces a GL-side packet as soon as the D-side of (4.9) is nonzero for *some* matching regular-supported family. We now show that this happens if $P_D(\rho)$ is finite. We also determine exactly what a failure would look like (Lemma 4.7.8). Since we cannot exclude such a failure for infinite packets, Theorem C is stated under a hypothesis (Hyp_D) that is parallel to the hypothesis (Hyp) of the converse.

Lemma 4.7.7 (Finite case)**.** *Let $\tau_1,\dots,\tau_k$ be pairwise inequivalent genuine irreducible representations of $\overline{G}_D(K_S) := \prod_{v\in S}\overline{G}_D(K_v)$ (modulo the identification of the μ_n's) with central character*

$\tilde{\theta}_S$, and let $m_1,\dots,m_k>0$. Then there are $\tilde{f}_{D,v}\in C_c^\infty(\overline{G}_D(K_v),\tilde{\theta}_v)_{\mathrm{gen}}$, $v\in S$, each vanishing near $\overline{Z}_v$, with $\sum_i m_i \operatorname{tr}\tau_i(\otimes_v \tilde{f}_{D,v})\neq 0$.

Proof. Each τ_i is finite-dimensional, since $\overline{G}_D(K_S)$ is compact modulo its center. Its character Θ_i is locally constant with $\Theta_i(z\tilde{g})=\tilde{\theta}_S(z)\Theta_i(\tilde{g})$, and $\operatorname{tr}\tau_i(\tilde{f})=\int_{\overline{G}_D(K_S)/\overline{Z}_S}\tilde{f}\,\Theta_i$. By Schur orthogonality on the compact group $\overline{G}_D(K_S)/\overline{Z}_S$ (with fixed central character), the Θ_i are linearly independent, so $\Theta:=\sum_i m_i\Theta_i\neq 0$. Since Θ is locally constant and the product R of the regular sets is dense and open, $\Theta(\tilde{x})\neq 0$ for some $\tilde{x}=(\tilde{x}_v)\in R$. Choose small open neighborhoods $N_v\ni\tilde{x}_v$ inside the regular set, on whose product Θ is constant, and let $\tilde{f}_{D,v}$ be the function supported on $\overline{Z}_vN_v$ with $\tilde{f}_{D,v}(z\tilde{y})=\tilde{\theta}_v(z)^{-1}$ for $z\in\overline{Z}_v$, $\tilde{y}\in N_v$ (well defined if N_v is small). Then $\sum_i m_i\operatorname{tr}\tau_i(\otimes\tilde{f}_{D,v})=\Theta(\tilde{x})\prod_v\operatorname{vol}(N_v)\neq 0$. The finiteness of the set $\{\tau_i\}$ is essential: shrinking the N_v raises the level, but no new terms appear. □

The example $\delta_e=\sum_\tau(\dim\tau)\Theta_\tau$ on a compact group shows that the finiteness assumption in Lemma 4.7.7 cannot simply be dropped. The next lemma determines exactly which coefficient functions cancel at a fixed level. When S has one element, the cancelling functions are the multiples of the Plancherel weight, as in this example. Since $|S|\geq 2$ under our assumptions, the actual condition is weaker: the coefficient function must be a sum of terms, each of which has a Plancherel factor at one place of S.

Lemma 4.7.8 (When cancellation occurs)**.** *For $v\in S$ let $U_v\subset D_v^\times$ be an open compact subgroup over which the cover splits by κ_v, with $\tilde{\theta}_v\circ\kappa_v$ trivial on $U_v\cap K_v^\times$. Let e_{U_v} be the central-character idempotent of $\overline{Z}_v\kappa_v(U_v)$, as in Lemma 4.7.5, and let $\mathscr{H}_v:=e_{U_v}*C_c^\infty(\overline{G}_D(K_v),\tilde{\theta}_v)_{\mathrm{gen}}*e_{U_v}$, which is finite-dimensional. Let $\mathscr{R}_v\subset\mathscr{H}_v$ be the subspace of functions vanishing near $\overline{Z}_v$, and let $\mathscr{T}_v$ be the finite set of genuine irreducible τ_v with central character $\tilde{\theta}_v$ and $\tau_v^{U_v}\neq 0$. For a function a on $\mathscr{T}:=\prod_{v\in S}\mathscr{T}_v$, put $\Lambda_a(\otimes_v f_v):=\sum_{\tau\in\mathscr{T}}a(\tau)\prod_v\operatorname{tr}\tau_v(f_v)$. Then Λ_a vanishes on $\bigotimes_v\mathscr{R}_v$ if and only if*

$$a\in\sum_{v\in S}d_v\otimes\mathbb{C}^{\mathscr{T}_{\neq v}},\qquad d_v(\tau_v):=\dim\tau_v,\qquad \mathscr{T}_{\neq v}:=\prod_{w\in S\smallsetminus\{v\}}\mathscr{T}_w.$$

Equivalently, a maps to 0 in $\bigotimes_{v\in S}\left(\mathbb{C}^{\mathscr{T}_v}/\mathbb{C}d_v\right)$. If S had one element, this would say that a is proportional to $\tau\mapsto\dim\tau$, i.e. to the Plancherel weight. For $|S|\geq 2$ it does not *say that a is proportional to the product weight $\prod_v d_v$. For example, with two places and $d_1=d_2=(1,1)$, the positive matrix $a=\left(\begin{smallmatrix}2&3\\4&5\end{smallmatrix}\right)=d_1\otimes(0,1)+(2,4)\otimes d_2$ cancels.*

Proof. Since the cover splits over U_v and $\tilde{\theta}_v\circ\kappa_v$ is trivial on $U_v\cap K_v^\times$, every element of $\mathscr{H}_v$ is determined on $\overline{Z}_v\kappa_v(U_v)$ by its value at 1. So $\mathscr{R}_v=\ker(\mathrm{ev}_{1,v})$, with $\mathrm{ev}_{1,v}(f)=f(1)$; indeed, a bi-$U_v$-invariant function vanishing at 1 vanishes on the open set $\overline{Z}_v\kappa_v(U_v)$, and the singular elements of $D_v^\times$ are central. The group $\overline{G}_D(K_v)/\overline{Z}_v$ is compact, so by Peter–Weyl (with fixed central character) $\mathscr{H}_v\cong\bigoplus_{\tau_v\in\mathscr{T}_v}\operatorname{End}(\tau_v^{U_v})$ is semisimple, the functionals $\operatorname{tr}\tau_v$ ($\tau_v\in\mathscr{T}_v$) form a basis of the trace functionals on $\mathscr{H}_v$, and Fourier inversion gives

$$\mathrm{ev}_{1,v}=\operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v)^{-1}\sum_{\tau_v\in\mathscr{T}_v}(\dim\tau_v)\operatorname{tr}\tau_v\quad\text{on }\mathscr{H}_v.$$

For finite-dimensional spaces, $\bigotimes_v\ker(\mathrm{ev}_{1,v})=\bigcap_v\ker(\mathrm{ev}_{1,v}\otimes\mathrm{id})$, so its annihilator in $(\bigotimes_v\mathscr{H}_v)^*$ is $\sum_v\mathbb{C}\,\mathrm{ev}_{1,v}\otimes\bigotimes_{w\neq v}\mathscr{H}_w^*$. Let $E_v:\mathscr{H}_v\to Z(\mathscr{H}_v)$ be the central projection (with kernel $[\mathscr{H}_v,\mathscr{H}_v]$). Then $\lambda\mapsto\lambda\circ(\otimes_vE_v)$ is a projection of $(\bigotimes\mathscr{H}_v)^*$ onto the span of products of trace functionals;

it fixes Λ_a and the $\mathrm{ev}_{1,v}$. Applying it to an expression $\Lambda_a = \sum_v \mathrm{ev}_{1,v} \otimes \mu_v$, we may assume that each μ_v is a combination of products of the $\operatorname{tr}\tau_w$. Expanding in the basis $\{\prod_v \operatorname{tr}\tau_v\}$ gives the stated condition, and conversely. □

Definition 4.7.9 (Hypothesis (Hyp_D))**.** We say that a packet $P_D(\rho)$ satisfies (Hyp_D) if there are $\tilde{f}_{D,v} \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ $(v \in S)$, each vanishing near $\overline{Z}_v$, with

$$\sum_{\overline{\pi}'_D \in P_D(\rho)} m(\overline{\pi}'_D) \prod_{v \in S} \operatorname{tr}\overline{\pi}'_{D,v}(\tilde{f}_{D,v}) \neq 0$$

(a sum with finitely many nonzero terms).

Example 4.7.10 (The hypothesis for a packet with one member)**.** Suppose that $P_D(\rho) = \{\overline{\pi}_D\}$ has one member, with multiplicity $m(\overline{\pi}_D) \geq 1$. Then the sum has one term. So (Hyp_D) asks for functions $\tilde{f}_{D,v}$, one for each $v \in S$, each vanishing near $\overline{Z}_v$, with $\operatorname{tr}\overline{\pi}_{D,v}(\tilde{f}_{D,v}) \neq 0$ for every $v \in S$. In this case the hypothesis is a local condition at each place of S. In the running example these are the two places $\mathfrak{p}_3$ and $\mathfrak{p}_7$. Proposition 4.7.11 below shows that such functions exist, and more generally that every finite nonempty packet satisfies (Hyp_D).

By Lemma 4.7.8, (Hyp_D) fails if and only if, at every level $U = \prod_{v \in S} U_v$, the function a_U on $\prod_v \mathscr{T}_v$ is of the form in that lemma. Here $a_U(\tau)$ is the total multiplicity of the members of $P_D(\rho)$ whose S-component is τ. So a failure requires that, at every level, the coefficient function of the packet at S be a sum of terms each having a Plancherel factor at some place of S. This is a strong but not obviously contradictory condition when $P_D(\rho)$ is infinite.

Proposition 4.7.11. *If $P_D(\rho)$ is finite and nonempty, then (Hyp_D) holds.*

Proof. The S-components of the members of $P_D(\rho)$ are pairwise inequivalent. Two members with isomorphic components at S have isomorphic components everywhere, and so are the same automorphic representation, counted once with its multiplicity. Apply Lemma 4.7.7. □

Theorem 4.7.12 (Theorem C)**.** *under (Hyp_D) Let $\overline{\pi}_D$ be an irreducible genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ with central character $\tilde{\theta}$, such that $\overline{\pi}_{D,w_0}$ is supercuspidal for some finite place $w_0 \notin S$. Let $\rho = \overline{\pi}_D^V \circ (\psi^V)^{-1}$ and $\sigma = \overline{\pi}_{D,w_0} \circ \psi_{w_0}^{-1}$, and assume that $P_D(\rho)$ satisfies (Hyp_D); this holds, for example, if $P_D(\rho)$ is finite (Proposition 4.7.11). Then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$ such that $\overline{\pi}_v \cong \overline{\pi}_{D,v} \circ \psi_v^{-1}$ for every $v \notin S$, and the packet identity (4.9) holds.*

Proof. By (Hyp_D) there are $\tilde{f}_{D,v}$ $(v \in S)$ vanishing near $\overline{Z}_v$ for which the left side of (4.9) is nonzero. By Theorem 4.4.7(a) they have matching $\tilde{f}_v$ supported on the good regular elliptic set. By Proposition 4.7.6 the right side is nonzero, so $P(\rho) \neq \emptyset$. Any $\overline{\pi} \in P(\rho)$ is cuspidal, with $\overline{\pi}^V \cong \rho$ and $\overline{\pi}_{w_0} \cong \sigma$, which is the assertion. □

Theorem 4.7.13 (Converse)**.** *under hypothesis (Hyp) Let $\overline{\pi}$ be a genuine cuspidal automorphic representation of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$, with $\overline{\pi}_{w_0}$ supercuspidal for some finite $w_0 \notin S$. Let $\rho = \overline{\pi}^V$, $\sigma = \overline{\pi}_{w_0}$. Assume*

(Hyp) there are $\tilde{f}_v$ $(v \in S)$, supported in the regular set of $\overline{G}_v$ with vanishing orbital integrals at good regular non-elliptic elements, such that $\sum_{\overline{\pi}' \in P(\rho)} m(\overline{\pi}') \prod_{v \in S} \operatorname{tr}\overline{\pi}'_v(\tilde{f}_v) \neq 0$.

Then there is $\overline{\pi}_D$ with $\overline{\pi}_{D,v} \cong \overline{\pi}_v \circ \psi_v$ for all $v \notin S$.

Proof. By Theorem 4.4.7(b), the $\tilde{f}_v$ in (Hyp) have matching $\tilde{f}_{D,v}$ vanishing near $\overline{Z}_v$. Proposition 4.7.6 applies to this family, and its right side is nonzero by (Hyp). Hence $P_D(\rho) \neq \emptyset$, and any $\overline{\pi}_D \in P_D(\rho)$ has the required components. □

Remark 4.7.14. (1) Hypothesis (Hyp) holds if $P(\rho) = \{\overline{\pi}\}$ and, for each $v \in S$, $\Theta_{\overline{\pi}_v}$ does not vanish identically on the good regular elliptic set: bump functions as in Lemma 4.7.7, supported near a good regular elliptic point, detect a single product of characters. The second condition holds when $\overline{\pi}_v$ is supercuspidal. Indeed, $\operatorname{tr}\overline{\pi}_v(\tilde{f}_{\overline{\pi}_v}) = 1$ for a matrix coefficient $\tilde{f}_{\overline{\pi}_v}$, and its orbital integrals vanish at regular non-elliptic elements [FK86, §13(i)], so only the elliptic terms of the Weyl integration formula contribute.

(2) (Hyp_D) and (Hyp) play symmetric roles. Each would follow from finiteness of the relevant packet. On the D side this is Proposition 4.7.11. On the GL side one needs, in addition, that the elliptic characters of the members of a finite packet are linearly independent, which is not automatic. Proving finiteness of near-equivalence packets for $\overline{G}_D(\mathbb{A}_K)$, for instance by bounding the level at S of the members of $P_D(\rho)$ in terms of ρ, would make Theorem C unconditional. This is goal (iv) of §4.1.3. For $n = 1$, strong multiplicity one for $D^\times$ [Bad08] gives $|P_D(\rho)| \leq 1$.

4.8 What is left after Chapter 4

This section lists the steps that need new input, and exactly where each would enter. They correspond to the goals in §4.1.3, where their status in Chapters 5 and 6 is recorded: (D1) is proved in Chapter 5; Chapter 6 proves (D2) at tame places where the corrected Flicker–Kazhdan correspondence is available (in particular when $(n, N_r) = 1$), as explained in Remark 4.8.1. The section depends only on the statements of §§4.3–4.7, and it introduces the labels (D1)–(D4), which Chapters 5–7 use.

(D1) **(C)** *Matching of arbitrary functions.* (a) For $\tilde{f}_D$ not vanishing near $\overline{Z}_v$: needed to remove the regular-support condition at $v \in S$ in Theorems 4.6.7–4.7.12. Inputs: the germ expansion of [FK86, §8] at central good elements, the restriction of germs to the elliptic set, and a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center (Proposition 4.4.8 and the Remark after it). (b) For $\tilde{f}$ with vanishing non-elliptic orbital integrals but not regular support (e.g. supercusp forms at $v \in S$): needed to put w_0 in S and to drop the regular-support condition in (Hyp). Input: a characterization of orbital integrals of genuine functions on $\overline{G}_v$ (the analog of [DKV84, A.3]), via [FK86, §8], [KP86, §1].

(D2) **(C)** *Local character relations.* For square-integrable genuine $\overline{\pi}_v$ there should be $\overline{\pi}_{D,v}$ with $\Theta_{\overline{\pi}_{D,v}} = (-1)^{r-1}\Theta_{\overline{\pi}_v} \circ \jmath$ on good regular elements, and conversely. Needed to turn (4.9) into a correspondence and to remove (Hyp). Inputs: isolating single representations in (4.9), using multiplicity one and rigidity for $\overline{G}(\mathbb{A}_K)$ ([FK86, §§27–28] as corrected by [Zo23]); the elliptic orthogonality relations [FK86, §§24–25]; and a canonical normalization of $dg_{D,v}$ relative to dg_v.

(D3) **(C)** *Finiteness of near-equivalence packets.* Needed to remove (Hyp_D) from Theorem 4.7.12, via Proposition 4.7.11. For the converse one needs in addition linear independence of the elliptic characters of the members of a finite GL-side packet (Remark 4.7.14).

(D4) **(C)** *Composing with [JC26a].* The lifts of [JC26a, Theorem 6.13] are unramified almost everywhere, with no supercuspidal component imposed. Applying Theorem 4.7.12 requires one supercuspidal $\overline{\pi}_{D,w_0}$. That means either extending [JC26a] to allow an arbitrary test function at one extra place inert in L (the local transfer [JC26a, Lemma 4.5] holds there), or removing the supercuspidal condition, by [Li14] or by (D1)(b).

Remark 4.8.1 ((D2) at places prime to n)**.** The character relation (D2) follows at a tame place $v \in S$ in the published range of the corrected square-integrable Flicker–Kazhdan correspondence, not merely from the unitarity of the division-side transform. Put $F = K_v$, and assume $p \nmid n$. A sufficient published hypothesis for the external correspondence is $(n, N_r) = 1$, where N_r is the least common multiple of the composite positive integers at most r (and $N_r = 1$ if none occur) [FK86, §26]; [Zo23, Proposition 3.6 and Definition 3.7].

Chapter 4, Proposition 4.2.4(b), identifies the restricted ambient cover, after relabeling μ_n, with the parameter-c_v Kazhdan–Patterson extension. Its diagonal cocycle is $\prod_{i<j}(a_i, b_j)_n(\det a, \det b)_n^{c_v}$, which is Zou's formula [Zo23, (2.1)] for one-dimensional blocks. The scalar cocycle exponent is $e_r = \binom{r}{2} + c_v r^2$; it is divisible by n because $2e_r = r(r-1+2rc_v)$, n is odd, and $n \mid r-1+2rc_v$. Consequently the scalar-section and central-character condition in Zou's correspondence agree with those of this chapter; in particular the associated ordinary scalar character is trivial on μ_n.

For odd n, the star map is intrinsically $x^* = \tilde{x}^n$ [FK86, §4]. Any change of extension isomorphism by a determinant character χ with values in μ_n disappears on star images. The canonical torus map of Theorem 4.3.10 satisfies $\jmath(zt_D^*) = z(jt)_{\mathrm{GL}}^*$ for $z \in \overline{Z}_v$. The discriminants on the two inner forms agree on matching elliptic elements. For a regular elliptic t the root sum in [Zo23, (3.2)] has exactly n equal contributions (the roots ζt, $\zeta \in \mu_n$), yielding the coefficient

$$b_D = \frac{n}{d_F(n)|n|_F^{r/2}} = |n|_F^{-(r-1)/2}, \qquad d_F(m) = m|m|_F^{-1/2}.$$

Here $d_r = \gcd(n, 2rc_v + r - 1) = n$; at a tame place $b_D = 1$ and the discriminant ratio $\Delta(t)/\Delta(t^n)$ is 1. Thus Corollary 6.7.8 composes its tame irreducible-character theorem (Theorem 6.7.6), ordinary local Jacquet–Langlands [DKV84], and the corrected Flicker–Kazhdan–Zou correspondence to obtain (D2), with the sign $(-1)^{r-1}$, on matching good regular elliptic elements. This is proved within the explicitly stated external-correspondence range (using the cited **(E)** inputs), and is *not* an unconditional assertion for every pair (n, r) with $p \nmid n$. At wild places, irreducible-character integrality of L_D remains open.

4.9 Conclusion: what the local theorem gives on its own

This section records what the local results of §§4.2–4.4 establish independently of the global argument. It proves nothing new and introduces no notation.

At a place v where D is ramified, the cover $\overline{G}_D(K_v)$ of [JC26a] and the Kazhdan–Patterson cover $\overline{G}(K_v)$ with parameter c_v are subgroups of one cover of $\mathrm{GL}(r, L_v)$. This identifies their centers and central characters without any choice. For every maximal subfield $E \subset D_v$, the covered tori on the two sides have identical commutator pairings and good elements, and Theorem 4.3.10 gives a *canonical* identification of their good parts. Consequently there is a canonical bijection between good regular elliptic conjugacy classes (Corollary 4.3.13). Matching of genuine test functions is

therefore well defined, with no choices beyond Haar measures. Theorem 4.4.7 then shows that matching is possible in both directions for functions supported on the regular set, for arbitrary (not necessarily cyclic) maximal subfields, by a self-contained argument (§4.4.6).

These statements are the local prerequisites for *any* trace-formula comparison between $\overline{G}_D$ and $\overline{G}$. They do not depend on the trace formula of [FK86], on finiteness of packets, or on multiplicity one. Proposition 4.4.8 also shows exactly where the regular-support condition has to be removed (germs at the center); Chapter 5 removes it. Globally, the packet identity of Proposition 4.7.6 is unconditional. What remains conditional is only the final non-cancellation step, isolated as (Hyp_D) and (Hyp).

Chapter 5

Matching of arbitrary functions and an improved global transfer

Overview. This is the second part of a study of a metaplectic Jacquet–Langlands transfer between the n-fold cover of $D^{\times}_{\mathbb{A}}$ constructed in [JC26a] and the Kazhdan–Patterson cover of $\mathrm{GL}(r,\mathbb{A}_K)$, where D is a central division algebra of prime degree r over a totally imaginary number field K containing μ_n, n odd and $(n,r)=1$. Chapter 4 proved matching of test functions supported on the regular set, and a transfer of packets under a non-cancellation hypothesis. Here we remove the regular-support condition. Near the center, an equivariant section of the cover, built from canonical n-th roots, identifies genuine functions on either cover with ordinary functions on the underlying linear group, with the same orbital integrals, compatibly with the identification of elliptic tori of Chapter 4 (*Theorem D*). Combined with Chapter 4 away from the center, this gives matching of arbitrary genuine test functions on the cover of $D^{\times}_{v}$, and of all genuine test functions with vanishing non-elliptic orbital integrals on the cover of $\mathrm{GL}(r,K_v)$ (*Theorem E*). Beyond the results imported from Chapter 4 and the standard structure theory of the covers, the new external inputs concern the linear groups: orbital integrals of the Steinberg pseudo-coefficient and the linear transfer of cuspidal functions, which we derive from results of Flicker–Kazhdan and Badulescu. As an application we prove *Theorem C′*, an improvement of Theorem C of Chapter 4: the packet identity holds with arbitrary matching test functions at all but one ramified place, and the non-cancellation hypothesis is replaced by a weaker one.

5.1 Introduction

This section states the main results, explains how they improve the results of Chapter 4, fixes conventions and labels, and lists what is left after this chapter, with what Chapter 6 has since done. It depends only on the statements of Chapter 4, and its notation is that of Chapter 4; all notation is collected in §3, and the results are listed with their labels in the Summary of results that follows this section. Here [JC26a] refers to the corrected version of [Jo88] (`metaplectic_kazhdan_revised12.tex`).

5.1.1 Setting

We use the global data of Chapter 4 without change.

The standing assumptions are those of Assumption 4.1.1: K is a totally imaginary number field containing μ_n, with n odd; L/K is cyclic of prime degree r with $(n,r)=1$; D is a central division algebra over K of dimension r^2 containing L; S is the set of places where D ramifies; and $c_v = r'c$ with $n \mid r-1+2rc_v$. Both the cover $\overline{G}_D(\mathbb{A}_K)$ of $D^\times_{\mathbb{A}}$ from [JC26a, §6] and the Kazhdan–Patterson cover $\overline{G}(\mathbb{A}_K)$ of $\mathrm{GL}(r,\mathbb{A}_K)$ with parameter c_v are subgroups of one cover of $\mathrm{GL}(r,\mathbb{A}_L)$ (§§4.2 and 4.5). We use the splitting conventions of Lemma 4.5.1, good elements, the canonical identification $\jmath$ of covered elliptic tori on good elements (Theorem 4.3.10 and Corollary 4.3.13), the normalized orbital integrals $F = \Delta\Phi$ (Definition 4.4.1), and matching with the sign $(-1)^{r-1}$ (Definition 4.4.4).

5.1.2 Main results

We state the three main results informally, with pointers to the precise statements.

Chapter 4 left open the matching of test functions that do not vanish near the center: by Proposition 4.4.8, the orbital integrals of a genuine function $\tilde{f}_D$ on $\overline{G}_D(K_v)$ near the center are a nonzero constant times Δ, so a matching function on $\overline{G}_v$ cannot be supported in the regular set. The key new tool is the following.

Theorem D (Propositions 5.3.4 and 5.3.7)**.** *Let v be a finite place. On a conjugation-invariant open and closed neighborhood V of the unipotent set, the map $g \mapsto \sigma(g) := \widetilde{g^{1/n}}^{\,n}$ (canonical n-th root, any lift, n-th power) is a well-defined continuous section of the cover. It is equivariant under conjugation, consists of good elements at regular points, agrees with the splittings over small open subgroups and over unipotent radicals, and is compatible with $\jmath$. Genuine functions supported on $p^{-1}(ZV)$ correspond bijectively to ordinary functions on G/Z supported on ZV, with the same orbital integrals, on both covers.*

Combined with Chapter 4 away from the center, Theorem D gives:

Theorem E (Theorem 5.5.8)**.** *Let $v \in S$. Every genuine test function on $\overline{G}_D(K_v)$ has a matching genuine test function on $\overline{G}_v$. Every genuine test function on $\overline{G}_v$ whose orbital integrals vanish at good regular non-elliptic elements has a matching function on $\overline{G}_D(K_v)$.*

The matching function in the first part of Theorem E is explicit: $\tilde{f} = \tilde{f}_D(1)\,\tilde{f}^{\mathrm{St}} + \tilde{f}_{\mathrm{reg}}$, where $\tilde{f}^{\mathrm{St}}$ is the transport of the linear Steinberg pseudo-coefficient (§5.5.2) and $\tilde{f}_{\mathrm{reg}}$ is supported on the good regular elliptic set. We also determine the germ near the center that any matching function must have (Theorem 5.5.14).

Three objects should be kept apart: the Steinberg pseudo-coefficient f_{St} on $\mathrm{GL}(r,F)$; the genuine function $\tilde{f}^{\mathrm{St}}$ on $\overline{G}_v$, its transport near the center (cut off to a neighborhood of $\overline{Z}_v$ and rescaled); and pseudo-coefficients of genuine irreducible representations of $\overline{G}_v$. The second is *not* an instance of the third (Remark 5.5.6); matching needs only its orbital integrals.

Theorem C′ (Theorems 5.6.7 and 5.6.9)**.** *Let $\overline{\pi}_D$ be a genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ with a supercuspidal component at a finite place $w_0 \notin S$. The packet identity of Chapter 4 holds with arbitrary matching test functions at all places of S except one, u, which may be chosen freely. If the D-side packet satisfies the non-cancellation hypothesis* (Hyp'_D) *(Definition 5.6.5), which*

is implied by the hypothesis (Hyp_D) *of Chapter 4, then there is a genuine cuspidal automorphic representation* $\overline{\pi}$ *of* $\overline{G}(\mathbb{A}_K)$ *with* $\overline{\pi}_v \cong \overline{\pi}_{D,v}$ *for all* $v \notin S$*. Conversely, if* $\overline{\pi}$ *is the only member of its* GL*-side packet and its components at the places of* S *are square-integrable, then* $\overline{\pi}$ *transfers to* $\overline{G}_D(\mathbb{A}_K)$.

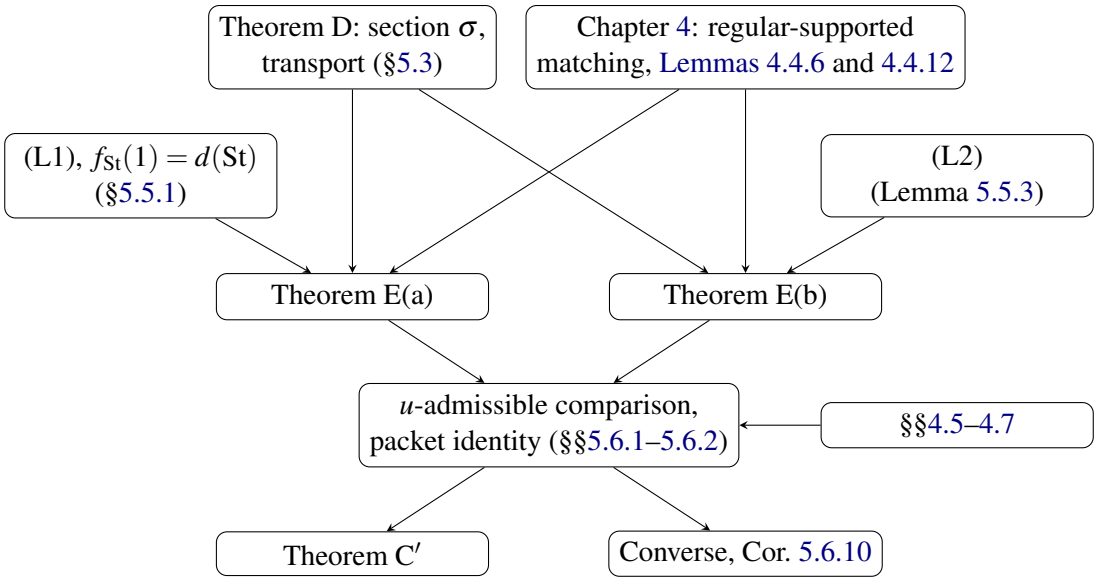
5.1.3 What changes from Chapter 4

We present every new global result as an improvement of a numbered result of Chapter 4. The following table is a guide for the reader of both chapters; each result in Chapter 5 that improves a result of Chapter 4 ends with a paragraph headed "What changes from Chapter 4".

Chapter 4	Chapter 5	New input
Theorem 4.4.7 (Theorem B: regular-supported matching)	Thm. 5.5.8 (Theorem E: arbitrary functions)	Theorem D; (L1), (L2)
Proposition 4.4.8 (orbital integrals near $\overline{Z}_v$, D side)	Thm. 5.5.14 (required germ, GL side)	Theorem D; measures
Lemma 4.4.12 (regular support)	Lemma 5.4.2 (arbitrary $\tilde{f}$)	Lemma 5.4.1
Definition 4.6.5, Theorem 4.6.7 (comparison)	Def. 5.6.1, Thm. 5.6.3	Theorem E
Proposition 4.7.6 (packet identity)	Prop. 5.6.4	Theorem E
Definition 4.7.9, Theorem 4.7.12 (Theorem C, (Hyp_D))	Def. 5.6.5, Thm. 5.6.7 (Theorem C′, (Hyp'_D))	Prop. 5.6.4
Theorem 4.7.13, Remark 4.7.14 (converse, (Hyp))	Thm. 5.6.9, Cor. 5.6.10	Theorem E(b); pseudo-coefficients
Lemma 4.7.8 (cancellation)	Prop. 5.6.11	choice of u
§4.8, items (D1)–(D4)	§5.7	–

5.1.4 Logical structure

The dependencies among the main results are as follows; arrows mean "is used in".



The germ results (Corollary 5.3.8, Lemma 5.5.10, Corollary 5.5.11, Theorem 5.5.14(b)) and the cancellation criterion (Proposition 5.6.11) are not used in the proofs of Theorems D, E and C′.

5.1.5 Conventions and labels

We keep the conventions and the labeling scheme of Chapter 4. They are collected, with the rest of the series' notation and the cocycle normalization, in the shared glossary of §3 (labels in §3.1, covers and the relabeling of μ_n in §10.1, Haar measures in §10.4).

All Hilbert symbols are n-th Hilbert symbols, and μ_n is relabeled once by $\zeta \mapsto \zeta^{r'}$ as in §4.1.4. Statements are marked by status, with proved-here the default and carrying no visible tag:

Proved here (no tag): a complete argument is included, possibly using results marked **(E)**. For global statements, "proved" means proved relative to the cited results of Chapter 4, which are restated in §5.2.3.

- **(E)** *Established elsewhere*: a precisely cited theorem, used as a black box. Several of these are results of Kazhdan [Kaz86] that Flicker and Kazhdan state for covers in [FK86]; we cite them in the form stated in [FK86].
- **(O)** *Proof outlined*: the steps are identified, but at least one remains to be written out or checked.
- **(C)** *Conjectural*: requires new mathematical input (later parts of the series).

A remark immediately after each **(E)**, **(O)** or **(C)** statement explains its status.

Statement	Where	Status
Compatible measures at S; Chapter 4 unaffected	§5.2.2	**(P)**
n-th roots; the section σ; compatibility with $\jmath$ (Theorem D)	§§5.3.1–5.3.2	**(P)**
Transport; regular elements near $\overline{Z}_v$ are good (Theorem D)	§5.3.3	**(P)**
Germ expansion at central good elements	Cor. 5.3.8	**(P)** using **(E)**
Orbital integrals of arbitrary genuine functions (r prime)	§5.4	**(P)**
Linear inputs (L1), (L2), $f_{\mathrm{St}}(1) = d(\mathrm{St})$	§5.5.1	**(P)** using **(E)**
The pair $(e_D, \tilde{f}^{\mathrm{St}})$	Prop. 5.5.5	**(P)**
Matching of arbitrary functions (Theorem E)	Thm. 5.5.8	**(P)**
Required germ near the center: orbital-integral form	Thm. 5.5.14(a)	**(P)**
Unipotent orbital integrals of functions cuspidal near 1; the trivial germ	Lemma 5.5.10, Cor. 5.5.11	**(P)** using **(E)**
Required germ: unipotent form (G1), (G2)	Thm. 5.5.14(b)	**(P)**
Comparison and packet identity with u-admissible pairs	§§5.6.1–5.6.2	**(P)** relative to Chapter 4
Theorem C′ and its converse	§5.6.3	**(P)** relative to Chapter 4, under (Hyp'_D), (Hyp')
Cancellation criterion for (Hyp'_D); finitely many S-types suffice	Prop. 5.6.11, Cor. 5.6.13	**(P)**
Local character relations; unconditional transfer	§5.7	**(C)** (character relations at tame places: Chapter 6)

The running example

The series' running example is $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$ (Example 1.3.1, and [JC26b]). Definition 5.2.1 fixes $\mathrm{vol}(D_v^\times / F^\times) = d(\mathrm{St})^{-1}$ at both places of S. Theorem E applies at both. At the wild place $\mathfrak{p}_3$, where the covered elliptic tori are non-abelian, matching near the center uses Theorem D. In Theorem C′ the place u is $\mathfrak{p}_3$ or $\mathfrak{p}_7$, and (Hyp'_D) is a condition on test functions at these two places.

5.1.6 Organization

Section 5.2 recalls what we use from Chapter 4 and fixes the Haar measures at the ramified places. Sections 5.3–5.5 contain the local theory: the section σ and the transport (§5.3), orbital integrals of arbitrary genuine functions away from the center (§5.4), and matching of arbitrary functions (§5.5). Section 5.6 proves Theorem C′. Section 5.7 lists what is left for Chapter 6. Section 3 collects the notation of the series and the notation specific to Chapter 5, and Section 2.2 discusses the work related to this Part; the literature of the whole series is reviewed in §2.1.

5.2 Recollections from Chapter 4, and Haar measures

This section recalls the notation of Chapter 4 that is used throughout, and fixes the Haar measures at the ramified places once and for all. It depends only on Chapter 4, whose results used here are restated in §5.2.3 as (I1)–(I11). It introduces the local notation P_g, A_g, Φ^z, the compatible measures of Definition 5.2.1, and the constant λ_v.

5.2.1 Local notation

We fix the local notation used in §§5.3–5.5.

Throughout §§5.3–5.5, v is a finite place, $F = K_v$, $\mathfrak{p}$ its maximal ideal, ϖ a uniformizer and p the residual characteristic. We write G for either $\mathrm{GL}(r,F)$ (embedded in $\mathscr{G}_v$ by ι_{GL}) or $D_v^\times$ (embedded by ι_D), $\overline{G} = p^{-1}(G) \subset \overline{\mathscr{G}}_v$, $Z = F^\times$ and $\overline{Z} = p^{-1}(Z)$, which is central and common to both covers (Lemma 4.2.3). For $g \in G$ let P_g be its characteristic polynomial (reduced characteristic polynomial if $G = D_v^\times$), $A_g := F[g]$ the commutative F-algebra it generates, and call the roots of P_g in $\overline{F}$ the eigenvalues of g. For a genuine character $\tilde{\theta}$ of $\overline{Z}$, $C_c^\infty(\overline{G},\tilde{\theta})_{\mathrm{gen}}$ is the space of smooth genuine functions with $\tilde{f}(z\tilde{g}) = \tilde{\theta}(z)^{-1}\tilde{f}(\tilde{g})$ and support compact modulo $\overline{Z}$. Orbital integrals Φ and $F = \Delta\Phi$ are those of Definition 4.4.1, with $\Delta(\gamma) = |D(\gamma)|^{1/2}|\det\gamma|^{-(r-1)/2}$; as in [FK86, §6], $D(\gamma)$ is formed from the distinct eigenvalues. We also use the variant $\Phi^z(\gamma,f) = \int_{Z\backslash G} f(g^{-1}\gamma g)\,dg$ of [FK86, §6] for regular elliptic γ; then $\Phi^z(\gamma,f) = \mathrm{vol}(T_\gamma/Z)\,\Phi(\gamma,f)$.

5.2.2 Haar measures at the ramified places

We fix compatible Haar measures at the places of S (Definition 5.2.1), used in the rest of the chapter. This choice is made once and for all; it simplifies the normalization at the identity and the formal-degree relations of §5.7(ii).

Measures are as in §4.4.1: dg on G/Z, dt on T/Z, and dg/dt on $T\backslash G$; at $v \in S$ they are specialized as follows.

Definition 5.2.1 (Compatible measures)**.** At each $v \in S$, let dg_v be any Haar measure on $\mathrm{GL}(r,F)/F^\times$, and let $d(\mathrm{St}) = d(\mathrm{St};dg_v)$ be the formal degree, with respect to dg_v, of the Steinberg representation of $\mathrm{GL}(r,F)$ (trivial central character). Normalize the Haar measure $dg_{D,v}$ on the compact group $D_v^\times/F^\times$ by

$$\mathrm{vol}(D_v^\times/F^\times; dg_{D,v}) = d(\mathrm{St};dg_v)^{-1}.$$

The measures dt on tori are arbitrary, as in Chapter 4 (matching does not depend on them).

Example 5.2.2 (How the normalization behaves)**.** Let $v \in S$ and fix dg_v. The formal degree of a square-integrable representation is inversely proportional to the Haar measure. So if dg_v is replaced by $c\,dg_v$ with $c > 0$, then $d(\mathrm{St})$ is replaced by $c^{-1}d(\mathrm{St})$, and Definition 5.2.1 replaces $dg_{D,v}$ by $c\,dg_{D,v}$. The two measures are tied together, and only their common scale is free. For instance, one may choose dg_v so that $d(\mathrm{St};dg_v) = 1$. Then $D_v^\times/F^\times$ has volume 1, and $dg_{D,v}$ is the probability Haar measure. In the running example this choice can be made at each of $\mathfrak{p}_3$ and $\mathfrak{p}_7$.

Remark 5.2.3. (1) §4.5.3 allows arbitrary dg_v and $dg_{D,v}$ at $v \in S$, provided matching and the trace formulas use the same ones. So every result of Chapter 4, local and global, holds with the measures of Definition 5.2.1, and we cite Chapter 4 without change.

(2) The formal degree of the trivial representation of $D_v^\times$ is $\mathrm{vol}(D_v^\times/F^\times)^{-1}$, and the trivial representation corresponds to St under the linear Jacquet–Langlands correspondence [Bad08, Theorem 2.2]. So Definition 5.2.1 says that this pair has equal formal degrees. We expect that it agrees with Kottwitz's normalization of measures on inner forms [Kot88], but nothing below uses this.

(3) For arbitrary measures, the constant $\lambda_v := d(\mathrm{St})\,\mathrm{vol}(D_v^\times/F^\times)$ would appear in the normalization at the identity (Theorem 5.5.14(c)). Definition 5.2.1 is exactly the condition $\lambda_v = 1$.

5.2.3 Statements used from Chapter 4

For the reader's convenience, and to make the dependence on Chapter 4 auditable, we restate the results of Chapter 4 that are used, in the form used. Numbers refer to Chapter 4. The labels (I1)–(I11) are used in the rest of the paper.

(I1) *Admissible functions* (§4.4.3). For $\bullet \in \{D, \mathrm{GL}\}$ and an elliptic torus of type i, a function h on the good regular elements $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ is admissible if it is (i) locally constant, (ii) satisfies $h(z\tilde{t}) = \tilde{\theta}_v(z)^{-1}h(\tilde{t})$ for $z \in \overline{Z}_v$, (iii) has support compact modulo $\overline{Z}_v$ with closure contained in the regular set, and (iv) is invariant under the normalizer of the torus.

(I2) *Lemma 4.4.6.* (b) $h \mapsto h \circ \jmath_i^{-1}$ is a bijection between admissible functions on the two sides. (c) Every admissible h equals $F(\cdot, \tilde{f}_i)$ on $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ for some $\tilde{f}_i \in C_c^\infty$ supported on the good regular elements conjugate into that torus; $F(\tilde{\delta}, \tilde{f}_i) = 0$ for regular $\tilde{\delta}$ not conjugate into it.

(I3) *Theorem 4.4.7.* (a) Every $\tilde{f}_D$ vanishing near $\overline{Z}_v$ has a matching $\tilde{f}$ supported on the good regular elliptic set. (b) Every $\tilde{f}$ supported in the regular set with vanishing orbital integrals at good regular non-elliptic elements has a matching $\tilde{f}_D$ vanishing near $\overline{Z}_v$.

(I4) *Proposition 4.4.8.* For $\tilde{f}_D$ there are a normal open compact $U \subset D_v^\times$ and a splitting κ_U with $F(z\kappa_U(t), \tilde{f}_D) = \Delta(t)\,\mathrm{vol}(T\backslash D_v^\times)\tilde{f}_D(z)$ for $z \in \overline{Z}_v$ and regular $t \in U$.

(I5) *Lemma 4.4.10(b)* (Harish-Chandra). For $C \subset T^{\mathrm{reg}}$ and $\Omega \subset G$ compact modulo Z, the set of $\dot{x} \in T\backslash G$ with $x^{-1}Cx \cap \Omega \neq \emptyset$ is compact. (Stated in Chapter 4 for any such Ω.)

(I6) *Lemma 4.4.12.* For regular-supported $\tilde{f}$, $F(\cdot, \tilde{f})$ on an elliptic torus is admissible. *Lemma 4.4.14.* Traces of regular-supported functions depend only on their good regular orbital integrals.

(I7) *Proposition 4.6.1.* The D-side trace formula: $\sum m(\overline{\pi}_D)\,\mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D) = \mathrm{vol}\cdot\tilde{f}_D(1) + \sum_{[\gamma]} c(\gamma)F(\iota(\gamma), \tilde{f}_D)$, the sum over rational regular classes modulo the center that are good everywhere; every non-central rational element of $G_D(K)$ is regular since r is prime. No condition on the test function at S.

(I8) *Proposition 4.6.2* (variant of [FK86, §18]). For $\tilde{f}$ with (i) a supercusp form at some finite place w_0 and (ii) support in the regular elliptic set at some finite place u, and no condition elsewhere: $\sum_{\mathrm{cuspidal}} m(\overline{\pi})\,\mathrm{tr}\,\overline{\pi}(\tilde{f}) = \sum_{[\gamma]} c(\gamma)F(\iota(\gamma), \tilde{f})$, the sum over rational regular elliptic classes good everywhere.

(I9) *Theorem 4.5.7 and Corollary 4.5.8.* If $\tilde{f}_{D,v}$ and $\tilde{f}_v$ match at every $v \in S$ (no support condition) and correspond via ψ_v outside S, then $F(\iota(\gamma), \tilde{f}_D) = F(\iota(j\gamma), \tilde{f})$, the local signs multiplying to $(-1)^{(r-1)|S|} = 1$ (if r is odd, $r-1$ is even; if $r = 2$, $|S|$ is even because the local invariants of D sum to 0 and K has no real places); and classes whose field does not embed in D contribute 0 whenever (M2) holds at all $v \in S$, since such a class is non-elliptic at some $v \in S$ (Albert–Brauer–Hasse–Noether).

(I10) *Lemma 4.7.2.* For fixed levels at the finite places (with any test functions bi-invariant under them) and fixed K_∞-types and infinitesimal character, only finitely many automorphic representations contribute on either side.

(I11) *Lemma 4.7.8 (proof).* For a level U_v at $v \in S$ (split by κ_v, with $\tilde{\theta}_v \circ \kappa_v$ trivial on $U_v \cap K_v^\times$), the corner algebra satisfies $\mathscr{H}_v \cong \bigoplus_{\tau_v \in \mathscr{T}_v} \mathrm{End}(\tau_v^{U_v})$, and $\mathrm{ev}_{1,v} = \mathrm{vol}(\overline{G}_D(K_v)/\overline{Z}_v)^{-1} \sum_{\tau_v} (\dim \tau_v) \operatorname{tr} \tau_v$ on $\mathscr{H}_v$; $\mathscr{R}_v = \ker \mathrm{ev}_{1,v}$.

5.3 Harmonic analysis near the center

This section proves Theorem D. We construct a canonical n-th root on a conjugation-invariant neighborhood of the unipotent set (§5.3.1), lift it to an equivariant section σ of the cover (§5.3.2), and use σ to transport genuine functions supported near $\overline{Z}$ to ordinary functions on G/Z, with the same orbital integrals (§5.3.3). As a corollary, the germ expansion of genuine orbital integrals at central good elements is the Shalika germ expansion of the linear group (§5.3.4). Everything in this section holds at every finite place, including those above n, and on both covers. The section depends on §§4.2–4.3 (the covers, scalar centrality and the identification j) and on the Shalika germ expansion for $\mathrm{GL}(r,F)$ [Sha72]. It introduces the neighborhoods V_M, the constants δ_n, B, M_0, M_1, the root $g^{1/n}$, the section σ and the transport $\mathscr{T}$.

5.3.1 n-th roots near the unipotent set

We define a canonical n-th root on a conjugation-invariant neighborhood of the unipotent set, and introduce Ω_M, V_M, δ_n, B, M_0 and $g^{1/n}$, which are used throughout. The point of using n-th roots is that the n-th power of a lift does not depend on the lift.

For $M \geq 1$ let Ω_M be the set of monic polynomials of degree r over F that are congruent to $(X-1)^r$ coefficientwise modulo $\mathfrak{p}^M$, and put

$$V_M := \{g \in G : \ P_g \in \Omega_M\}.$$

Absolute values. In this subsection $q = |\mathscr{O}/\mathfrak{p}|$, $e = v_F(p)$ is the absolute ramification index, and $|x|_F = q^{-v_F(x)}$ is the normalized absolute value of F, extended uniquely to $\overline{F}$; on $\mathscr{O}$-algebras we use the corresponding sup norm $\|\cdot\|$ on coordinates. (In terms of the p-adic absolute value $|\cdot|_p$ with $|p|_p = p^{-1}$, one has $|x|_F = |x|_p^{[F:\mathbb{Q}_p]}$.) Put

$$\delta_n := \begin{cases} 1, & p \nmid n, \\ q^{-e/(p-1)}, & p \mid n, \end{cases} \qquad B := |n|_F^{-1} q^{e/(p-1)}.$$

Lemma 5.3.1. *(a) V_M is open, closed and stable under conjugation, and so is ZV_M. Every eigenvalue λ of $g \in V_M$ satisfies $|\lambda-1|_F \le |\varpi|_F^{M/r}$.*

(b) If $u \in \overline{F}$, $u^n = 1$ and $|u-1|_F < \delta_n$, then $u = 1$.

(c) $|\binom{1/n}{k}|_F \le B^k$ for all $k \ge 0$. There is M_0 such that for $M \ge M_0$ and $g \in V_M$ the series $g^{1/n} := \sum_{k\ge 0}\binom{1/n}{k}(g-1)^k$ converges in A_g. Its sum is the unique $h \in A_g$ with $h^n = g$ whose eigenvalues μ satisfy $|\mu-1|_F < \delta_n$.

(d) For $M \ge M_0$: $(xgx^{-1})^{1/n} = xg^{1/n}x^{-1}$ for $x \in G$; the map $g \mapsto g^{1/n}$ is continuous on V_M; and $(ag)^{1/n} = a^{1/n}g^{1/n}$ whenever $a \in F^\times$ and $a, g, ag \in V_{M_0}$.

Proof. (a) The map $g \mapsto P_g$ is continuous and Ω_M is open and closed, so V_M is open and closed; it is conjugation-stable because P_g is. If $\mu = \lambda - 1$, then μ is a root of $P_g(X+1) = X^r + \sum_{i<r} c_i X^i$ with $|c_i|_F \le |\varpi|_F^M$. If $|\mu|_F > 1$, then $|\mu|_F^r \le |\varpi|_F^M |\mu|_F^{r-1}$ gives $|\mu|_F < 1$, a contradiction; so $|\mu|_F \le 1$ and $|\mu|_F^r \le |\varpi|_F^M$. For ZV_M: openness is clear. If $a_k g_k \to g$ with $g_k \in V_M$, then $\det g_k \in \mathscr{O}^\times$ (its reduced norm, when $G = D_v^\times$), so $|a_k|_F^r \to |\det g|_F$ and the a_k lie in a compact subset of $F^\times$; passing to a convergent subsequence $a_k \to a$, we get $g_k \to a^{-1}g \in V_M$, so $g \in ZV_M$.

(b) If $u \ne 1$, then u is a root of unity of order $m > 1$ dividing n. If $p \nmid m$, the reduction of u is a nontrivial root of unity in the residue field, so $|u-1|_F = 1$. If $p \mid m$, some power u^{m/p^a} is a primitive p^a-th root of unity ζ with $a \ge 1$, and $|u-1|_F \ge |\zeta-1|_F \ge |\zeta_p - 1|_F = |p|_F^{1/(p-1)} = q^{-e/(p-1)}$. (The first inequality holds because $u^j - 1 = (u-1)(u^{j-1} + \cdots + 1)$ with the second factor integral; the second because $|\zeta - 1|_F = |p|_F^{1/(p^{a-1}(p-1))}$ for ζ of exact order p^a.)

(c) Since $\binom{1/n}{k} = n^{-k}\prod_{j<k}(1-jn)/k!$ and $v_p(k!) \le k/(p-1)$, we have $|\binom{1/n}{k}|_F \le |n|_F^{-k}|k!|_F^{-1} \le |n|_F^{-k} q^{ek/(p-1)} = B^k$. Put $y = g - 1$. Since P_g has integral coefficients, y is integral over $\mathscr{O}$, so $\mathscr{O}[y] \subset A_g$ is a finitely generated, hence complete, $\mathscr{O}$-module. By Cayley–Hamilton $y^r = -\sum_{i<r} c_i y^i \in \varpi^M \mathscr{O}[y]$, so $y^{kr+i} \in \varpi^{kM}\mathscr{O}[y]$. So the series converges in $\mathscr{O}[y]$ as soon as $|\varpi|_F^M B^r < 1$. Then $h^n = g$, by the identity $(\sum_k \binom{1/n}{k} X^k)^n = 1 + X$ in $\mathbb{Q}[[X]]$ and absolute convergence. Each F-algebra map $\chi : A_g \to \overline{F}$ is continuous, and $\chi(g)$ runs over the eigenvalues of g. Hence the eigenvalues of h are the numbers $b(\lambda - 1)$, where $b(x) = \sum_k \binom{1/n}{k} x^k$ and $|\lambda - 1|_F \le |\varpi|_F^{M/r}$. Since $|b(x) - 1|_F \le \max_{k\ge 1}(B|x|_F)^k$, enlarging M_0 gives $|b(\lambda-1) - 1|_F < \delta_n$. For uniqueness, if h' is another such root, then $u := h'h^{-1} \in A_g$ satisfies $u^n = 1$, and each eigenvalue of u is within δ_n of 1 (ultrametric inequality, since the eigenvalues of h are units). By (b) all eigenvalues of u equal 1. Since $X^n - 1$ is separable, u is semisimple, so $u = 1$.

(d) Equivariance and multiplicativity follow from the uniqueness in (c): $xg^{1/n}x^{-1} \in A_{xgx^{-1}}$ is an n-th root of xgx^{-1} with the right eigenvalues, and $a^{1/n}g^{1/n} \in A_g = A_{ag}$ is an n-th root of ag whose eigenvalues are products of numbers within δ_n of 1. For continuity, let $g_0 \in V_M$. Suppose first that $G = \mathrm{GL}(r,F)$, or $G = D_v^\times$ with D_v split, identified with $\mathrm{GL}(r,F)$ by some $D_v \cong M_r(F)$. Then $\Lambda := \mathscr{O}[g_0]\mathscr{O}^r$ is a g_0-stable lattice (finitely generated since g_0 is integral), and g_0^{-1} is also integral since $\det g_0 \in \mathscr{O}^\times$; so after conjugation $g_0 \in \mathrm{GL}(r,\mathscr{O})$. On the open set $V_M \cap \mathrm{GL}(r,\mathscr{O})$ we have $y \in M_r(\mathscr{O})$ and $\|y^r\| \le |\varpi|_F^M$, so the series converges uniformly there. (Note that $V_M \not\subset \mathrm{GL}(r,\mathscr{O})$: for instance $\left(\begin{smallmatrix} 1 & \\ 0 & \varpi^{-N}_1 \end{smallmatrix}\right) \in V_M$ for all M, N. This is why we conjugate.) If D_v is a division algebra, then $V_M \subset \mathscr{O}_D^\times$, since an element of D_v with integral reduced characteristic polynomial lies in the

maximal order $\mathscr{O}_D$, and the same estimate holds in $\mathscr{O}_D$. Conjugation is a homeomorphism, so continuity holds everywhere. □

5.3.2 The section σ

We now lift the n-th root to the cover, defining the section σ (and σ_D, σ_{GL}) used in the rest of the chapter. The resulting section is equivariant, consists of good elements, and is compatible with every splitting in sight and with the identification $\jmath$ of Chapter 4.

Definition 5.3.2 (The section σ)**.** For $M \geq M_0$ and $g \in V_M$ put $\sigma(g) := \tilde{h}^n$, where $h = g^{1/n}$ and $\tilde{h}$ is any lift of h to $\overline{G}$. We write σ_D and σ_{GL} when the group must be specified.

Example 5.3.3 (The section at the identity and on scalars)**.** Let $M \geq M_0$. The identity lies in V_M and $1^{1/n} = 1$. Every lift of 1 is an element $\zeta \in \mu_n$, and $\zeta^n = 1$, so $\sigma(1)$ is the identity of $\overline{G}$. Next let $a \in F^\times$ be a scalar in V_M. By Lemma 5.3.1(c), $h = a^{1/n}$ is the n-th root of a in F with $|h-1|_F < \delta_n$. So $\sigma(a) = \tilde{h}^n$ for any lift $\tilde{h}$ of h, and it lies in $\overline{Z}$ because h is a scalar. In the running example at $\mathfrak{p}_7$ we have $n = 3$, $F = \mathbb{Q}_7$ and $\delta_3 = 1$. For a scalar $a \in V_M$ the root h is the cube root of a that is congruent to 1 modulo 7. The other two cube roots of a in $\mathbb{Q}_7$ are $\zeta_3 h$ and $\zeta_3^2 h$, and they are not congruent to 1 modulo 7.

Since two lifts of h differ by $\zeta \in \mu_n$ and $\zeta^n = 1$, $\sigma(g)$ is well defined.

Proposition 5.3.4. *Let $M \geq M_0$.*

(a) $\sigma : V_M \to \overline{G}$ *is a continuous section of p, and $\sigma(xgx^{-1}) = \tilde{x}\sigma(g)\tilde{x}^{-1}$ for all $x \in G$ and any lift $\tilde{x}$.*

(b) For every $g \in V_M$, $\sigma(g)$ commutes with $p^{-1}(\mathrm{Cent}_G(g))$. In particular, $\sigma(\gamma)$ is good for every regular $\gamma \in V_M$, and $\sigma(t) \in Z(\overline{T})$ for every $t \in T \cap V_M$ and every maximal torus T.

(c) If $a \in F^\times$ and $a, g, ag \in V_{M_0}$, then $\sigma(a) \in \overline{Z}$ and $\sigma(ag) = \sigma(a)\sigma(g)$.

(d) Let $W \subseteq V_{M_0}$ be an open compact subgroup over which p splits by a homomorphism κ. Then there is an open compact subgroup $U \subseteq W \cap V_M$ with $u^{1/n} \in W$ for all $u \in U$, and $\sigma|_U = \kappa|_U$. In particular, the splitting κ_U of Proposition 4.4.8, is $\sigma_D|_U$ (there $U = U_k$, $W = U_{k'}$, and $\kappa_U(t) = \kappa(s)^n$ with $s = t^{1/n} \in U_{k'}$).

(e) If u is unipotent and lies in the unipotent radical N of a parabolic subgroup of $\mathrm{GL}(r,F)$, then $\sigma(u) = s_N(u)$, where s_N is the unique splitting of p over N.

(f) (Compatibility of the two sides.) *For $a \in F^\times \cap V_{M_0}$, $\sigma_D(a) = \sigma_{\mathrm{GL}}(a)$ in $\overline{Z}$. Let $E \subset D_v$ be a maximal subfield, $j : E \hookrightarrow M_r(F)$ an embedding, and $\jmath_{E,j} : Z(\overline{E}_D) \to Z(\overline{E}_{\mathrm{GL}})$ the canonical identification of Theorem 4.3.10. Then $\jmath_{E,j}(z\,\sigma_D(t)) = z\,\sigma_{\mathrm{GL}}(j(t))$ for $z \in \overline{Z}$ and $t \in E^\times \cap V_M$.*

Proof. (a) Near any point, $g \mapsto g^{1/n}$ is continuous (Lemma 5.3.1(d)) and p has continuous local sections, so $\tilde{h}$ may be chosen to depend continuously on g locally. Equivariance: $\tilde{x}\tilde{h}\tilde{x}^{-1}$ is a lift of $xhx^{-1} = (xgx^{-1})^{1/n}$, so $\tilde{x}\tilde{h}^n\tilde{x}^{-1} = (\tilde{x}\tilde{h}\tilde{x}^{-1})^n = \sigma(xgx^{-1})$.

(b) If $c \in \mathrm{Cent}_G(g)$, then c commutes with $A_g \ni h$, so for a lift $\tilde{c}$ we have $\tilde{c}\tilde{h}\tilde{c}^{-1} = \zeta\tilde{h}$ with $\zeta \in \mu_n$, and therefore $\tilde{c}\tilde{h}^n\tilde{c}^{-1} = \zeta^n\tilde{h}^n = \tilde{h}^n$.

(c) $a^{1/n} \in F^\times$, so its lifts lie in $\overline{Z}$, which is central. Hence $\sigma(ag) = (\widetilde{a^{1/n}}\tilde{h})^n = \widetilde{a^{1/n}}^n \tilde{h}^n = \sigma(a)\sigma(g)$, using Lemma 5.3.1(d).

(d) The root map $u \mapsto u^{1/n}$ is continuous at 1 and fixes 1, so some open compact subgroup $U \subseteq W \cap V_M$ satisfies $u^{1/n} \in W$ for $u \in U$. For such u, $\kappa(u^{1/n})$ is a lift of $u^{1/n}$, so $\sigma(u) = \kappa(u^{1/n})^n = \kappa(u)$, since κ is a homomorphism. We do *not* assume $u^{1/n} \in U$: when $p \mid n$ this generally fails for the standard filtration subgroups, since a p-th root of an element of $1+\mathfrak{p}^m$ lies only in a larger principal-unit group.

(e) $u^{1/n} = \sum_k \binom{1/n}{k}(u-1)^k$ is a finite sum, and it lies in N. Since s_N is a homomorphism, $\sigma(u) = s_N(u^{1/n})^n = s_N(u)$.

(f) The first assertion holds because $\iota_D(a) = \iota_{\mathrm{GL}}(a)$ and $a^{1/n} \in F^\times$, so both sides are the n-th power of a lift of the same element of $\mathscr{G}_v$. For the second, put $s = t^{1/n} \in F[t] \subseteq E$. Then $j(s) = j(t)^{1/n}$, by uniqueness in Lemma 5.3.1(c), since j is an F-algebra embedding and preserves eigenvalues. Let $\varphi \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$. It is a group isomorphism over $\iota_{\mathrm{GL}} j \iota_D^{-1}$, so $\varphi(\sigma_D(t)) = \varphi(\tilde{s})^n$, and $\varphi(\tilde{s})$ is a lift of $j(s)$. Hence $\varphi(\sigma_D(t)) = \sigma_{\mathrm{GL}}(j(t))$. By (b), $\sigma_D(t) \in Z(\overline{E}_D)$, where φ restricts to $\jmath_{E,j}$; and φ is the identity on $\overline{Z}$. □

Remark 5.3.5. (1) Proposition 5.3.4 holds equally for $v \mid n$; only M_0 grows. So the distinction between $v \mid n$ and $v \nmid n$, which governs the structure of the covered tori (Proposition 4.3.5), plays no role near the center.

(2) Part (e) says that σ extends the canonical lifting of unipotent elements. So the unipotent orbital integrals $\Phi(z\sigma(u), \tilde{f}) := \int_{G_u \backslash G} \tilde{f}(\tilde{x}^{-1} z \sigma(u) \tilde{x})\, d\dot{x}$ are well defined by (b), and they are the unipotent orbital integrals of [FK86, §8], where unipotent elements are lifted by the splitting s of [FK86, §2] (see Remark 5.3.9(1) for the normalization of F).

5.3.3 Transport to the linear group

We show that genuine functions supported in $p^{-1}(ZV_M)$ are the same thing as linear functions on G/Z supported in ZV_M, with the same orbital integrals. This holds on both sides and is compatible with $\jmath$. The constant M_1 and the transport $\mathscr{T}$ (and $\mathscr{T}_D$) defined here are used in §5.5.

Lemma 5.3.6 (Choice of M)**.** *Let $\tilde{\theta}$ be a genuine character of $\overline{Z}$. There is $M_1 \geq M_0$ such that for $M \geq M_1$:*

(a) if $a \in F^\times$, $g \in V_M$ and $ag \in V_M$, then $a \in 1+\mathfrak{p}^{\lceil M/r \rceil} \subseteq V_{M_0}$;

(b) $\tilde{\theta}(\sigma(a)) = 1$ for all $a \in 1+\mathfrak{p}^{\lceil M/r \rceil}$.

Proof. (a) If λ is an eigenvalue of g, then $a\lambda$ is one of ag, and Lemma 5.3.1(a) gives $|a-1|_F = |a\lambda - \lambda|_F / |\lambda|_F \leq |\varpi|_F^{M/r}$. A scalar a with $|a-1|_F \leq |\varpi|_F^m$ has $P_a = (X-a)^r \in \Omega_m$. (b) By Proposition 5.3.4(c), $a \mapsto \tilde{\theta}(\sigma(a))$ is a continuous character of $1+\mathfrak{p}^{M_0}$, so it is trivial on $1+\mathfrak{p}^m$ for m large. □

Measure convention. For $g \in G$ let G_g be its centralizer. If $g \in V_M$, then $\sigma(g)$ commutes with $p^{-1}(G_g)$ (Proposition 5.3.4(b)), so the genuine orbital integral $\Phi(z\sigma(g), \tilde{f}) = \int \tilde{f}(\tilde{x}^{-1} z \sigma(g) \tilde{x})\, d\dot{x}$ is an integral over the same quotient space $G_g \backslash G$ as the linear orbital integral at $p(z)g$. Throughout, the two are computed with the *same* quotient measure on $G_g \backslash G$: for regular g, the measure dg/dt

of §4.4.1; for unipotent g, a fixed G-invariant measure on the class of g, used both in the Shalika expansion (Corollary 5.3.8) and in the Richardson formula of Lemma 5.5.10. "Same orbital integrals" below always means equality for these measures.

Fix $M \geq M_1$, and let $C_c^\infty(\overline{G},\tilde{\theta})_{\mathrm{gen},M}$ be the space of $\tilde{f} \in C_c^\infty(\overline{G},\tilde{\theta})_{\mathrm{gen}}$ supported in $p^{-1}(ZV_M)$, and $C_c^\infty(G/Z)_M$ the space of $\varphi \in C_c^\infty(G/Z)$ supported in ZV_M.

Proposition 5.3.7 (Transport)**.** *(a) Every element of $p^{-1}(ZV_M)$ is of the form $z\sigma(g)$ with $z \in \overline{Z}$ and $g \in V_M$. Every regular element of $p^{-1}(ZV_M)$ is good.*

(b) The formula $(\mathscr{T}\tilde{f})(p(z)g) := \tilde{\theta}(z)\,\tilde{f}(z\sigma(g))$ ($z \in \overline{Z}$, $g \in V_M$) defines a linear bijection

$$\mathscr{T} : C_c^\infty(\overline{G},\tilde{\theta})_{\mathrm{gen},M} \xrightarrow{\sim} C_c^\infty(G/Z)_M,$$

with inverse $(\mathscr{T}^{-1}\varphi)(z\sigma(g)) = \tilde{\theta}(z)^{-1}\varphi(p(z)g)$ on $p^{-1}(ZV_M)$ and 0 elsewhere.

(c) For $z \in \overline{Z}$ and regular $\gamma \in V_M$,

$$\Phi(z\sigma(\gamma),\tilde{f}) = \tilde{\theta}(z)^{-1}\,\Phi(p(z)\gamma,\mathscr{T}\tilde{f}), \qquad F(z\sigma(\gamma),\tilde{f}) = \tilde{\theta}(z)^{-1}\,F(p(z)\gamma,\mathscr{T}\tilde{f}),$$

where on the right Φ and $F = \Delta\Phi$ are the linear orbital integrals with the same measures. More generally, the first identity holds for every $\gamma \in V_M$, with Φ the integral over $G_\gamma\backslash G$ (for instance, for unipotent γ).

(d) For a function β on G that is constant on conjugacy classes and on Z-cosets, with $\beta \cdot 1_{ZV_M}$ locally constant, $\mathscr{T}\big((\beta\circ p)\tilde{f}\big) = \beta \cdot \mathscr{T}\tilde{f}$.

Proof. (a) If $\tilde{x} \in p^{-1}(ZV_M)$, write $p(\tilde{x}) = ag$ with $g \in V_M$, and choose a lift z of a; then $\tilde{x} = \zeta z\sigma(g)$ for some $\zeta \in \mu_n \subset \overline{Z}$. A regular element $z\sigma(\gamma)$ is good by Proposition 5.3.4(b), since z is central.

(b) *Independence of the lift z of $a = p(z)$:* replacing z by ζz multiplies $\tilde{\theta}(z)$ by ζ and $\tilde{f}(z\sigma(g))$ by ζ^{-1}. *Independence of the factorization:* if $ag = a'g'$ with $g, g' \in V_M$, then $g' = cg$ with $c = a/a' \in 1+\mathfrak{p}^{\lceil M/r\rceil}$ (Lemma 5.3.6(a)), so $\sigma(g') = \sigma(c)\sigma(g)$ (Proposition 5.3.4(c)). With z' a lift of a', $z := z'\sigma(c)$ is a lift of a, and

$$\tilde{\theta}(z')\tilde{f}(z'\sigma(g')) = \tilde{\theta}(z')\tilde{f}(z\sigma(g)) = \tilde{\theta}(\sigma(c))^{-1}\tilde{\theta}(z)\tilde{f}(z\sigma(g)) = \tilde{\theta}(z)\tilde{f}(z\sigma(g)),$$

by Lemma 5.3.6(b). So $\mathscr{T}\tilde{f}$ is a well-defined function on ZV_M, extended by 0. It is Z-invariant by construction, locally constant because σ is continuous and $\tilde{f}$ is locally constant, and supported in the set ZV_M, which is open and closed (Lemma 5.3.1(a)); its support is compact modulo Z because that of $\tilde{f}$ is compact modulo $\overline{Z}$. The inverse is well defined by the same computation, and it is smooth because $p^{-1}(ZV_M)$ is open and closed.

(c) For $x \in G$, $\tilde{x}^{-1}z\sigma(\gamma)\tilde{x} = z\sigma(x^{-1}\gamma x)$ by Proposition 5.3.4(a). Hence $\tilde{f}(\tilde{x}^{-1}z\sigma(\gamma)\tilde{x}) = \tilde{\theta}(z)^{-1}(\mathscr{T}\tilde{f})(x^{-1}p(z)\gamma x)$. Integrate over $T_\gamma\backslash G$, and note that $\Delta(p(z)\gamma) = \Delta(\gamma)$. For non-regular $\gamma \in V_M$ the same computation applies over $G_\gamma\backslash G$, since $\sigma(\gamma)$ commutes with $p^{-1}(G_\gamma)$ (Proposition 5.3.4(b)).

(d) is immediate from the definitions. □

5.3.4 The germ expansion at central good elements

We record what the transport says about germs. This is not needed for the matching theorem, but it identifies the germs of [FK86, §8] and [KP86, §1] at central elements.

Corollary 5.3.8 (Germ expansion at central good elements)**.** *(using **(E)**) Let $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ with $G = \mathrm{GL}(r,F)$, and let $z \in \overline{Z}$. There is a neighborhood V' of 1 such that for regular $\gamma \in V'$,*

$$\Phi(z\sigma(\gamma), \tilde{f}) = \sum_{\mathscr{O}} \Gamma_{\mathscr{O}}(\gamma)\, \Phi(z\sigma(u_{\mathscr{O}}), \tilde{f}),$$

where $\mathscr{O}$ runs over the unipotent conjugacy classes of $\mathrm{GL}(r,F)$ (partitions of r), $u_{\mathscr{O}} \in \mathscr{O}$, and the $\Gamma_{\mathscr{O}}$ are the Shalika germs of $\mathrm{GL}(r,F)$ for the measures fixed above. On the D side the only unipotent class is $\{1\}$, and the expansion is Proposition 4.4.8.

Proof. Apply Proposition 5.3.7 to $\tilde{f} \cdot 1_{p^{-1}(ZV_M)}$, and the Shalika germ expansion [Sha72], [HC70] **(E)** to $\mathscr{T}(\tilde{f} \cdot 1_{p^{-1}(ZV_M)})$ at 1. Since that function is Z-invariant, its expansion at $p(z)$ is its expansion at 1. By Proposition 5.3.7(c), applied also to the unipotent orbits, the factors $\tilde{\theta}(z)^{\pm 1}$ cancel. □

Remark 5.3.9. (1) [FK86, §8, Theorem] and [KP86, Theorem 1.4] (credited there to Shalika, Harish-Chandra, Vignéras and Kazhdan–Patterson) give a germ expansion of genuine orbital integrals at every semisimple element, with germs indexed by the good elements above it, and a converse characterizing orbital-integral functions [KP86, Theorem 1.7]. The germs are shown to exist; their structure is not determined there ([KP86, §1, Remark 2]). Corollary 5.3.8 identifies them, at central good elements, with the Shalika germs of $\mathrm{GL}(r,F)$. In comparing normalizations, note that [FK86] normalize $F = \Delta\Phi$ with Δ formed from distinct eigenvalues, so that at $z\sigma(u)$ with z central $F = |\det z|^{-(r-1)/2}\Phi$.

(2) The principle that orbital integrals on covers near a semisimple element reduce to the Lie algebra of its centralizer is part of W.-W. Li's semisimple descent [Li12, §4], and we make no claim of priority for Corollary 5.3.8. What we use is the explicit transport of Proposition 5.3.7, which is compatible with *both* covers and with $\jmath$ (Proposition 5.3.4(f)).

5.4 Orbital integrals of arbitrary genuine functions

This section extends the facts about orbital integrals on elliptic tori in §4.4.6, from regular-supported functions to arbitrary genuine functions. For r prime, the only singular elements near which the elliptic regular set accumulates are central (Lemma 5.4.1), so once the behavior near $\overline{Z}$ is removed, the elliptic orbital integrals of an arbitrary genuine function are admissible in the sense of §4.4.3 (Lemma 5.4.2). The section depends on §4.4.6, and introduces no new notation.

Lemma 5.4.1 (Singular elements, r prime)**.** *Let $G = \mathrm{GL}(r,F)$ with r prime. A non-central singular semisimple element s has a neighborhood containing no regular elliptic element. Hence, for every elliptic maximal torus T, every point of $T \smallsetminus Z$ is regular, and $(T \smallsetminus ZV_M)/Z$ is a compact subset of T^{reg}/Z.*

Proof. Write $P_s = \prod_i Q_i^{m_i}$ with distinct monic irreducible Q_i. If there were only one factor, $P_s = Q^m$ with $m \deg Q = r$, so $m = 1$ (then P_s is separable and s is regular) or $m = r$ and $\deg Q = 1$ (then the

semisimple element s has a single eigenvalue in F and is central). Both are excluded, so $P_s = AB$ with A, B coprime and of positive degree. By Hensel's lemma, every monic polynomial close enough to P_s factors as $A'B'$ with $\deg A' = \deg A$ and $\deg B' = \deg B$. So elements close to s have reducible characteristic polynomials and are not elliptic. The second statement holds because the singular elements of T are central, T/Z is compact, and $T \smallsetminus ZV_M$ is closed. □

Lemma 5.4.2 (Extension of Lemma 4.4.12)**.** *(using **(E)**) Let $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ be arbitrary, T an elliptic maximal torus, and $h = F(\cdot, \tilde{f})$ on $\overline{T}^{g,\mathrm{reg}}$. Then h satisfies conditions (i), (ii) and (iv) of §4.4.3. If moreover $h = 0$ on $p^{-1}(ZV_M) \cap \overline{T}^{g,\mathrm{reg}}$ for some M, then h is admissible.*

Proof. The proofs of (ii) and (iv) in Lemma 4.4.12, do not use the support of $\tilde{f}$. The proof of (i) uses Lemma 4.4.10(b), with $\Omega = p(\operatorname{supp} \tilde{f})$; that lemma holds for every Ω compact modulo Z [HC70] **(E)**. For (iii): $\operatorname{supp} h$ is closed in the open and closed subgroup $Z(\overline{T})$ and contained in $p^{-1}(T \smallsetminus ZV_M)$, whose image in T/Z is compact and regular by Lemma 5.4.1. □

Remark 5.4.3. Lemma 5.4.1 is the only place where primality of r enters the local theory of this chapter (besides its use in Chapter 4). For composite r there are non-central singular semisimple elements with elliptic centralizers, of the form $\mathrm{GL}(m, E')$ with $m[E' : F] = r$ and $m > 1$. Germ expansions there exist by [FK86, §8], but they are not explicit, and the method of §5.3 would have to be extended to covers of $\mathrm{GL}(m, E')$.

5.5 Matching of arbitrary functions

This section proves Theorem E. We first collect the inputs about the linear groups (§5.5.1). We then construct the genuine analog of the pair (constant function on $D_v^\times$, Steinberg pseudo-coefficient on $\mathrm{GL}(r, F)$) near the center (§5.5.2), prove matching in both directions (§5.5.3), and determine the germ near the center that every matching function must have (§5.5.4). Throughout, $v \in S$, $F = K_v$, and $\tilde{\theta}_v$ is a genuine character of $\overline{Z}_v$, common to both covers. The section depends on §§5.3–5.4, on §4.4, and on the linear results of [FK86], [Kaz86] and [Bad08] cited in §5.5.1. It introduces the linear inputs (L1), (L2), the functions f_{St}, e_D, $\tilde{f}^{\mathrm{St}}$, and the conditions (Ell), (Non), (G1), (G2).

5.5.1 Inputs about the linear groups

We state the two linear inputs (L1) and (L2), which are used throughout this section, prove the value of the Steinberg pseudo-coefficient at the identity, and derive (L2) from statements in [FK86] and [Bad08].

Let $G = \mathrm{GL}(r, F)$.

(L1) **(E)** There is $f_{\mathrm{St}} \in C_c^\infty(G/Z)$, a pseudo-coefficient of the Steinberg representation, with $\operatorname{tr} \mathrm{St}(f_{\mathrm{St}}) = 1$, $\operatorname{tr} \pi'(f_{\mathrm{St}}) = 0$ for every irreducible tempered $\pi' \ncong \mathrm{St}$ with trivial central character, $\Phi(\gamma, f_{\mathrm{St}}) = (-1)^{r-1} \operatorname{vol}(T_\gamma/Z)^{-1}$ for regular elliptic γ, and $\Phi(\gamma, f_{\mathrm{St}}) = 0$ for regular non-elliptic γ.

(L2) (Lemma 5.5.3) Every $\varphi \in C_c^\infty(G/Z)$ with $\Phi(\gamma, \varphi) = 0$ for all regular non-elliptic γ has a matching $\varphi_D \in C_c^\infty(D_v^\times/Z)$.

Source of (L1). [FK86, §25.1, Theorem], from [Kaz86, Theorem K], is stated for every square-integrable representation of the cover, and so, with $n = 1$ (the convention of [FK86, §8, Remark (3)]), for $\mathrm{GL}(r,F)$. It gives $\Phi^z(\gamma, f_{\mathrm{St}}) = \overline{\Theta_{\mathrm{St}}(\gamma)}$ on the regular elliptic set and $\Phi^z = 0$ at regular non-elliptic elements, together with the trace properties. Since $\Phi^z = \mathrm{vol}(T_\gamma/Z)\Phi$ and $\Theta_{\mathrm{St}} = (-1)^{r-1}$ on the regular elliptic set, this is (L1). The value of Θ_{St} follows from [Bad08, Theorem 2.2] with $n = 1$, $d = r$, since the trivial representation of $D_v^\times$ corresponds to St.

Remark 5.5.1. *From compact support to trivial central character.* The linear statements (L1), (L2) are usually formulated for $C_c^\infty(G)$ with measures on G and T. Choose $b \in C_c^\infty(F^\times)$ supported in small neighborhoods of representatives of $F^\times/F^{\times r}$ (a finite group), normalized so that $\int_{F^\times} b(z^r y)\,dz = 1$ for all y. For $\varphi \in C_c^\infty(G/Z)$, the function $\varphi' := \varphi \cdot (b \circ \det)$ (respectively $b \circ \mathrm{Nrd}$) lies in $C_c^\infty(G)$, satisfies $\int_Z \varphi'(zg)\,dz = \varphi(g)$, and has orbital integrals $b(\det\gamma)$ times those of φ. In particular φ' is cuspidal if φ is. Since averaging over Z commutes with matching ($z\gamma \leftrightarrow z\gamma'$, $\Delta(z\gamma) = \Delta(\gamma)$), matching statements for C_c^∞ imply the same statements for $C_c^\infty(\cdot/Z)$.

Lemma 5.5.2 (Value of f_{St} at the identity)**.** *(using **(E)**) With f_{St} as in (L1), $f_{\mathrm{St}}(1) = d(\mathrm{St})$, the formal degree of* St *with respect to dg.*

Proof. By Harish-Chandra's Plancherel formula for G/Z with trivial central character [Wal03, Théorème VIII.1.1, in the form (3) of §VIII.1] (in explicit form for $\mathrm{GL}(n)$, see [AP05]), for $f \in C_c^\infty(G/Z)$,

$$f(1) = \sum_\pi d(\pi)\,\mathrm{tr}\,\pi(f) + \sum_{P \neq G} \int \mathrm{tr}\left(\mathrm{Ind}_P^G \sigma\right)(f)\,d\mu_P(\sigma),$$

where π runs over square-integrable representations with trivial central character, and the second sum runs over proper standard parabolic subgroups $P = MN$, with σ running over unitary twists of square-integrable representations of M and μ_P absolutely continuous Plancherel measures. For $\mathrm{GL}(r,F)$ every $\mathrm{Ind}_P^G\sigma$ here is irreducible and tempered (the segments attached to unitary twists of square-integrable representations are pairwise unlinked, so [Zel80, Theorem 9.7(a)] applies), and it is not square-integrable when $P \neq G$, so it is not St. By [FK86, §25.1, Theorem] with $n = 1$, $\mathrm{tr}\,\pi'(f_{\mathrm{St}}) = 0$ for every irreducible tempered $\pi' \ncong \mathrm{St}$, and $\mathrm{tr}\,\mathrm{St}(f_{\mathrm{St}}) = 1$. Hence only the term $\pi = \mathrm{St}$ survives, and $f_{\mathrm{St}}(1) = d(\mathrm{St})$. □

Lemma 5.5.3 (The linear input (L2))**.** *(using **(E)**) Let $G = \mathrm{GL}(r,F)$, r prime, and let $\varphi \in C_c^\infty(G/Z)$ satisfy $\Phi(\gamma,\varphi) = 0$ for all regular non-elliptic γ. Then there is $\varphi_D \in C_c^\infty(D_v^\times/Z)$ with $F(t,\varphi_D) = (-1)^{r-1}F(j(t),\varphi)$ for every regular $t \in D_v^\times$ and every embedding j of $F[t]$ in $M_r(F)$.*

Proof. Step 1 (finitely many discrete series). Let φ be bi-invariant under an open compact subgroup K'. Then $\mathrm{tr}\,\pi(\varphi) \neq 0$ only if $\pi^{K'} \neq 0$. There are only finitely many square-integrable π with trivial central character and $\pi^{K'} \neq 0$: by Harish-Chandra's finiteness theorem [Wal03, Théorème VIII.1.2] **(E)** these form finitely many orbits under twisting by unramified characters $|\det|^s$, and in each orbit only finitely many members have trivial central character (the twist multiplies the central character by $|z|^{rs}$). Let Π be this finite set, and for $\pi \in \Pi$ let f_π be a pseudo-coefficient as in [FK86, §25.1, Theorem] with $n = 1$ **(E)**. Put

$$\psi := \varphi - \sum_{\pi \in \Pi} \mathrm{tr}\,\pi(\varphi)\,f_\pi.$$

Step 2 (ψ has vanishing tempered traces). Let π' be irreducible and tempered, with trivial central character. If π' is square-integrable, then $\mathrm{tr}\,\pi'(\psi) = \mathrm{tr}\,\pi'(\varphi) - \mathrm{tr}\,\pi'(\varphi) = 0$ by the defining

property of the f_π (and $\operatorname{tr}\pi'(\varphi)=0$ if $\pi'\notin\Pi$). Otherwise $\pi'=\operatorname{Ind}_P^G\sigma$ with $P\neq G$ and σ a unitary twist of a square-integrable representation of M (the classification of tempered representations of $\mathrm{GL}(r,F)$; [Zel80, §9]), and $\operatorname{tr}\pi'(f_\pi)=0$ by [FK86, §25.1]. Also $\operatorname{tr}\pi'(\varphi)=0$: by the Weyl integration formula (characters are locally integrable [FK86, §13]), $\operatorname{tr}\pi'(\varphi)$ is a sum over conjugacy classes of maximal tori of integrals of $\Theta_{\pi'}\cdot\Phi(\cdot,\varphi)$. On non-elliptic tori $\Phi(\cdot,\varphi)=0$ by hypothesis, and on elliptic tori $\Theta_{\pi'}=0$, since the character of a representation induced from a proper parabolic subgroup vanishes on the regular elliptic set [vD72] **(E)**.

Step 3 (density). We use [FK86, §27.3, Proposition] **(E)** with $n=1$. For $n=1$ the cover is G itself, $x^*=x$, every representation is genuine and relevant, and the correspondence of [FK86, §26] is the identity (the character relation (26.1) becomes $\Theta_{\tilde\pi}=\Theta_\pi$); "matching" means equality of $F(x,\cdot)$ at regular x. The proposition then reads: *if $f,f'\in C_c^\infty(G/Z)$ satisfy $\operatorname{tr}\pi(f)=\operatorname{tr}\pi(f')$ for every irreducible tempered π with trivial central character, then $F(x,f)=F(x,f')$ for every regular x.* Apply it with $f=\psi$ and $f'=0$, using Step 2. (For covers in general, the corresponding density of tempered characters is proved in [Li12].) So $\Phi(\gamma,\varphi)=\sum_{\pi\in\Pi}\operatorname{tr}\pi(\varphi)\Phi(\gamma,f_\pi)$ for all regular γ, and on the regular elliptic set

$$\Phi^z(\gamma,\varphi)=\sum_{\pi\in\Pi}c_\pi\overline{\Theta_\pi(\gamma)},\qquad c_\pi:=\operatorname{tr}\pi(\varphi).$$

Step 4 (transfer). By [Bad08, Theorem 2.2] with $n=1$, $d=r$ **(E)**, there is a bijection $\pi\mapsto C(\pi)$ from essentially square-integrable representations of $\mathrm{GL}(r,F)$ to irreducible representations of $D_v^\times$, preserving central characters, with $\Theta_{C(\pi)}(t)=(-1)^{r-1}\Theta_\pi(j(t))$ for regular t. For an irreducible representation τ of $D_v^\times$ with trivial central character, put $f_\tau:=\operatorname{vol}(D_v^\times/Z)^{-1}\overline{\Theta_\tau}\in C_c^\infty(D_v^\times/Z)$. Since f_τ is conjugation invariant, $\Phi^z(t,f_\tau)=\operatorname{vol}(D_v^\times/Z)f_\tau(t)=\overline{\Theta_\tau(t)}$. Put $\varphi_D:=\sum_{\pi\in\Pi}c_\pi f_{C(\pi)}$. Then

$$\Phi^z(t,\varphi_D)=\sum_\pi c_\pi\overline{\Theta_{C(\pi)}(t)}=(-1)^{r-1}\sum_\pi c_\pi\overline{\Theta_\pi(j(t))}=(-1)^{r-1}\Phi^z(j(t),\varphi).$$

Divide by $\operatorname{vol}(T/Z)$, which is the same on both sides because the same dt is used on T and $j(T)$ (§4.4.1), and multiply by $\Delta(t)=\Delta(j(t))$. The measure $dg_{D,v}$ enters only through $\operatorname{vol}(D_v^\times/Z)$ in f_τ and in $\Phi^z(t,\cdot)$, and cancels.

On the complex conjugate. The identity $\Phi^z(\gamma,f_\pi)=\overline{\Theta_\pi(\gamma)}$ is the normalization compatible with $\operatorname{tr}\pi(f_\pi)=1$: by the Weyl integration formula, for f with vanishing non-elliptic orbital integrals, $\operatorname{tr}\pi'(f)=\sum_T(|W_T|\operatorname{vol}(T/Z))^{-1}\int_{T/Z}\Delta(t)^2\Theta_{\pi'}(t)\Phi^z(t,f)\,dt$, which for $\Phi^z(\cdot,f)=\overline{\Theta_\pi}$ is the elliptic inner product $\langle\Theta_{\pi'},\Theta_\pi\rangle$ of [FK86, §24], equal to $\delta_{\pi\pi'}$ for square-integrable π' [FK86, §25]. In any case the argument is insensitive to this convention: replacing $\overline{\Theta}$ by Θ throughout Steps 3 and 4 changes nothing, because the transfer identity of Step 4 has real coefficients $(-1)^{r-1}$. □

Remark 5.5.4. Lemma 5.5.3, combined with Remark 5.5.1, is the converse half of the linear transfer of functions of [DKV84]. We include the proof because [Bad08, §4.6] states only the regular-supported case, and because it uses only results whose statements we have checked in [FK86] and [Bad08].

5.5.2 The Steinberg-type function

We construct the genuine analog, near the center, of the pair (constant function on $D_v^\times$, Steinberg pseudo-coefficient). It exists as a pair of functions with matching orbital integrals; it is not a pair of

pseudo-coefficients of single genuine representations. The functions e_D and $\tilde{f}^{\mathrm{St}}$ defined here enter Theorem E.

Fix $M \geq M_1$ (for $\tilde{\theta}_v$, on both sides) and put

$$e_D := \mathscr{T}_D^{-1}\big(1_{ZV_{D,M}}\big) \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}},$$
$$\tilde{f}^{\mathrm{St}} := d(\mathrm{St})^{-1}\,\mathscr{T}^{-1}\big(f_{\mathrm{St}} \cdot 1_{ZV_M}\big) \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}},$$

with f_{St} as in (L1). Here $1_{ZV_{D,M}} \in C_c^\infty(D_v^\times/Z)$ because $D_v^\times/Z$ is compact and $ZV_{D,M}$ is open and closed. Concretely, $e_D(z\sigma_D(g)) = \tilde{\theta}_v(z)^{-1}$ for $g \in V_{D,M}$, and $e_D = 0$ off $p^{-1}(ZV_{D,M})$. So $e_D(1) = 1$.

Proposition 5.5.5. *(using (L1)) The functions e_D and $\tilde{f}^{\mathrm{St}}$ match. Explicitly, $\tilde{f}^{\mathrm{St}}(1) = 1 = e_D(1)$, and at good regular elements*

$$F(z\sigma(\gamma), \tilde{f}^{\mathrm{St}}) = \begin{cases} (-1)^{r-1}\tilde{\theta}_v(z)^{-1}\Delta(\gamma)\big(d(\mathrm{St})\,\mathrm{vol}(T_\gamma/Z)\big)^{-1} & \gamma \in V_M \text{ regular elliptic}, \\ 0 & \gamma \in V_M \text{ regular non-elliptic}, \end{cases}$$

and $F(\cdot, \tilde{f}^{\mathrm{St}}) = 0$ at good regular elements outside $p^{-1}(ZV_M)$.

Proof. By Proposition 5.3.7(c),(d), $F(z\sigma(\gamma), \tilde{f}^{\mathrm{St}}) = d(\mathrm{St})^{-1}\tilde{\theta}_v(z)^{-1}1_{V_M}(\gamma)\Delta(\gamma)\Phi(\gamma, f_{\mathrm{St}})$, and (L1) gives the displayed values. On the D side, $e_D(\tilde{x}^{-1}z\sigma_D(t)\tilde{x}) = \tilde{\theta}_v(z)^{-1}1_{V_{D,M}}(t)$, so $F(z\sigma_D(t), e_D) = \tilde{\theta}_v(z)^{-1}1_{V_{D,M}}(t)\Delta(t)\,\mathrm{vol}(D_v^\times/Z)/\,\mathrm{vol}(T/Z)$, and $\mathrm{vol}(D_v^\times/Z) = d(\mathrm{St})^{-1}$ by Definition 5.2.1. Outside $p^{-1}(ZV_{D,M})$ and $p^{-1}(ZV_M)$ both orbital integrals vanish, and $\jmath$ preserves these sets, since it preserves characteristic polynomials and is the identity on $\overline{Z}_v$. Now compare, using Proposition 5.3.4(f), the equality $1_{V_{D,M}}(t) = 1_{V_M}(j(t))$, and the sign $(-1)^{r-1}$ in (M1). □

Remark 5.5.6 (Not a pseudo-coefficient of one genuine representation)**.** $\tilde{f}^{\mathrm{St}}$ is defined by orbital integrals, not by representations, and matching needs nothing more. Under the character relations (D2) of Chapter 4 (proved at tame places where the corrected Flicker–Kazhdan correspondence is available, for example when $(n, N_r) = 1$; see §5.7(ii)), $\mathrm{tr}\,\overline{\pi}(\tilde{f}^{\mathrm{St}}) = \mathrm{tr}\,\overline{\pi}_D(e_D)$ for corresponding pairs. Now $\mathrm{tr}\,\tau(e_D) = \int_{\overline{G}_D(K_v)/\overline{Z}_v} e_D\,\Theta_\tau$, which is the integral of $g \mapsto \Theta_\tau(\sigma_D(g))$ over $ZV_{D,M}/Z$, is in general nonzero for more than one genuine irreducible τ (Fourier inversion on the compact group $\overline{G}_D(K_v)/\overline{Z}_v$ expresses e_D in terms of the characters of all τ of level contained in $V_{D,M}$). So $\tilde{f}^{\mathrm{St}}$ should be a combination of pseudo-coefficients of several genuine square-integrable representations, not the pseudo-coefficient of a single "genuine Steinberg" representation. Accordingly, the phrase "a genuine analog of a Steinberg pseudo-coefficient" in §4.4.5, should be read as "a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center".

Remark 5.5.7 (No genuine characters when $v \mid n$)**.** If $v \mid n$, then $\overline{G}_D(K_v)$ has no genuine one-dimensional representation. By Proposition 4.3.5(b), some covered elliptic torus $\overline{E}_D$ is non-abelian, so there are commuting $x, y \in E^\times$ whose lifts have commutator $\zeta \neq 1$. A one-dimensional genuine χ would give $1 = \chi([\tilde{x}, \tilde{y}]) = \zeta$. So the linear pair (trivial representation of $D^\times$, St) has no direct analog among genuine representations. The pair $(e_D, \tilde{f}^{\mathrm{St}})$ is its analog among functions.

5.5.3 The matching theorem

We now prove Theorem E. The D-side function is split into a multiple of e_D and a function vanishing near the center; the GL-side function is split by a cut-off into a piece near the center and a piece away from it.

Theorem 5.5.8 (Matching of arbitrary functions)**.** *(using (L1), (L2)) Let $v \in S$.*

(a) Every $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ has a matching $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$. One may take

$$\tilde{f} = \tilde{f}_D(1)\, \tilde{f}^{\mathrm{St}} + \tilde{f}_{\mathrm{reg}},$$

with $\tilde{f}_{\mathrm{reg}}$ supported on the good regular elliptic set. This uses only (L1).

(b) Every $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ with $F(\tilde{\delta}, \tilde{f}) = 0$ for all good regular non-elliptic $\tilde{\delta}$ has a matching $\tilde{f}_D$. This uses (L2).

Proof. (a) By construction $e_D(z) = \tilde{\theta}_v(z)^{-1}$ on $\overline{Z}_v$, so $\tilde{f}_D - \tilde{f}_D(1)e_D$ vanishes on $\overline{Z}_v$. Being smooth with support compact modulo $\overline{Z}_v$, it is right invariant under some $\kappa(U)$, so it vanishes on the neighborhood $\overline{Z}_v\kappa(U)$ of $\overline{Z}_v$. By Theorem 4.4.7(a), it has a matching $\tilde{f}_{\mathrm{reg}}$ supported on the good regular elliptic set. By Proposition 5.5.5, e_D matches $\tilde{f}^{\mathrm{St}}$. Matching is linear.

(b) Fix $M \geq M_1$ and write $\tilde{f} = \tilde{f}_1 + \tilde{f}_2$ with $\tilde{f}_1 = \tilde{f} \cdot 1_{p^{-1}(ZV_M)}$. Both pieces are smooth (the set is open and closed), and both have vanishing non-elliptic orbital integrals, since the cut-off is constant on conjugacy classes (Proposition 5.3.7(d)).

The piece near the center. $\varphi := \mathscr{T}\tilde{f}_1 \in C_c^\infty(G/Z)_M$ has vanishing regular non-elliptic orbital integrals, by Proposition 5.3.7(c) and because orbital integrals of φ outside ZV_M vanish. By (L2) and Remark 5.5.1, there is a matching $\varphi_D \in C_c^\infty(D_v^\times/Z)$. Since the elliptic orbital integrals of φ vanish off ZV_M, $\varphi_D \cdot 1_{ZV_{D,M}}$ still matches φ, and it lies in $C_c^\infty(D_v^\times/Z)_M$. By Propositions 5.3.7(c) and 5.3.4(f), $\tilde{f}_{D,1} := \mathscr{T}_D^{-1}(\varphi_D 1_{ZV_{D,M}})$ matches $\tilde{f}_1$. (The good regular elements of $p^{-1}(ZV_{D,M})$ correspond under $\jmath$ to the good regular elliptic elements of $p^{-1}(ZV_M)$, as in the proof of Theorem 5.5.14(a).)

The piece away from the center. By Lemma 5.4.2, for each i the function $F(\cdot, \tilde{f}_2)$ on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$ is admissible, since it vanishes on $p^{-1}(ZV_M)$. Transport by $\jmath_i^{-1}$, multiply by $(-1)^{r-1}$, realize on $\overline{G}_D(K_v)$ by Lemma 4.4.6(b),(c), and sum over i, exactly as in the proof of Theorem 4.4.7(b). This gives $\tilde{f}_{D,2}$, and (M2) for $\tilde{f}_2$ is the hypothesis.

Then $\tilde{f}_D := \tilde{f}_{D,1} + \tilde{f}_{D,2}$ matches $\tilde{f}$. □

Remark 5.5.9. (1) Theorem 5.5.8(b) applies to supercusp forms at $v \in S$, whose regular non-elliptic orbital integrals vanish (Selberg's principle; [FK86, §13(i)]) **(E)**. This is item (D1)(b) of Chapter 4.

(2) An alternative proof of (b), using (L1) and Lemma 5.5.10 instead of (L2), is noted after Theorem 5.5.14.

(3) Theorem 5.5.8 does not by itself give $\mathrm{tr}\,\overline{\pi}(\tilde{f}) = \mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D)$ for single representations; that is goal (ii) (§5.7). It does strengthen the global results of Chapter 4, since arbitrary test functions may now be used at the places of $S \smallsetminus \{u\}$ (§5.6).

5.5.4 The germ near the center

We determine the orbital integrals near $\overline{Z}_v$ that any function matching a given $\tilde{f}_D$ must have. The point is that $\tilde{f}_D$ has only one number's worth of germ at the whole center, namely $\tilde{f}_D(1)$. The conditions (Ell), (Non), (G1), (G2) and the germs $\Gamma_{\mathscr{O}}$ appear here.

Let $v \in S$ and $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$. Since $\tilde{f}_D(z\tilde{g}) = \tilde{\theta}_v(z)^{-1}\tilde{f}_D(\tilde{g})$, we have $\tilde{f}_D(z) = \tilde{\theta}_v(z)^{-1}\tilde{f}_D(1)$ on $\overline{Z}_v$. So Proposition 4.4.8, together with Proposition 5.3.4(d), reads

$$F(z\sigma_D(t), \tilde{f}_D) = \tilde{\theta}_v(z)^{-1}\, \tilde{f}_D(1)\, \frac{\Delta(t)}{d(\mathrm{St})\ \mathrm{vol}(T/Z)} \qquad (t \in T \cap V_{D,M} \text{ regular}), \tag{5.1}$$

for M large (depending on $\tilde{f}_D$), every maximal subfield E, $T = E^\times$, and $z \in \overline{Z}_v$. We used $\mathrm{vol}(T\backslash D_v^\times) = \mathrm{vol}(D_v^\times/Z)/\mathrm{vol}(T/Z)$ and Definition 5.2.1. Note that (5.1) is uniform over all types of maximal subfields, cyclic or not.

We first prove two facts about the linear group. Let $G = \mathrm{GL}(r,F)$ (here $r \geq 2$ need not be prime). Shalika germs $\Gamma_{\mathscr{O}}$ are normalized so that $\Phi(\gamma,f) = \sum_{\mathscr{O}} \Gamma_{\mathscr{O}}(\gamma)\Phi(u_{\mathscr{O}},f)$ for regular γ near 1, with $\Phi(1,f) = f(1)$, unnormalized orbital integrals Φ, and fixed invariant measures on the unipotent classes [Sha72].

Lemma 5.5.10 (Functions cuspidal near the identity)**.** *(using **(E)**) Let $\varphi \in C_c^\infty(G/Z)$, and let Ω be a conjugation-invariant neighborhood of 1 in G. If $\Phi(\gamma,\varphi) = 0$ for every regular non-elliptic $\gamma \in \Omega$, then $\Phi(u,\varphi) = 0$ for every non-trivial unipotent $u \in G$. Consequently $\Phi(\gamma,\varphi) = \Gamma_{\{1\}}(\gamma)\,\varphi(1)$ for regular γ near 1.*

Proof. Step 1 (compact support). Choose $\eta \in C_c^\infty(F^\times)$ equal to 1 near 1, and put $f(g) := \varphi(g)\eta(\det g)$. Then $f \in C_c^\infty(G)$, since $z \mapsto z^r$ is proper on $F^\times$. The unipotent orbital integrals of f and φ agree, because $\det u = 1$ on unipotent classes; and the regular orbital integrals agree near 1, because $\eta \circ \det = 1$ on the conjugacy class of γ when $\det\gamma$ is near 1. So we may replace φ by f and Ω by a smaller neighborhood.

Step 2 (descent). Let $P = MN$ be a proper standard parabolic subgroup, and $f^P(m) = \delta_P(m)^{1/2}\int_K\int_N f(k^{-1}mnk)\,dn\,dk$ the constant term, with $K = \mathrm{GL}(r,\mathscr{O})$. By [FK86, §7, Proposition] with $n = 1$ **(E)**, $F(x,f) = F^M(x,f^P)$ for every $x \in M$ that is regular in G. Here $F = \Delta\Phi$ and $F^M = \Delta_M\Phi^M$ are normalized orbital integrals, in the conventions of §5.2.1, and [FK86] define the constant term with the same factor $\delta_P^{1/2}$ as here. Since $\Delta(x)$ and $\Delta_M(x)$ are nonzero at such x, $\Phi(x,f) = 0$ if and only if $\Phi^M(x,f^P) = 0$. Compare the parabolic descent statement [KT26, Proposition 2.39]; its differently normalized descent function differs from ours by factors that do not vanish (discriminants, and $\delta_P(x) = 1$ for x near 1), so it gives the same vanishing assertion. Such x are non-elliptic in G, since their centralizer lies in M. So $\Phi^M(x,f^P) = 0$ for $x \in M \cap \Omega$ regular in G. The elements of M that are regular in G are dense in the M-regular set near 1 (scale the blocks by distinct central scalars close to 1), and $\Phi^M(\cdot,f^P)$ is locally constant on the M-regular set. Hence $\Phi^M(x,f^P) = 0$ for all M-regular x near 1.

Step 3 (independence of germs on M). By the Shalika expansion on $M \simeq \prod_i \mathrm{GL}(n_i,F)$ [Sha72] **(E)**, $0 = \sum_\nu \Gamma_\nu^M(x)\Phi^M(u_\nu,f^P)$ for M-regular x near 1, the sum over unipotent classes ν of M. The germs Γ_ν^M are products of the germs of the factors, and the germs of each $\mathrm{GL}(n_i,F)$ are linearly independent as germs at 1 on the regular set ([KP86, Proposition 1.6] with $n = 1$; [HC78, Lemma 24]) **(E)**. Products of linearly independent families on the factors are linearly independent on the product. So every $\Phi^M(u_\nu,f^P)$ vanishes; in particular $f^P(1) = 0$.

Step 4 (Richardson classes). Let u be a non-trivial unipotent element with Jordan type $\lambda \neq (1^r)$, and let $P_\lambda = M_\lambda N_\lambda$ be the standard parabolic subgroup whose Levi blocks have the sizes of the transposed partition λ^t. It is proper, and the class of u is the Richardson class of P_λ, the dense orbit in N_λ. We claim that $\Phi(u,f) = a_\lambda f^{P_\lambda}(1)$ with $a_\lambda > 0$ depending only on the normalizations of measures. On the Lie algebra this is [How74, Proposition 5] **(E)**: for $\phi \in C_c^\infty(\mathfrak{gl}_r(F))$ and the nilpotent orbit $\mathscr{O}_\lambda$, the orbital integral is $\int_{\mathfrak{u}_\lambda}\int_K \phi(\mathrm{Ad}(k)Y)\,dk\,dY$, where $\mathfrak{u}_\lambda$ is the nilradical of the parabolic subalgebra stabilizing the flag defined by an element of $\mathscr{O}_\lambda$ (that is, of $\mathrm{Lie}\,P_\lambda$, up to conjugacy). The map $Y \mapsto \exp Y = \sum_{k<r} Y^k/k!$ is a G-equivariant homeomorphism from the nilpotent cone onto the unipotent set, carrying $\mathscr{O}_\lambda$ onto the class of u and $\mathfrak{u}_\lambda$ onto N_λ, with

$\exp_* dY = dn$ on $\mathfrak{u}_\lambda$ (Haar measures on a unipotent group correspond under exp). Choose $\phi \in C_c^\infty(\mathfrak{gl}_r(F))$ with $\phi(Y) = f(\exp Y)$ for nilpotent Y (possible since the nilpotent cone is closed, exp is a homeomorphism onto the closed unipotent set, and so $f \circ \exp$ is locally constant with compact support on the cone; such functions extend to $C_c^\infty(\mathfrak{gl}_r(F))$). Howe's identity involves ϕ only at nilpotent elements. Then both sides of Howe's identity for ϕ are the corresponding integrals for f, which gives the claim. By Step 3, $\Phi(u, f) = 0$.

The last assertion follows from the Shalika expansion of φ at 1. □

Corollary 5.5.11 (The trivial germ)**.** *(using (L1)) For regular γ in a neighborhood of* 1*,*

$$\Gamma_{\{1\}}(\gamma) = \begin{cases} (-1)^{r-1}\big(d(\mathrm{St})\,\mathrm{vol}(T_\gamma/Z)\big)^{-1} & \gamma \textit{ elliptic}, \\ 0 & \gamma \textit{ non-elliptic}. \end{cases}$$

Proof. By (L1), f_{St} satisfies the hypothesis of Lemma 5.5.10 with $\Omega = G$. Hence $\Phi(\gamma, f_{\mathrm{St}}) = \Gamma_{\{1\}}(\gamma) f_{\mathrm{St}}(1) = \Gamma_{\{1\}}(\gamma) d(\mathrm{St})$ near 1 (Lemma 5.5.2). Compare with the values of $\Phi(\gamma, f_{\mathrm{St}})$ in (L1). □

Remark 5.5.12. Corollary 5.5.11 agrees with Rogawski's theorem on the trivial germ on elliptic tori [Rog81] and with Harish-Chandra's vanishing of the trivial germ on non-elliptic tori [HC78]. Our proof derives it from (L1) and the Plancherel formula instead.

Example 5.5.13 (Continuing Example 1.3.1: the trivial germ at $\mathfrak{p}_7$)**.** Take $v = \mathfrak{p}_7$, the tame place of Example 1.3.1, so $r = 2$ and $F = K_v = \mathbb{Q}_7$, with $D_v \cong (-1,7)_{\mathbb{Q}_7}$ a division quaternion algebra. Since $r = 2$, Corollary 5.5.11 gives the sign $(-1)^{r-1} = -1$: on every elliptic torus,

$$\Gamma_{\{1\}}(\gamma) = -\big(d(\mathrm{St})\,\mathrm{vol}(T_\gamma/Z)\big)^{-1} < 0,$$

and $\Gamma_{\{1\}}(\gamma) = 0$ on the split torus. The maximal tori of $\mathrm{GL}(2, \mathbb{Q}_7)$ fall into exactly two types: the split torus $F^\times \times F^\times$, and an elliptic torus $E^\times$ for each of the three quadratic extensions $E/\mathbb{Q}_7$, namely the unramified $\mathbb{Q}_7(\sqrt{u})$ (for u a nonsquare unit) and the two ramified extensions $\mathbb{Q}_7(\sqrt{7})$, $\mathbb{Q}_7(\sqrt{7u})$.

Which of these four tori are *covered*, in the sense of corresponding to a maximal subfield of D_v? For a central division algebra D of prime degree r over a local field F, a degree-r field extension E/F embeds in D exactly when E splits D, i.e. when $\mathrm{inv}(D \otimes_F E) = 0$ in $\mathrm{Br}(F) \cong \mathbb{Q}/\mathbb{Z}$. Here $\mathrm{inv}(D) = 1/2$, and restriction to a degree-2 extension multiplies the invariant by 2, so $\mathrm{inv}(D \otimes_F E) = 2 \cdot \frac{1}{2} = 0$ for *every* quadratic $E/\mathbb{Q}_7$ ($r = 2$ is exactly why this works for all three at once; compare the obstruction in Remark 6.3.2, which is the same computation with $r \neq 2$). So all three quadratic extensions embed in D_v, and none of them is left over: the covered tori are precisely the three elliptic ones, and the split torus is exactly the one type that is never covered.

Consequently every nonvanishing value of the trivial germ lives on a covered torus, and the one torus where it vanishes is the one torus with no D-side counterpart. This is the local picture behind Theorem 5.5.14: near the identity, a function matching $\tilde{f}_D$ must reproduce the (same, negative) germ coefficient on each of the three quadratic directions and vanish identically on the split direction, so the germ condition is already concentrated exactly where the two sides can see each other.

Theorem 5.5.14 (The required germ)**.** *Let $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$, and let $M \geq M_1$ be large enough for* (5.1) *on both sides.*

(a) *$\tilde{f}$ satisfies (M1) and (M2) of Definition 4.4.4, with $\tilde{f}_D$ at all good regular elements of $p^{-1}(ZV_M)$ if and only if*

$$F(z\sigma(\gamma),\tilde{f}) = (-1)^{r-1}\,\tilde{\theta}_v(z)^{-1}\,\tilde{f}_D(1)\,\frac{\Delta(\gamma)}{d(\mathrm{St})\,\mathrm{vol}(T_\gamma/Z)} \quad (\gamma \in V_M \text{ reg. elliptic}), \tag{Ell}$$

$$F(z\sigma(\gamma),\tilde{f}) = 0 \quad (\gamma \in V_M \text{ reg. non-elliptic}), \tag{Non}$$

for all $z \in \overline{Z}_v$. The right side of (Ell) is the same multiple of $\Delta(\gamma)/\mathrm{vol}(T_\gamma/Z)$ for every elliptic torus.

(b) (Germ form.) There is $M' \geq M$, depending on $\tilde{f}$, such that (Ell) and (Non) hold for all regular $\gamma \in V_{M'}$ if and only if

(G1) *$\Phi(z\sigma(u),\tilde{f}) = 0$ for every non-trivial unipotent u and every $z \in \overline{Z}_v$, and*

(G2) *$\tilde{f}(1) = \tilde{f}_D(1)$.*

(c) For arbitrary Haar measures at v, as allowed in Chapter 4, (G2) would read $\tilde{f}(1) = \lambda_v \tilde{f}_D(1)$ with $\lambda_v = d(\mathrm{St})\,\mathrm{vol}(D_v^\times/Z)$ (Remark 5.2.3(3)).

Proof. (a) Every good regular element of $p^{-1}(ZV_{D,M}) \subset \overline{G}_D(K_v)$ is conjugate to some $z\sigma_D(t)$ with $t \in T \cap V_{D,M}$ regular, for some maximal subfield E, $T = E^\times$ (Proposition 5.3.7(a)). By Proposition 5.3.4(f), $\jmath$ maps its class to that of $z\sigma_{\mathrm{GL}}(j(t))$, and $j(t) \in V_M$ because $P_{j(t)} = P_t$. Conversely, every good regular elliptic element of $p^{-1}(ZV_M) \subset \overline{G}_v$ arises in this way (Corollary 4.3.13). So (M1) at these elements is (Ell), by (5.1) and $\mathrm{vol}(T_{j(t)}/Z) = \mathrm{vol}(T/Z)$ (§4.4.1). (M2) at these elements is (Non).

(b) Put $\varphi = \mathscr{T}(\tilde{f}1_{p^{-1}(ZV_M)})$, so that $\varphi(1) = \tilde{f}(1)$. By Proposition 5.3.7(c), extended to non-regular elements, $\Phi(z\sigma(u),\tilde{f}) = \tilde{\theta}_v(z)^{-1}\Phi(p(z)u,\varphi) = \tilde{\theta}_v(z)^{-1}\Phi(u,\varphi)$ for unipotent u, since φ is Z-invariant. So (G1) says that $\Phi(u,\varphi) = 0$ for all non-trivial unipotent u.

Sufficiency. By Corollary 5.3.8 and (G1),

$$\Phi(z\sigma(\gamma),\tilde{f}) = \Gamma_{\{1\}}(\gamma)\tilde{f}(z) = \tilde{\theta}_v(z)^{-1}\Gamma_{\{1\}}(\gamma)\tilde{f}(1)$$

for regular γ near 1. Corollary 5.5.11 and (G2) give (Ell) and (Non) after multiplying by $\Delta(\gamma)$, near $\overline{Z}_v$.

Necessity. By (Non) and Proposition 5.3.7(c), $\Phi(\gamma,\varphi) = 0$ for regular non-elliptic $\gamma \in ZV_M$. Lemma 5.5.10 with $\Omega = ZV_M$ gives (G1). Then, as in the sufficiency part, $F(z\sigma(\gamma),\tilde{f}) = \tilde{\theta}_v(z)^{-1}\Delta(\gamma)\Gamma_{\{1\}}(\gamma)\tilde{f}(1)$ near $\overline{Z}_v$. Comparing with (Ell) at regular elliptic γ near 1, where $\Gamma_{\{1\}}(\gamma) \neq 0$ by Corollary 5.5.11, gives (G2).

(c) Repeat the computation in (b) with $\mathrm{vol}(D_v^\times/Z)$ in place of $d(\mathrm{St})^{-1}$ in (5.1) and (Ell). □

Remark 5.5.15. For matching, only part (a) of Theorem 5.5.14 is used. Part (b) is included because it states the result in terms of the germ expansion of Corollary 5.3.8, and because (G2), with the measures of Definition 5.2.1, is exactly the familiar linear normalization $f(1) = f_D(1)$. Part (b) is not used elsewhere. Lemma 5.5.10 also gives an alternative proof of Theorem 5.5.8(b) near the center, without (L2): subtract $\tilde{f}(1)\tilde{f}^{\mathrm{St}}$, after which the orbital integrals vanish near $\overline{Z}_v$ by Lemma 5.5.10 and Corollary 5.5.11.

5.6 Theorem C′: the global transfer improved

This section deduces Theorem C′ from Theorem E. We relax Chapter 4's notion of an admissible pair of global test functions so that arbitrary matching functions are allowed at all places of S but one (§5.6.1), check that the comparison of trace formulas and the packet identity of Chapter 4 still hold (§5.6.2), and prove Theorem C′ and its converse (§5.6.3). Finally we describe exactly when the new non-cancellation hypothesis can fail (§5.6.4). Notation is that of §§4.5–4.7. The section depends on Theorem E and on §§4.5–4.7, in the form (I7)–(I11). It introduces u-admissible pairs and the hypotheses (Hyp'_D) and (Hyp').

5.6.1 u-admissible pairs and the comparison

We define the class of global test functions used in this section (u-admissible pairs, Definition 5.6.1) and show that Chapter 4's comparison of trace formulas applies to it.

Definition 5.6.1 (u-admissible pairs)**.** Fix a finite place $w_0 \notin S$, a place $u \in S$, and a finite set $S_D \supseteq S \cup \{w_0\}$ as in §4.5.3. A pair $(\tilde{f}_D, \tilde{f})$ of factorizable genuine test functions with central characters $\tilde{\theta}^{-1}$ is *u-admissible* if:

(a) at u: $\tilde{f}_{D,u}$ vanishes near $\overline{Z}_u$, $\tilde{f}_u$ is supported on the regular elliptic set, and they match;

(b) at $v \in S \smallsetminus \{u\}$: $\tilde{f}_{D,v}$ and $\tilde{f}_v$ match (no support condition);

(c) for $v \notin S$: conditions (b) and (c) of Definition 4.6.5.

Example 5.6.2 (Continuing Example 1.3.1: a u-admissible pair that is not admissible)**.** In the running example $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$. Take $u = \mathfrak{p}_7$ and take w_0 above 13, as in Example 4.6.6. At $\mathfrak{p}_7$ choose $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$, with a matching $\tilde{f}_u$ supported on the good regular elliptic set (Theorem 4.4.7(a)). At $\mathfrak{p}_3$ choose any genuine test function $\tilde{f}_{D,\mathfrak{p}_3}$, for instance one with $\tilde{f}_{D,\mathfrak{p}_3}(1) \neq 0$, and let $\tilde{f}_{\mathfrak{p}_3}$ be a matching function. It exists by Theorem 5.5.8(a). Complete the pair outside S as in Example 4.6.6. The result is u-admissible. It is not admissible in the sense of Definition 4.6.5, because $\tilde{f}_{D,\mathfrak{p}_3}$ does not vanish near $\overline{Z}_{\mathfrak{p}_3}$. In the other direction, every admissible pair is u-admissible.

By Theorem 5.5.8(a) and Theorem 4.4.7(a), every family $(\tilde{f}_{D,v})_{v\in S}$ with $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$ is the D-side of a u-admissible pair.

Theorem 5.6.3. *For every u-admissible pair,*

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D)\,\mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D) = \sum_{\overline{\pi}\ \text{cuspidal}} m(\overline{\pi})\,\mathrm{tr}\,\overline{\pi}(\tilde{f}),$$

both sides converging absolutely.

Proof. We check the proof of Theorem 4.6.7, item by item, using the statements (I7)–(I9) of §5.2.3.

(1) *The D side.* By (I7), with no condition at S, the geometric side is $\mathrm{vol}\cdot\tilde{f}_D(1)$ plus a sum over regular classes; the only non-regular rational elements are central, since r is prime. The term $\tilde{f}_D(1) = \prod_v \tilde{f}_{D,v}(1)$ vanishes because $\tilde{f}_{D,u}$ vanishes near $\overline{Z}_u$.

(2) *The* GL *side.* The hypotheses of (I8) are a supercusp form at w_0 and regular elliptic support at the single finite place u; Definition 5.6.1(a),(c) provides both, and (I8) imposes nothing at $S \smallsetminus \{u\}$. So the geometric side is a finite sum over regular elliptic rational classes good everywhere, and the spectral side is the cuspidal one.

(3) *Classes whose field does not embed in D.* Such a class is non-elliptic at some $v \in S$ and contributes 0 by (I9), which uses only (M2) at that v; (M2) holds for every matching pair, whatever its support.

(4) *Corresponding classes, signs and measures.* By (I9), which needs only matching at $v \in S$, $F(\iota(\gamma), \tilde{f}_D) = F(\iota(j\gamma), \tilde{f})$; the product of the local signs is 1. The constants $c(\gamma)$ and the global measures are those of §4.5.3, which allows any dg_v, $dg_{D,v}$ at $v \in S$; in particular Definition 5.2.1 (Remark 5.2.3(1)).

(5) *Places outside S* are as in Definition 4.6.5(b),(c).

Summing gives the identity; absolute convergence is that of (I7) and (I8). □

What changes from Chapter 4. Definition 4.6.5(a), required regular support at every place of S. Only the place u is needed, for the simple trace formula and to kill the identity term; at the other places of S, Theorem E supplies matching partners of arbitrary functions.

5.6.2 The packet identity

We extend Chapter 4's packet identity to u-admissible families.

Proposition 5.6.4. *Let ρ, σ, $P_D(\rho)$ and $P(\rho)$ be as in Proposition 4.7.6, and let $u \in S$. Let $(\tilde{f}_{D,v}, \tilde{f}_v)_{v \in S}$ be matching pairs with $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$ and $\tilde{f}_u$ supported in the regular set. Then only finitely many terms of each side of*

$$\sum_{\overline{\pi}_D \in P_D(\rho)} m(\overline{\pi}_D) \prod_{v \in S} \operatorname{tr} \overline{\pi}_{D,v}(\tilde{f}_{D,v}) = \sum_{\overline{\pi} \in P(\rho)} m(\overline{\pi}) \prod_{v \in S} \operatorname{tr} \overline{\pi}_v(\tilde{f}_v) \tag{5.2}$$

are nonzero, and the identity holds.

Proof. The proof of Proposition 4.7.6, applies with three changes. (1) The reduction to elliptically supported $\tilde{f}_u$, via Lemma 4.4.14, is made only at u, where $\tilde{f}_u$ is regular-supported. (2) The uniform finiteness step is Lemma 4.7.2 ((I10)), applied at a level $U_S \times U_{w_0} \times U^V$ under which all test functions are bi-invariant. It needs only that every $\tilde{f}_{D,v}$ and $\tilde{f}_v$ ($v \in S$) is bi-invariant under some open compact subgroup over which the cover splits; this holds for every smooth function compactly supported modulo the center, regular-supported or not, and Lemma 4.7.2, makes no other assumption on the test functions at S. (3) Theorem 5.6.3 replaces Theorem 4.6.7. The separation by archimedean infinitesimal characters and by corner algebras away from $V = S \cup \{w_0\}$ does not involve the test functions at S. □

What changes from Chapter 4. In Proposition 4.7.6, all functions at S were regular-supported. Now only the one at u is. In particular, at $v \in S \smallsetminus \{u\}$ one may use the central-character idempotents e_{U_v}, pseudo-coefficients, or supercusp forms (on the GL side, via Theorem 5.5.8(b)).

5.6.3 Theorem C′ and its converse

We state the new non-cancellation hypotheses (Hyp'_D) and (Hyp') (Definition 5.6.5), which are used again in §5.6.4 and §5.7, and prove the transfer in both directions.

Definition 5.6.5 (Hypotheses (Hyp'_D), (Hyp'))**.** A packet $P_D(\rho)$ satisfies (Hyp'_D) if there are $u \in S$ and $\tilde{f}_{D,v} \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ $(v \in S)$, with $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$ and the others arbitrary, such that

$$\sum_{\overline{\pi}'_D \in P_D(\rho)} m(\overline{\pi}'_D) \prod_{v \in S} \operatorname{tr} \overline{\pi}'_{D,v}(\tilde{f}_{D,v}) \neq 0.$$

A packet $P(\rho)$ satisfies (Hyp') if there are $u \in S$ and $\tilde{f}_v \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ $(v \in S)$, all with vanishing orbital integrals at good regular non-elliptic elements, with $\tilde{f}_u$ supported in the regular set, such that $\sum_{\overline{\pi}' \in P(\rho)} m(\overline{\pi}') \prod_{v \in S} \operatorname{tr} \overline{\pi}'_v(\tilde{f}_v) \neq 0$.

Example 5.6.6 (Continuing Example 1.3.1: what the weaker hypothesis allows)**.** In the running example take $u = \mathfrak{p}_7$. Then (Hyp'_D) asks for a function $\tilde{f}_{D,\mathfrak{p}_7}$ vanishing near the center and an arbitrary function $\tilde{f}_{D,\mathfrak{p}_3}$ such that the displayed sum is nonzero. In (Hyp_D) both functions must vanish near the center. So the functions of Example 4.7.10 can be used here as well, and a packet with one member satisfies (Hyp'_D). The new freedom is at $\mathfrak{p}_3$, where one may now use, for instance, the idempotents e_{U_v} of §5.6.2.

The test functions allowed in Chapter 4's (Hyp_D) (Definition 4.7.9) and (Hyp) (Theorem 4.7.13) are among those allowed here, so $(\mathrm{Hyp}_D) \Rightarrow (\mathrm{Hyp}'_D)$ and $(\mathrm{Hyp}) \Rightarrow (\mathrm{Hyp}')$. We do not know whether these implications are strict.

Theorem 5.6.7 (Theorem C′)**.** *under* (Hyp'_D) *Let $\overline{\pi}_D$ be an irreducible genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ with central character $\tilde{\theta}$, such that $\overline{\pi}_{D,w_0}$ is supercuspidal for some finite place $w_0 \notin S$. Let $\rho = \overline{\pi}_D^V \circ (\psi^V)^{-1}$ and $\sigma = \overline{\pi}_{D,w_0} \circ \psi_{w_0}^{-1}$, and assume that $P_D(\rho)$ satisfies (Hyp'_D). Then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$ such that $\overline{\pi}_v \cong \overline{\pi}_{D,v} \circ \psi_v^{-1}$ for every $v \notin S$, and (5.2) holds for all families as in Proposition 5.6.4.*

Proof. Take u and $(\tilde{f}_{D,v})_{v \in S}$ as in (Hyp'_D). At u, Theorem 4.4.7(a), gives a matching $\tilde{f}_u$ supported on the good regular elliptic set; at $v \neq u$, Theorem 5.5.8(a) gives matching $\tilde{f}_v$. By Proposition 5.6.4 the right side of (5.2) is nonzero, so $P(\rho) \neq \emptyset$. Any $\overline{\pi} \in P(\rho)$ is as required. □

Remark 5.6.8 (Two roles of u)**.** In Proposition 5.6.4 the place $u \in S$ is an arbitrary *chosen* test place: the packet identity holds for every choice of u, with the corresponding families. In (Hyp'_D) the place u is *existential*: it suffices that for at least one $u \in S$ some admissible test family gives a nonzero packet sum. Theorem C′ uses that u in Proposition 5.6.4.

What changes from Chapter 4. Theorem 4.7.12, assumed (Hyp_D), which allows only test functions vanishing near the center at *every* place of S. Theorem C′ needs this at one place only. Proposition 5.6.11 describes what a failure of (Hyp'_D) would require.

Theorem 5.6.9 (Converse)**.** *under* (Hyp') *Let $\overline{\pi}$ be a genuine cuspidal automorphic representation of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$, with $\overline{\pi}_{w_0}$ supercuspidal for some finite $w_0 \notin S$, and let $\rho = \overline{\pi}^V$, $\sigma = \overline{\pi}_{w_0}$. If $P(\rho)$ satisfies (Hyp'), there is $\overline{\pi}_D$ with $\overline{\pi}_{D,v} \cong \overline{\pi}_v \circ \psi_v$ for all $v \notin S$.*

Proof. Take u and $(\tilde{f}_v)_{v\in S}$ as in (Hyp'). At u, Theorem 4.4.7(b), gives a matching $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$; at $v \neq u$, Theorem 5.5.8(b) gives matching $\tilde{f}_{D,v}$. Proposition 5.6.4 applies, and its right side is nonzero. Hence $P_D(\rho) \neq \emptyset$. □

Corollary 5.6.10. *(using (E)) Suppose $P(\rho) = \{\overline{\pi}\}$ (with any multiplicity) and $\overline{\pi}_v$ is square-integrable for every $v \in S$. Then (Hyp') holds, and $\overline{\pi}$ transfers to $\overline{G}_D(\mathbb{A}_K)$ as in Theorem 5.6.9.*

Proof. Fix $u \in S$. For $v \in S \smallsetminus \{u\}$ let $\tilde{f}_v$ be a pseudo-coefficient of $\overline{\pi}_v$ [FK86, §25.1, Theorem] **(E)**; its orbital integrals vanish at regular non-elliptic elements, and $\operatorname{tr}\overline{\pi}_v(\tilde{f}_v) = 1$. At u, the character $\Theta_{\overline{\pi}_u}$ is not identically zero on the good regular elliptic set, by the orthonormality of characters of square-integrable genuine representations on the elliptic set [FK86, §25], from [Kaz86, Theorem K] **(E)**. As in Remark 4.7.14(1), a bump function $\tilde{f}_u$ supported near a good regular elliptic point where $\Theta_{\overline{\pi}_u} \neq 0$ has $\operatorname{tr}\overline{\pi}_u(\tilde{f}_u) \neq 0$ and vanishing non-elliptic orbital integrals. The sum in (Hyp') is $m(\overline{\pi})\prod_v \operatorname{tr}\overline{\pi}_v(\tilde{f}_v) \neq 0$. □

What changes from Chapter 4. Remark 4.7.14(1), obtained (Hyp) for a single $\overline{\pi}$ only when its components at S are supercuspidal (or, at each place, have character not identically zero on the elliptic set, using regular-supported functions everywhere). Now square-integrable components suffice, because pseudo-coefficients, which are not regular-supported, can be used at all places of S but one.

5.6.4 When can (Hyp'_D) fail?

We describe the coefficient functions of packets for which (Hyp'_D) fails. This sharpens Lemma 4.7.8.

For a packet $P_D(\rho)$ let $m(\tau)$ be the total multiplicity of its members whose component at S is $\tau = \otimes_{v\in S}\tau_v$, a genuine irreducible representation of $\prod_{v\in S}\overline{G}_D(K_v)$ with central character $\tilde{\theta}_S$.

Proposition 5.6.11. (Hyp'_D) *fails for $P_D(\rho)$ if and only if there is $c \geq 0$ with $m(\tau) = c\prod_{v\in S}\dim\tau_v$ for every τ. Here $c = 0$ exactly when $P_D(\rho)$ is empty. The constant c need not be an integer.*

Proof. Step 1 (fixed level). Call a family $U = (U_v)_{v\in S}$ of open compact subgroups a *level* if each U_v is as in Lemma 4.7.8 (split by κ_v, with $\tilde{\theta}_v \circ \kappa_v$ trivial on $U_v \cap K_v^\times$). Let $\mathscr{H}_v$, $\mathscr{R}_v$, $\mathscr{T}_v$ be as there. Levels form a directed family (any two contain a common level) that is cofinal among open compact subgroups, and every $\tilde{f}_{D,v}$ lies in $\mathscr{H}_v$ for some level, since it is bi-invariant under small enough split subgroups. At a fixed level, $\operatorname{tr}\overline{\pi}'_{D,v}(\tilde{f}_{D,v}) = 0$ unless $\overline{\pi}'_{D,v} \in \mathscr{T}_v$. The members of $P_D(\rho)$ with a given component τ at S have all their components fixed, so their total multiplicity $m(\tau)$ is finite. Hence on $\bigotimes_v \mathscr{H}_v$ the packet sum of Definition 5.6.5 equals

$$\Lambda_U\left(\textstyle\bigotimes_v f_v\right) = \sum_{\tau\in\prod_v \mathscr{T}_v} m(\tau)\prod_{v\in S}\operatorname{tr}\tau_v(f_v),$$

a finite sum. So (Hyp'_D) fails if and only if, for every level U and every $u \in S$, Λ_U vanishes on $\mathscr{R}_u \otimes \bigotimes_{v\neq u}\mathscr{H}_v$.

Step 2 (coordinates). By the proof of Lemma 4.7.8, $\mathscr{H}_v \cong \bigoplus_{\tau_v\in\mathscr{T}_v}\operatorname{End}(\tau_v^{U_v})$, so the map $\mathscr{H}_v \to E_v := \mathbb{C}^{\mathscr{T}_v}$, $f \mapsto (\operatorname{tr}\tau_v(f))_{\tau_v}$, is surjective; and $\mathrm{ev}_{1,v} = \operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v)^{-1}\sum_{\tau_v}(\dim\tau_v)\operatorname{tr}\tau_v$ on $\mathscr{H}_v$, with $\mathscr{R}_v = \ker\mathrm{ev}_{1,v}$. So the image of $\mathscr{R}_v$ in E_v is the hyperplane $d_v^\perp$, where $d_v = (\dim\tau_v)_{\tau_v}$ and $\perp$ is taken for the bilinear pairing $\langle x,y\rangle = \sum x_\tau y_\tau$. Write m_U for the restriction of m to

$\prod_v \mathscr{T}_v$, an element of $E := \bigotimes_v E_v$; then $\Lambda_U(\otimes f_v) = \langle m_U, \otimes_v (\operatorname{tr}\tau_v(f_v))_{\tau_v}\rangle$. Hence Λ_U vanishes on $\mathscr{R}_u \otimes \bigotimes_{v\neq u}\mathscr{H}_v$ if and only if $m_U \in \mathbb{C}d_u \otimes \bigotimes_{v\neq u} E_v$. If every $\mathscr{T}_v$ is nonempty, so that $d_v \neq 0$, these conditions for all $u \in S$ together say that $m_U \in \mathbb{C}\bigotimes_v d_v$ (use a basis of each E_v whose first vector is d_v). That is, $m(\tau) = c_U \prod_v \dim \tau_v$ for $\tau \in \prod_v \mathscr{T}_v$, and $c_U \geq 0$ since $m \geq 0$. If some $\mathscr{T}_v$ is empty, there is no condition at that level.

Step 3 (independence of the level). If $U' \subseteq U$ componentwise, then $\mathscr{T}_v(U) \subseteq \mathscr{T}_v(U')$, and both identities hold on $\prod_v \mathscr{T}_v(U)$; since all $\dim \tau_v > 0$, $c_{U'} = c_U$ whenever $\prod_v \mathscr{T}_v(U)$ is nonempty. Any two levels contain a common smaller level, and every τ has nonzero fixed vectors at some level. So there is one $c \geq 0$ with $m(\tau) = c\prod_v \dim \tau_v$ for all τ.

Step 4 (converse and the constant). If $m = c\prod_v \dim \tau_v$, then at every level $\Lambda_U(\otimes f_v) = c\prod_v \operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v) f_v(1)$, which vanishes when $f_u \in \mathscr{R}_u$; so (Hyp'_D) fails. If $P_D(\rho)$ is nonempty, some $m(\tau) \geq 1$, so $c = m(\tau)/\prod_v \dim \tau_v \in \mathbb{Q}_{>0}$; if it is empty, $m = 0$ and $c = 0$. □

Remark 5.6.12. The linear algebra in the proof does not use $|S| \geq 2$; when $|S| = 1$, (Hyp'_D) coincides with (Hyp_D), and the proposition sharpens Lemma 4.7.8, in that case too. The weights are $\dim \tau_v$, not $\dim \tau_v^{U_v}$: it is the dimension of the whole representation that occurs in Fourier inversion at the identity. A proof of Proposition 5.6.11 along the same lines was written independently by ChatGPT (see the Acknowledgments).

Thus a failure of (Hyp'_D) would mean that the packet contains members with *every* possible component at S, with multiplicities proportional to the Plancherel weights. In Chapter 4, failure of (Hyp_D) allowed any coefficient function in the larger space $\sum_{v\in S} d_v \otimes \mathbb{C}^{\prod_{w\neq v}\mathscr{T}_w}$. Removing this last possibility is goal (iv) of Chapter 6. The following consequence reduces it to a finiteness statement.

Corollary 5.6.13. *If $P_D(\rho)$ is nonempty and its members have only finitely many distinct components at S, then (Hyp'_D) holds. The same conclusion holds if the levels at S of the members of $P_D(\rho)$ are bounded, that is, if there is a level U such that every member has nonzero U_v-fixed vectors at each $v \in S$ (level, for $D_v^\times$, is the Congruence filtration and level entry of §3.3; here it is the same notion for the group $\overline{G}_D(K_v)$).*

Proof. For each $v \in S$ the group $\overline{G}_D(K_v)/\overline{Z}_v$ is compact and infinite (it contains the infinite profinite group $\mathscr{O}_{D_v}^\times/\mathscr{O}_v^\times$). By the Peter–Weyl theorem for the twisted convolution algebra with central character $\tilde{\theta}_v$, the infinite-dimensional space $L^2(\overline{G}_D(K_v), \tilde{\theta}_v)$ is the Hilbert direct sum of the finite-dimensional spaces $\tau_v \otimes \tau_v^\vee$, so there are infinitely many genuine irreducible τ_v with central character $\tilde{\theta}_v$. If (Hyp'_D) failed, Proposition 5.6.11 with $c > 0$ (the packet being nonempty) would give $m(\tau) \geq c\prod_v \dim \tau_v > 0$ for every one of the infinitely many $\tau = \otimes_v \tau_v$, so the packet would have infinitely many distinct components at S. For the second statement: only finitely many τ have nonzero U-fixed vectors (Lemma 4.7.8: each $\mathscr{T}_v$ is finite). □

Remark 5.6.14. Positivity alone cannot rule out Plancherel-proportional multiplicities: the regular representation $L^2(H) \simeq \widehat{\bigoplus}_\tau (\dim \tau)\,\tau$ of a compact group H has exactly this shape. Corollary 5.6.13 shows that what is needed is a bound on the levels at S in a fixed near-equivalence class. This observation is due to ChatGPT (see the Acknowledgments).

5.7 What is left after Chapter 5

This section lists the goals of §4.1.3, that are not achieved here, with what we now know about each, including what Chapter 6 has since proved. Unless stated otherwise, all are **(C)**. The numbering (ii)–(v) follows §4.1.3, which refers to it; the section introduces no new notation.

(ii) *Local character relations.* A metaplectic local Jacquet–Langlands correspondence: a bijection $\overline{\pi}_v \leftrightarrow \overline{\pi}_{D,v}$ between genuine square-integrable representations of $\overline{G}_v$ and genuine irreducible representations of $\overline{G}_D(K_v)$, with $\Theta_{\overline{\pi}_{D,v}} = (-1)^{r-1}\Theta_{\overline{\pi}_v} \circ \jmath$ on good regular elements and equal formal degrees for the measures of Definition 5.2.1. The local inputs are now located: local integrability and character expansions [FK86, §13], [Li12, Théorème 4.3.2]; pseudo-coefficients [FK86, §25.1]; elliptic orthonormality [FK86, §25]; and Theorem E. The main difficulty is globalization: a pseudo-coefficient isolates a representation only among *tempered* ones, while a global cuspidal spectrum may have non-tempered unitary components at a place of S. A restricted correspondence, for example for supercuspidal $\overline{\pi}_v$, is the natural first target.

Status after Chapter 6. Chapter 6 constructs a unitary transform L_D between ordinary invariant class functions on $D_v^\times$ and genuine ones on $\overline{G}_D(K_v)$, with constant $b_D = |n|_{K_v}^{-(r-1)/2}$, and proves that it bijects irreducible characters, preserving dimensions, when the residue characteristic of v is prime to n (Theorem 6.7.6). Zou's constant in the Flicker–Kazhdan relation [Zo23, (3.2)] equals b_D for the cover with parameter c_v, because $\gcd(n, 2rc_v + r - 1) = n$. Corollary 6.7.8 gives the bijection and the character relation by composing L_D, [DKV84], and [Zo23, Proposition 3.6], *provided* the corrected square-integrable Flicker–Kazhdan correspondence is available for the chosen parameters. A sufficient published hypothesis is $(n, N_r) = 1$, where N_r is the least common multiple of the composite positive integers at most r (with $N_r = 1$ if none occur). Chapter 6 now checks the cocycle, star map, central-character and root-sum conventions against Remark 4.8.1. The result is established in this stated tame range using the cited **(E)** inputs, not for arbitrary (n, r) merely because $v \nmid n$. At places dividing n, irreducible-character integrality of L_D remains open. The formal-degree statement is consistent with these results. For the measures of Definition 5.2.1, integrality gives $d(\overline{\pi}_{D,v}) = c_D\, d(\pi_v)$ with $c_D = |n|_{K_v}^{-(r^2-1)/2}$ ($c_D = 1$ at tame places). Zou's relation near the identity, together with Corollaries 5.3.8 and 5.5.11, gives $d(\overline{\pi}_v) = c_D\, d(\pi_v)$, provided one has the genuine analog of the relation between the leading germ and the formal degree. Here π_v is the common linear partner.

(iii) *An unconditional global correspondence.* Remove the auxiliary place w_0, either with W.-W. Li's invariant trace formula for covers [Li14] (which assumes an invariant Paley–Wiener theorem at the archimedean places; here these are complex and the covers split there), or by placing a supercusp form at a place of S, which Theorem E(b) now permits on the matching side but which requires (ii) on the spectral side. Then refine the packet identity to individual representations. Multiplicity one and rigidity for $\overline{G}(\mathbb{A}_K)$ are known only for representations lifting to cuspidal representations of $\mathrm{GL}(r, \mathbb{A}_K)$ with elliptic components at three places [FK86, Theorem 28]; the argument should not assume more.

(iv) *Removing* (Hyp'_D) *and* (Hyp'). By Proposition 5.6.11, it suffices to rule out packets whose S-multiplicities are proportional to the Plancherel weights, and by Corollary 5.6.13 it suffices

to bound the level at S of the members of $P_D(\rho)$. One route would be a finite-to-one lifting of $P_D(\rho)$ to automorphic representations of $D^\times_{\mathbb{A}}$, and then to $\mathrm{GL}(r, \mathbb{A}_K)$, where strong multiplicity one gives rigidity; another is to use (ii). *Status:* open.

(v) *Completing [JC26a, Remark 10.3].* Compose the lifting of [JC26a] with the transfer, and compare with the Flicker–Kazhdan correspondence, using the corrections of [Zo23] to [FK86, §26.2] and [FK86, Lemma 27.1]. *Status:* open.

Chapter 6

The division-algebra side and its local character transform

Overview. Let D/F be a central division algebra of prime degree r and $\widetilde{D^\times}$ the n-fold cover (n odd) specified in Chapter 4, where $(n,r)=1$. We construct and normalize a unitary transform L_D of genuine and ordinary invariant class functions, with local coefficient $|n|_F^{-(r-1)/2}$. When the residue characteristic is prime to n, a unique splitting over the reduced-norm-one group, together with an Adams operation, proves that L_D bijects irreducible characters and preserves dimensions. At places dividing n we obtain a nonsplitting and divisibility obstruction. We give the normalized orbital-integral transfer and its trace-pairing identity. Globally we prove an almost-everywhere local-to-global principle for nth powers modulo scalars, and explain the relevant rational-orbit and measure considerations from Kottwitz's work. The tame character relation with the cover of GL_r follows, within its published correspondence range, by an explicit cocycle, star-map, central-character and root-sum check. The wild-place character correspondence and the adelic trace-formula comparison remain open.

6.1 Introduction and results

The classical local Jacquet–Langlands correspondence identifies irreducible representations of $D^\times$ with essentially square-integrable representations of $\mathrm{GL}_r(F)$ in the appropriate central-character families [DKV84]. It does not produce a correspondence between ordinary representations of $D^\times$ and genuine representations of its metaplectic cover: pullback from the linear group is not genuine. This chapter studies the division-algebra side directly and provides local results independent of a conjectural global trace-formula comparison.

This section places the chapter in the series, records the status of its results, and introduces the running example. It depends only on the statements of Chapters 4 and 5; references to them are to Chapter 4, and Chapter 5. The notation of the series and of this chapter is collected in §3, the results are listed with their labels in the Summary of results that follows this section, and related work is discussed in §2.3.

Position within the series. Chapter 4 constructs the specified covers, computes their good elements, and proves matching for regular-supported functions; Chapter 5 proves matching of

arbitrary functions (its Theorem E), identifies the germs at the center, and proves an improved global transfer (its Theorem C′). Here the central object is the transform L_D on the compact-mod-center group $D^\times$. The principal issues and their status are:

Topic	Assertion	Status
Covering theory	Scalar centrality and good-locus description from Chapter 4	(E)
Local harmonic analysis	Unitary L_D and explicit coefficient b_D	(P)
Tame representations	Splitting over D^1 and irreducible-character bijection for $p \nmid n$	(P)
Wild representations	Nonsplitting and necessary divisibility constraints	(P)
	Integrality of L_D at places dividing n	(C)
Character relation (D2) of Chapter 4	Tame places, with the Flicker–Kazhdan–Zou theorem available (in particular when $(n, N_r) = 1$); cocycle and star-map compatibility checked below (Corollary 6.7.8)	(P)
Test functions	Normalized local orbital-integral transfer and trace adjunction	(P)
Global arithmetic	Local-to-global powers modulo scalars outside any finite set	(P)
Global comparison	Good-element matching, adelic measures, spectral separation	(C)

The labels are those of the whole series (§3.1). As in Chapters 4 and 5, **(P)** means proved here (possibly using results marked **(E)**), **(E)** means established elsewhere (including Chapters 4 and 5) and used as a black box, **(O)** means that a proof is outlined but not fully written out or checked, and **(C)** means conjectural: new mathematical input is required. These statuses concern the precise stated hypotheses; the local claims rely on the specific covering cocycle of Chapter 4, not on an arbitrary central extension.

The two distinct maps in the chapter should be kept separate: the unitary operator on invariant L^2 class functions, and a bijection of individual irreducible characters. The first exists at all division places, whereas the second is established here only when $p \nmid n$. The Adams operation provides the bridge in that case.

After the local results, we review the part of Kottwitz's arithmetic framework needed for rational conjugacy and inner forms, then prove a Hasse principle for the global power quotient. The final section states what remains for an adelic metaplectic Jacquet–Langlands comparison. Section 3 collects the notation and glossary, and Section 2.3 discusses related work; the literature of the whole series is reviewed in §2.1.

The running example

In the series' running example, $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$ (Example 1.3.1, and [JC26b]), the two places of S illustrate the two regimes of this chapter. At $\mathfrak{p}_7$, which does not divide n, $b_D = c_D = 1$, the discriminant ratio is identically 1 (Lemma 6.7.4), and L_D is a dimension-preserving bijection of irreducible characters (Theorem 6.7.6). At $\mathfrak{p}_3$, where $K_v = \mathbb{Q}_3(\zeta_3)$ and $|3|_v = 1/9$, one has $b_D = 3$ and $c_D = 27$. There the cover does not split, and genuine irreducibles have dimension divisible by 3 (Proposition 6.8.1). Integrality of L_D would force $L_D\mathbf{1}$ to be irreducible of dimension 27 (Remark 6.5.5). Whether it is remains open; see Chapter 7.

6.2 Local setting and good elements

We fix the local field, central-character conventions, and the covering-theoretic theorem imported from Chapter 4. The distinction between a good element of the cover and an ordinary regular element is essential to the definition of the transform. The section depends on §§4.2–4.3 (the cover, scalar centrality, good elements). It introduces F, D, H, Z, $\widetilde{H}$, $\widetilde{Z}$, the characters $\tilde{\theta}$ and θ, the star map, the discriminant Δ and good regular elements. Let F be a finite extension of $\mathbb{Q}_p$ containing μ_n, where $n \geq 1$ is odd. Let $r \geq 2$ be prime and $(n,r) = 1$. Let D/F be a central division algebra of degree r, $H = D^\times$, and $Z = F^\times$ its scalar center. We use the normalized local absolute value $|\cdot|_F$ ($|\varpi_F|_F = q_F^{-1}$).

Let

$$1 \longrightarrow \mu_n \longrightarrow \widetilde{H} \xrightarrow{p} H \longrightarrow 1 \tag{6.1}$$

be a locally compact topological central extension. Throughout we use *the specified cocycle of Chapter 4*, rather than an arbitrary central extension. In its parameter convention, $n \mid r - 1 + 2rc_v$; under that condition the inverse image $\widetilde{Z} = p^{-1}(Z)$ is central (Lemma 4.2.3). The cocycle condition is essential; a general n-fold extension need not have central scalar preimage. References to Chapter 4 below refer to its stated cocycle conventions.

In the notation of Chapters 4 and 5, $\widetilde{H} = \overline{G}_D(F)$, $\widetilde{Z} = \overline{Z}$, and $\overline{G}_v$ denotes the Kazhdan–Patterson cover of $\mathrm{GL}_r(F)$ with parameter c_v; the normalized orbital integral $J = \Delta O$ of §6.9 is the function $F = \Delta\Phi$ of Definition 4.4.1, for the measures used here. Fix a unitary character $\widetilde{\theta} : \widetilde{Z} \to \mathbb{C}^\times$ whose restriction to μ_n is faithful. Write $\theta(a) = \widetilde{\theta}(a^*)$ for $a \in Z$, where the star map is defined next. For invariant functions we impose the central equivariance conventions $f(ah) = \theta(a)f(h)$ and $\widetilde{f}(zh) = \widetilde{\theta}(z)\widetilde{f}(h)$.

Definition 6.2.1 (The star map)**.** For $h \in H$ choose any lift $\widetilde{h} \in \widetilde{H}$ and set

$$h^* = \widetilde{h}^n \in \widetilde{H}. \tag{6.2}$$

The definition is independent of the lift. It is conjugation equivariant, and for $a \in Z$ and $h \in H$ one has $(ah)^* = a^* h^*$.

Example 6.2.2 (Continuing Example 1.3.1: the star map at $\mathfrak{p}_3$)**.** Take $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, so that H is the unit group of the quaternion division algebra over F and $\widetilde{H}$ is its 3-fold cover. The three lifts of $h \in H$ are $\widetilde{h}$, $\zeta\widetilde{h}$ and $\zeta^2\widetilde{h}$, where ζ generates μ_3. All three have the same cube, because μ_3 is central and $\zeta^3 = 1$. That common cube is h^*, and it lies above h^3. For instance, 1^* is the identity of $\widetilde{H}$, and $(ah)^* = a^* h^*$ for a scalar a. So the star map lifts every cube h^3 in a canonical way, although no lift of h itself has been singled out.

Proof. Another lift is $\zeta\widetilde{h}$ with $\zeta \in \mu_n$, whose nth power is $\widetilde{h}^n$. Conjugation equivariance follows by lifting the conjugating element. Elements of $\widetilde{Z}$ are central by assumption, hence $(\widetilde{a}\widetilde{h})^n = \widetilde{a}^n\widetilde{h}^n$. In particular $a \mapsto a^*$ is a homomorphism from Z to $\widetilde{Z}$. (For a maximal torus it is also a homomorphism, because n is odd; see below.) □

Every noncentral $h \in D^\times$ generates a maximal subfield $E = F(h)$, of degree r. Set $T = E^\times$; it is its own centralizer in H. A regular element of H means a noncentral element. The corresponding

Weyl discriminant is

$$\Delta(h) = \left|\det\big(1 - \mathrm{Ad}(h) \mid \mathfrak{h}/\mathfrak{t}\big)\right|_F^{1/2} = \prod_{i<j} \left|\frac{(\lambda_i - \lambda_j)^2}{\lambda_i \lambda_j}\right|_F^{1/2}, \tag{6.3}$$

where $\lambda_1, \ldots, \lambda_r$ are the conjugate eigenvalues in a splitting field, and the product is interpreted with the normalized absolute value.

Definition 6.2.3 (Good regular element)**.** A regular $\widetilde{h} \in \widetilde{H}$ is good if it commutes with $p^{-1}(Z_H(p(\widetilde{h})))$. The definition is equivalent to the usual statement that the commutator character on the preimage of the regular centralizer is trivial. Every h^* is good, since its commutators are nth powers in μ_n.

Example 6.2.4 (Continuing Example 1.3.1: good and non-good elements)**.** At the tame place $\mathfrak{p}_7$ the preimage of every maximal subfield is abelian (Proposition 4.3.5(b)), so every regular element of $\widetilde{H}$ is good. At the wild place $\mathfrak{p}_3$ this fails. There a regular $\widetilde{h}$ over $h \in E^\times$ is good exactly when $h \in F^\times E^{\times 3}$ (Proposition 4.3.5(a)), and these elements have index 9 in $E^\times$ (Example 4.3.7). Write $D = E_0 \oplus E_0 j$ with $j^2 = \varpi$ a uniformizer of F, as in Chapter 8. The lifts of j are good, because $j = \varpi^{-1} j^3$ lies in $F^\times E^{\times 3}$ for $E = F(j)$. The lifts of $1 + j$ are not good. Indeed, by Lemma 8.8.1(e) a good element of $1 + j\mathcal{O}_D$ has depth exactly 3 or at least 6 modulo scalars, and $1 + j$ has depth 1.

Remark 6.2.5. If $\widetilde{h}$ is regular and not good, every genuine conjugation-invariant function $\widetilde{f}$ vanishes at $\widetilde{h}$: some lift $\widetilde{t}$ centralizing $p(\widetilde{h})$ has $\widetilde{t}\widetilde{h}\widetilde{t}^{-1} = \zeta\widetilde{h}$, with $\zeta \neq 1$; invariance and genuine central equivariance then force $\widetilde{f}(\widetilde{h}) = 0$.

Proposition 6.2.6 (**(E)** Good lifts on the division side (Chapter 4))**.** *For each maximal subfield E/F of D, the good elements of $p^{-1}(E^\times)$ are precisely zt^* with $z \in \widetilde{Z}$ and $t \in E^\times$. Equivalently, modulo $\widetilde{Z}$ the good subset corresponds to $(E^\times/F^\times)^n$. These are the good-element and commutator results established in Lemma 4.2.3 and Proposition 4.3.5(a); the local index and preservation of regularity are treated there in Proposition 4.3.5(b),(d).*

6.3 Power maps on elliptic tori

The local index of the power map and the ratio of Weyl discriminants control the normalization of L_D. Both calculations use the coprimality of n and r, including at wild places. The section depends on §6.2 and introduces the discriminant ratio $\Delta(t)/\Delta(t^n)$.

Lemma 6.3.1 (No new singular elements)**.** *If $h \in H$ is noncentral, h^n is noncentral. In particular the image of a regular element under the star map is regular.*

Proof. Suppose $h^n \in F^\times$. The field $L = F(h)$ is an intermediate field of a maximal subfield of D, hence $[L : F]$ divides r. Because F contains μ_n and $h^n \in F^\times$, Kummer theory shows $[L : F]$ divides n. The coprimality $(n, r) = 1$ therefore gives $[L : F] = 1$, contrary to the assumption that h is noncentral. □

Remark 6.3.2 (Why $(n,r)=1$ cannot be dropped)**.** The proof isolates exactly where coprimality is used: $[L:F]$ must divide both r (it sits inside a degree-r extension) and n (Kummer theory, since h^n is a scalar), and $(n,r)=1$ is what forces the common divisor down to 1. Drop the hypothesis and the lemma is false, by an explicit example at a wild place of the running example's own field.

Take $F = K_{\mathfrak{p}_3} = \mathbb{Q}_3(\zeta_3)$, the local field of §1.3, but now with $n = r = 3$ instead of $n = 3, r = 2$ (so $(n,r) = 3 \neq 1$). Fix a central division algebra D/F of degree 3 (one exists over every local field, for every degree). A uniformizer ϖ of F has valuation $v(\varpi) = 1$, not divisible by 3, so ϖ is never a cube in $F^\times$; hence $L = F(\varpi^{1/3})$ is a genuine, totally ramified, cyclic cubic Kummer extension of F (cyclic because $\mu_3 \subset F$), and it embeds as a maximal subfield of D. Its generator $h = \varpi^{1/3}$ is noncentral, since $[F(h):F] = 3 \neq 1$, but

$$h^n = h^3 = \varpi \in F^\times,$$

a scalar. So the star map sends the regular element h to a central, hence singular, element: the exact failure the lemma rules out when $(n,r) = 1$.

This is the same arithmetic as the elementary fact that "multiply by n" on $\mathbb{Z}/r\mathbb{Z}$ is a bijection exactly when $\gcd(n,r) = 1$: here the power map on $E^\times/F^\times$ for a degree-r field E plays the role of that multiplication, and losing injectivity on the subgroup generated by h is losing it on all of $E^\times/F^\times$ by Lemma 6.3.3 below. The failure is not cosmetic: Definition 6.4.1 of L_D is only written down on the regular set, exactly because the formula needs $t^* = t^n$ to again be regular; at $n = r$, the computation above produces a regular t whose star t^* is central, so L_D as defined would have to assign a value to zt^* coming from every regular class with that same central image, which is inconsistent unless those classes already agree. The construction of §6.4 simply does not get off the ground without $(n,r) = 1$.

Lemma 6.3.3 (Degree of the power map)**.** *For any degree-r extension E/F embedded in D, the map*

$$P_n : E^\times/F^\times \longrightarrow E^\times/F^\times, \qquad [t] \longmapsto [t^n]$$

is injective and has open image of index

$$I_F(n,r) = \left[E^\times/F^\times : (E^\times/F^\times)^n\right] = |n|_F^{-(r-1)}. \tag{6.4}$$

The index is independent of E.

Proof. If $t^n \in F^\times$, Kummer theory over a field containing μ_n gives $[F(t):F] \mid n$. Since $F(t) \subset E$ and $[E:F] = r$, this degree also divides r; hence $t \in F^\times$, proving injectivity.

The quotient $A = E^\times/F^\times$ is a compact abelian p-adic Lie group. Its p-adic analytic dimension is $d = ([E:\mathbb{Q}_p] - [F:\mathbb{Q}_p]) = (r-1)[F:\mathbb{Q}_p]$. The structure theorem for compact abelian p-adic Lie groups writes $A \simeq \mathbb{Z}_p^d \times C$ for a finite abelian group C. The injectivity already proved implies $A[n] = \{1\}$, so multiplication by n is an automorphism of C. On $\mathbb{Z}_p^d$ it has index $p^{v_p(n)d}$. Since $|n|_F = p^{-[F:\mathbb{Q}_p]v_p(n)}$, this is exactly $|n|_F^{-(r-1)}$. (The argument also handles $p \mid n$.) □

Lemma 6.3.4 (Discriminant ratio)**.** *For noncentral h, the ratio*

$$R(h) = \frac{\Delta(h)}{\Delta(h^n)}$$

is locally constant and nowhere zero. It extends uniquely to a locally constant function on all of H, taking the value

$$R(a) = |n|_F^{-r(r-1)/2} \qquad (a \in F^\times). \tag{6.5}$$

Proof. By Lemma 6.3.1, both discriminants are nonzero at every noncentral element. They are continuous functions with values in the discrete multiplicative group $q_F^{\frac{1}{2}\mathbb{Z}}$ and hence locally constant on this open set.

For h sufficiently close to a scalar $a \in F^\times$, all its eigenvalues are close to a in a finite splitting extension. Factoring

$$\lambda_i^n - \lambda_j^n = (\lambda_i - \lambda_j) \sum_{k=0}^{n-1} \lambda_i^{n-1-k} \lambda_j^k,$$

the final factor approaches na^{n-1} and thus has constant absolute value $|na^{n-1}|$ for h in a sufficiently small neighborhood. The factors involving $\lambda_i \lambda_j$ in the normalized discriminant cancel the $|a|^{n-1}$ powers. Each of the $r(r-1)/2$ unordered pairs consequently contributes $|n|_F^{-1}$ to $R(h)$. This proves the formula and locally constant extension; uniqueness follows from density of the regular set. □

6.4 The normalized local class-function transform

We first transfer *invariant functions*; the problem of sending individual irreducible characters to irreducible characters is addressed separately. The transform is normalized using quotient probability measures. The section depends on §§6.2–6.3. It introduces the probability measures, the Hilbert spaces $\mathscr{H}_\theta(H)$ and $\mathscr{H}_{\widetilde{\theta}}(\widetilde{H})$, the transform L_D and the constant b_D. We equip H/Z with Haar probability measure. Because $\widetilde{Z} = p^{-1}(Z)$, the quotient $\widetilde{H}/\widetilde{Z}$ maps isomorphically to H/Z; we transport the probability measure through this isomorphism. Each elliptic torus quotient T/Z likewise has its probability Haar measure. These conventions are the relevant ones for *local unitarity*; matching geometric distributions globally may require additional, separately stated orbital-integral conventions.

Write $\mathscr{H}_\theta(H)$ for the Hilbert space of L^2 conjugation-invariant functions on H with θ-equivariance under Z. Similarly, write $\mathscr{H}_{\widetilde{\theta}}(\widetilde{H})$ for the genuine invariant functions with $\widetilde{\theta}$-equivariance under $\widetilde{Z}$. The inner products are taken over H/Z and $\widetilde{H}/\widetilde{Z}$, respectively.

Definition 6.4.1 (The division-side transform)**.** Using Proposition 6.2.6, define on regular elements

$$\boxed{(L_D\varphi)(zt^*) = |n|_F^{-(r-1)/2}\, \widetilde{\theta}(z) \frac{\Delta(t)}{\Delta(t^n)}\, \varphi(t), \quad z \in \widetilde{Z},\ t \in T^{\mathrm{reg}},} \tag{6.6}$$

and let $L_D\varphi = 0$ at nongood regular elements. The value at singular elements is understood by continuous extension whenever φ is smooth.

Example 6.4.2 (Continuing Example 1.3.1: the transform at the two ramified places)**.** At $\mathfrak{p}_7$ we have $|3|_F = 1$, and the discriminant ratio is identically 1 (Lemma 6.7.4). So (6.6) reduces to $(L_D\varphi)(zt^*) = \widetilde{\theta}(z)\varphi(t)$. At $\mathfrak{p}_3$ we have $|3|_F^{-1/2} = 3$, and the transform rescales. Take $\varphi = \mathbf{1}$, with the central-character normalization of Example 8.7.1, and let $E = F(j)$ with $j^2 = \varpi$ a uniformizer. By (8.15) the ratio $\Delta(t)/\Delta(t^3)$ equals 1, 3 and 9 at $t = j$, $1+j$ and $1+\varpi j$. So $(L_D\mathbf{1})(t^*)$ equals 3, 9 and 27 at these three elements. The value 27 is also the value $c_D \dim \mathbf{1}$ of $L_D\mathbf{1}$ at the identity.

Lemma 6.4.3 (Well-definedness and smoothness)**.** *The expression* (6.6) *is independent of the choice of z,t and of the maximal subfield $E = F(t)$. It sends locally constant invariant functions to locally constant invariant functions.*

Proof. For a good regular element, the regular centralizer is uniquely $E^\times$. If $zt^* = z'(t')^*$ then t'^n and t^n differ by a scalar. Since their centralizers are E and $t', t \in E^\times$, the injectivity in Lemma 6.3.3 gives $t' = at$ with $a \in F^\times$. As $(at)^* = a^* t^*$ and $\Delta(at) = \Delta(t)$, the formula changes by the same factor $\theta(a) = \widetilde{\theta}(a^*)$ on either side. Conjugation equivariance follows from Definition 6.2.1.

On a regular semisimple conjugacy chart, the image of the nth-power map on T/Z is an open-and-closed subgroup by Lemma 6.3.3. Thus the prescribed function is locally constant on both the good and nongood regular loci, using Lemma 6.3.4. A sufficiently small neighborhood of any central element consists of elements represented by zt^*: the local nth-root map exists by the nonarchimedean inverse function theorem, since n is nonzero in F. The same discriminant ratio has the constant value of (6.5) there. Hence the function extends locally constantly across the center. Since r is prime there are no other nonregular elements. □

Theorem 6.4.4 (Unitarity and explicit constant)**.** *For the specified cover of Chapter 4, L_D extends uniquely to a unitary isomorphism*

$$L_D : \mathscr{H}_\theta(H) \xrightarrow{\sim} \mathscr{H}_{\widetilde{\theta}}(\widetilde{H}). \tag{6.7}$$

The positive coefficient $b_D = |n|_F^{-(r-1)/2}$ in (6.6) *is uniquely determined by unitarity for the quotient probability measures used here.*

Proof. The Weyl integration formula, applied to $|\varphi|^2$, reduces the norm on H/Z to a sum over elliptic maximal tori T/Z of weighted integrals

$$\frac{1}{|W(H,T)|} \int_{T/Z} \Delta(t)^2 |\varphi(t)|^2 \, dt.$$

There is no need for a separate Weyl integration theorem on the covering group: since $\widetilde{H}/\widetilde{Z} \simeq H/Z$, the function $|\widetilde{f}|^2$ is an honest conjugation-invariant function on H/Z, and the ordinary Weyl formula applies directly. By Remark 6.2.5 and Proposition 6.2.6, a genuine invariant function vanishes off the open-and-closed subset $(T/Z)^n$ of each torus quotient.

The power map $P_n : T/Z \to (T/Z)^n$ is a topological group isomorphism. Since T/Z has probability Haar measure and $(T/Z)^n$ has index $I = |n|_F^{-(r-1)}$ in T/Z, for every integrable function Q on this subgroup,

$$\int_{(T/Z)^n} Q(x)\, dx = \frac{1}{I} \int_{T/Z} Q(t^n)\, dt. \tag{6.8}$$

Consequently, with a provisional coefficient $b > 0$, the covered Weyl integral equals

$$\frac{b^2}{I\,|W(H,T)|} \int_{T/Z} \Delta(t)^2 |\varphi(t)|^2 \, dt.$$

The index is independent of T, so the equality of norms for all class functions is equivalent to $b^2 = I$. This gives $b = b_D$.

To see surjectivity, start with a smooth genuine invariant function on the cover. It vanishes off the good regular set. By the uniqueness in Lemma 6.4.3, the formula defining L_D can be inverted on every good regular element. The inverse has the same local smoothness properties, by the inverse of P_n on its open image. This proves surjectivity on a dense subspace and therefore the unitary isomorphism on completions. □

Remark 6.4.5 (Local versus global normalization; comparison with Zou)**.** The coefficient $b_D = |n|_F^{-(r-1)/2}$ is determined by the power-map Jacobian and the quotient probability measures in Theorem 6.4.4. A common compatible rescaling of Haar measures does not alter the ratio, but the proposed global trace formula requires independent, explicit adelic normalizations.

Zou's corrected local Flicker–Kazhdan character relation [Zo23, (3.2) and Remark 3.3] has elliptic coefficient

$$\frac{n}{d_F(d_r)|n|_F^{r/2}}, \qquad d_F(m) = m|m|_F^{-1/2}, \qquad d_r = \gcd(n, 2rc + r - 1).$$

Under Chapter 4's scalar-centrality congruence one has $d_r = n$, and this coefficient simplifies algebraically to b_D. To identify the *character transforms*, however, one must also identify the two cocycles, their star maps, the central-character conventions, and the discriminant and orbital-measure normalizations. The equality of numerical coefficients alone is not such an identification; see Corollary 6.7.8.

6.5 Irreducible characters and the integrality obstruction

The unitary transform has a matrix relative to irreducible characters, but unitarity alone does not force that matrix to be integral. We record explicit arithmetic tests that any true character correspondence must pass. The section depends on §6.4 and introduces the character matrix of L_D and the constant c_D. Since H/Z is compact, every smooth irreducible representation with a fixed central character is finite dimensional. Its character is a locally constant invariant function on H. The same holds on the cover.

Corollary 6.5.1 (A unitary character matrix)**.** *Let ρ range over $\mathrm{Irr}_\theta(H)$ and τ over $\mathrm{Irr}_{\tilde\theta}(\widetilde{H})$. For each ρ there is a* finite *expansion*

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau. \tag{6.9}$$

The matrix $A = (a_{\rho\tau})$ is unitary (in the sense of the associated operator on ℓ^2), and its rows and columns are finite.

Proof. Schur orthogonality for compact groups modulo the fixed center gives orthonormal bases of irreducible characters for the two invariant L^2 spaces. Theorem 6.4.4 therefore makes A unitary. Local constancy and compactness modulo center show that each smooth invariant function is invariant under an open normal subgroup of the compact quotient, so it factors through a finite quotient (or through a corresponding finite-dimensional projective quotient with fixed central character). Its irreducible-character expansion is therefore finite. Applying this observation to the inverse transform gives finite columns as well. □

Corollary 6.5.2 (What integral coefficients would imply)**.** *Suppose the coefficients $a_{\rho\tau}$ in (6.9) are integers for every ρ,τ. Then L_D maps each irreducible character to a unique irreducible character with positive sign, and it induces a bijection*

$$\mathrm{Irr}_\theta(H) \xrightarrow{\sim} \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H}).$$

More precisely, if $L_D\Theta_\rho = \Theta_\tau$, then

$$\boxed{\dim\tau = |n|_F^{-(r^2-1)/2}\dim\rho.} \tag{6.10}$$

Proof. An integral vector of squared ℓ^2-norm one has exactly one nonzero entry, equal to $+1$ or -1. The sign cannot be negative because $\Theta_\rho(1) = \dim\rho > 0$ and the continuous value at the identity prescribed by (6.6) is

$$(L_D\Theta_\rho)(1) = b_D\,|n|_F^{-r(r-1)/2}\dim\rho > 0.$$

The same argument applied to L_D^{-1} yields bijectivity. Substituting $b_D = |n|_F^{-(r-1)/2}$ gives (6.10). □

Remark 6.5.3 (A concrete wild-place check)**.** Take $r=2$, $n=3$, $F=\mathbb{Q}_3(\zeta_3)$. Then $|3|_F = 3^{-2} = 1/9$, so $I=9$, $b_D=3$, and formula (6.10) predicts $\dim\tau = 27\dim\rho$ *if* integral irreducible transfer holds. This is a sharp, falsifiable numerical consequence of the proposed correspondence; it is not evidence that the correspondence is already proved.

Lemma 6.5.4 (Integral local scaling constants)**.** *If $p \mid n$ and $\mu_n \subset F$, with n odd as assumed, then*

$$b_D = |n|_F^{-(r-1)/2}, \qquad c_D = |n|_F^{-(r^2-1)/2}$$

are positive integers.

Proof. Write $p^k \| n$ and $d = [F:\mathbb{Q}_p]$. Since $\mu_{p^k} \subset F$ and p is odd, $p-1$ divides d; in particular d is even. For the normalized absolute value, $|n|_F^{-1} = p^{kd}$ is a perfect square. Both exponents above therefore give integral nonnegative powers of p. □

Remark 6.5.5 (A falsifiable wild-place divisibility test)**.** Suppose the matrix of Corollary 6.5.1 is integral at a wild place, so the transform is a bijection on irreducible characters. Every genuine $\tau \in \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})$ must then satisfy $c_D \mid \dim\tau$, since $\dim\tau = c_D\dim\rho$ for its unique preimage ρ. In the example $(r,n,F) = (2,3,\mathbb{Q}_3(\zeta_3))$, $c_D = 27$, while the covered-torus argument of Proposition 6.8.1 yields only the weaker factor $b_D = 3$ in the dimension constraints. A genuine irreducible of dimension not divisible by 27 would disprove *integrality of this specific transform L_D*, but would not disprove the existence of some other wild-place correspondence. The existence of such an irreducible is an open computational test, not an asserted example.

Remark 6.5.6 (Why unitarity is not integrality)**.** A unitary operator need not send an orthonormal basis to itself. Even if its matrix is locally finite, the entries need not be integral. A trace-formula or representation-theoretic positivity argument is needed to show $a_{\rho\tau} \in \mathbb{Z}$. The remaining genuinely unresolved local step is the integrality statement at *wild places*, where $p \mid n$; see Section 6.8. It should not be concealed under the phrase "unitary character transfer."

6.6 Adams operations and finite-group characters

We recall the K-theoretic origin of the Adams operations and explain why they preserve irreducibility in certain tame finite-group situations. The section depends on nothing earlier in the chapter and introduces the Adams operations ψ^n and virtual characters.

Background from K-theory and finite-group characters

The Adams operations (after J. F. Adams) originated in algebraic topology, where complex topological K-theory studies the Grothendieck group of complex vector bundles. For a finite group Q, a Q-equivariant complex vector bundle over a point is simply a finite-dimensional complex representation of Q. Consequently

$$K_Q^0(\mathrm{pt}) \cong R(Q),$$

the representation (or character) ring [Seg68]. The operations $\lambda^j[V] = [\bigwedge^j V]$ give this ring its λ-ring structure. The Adams operations ψ^k are the corresponding Newton-polynomial operations; for instance

$$\psi^1(V) = V, \qquad \psi^2(V) = V \otimes V - 2\bigwedge^2 V \quad \text{in } R(Q).$$

This description connects Adams's topological construction [Ada62] with Atiyah's systematic account of power operations [Ati66].

To see the character-theoretic formula, restrict V to the cyclic subgroup generated by $g \in Q$. On this subgroup V decomposes as a sum of eigenlines with eigenvalues α_i; the defining property $\psi^k(L) = L^{\otimes k}$ on line representations gives

$$\boxed{\chi_{\psi^k(V)}(g) = \sum_i \alpha_i^k = \chi_V(g^k).} \tag{6.11}$$

The operation is integral on the Grothendieck ring, but need not preserve the positive cone of actual characters. For example, if V is the standard two-dimensional irreducible representation of S_3, then $\psi^2(V) = \mathbf{1} - \mathrm{sgn} + V$, a virtual character. If $\gcd(k, |Q|) = 1$, however, raising the root-of-unity eigenvalues to the kth power is a cyclotomic Galois automorphism; it permutes irreducible characters. Section 6.7 exploits a more delicate version of this fact with a *fixed central character*, so that a central n-primary subgroup can be retained even when the full finite quotient has order divisible by n. This is why the Adams argument establishes integrality at tame places but not at wild ones.

For a genuine finite-dimensional representation τ of $\widetilde{H}$, let

$$\Psi_n(\Theta_\tau)(h) = \Theta_\tau(h^*) = \mathrm{tr}\,\tau(\widetilde{h}^n).$$

The nth Adams operation in the complex representation ring of $\widetilde{H}$ is a virtual character with integer coefficients. Since μ_n acts on τ by scalars, its nth Adams operation is trivial on μ_n and descends to a virtual character of H. Thus Ψ_n is integral and has central character θ.

Proposition 6.6.1 (Adams pullback and its weight)**.** *For $\tau \in \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})$, the class function $\Psi_n(\Theta_\tau)$ is a virtual character in the integral representation ring $R_\theta(H)$. On regular $t \in H$,*

$$\Psi_n(\Theta_\tau)(t) = |n|_F^{-(r-1)/2}\, R(t)\, \big(L_D^{-1}\Theta_\tau\big)(t). \tag{6.12}$$

Proof. After a suitable unramified twist, a given finite-dimensional smooth representation has finite image, and hence factors through a finite quotient Q of $\widetilde{H}$. The usual nth Adams operation is a virtual character of Q. For $\zeta \in \mu_n$ it satisfies $\psi^n\Theta(\zeta g) = \Theta((\zeta g)^n) = \Theta(g^n)$, so linear independence of irreducible characters shows that all its constituents have trivial μ_n-action. Thus it descends to H, with the required central character. Undoing the unramified twist preserves this assertion (it changes under the nth power in the expected way). The displayed formula is the inverse of (6.6) at $z = 1$. □

In general the nonconstant weight $R(t)^{-1}$ prevents us from deducing integrality of L_D from integrality of the Adams operation alone. In Section 6.7, however, we prove $R \equiv 1$ when $p \nmid n$, and then obtain a complete tame-case character correspondence. At wild places, the discriminant weight is generally nontrivial.

6.7 The tame case: splitting and irreducible character integrality

In this section assume that the residue characteristic p does not divide n. We use the special cover of Chapter 4, including the centrality of $\widetilde{Z}$ and its good-element calculation. Write $H^1 = \ker(\mathrm{Nrd} : H \to F^\times)$, and put $U_D^1 = 1 + \mathfrak{P}_D$ for the principal units in the maximal order of D. The term *tame* here means $p \nmid n$; it does not impose a condition on the ramification of a maximal subfield of D. The section depends on §§6.2–6.6, on Chapter 4, and, for Corollary 6.7.8, on [DKV84] and [Zo23]. It introduces H^1 and U_D^1.

Lemma 6.7.1 (Prime-to-n finite quotients)**.** *All finite continuous quotients of H^1 and H/Z have order relatively prime to n. In particular both groups have continuous cohomology $H^i(H^1, \mu_n) = 0$ for $i = 1, 2$, where H^1 acts trivially on μ_n.*

Proof. Write $q = |\mathcal{O}_F/\mathfrak{p}_F|$. The standard residue-field quotient of the norm-one group embeds in

$$\ker\bigl(N_{\mathbb{F}_{q^r}/\mathbb{F}_q} : \mathbb{F}_{q^r}^\times \longrightarrow \mathbb{F}_q^\times\bigr), \qquad M := \frac{q^r - 1}{q - 1};$$

the kernel of reduction is pro-p. In fact the displayed finite group is the full residue quotient, but divisibility of its order by M is all that is needed. Since $\mu_n \subset F$ and $p \nmid n$, $q \equiv 1 \pmod{n}$, and therefore $M \equiv r \pmod{n}$. Consequently $\gcd(M, n) = 1$ by $\gcd(r, n) = 1$, and no finite quotient of H^1 can have order divisible by a prime factor of n.

For H/Z, the image of U_D^1 is pro-p, and the residual quotient has an extension with one factor of order dividing M and another of order r (the valuation modulo scalar valuations). Thus every finite quotient of H/Z has order prime to n as well. For a profinite group all of whose finite quotients have order prime to n, continuous cohomology with trivial finite μ_n-coefficients vanishes in positive degrees: every cocycle factors through a finite quotient, where averaging is available. □

Proposition 6.7.2 (Unique splitting over the norm-one group)**.** *The restriction of the cover to H^1 admits a unique continuous homomorphic splitting $s : H^1 \to \widetilde{H}$. Its image is normal in $\widetilde{H}$. The quotient*

$$\mathscr{A} = \widetilde{H}/s(H^1)$$

is an abelian locally compact group, fitting into a central extension $1 \to \mu_n \to \mathscr{A} \to F^\times \to 1$. *For each prescribed faithful character* ε *of* μ_n *there exists* some *continuous unitary character* $\nu : \widetilde{H} \to \mathbb{C}^\times$ *with* $\nu|_{\mu_n} = \varepsilon$ *and* $\nu|_{s(H^1)} = 1$.

Proof. Vanishing of $H^2(H^1, \mu_n)$ in Lemma 6.7.1 provides a continuous splitting, while vanishing of H^1 gives uniqueness. Conjugating a splitting by a lift of an element of H produces another splitting of the normal group H^1; uniqueness therefore proves normality.

The reduced norm identifies H/H^1 with $F^\times$. The commutator of two elements of $\mathscr{A}$ gives a continuous alternating bicharacter $c : F^\times \times F^\times \to \mu_n$. The preimage of scalar elements is central in $\widetilde{H}$, and the reduced norm of a scalar a is a^r. Hence $c(a^r, b) = 1$. By bilinearity $c(a,b)^r = 1$, while $c(a,b)^n = 1$ since its image lies in μ_n. Coprimality of r, n implies $c = 1$, so $\mathscr{A}$ is abelian. Every continuous unitary character of a closed subgroup of a locally compact abelian group extends to such a character of the whole group. Apply this to the finite closed subgroup $\mu_n \subset \mathscr{A}$ and compose with $\widetilde{H} \to \mathscr{A}$. □

Remark 6.7.3 (Twisting is not the character transfer)**.** A choice of ν produces an elementary dimension-preserving bijection between irreducible ordinary representations and genuine irreducible representations, via $\rho \mapsto \nu \otimes (\rho \circ p)$. This statement concerns the *full* genuine dual, with central characters adjusted by ν. It is not, by itself, the assertion that the canonical transform L_D realizes this particular bijection. Moreover a prescribed $\widetilde{\theta}$ need not equal $\nu|_{\widetilde{Z}}$; keeping track of this difference is necessary.

Lemma 6.7.4 (Tame discriminant identity)**.** *For every* $h \in H \setminus Z$ *one has*

$$\boxed{\Delta(h^n) = \Delta(h), \qquad R(h) = 1.}$$

The locally constant extension of R *to the scalar center is also identically one.*

Proof. The compact profinite group $K = H/Z$ has all finite quotients of order prime to n by Lemma 6.7.1. On each finite quotient Q the power map $x \mapsto x^n$ is bijective (its inverse is $x \mapsto x^m$ for any m with $mn \equiv 1 \pmod{\exp Q}$). Consequently $P_n : K \to K$ is a homeomorphism which preserves probability Haar measure: it permutes the cosets of every open normal subgroup. It is an analytic local diffeomorphism, since its derivative is invertible at regular points by Lemma 6.3.1, and at scalars it is multiplication by $n \neq 0$ on the Lie algebra.

At a regular element h, write $h\exp(X)$ for a local left-trivialized chart. The derivative of the power map on $\mathfrak{h}/\mathfrak{z}$ is

$$(dP_n)_h = \sum_{k=0}^{n-1} \mathrm{Ad}(h)^{-k}.$$

On the $r-1$ dimensional subspace $\mathfrak{t}/\mathfrak{z}$, $\mathrm{Ad}(h)$ acts trivially, so these directions contribute $|n|_F^{r-1}$ to the absolute determinant. Over a splitting field, the complementary root directions have eigenvalues $u_{ij} = \lambda_i/\lambda_j$ for $i \neq j$ under $\mathrm{Ad}(h)$. The corresponding eigenvalues of the derivative are

$$\sum_{k=0}^{n-1} u_{ij}^{-k} = \frac{1 - u_{ij}^{-n}}{1 - u_{ij}^{-1}},$$

which are nonzero by Lemma 6.3.1. Taking the product over $i \neq j$ (equivalently, the F-determinant) yields

$$|\det(dP_n)_h|_F = |n|_F^{r-1} \prod_{i \neq j} \left| \frac{1 - u_{ij}^{-n}}{1 - u_{ij}^{-1}} \right|_F = |n|_F^{r-1} \frac{\Delta(h^n)^2}{\Delta(h)^2}.$$

Here the middle product is interpreted with the appropriate extension of the normalized absolute value to a splitting field. The p-adic analytic change-of-variables theorem [Loe08, Proposition 1.2.1] applies to the local analytic diffeomorphism P_n. The probability Haar measure is associated locally to a nonvanishing left-invariant top-degree differential form, so preservation of Haar measure implies $|\det(dP_n)_h|_F = 1$ almost everywhere. The absolute determinant is locally constant and nonzero on the regular set, hence is 1 at every regular h. Since $|n|_F = 1$, the discriminant identity follows. The value at scalars is 1 by Lemma 6.3.4. □

Lemma 6.7.5 (Adams operations with fixed central character)**.** *Take $G = D^\times$ and $Z_G = F^\times$, and assume $p \nmid n$. For every smooth unitary character δ of Z_G, the nth Adams operation gives a bijection*

$$\psi^n : \mathrm{Irr}_\delta(G) \xrightarrow{\sim} \mathrm{Irr}_{\delta^n}(G), \qquad \Theta_{\psi^n \sigma}(g) = \Theta_\sigma(g^n),$$

with central characters interpreted on the same scalar center.

Proof. First consider a finite group Q with central subgroup C and $|Q/C|$ prime to n. Let π be the set of primes dividing n, and let Q_π be the π-primary subgroup of the finite abelian group C. Since $|Q/C|$ is prime to n, we have $|Q : Q_\pi| = |Q/C|\,|C : Q_\pi|$ prime to n. Thus Q_π is a *normal Hall π-subgroup* of Q (rather than a single-prime Sylow subgroup); it is central because $Q_\pi \subset C$. Schur–Zassenhaus provides a complement $Q_{\pi'}$, and centrality gives $Q = Q_\pi \times Q_{\pi'}$, with $|Q_{\pi'}|$ prime to n. Fixing the central character on C fixes the one-dimensional representation of Q_π. On $Q_{\pi'}$ the operation ψ^n is the cyclotomic Galois automorphism $\zeta \mapsto \zeta^n$ of character values; since n is prime to $|Q_{\pi'}|$, it permutes irreducible characters. On Q_π it raises the fixed central character to its nth power. Therefore ψ^n bijects irreducibles of central character δ with those of central character δ^n. This is a statement *at a fixed* δ: the operation need not be injective on a union of different central-character fibres, since distinct characters whose quotient has order dividing n can have the same nth power.

For $G = D^\times$, given δ choose a unitary unramified character $\eta = \xi \circ \mathrm{Nrd}$ of G so that $\delta\eta|_Z$ is trivial on a chosen scalar uniformizer ϖ_F (choose an rth root of the inverse scalar value). Then every smooth irreducible with central character $\delta\eta|_Z$ becomes a smooth representation of the compact profinite group $G/\langle\varpi_F\rangle$; it factors through a finite quotient, and its image central subgroup has quotient of order prime to n. For the reverse direction, choose the open normal subgroup through which a given representation factors small enough that $\delta\eta|_Z$ is trivial on its intersection with the center; this is possible because the center modulo $\langle\varpi_F\rangle$ is a closed subgroup of the compact profinite quotient. Then both the specified central character and its nth power factor through the same finite central subgroup. The preceding finite-group argument applies to these representations. Undo the twist, observing that $\psi^n(\eta \otimes \sigma) = \eta^n \otimes \psi^n\sigma$. The same finite-quotient argument applied in reverse proves surjectivity. Note that this argument keeps the central π-part of each finite quotient: an unramified twist cannot generally remove n-torsion coming from the units of Z. □

Theorem 6.7.6 (Tame irreducible character transfer)**.** *Assume $p \nmid n$ and the covering data of Chapter 4. For each unitary genuine central character $\widetilde{\theta}$ on $\widetilde{Z}$, put $\theta(a) = \widetilde{\theta}(a^*)$. Then L_D maps*

irreducible characters bijectively onto genuine irreducible characters,

$$\boxed{L_D : \{\Theta_\rho : \rho \in \mathrm{Irr}_\theta(H)\} \xrightarrow{\sim} \{\Theta_\tau : \tau \in \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})\}.}$$

Every representation on the right corresponding to ρ has $\dim \tau = \dim \rho$. In particular the character matrix of Corollary 6.5.1 is a permutation matrix in the tame case.

Proof. Choose the genuine one-dimensional character ν given by Proposition 6.7.2, with $\nu|_{\mu_n} = \widetilde{\theta}|_{\mu_n}$. Put

$$\chi_\nu(h) = \nu(\widetilde{h})^n \quad (h \in H), \qquad \delta(a) = \frac{\widetilde{\theta}(\widetilde{a})}{\nu(\widetilde{a})} \quad (a \in F^\times).$$

These are well-defined ordinary characters; in particular χ_ν is defined on all of H, while δ is a character of Z. By the definition of the star map,

$$\theta(a) = \chi_\nu(a)\,\delta(a)^n.$$

For $\rho \in \mathrm{Irr}_\theta(H)$, the representation $\chi_\nu^{-1} \otimes \rho$ has central character δ^n. By Lemma 6.7.5 there is a unique $\sigma \in \mathrm{Irr}_\delta(H)$ such that

$$\Theta_\rho(t) = \chi_\nu(t)\,\Theta_\sigma(t^n) \qquad (t \in H).$$

Define the genuine irreducible representation $\tau = \nu \otimes (\sigma \circ p)$. It has central character $\widetilde{\theta}$, and for regular t

$$\Theta_\tau(t^*) = \nu(t^*)\,\Theta_\sigma(t^n) = \chi_\nu(t)\,\Theta_\sigma(t^n) = \Theta_\rho(t).$$

By Lemma 6.3.3 the power map is an automorphism of each compact torus $E^\times/F^\times$ when $p \nmid n$. Thus every regular element of $\widetilde{H}$ is of the form zt^* for some $z \in \widetilde{Z}$ and regular t. The same equality holds on all regular elements by central equivariance. Since $b_D = 1$ and $R = 1$ by Lemma 6.7.4, this is exactly $L_D\Theta_\rho = \Theta_\tau$. The map is bijective by the corresponding bijection for ψ^n and by tensoring with ν. Finally tensoring with one-dimensional characters and the finite-group Galois operation preserve dimension. □

Remark 6.7.7 (Independence of the auxiliary genuine character)**.** Although the construction in the proof chooses a genuine one-dimensional character ν, the resulting correspondence is independent of that choice. Indeed, L_D was defined solely from $\widetilde{\theta}$, the star map, and the normalized discriminant. For each ρ, the character $L_D\Theta_\rho$ determines a unique irreducible genuine representation of $\widetilde{H}$; this follows from orthogonality (or linear independence) of irreducible characters. Consequently any permissible choice of ν produces the same isomorphism class τ. This is a *canonical correspondence relative to the fixed covering datum and central characters*, not an assertion that ν itself is canonical.

Compatibility audit preceding Corollary 6.7.8. We distinguish the intrinsically defined nth power of a lift from any choice of cocycle section. The following facts are the exact compatibilities needed to compose the two local correspondences; they are consequences of Chapter 4 and the formulas in [Zo23, §§2.2, 3.1], not additional assumptions.

Write $F = K_v$ and $c_v = r'c$ as in Chapter 4, with $rr' \equiv 1 \pmod{n}$. Proposition 4.2.4(b), identifies the restriction of the ambient cover, after the prescribed relabeling of μ_n, with the Kazhdan–Patterson

extension whose parameter is c_v. In the standard parameter-zero section the cocycle on the diagonal torus is $\sigma_F^{(0)}(a,b)=\prod_{i<j}(a_i,b_j)_n$; adding the determinant factor gives

$$\sigma_F^{(c_v)}(a,b)=\prod_{i<j}(a_i,b_j)_n\left(\prod_i a_i,\prod_j b_j\right)_n^{c_v}. \tag{6.13}$$

This is precisely Zou's block formula [Zo23, (2.1)] in the one-dimensional-block case. In particular the two models determine the same extension class and the same centrality parameter

$$d_r=\gcd(n,r-1+2rc_v)=n.$$

Here the claimed identification is of *extensions*; Chapter 4 does not claim a preferred equality of two cocycles on all pairs of nondiagonal matrices.

On scalar matrices aI_r,bI_r, equation (6.13) becomes

$$\sigma_F^{(c_v)}(aI_r,bI_r)=(a,b)_n^{e_r},\qquad e_r=\binom{r}{2}+c_vr^2,\qquad 2e_r=r(r-1+2rc_v). \tag{6.14}$$

Since n is odd and divides $r-1+2rc_v$, it divides e_r. Consequently the *standard* Kazhdan–Patterson section is multiplicative on scalar matrices, and for $a\in F^\times$ its nth power satisfies $(s(aI_r))^n=s(a^nI_r)$. In particular if $\zeta\in\mu_n(F)$, then $(\zeta I_r)^*=1$. Thus our scalar character $\theta(a)=\widetilde{\theta}(a^*)$ satisfies $\theta|_{\mu_n}=1$ and is exactly Zou's required relation $\widetilde{\omega}(s(a^nI_r))=\omega(a)$, after transporting $\widetilde{\theta}$ through an isomorphism of covers.

Flicker–Kazhdan define the star map for odd n by $x^*=\widetilde{x}^n$ [FK86, §4]; this agrees with (6.2) and is natural under *every* isomorphism of central extensions fixing μ_n. Two identifications of the ambient GL_r cover with the Kazhdan–Patterson model differ by $\chi\circ\det$ for a character $\chi:F^\times\to\mu_n$ (Proposition 4.2.4(c)). Their values agree on star images, since $\chi(\det(x^n))=1$. At arbitrary scalar translates one transports the central character through the same identification. Hence this choice does not alter the eventual identity for genuine characters of the original ambient cover.

Finally, Theorem 4.3.10(b), identifies the covered elliptic tori by conjugation inside the common ambient cover and is the identity on their common central subgroup. It therefore satisfies

$$\jmath(t_D^*)=(j(t))_{\mathrm{GL}}^*,\qquad \jmath(zt_D^*)=z\,(j(t))_{\mathrm{GL}}^*\quad(z\in\widetilde{Z}). \tag{6.15}$$

The Weyl discriminants on $D^\times$ and $\mathrm{GL}_r(F)$ are the same expression in the conjugate eigenvalues of t; see (6.3) and Definition 4.4.1. These are character *values*, not global orbital integrals: no comparison of arbitrary quotient Haar measures is being inferred.

Corollary 6.7.8 (Tame character relation in the available Flicker–Kazhdan range)**.** *Let $F=K_v$ be a finite place at which D is division, $p\nmid n$, and the standing odd-n and $(n,r)=1$ assumptions hold. Fix the common faithful genuine central character $\widetilde{\theta}$ of Chapter 4 and let $\theta(a)=\widetilde{\theta}(a^*)$. Assume that the square-integrable Flicker–Kazhdan correspondence for the parameter-c_v Kazhdan–Patterson cover is available in the corrected form of [Zo23, Proposition 3.6 and Definition 3.7]. A sufficient published range for this external input is $(n,N_r)=1$, where N_r is the least common multiple of the composite positive integers at most r, with $N_r=1$ when there are none [FK86, §26].*

Then the division-side tame transfer L_D of Theorem 6.7.6, the classical local Jacquet–Langlands correspondence, and the corrected metaplectic correspondence compose to a bijection between the

relevant genuine irreducible representations $\widetilde{\pi}_D$ of $\widetilde{D^\times}$ and genuine essentially square-integrable representations $\widetilde{\pi}$ of $\widetilde{G}_v$ with the specified central character. On matching regular good elements,

$$\boxed{\Theta_{\widetilde{\pi}_D}(\widetilde{\gamma}) = (-1)^{r-1}\Theta_{\widetilde{\pi}}(\jmath(\widetilde{\gamma})).} \tag{6.16}$$

Thus relation (D2) of Chapter 4 is proved in this stated tame range, conditional only *on the cited external Flicker–Kazhdan theorem and the established local results of Chapters 4–6; it is not asserted for all (n,r).*

Proof. 1. Matching characters on the division side. By Theorem 6.7.6 there is a unique $\rho \in \mathrm{Irr}_\theta(D^\times)$ for which $L_D\Theta_\rho = \Theta_{\widetilde{\pi}_D}$. At $p \nmid n$ we have $b_D = 1$ and $\Delta(t^n) = \Delta(t)$ (Lemma 6.7.4); thus

$$\Theta_{\widetilde{\pi}_D}(z t_D^*) = \widetilde{\theta}(z)\Theta_\rho(t) \qquad (t \in E^\times \setminus F^\times, z \in \widetilde{Z}). \tag{6.17}$$

Every regular good element has this form (Proposition 4.3.5(a)), and the power map is injective on $E^\times/F^\times$ by Lemma 6.3.3.

2. Classical Jacquet–Langlands. Let $\pi = \mathrm{JL}(\rho)$, with the same ordinary scalar character θ. The usual elliptic-character relation is

$$\Theta_\rho(t) = (-1)^{r-1}\Theta_\pi(j(t)), \tag{6.18}$$

where $j : E \hookrightarrow M_r(F)$ and the discriminant normalization is the common one above [DKV84]. The character θ is trivial on μ_n by (6.14), so π lies in the permitted linear central-character family in Zou's correspondence.

3. Explicit evaluation of Zou's root sum. Let $x = j(t)$ be regular elliptic. If $y^n = x^n$ in $\mathrm{GL}_r(F)$, then y commutes with x^n. As t^n is regular (Lemma 6.3.1), its centralizer is $j(E^\times)$; therefore $y \in j(E^\times)$ and $(yx^{-1})^n = 1$. Since $\mu_n(E) = \mu_n(F)$, the solutions are exactly the n scalar multiples $y = \zeta x$ with $\zeta \in \mu_n$. For every one, $\Delta(y) = \Delta(x)$ and $\Theta_\pi(y) = \Theta_\pi(x)$ because $\theta(\zeta) = 1$. Equation (6.14) gives $(\zeta I_r)^* = 1$, so the phase $\varepsilon(x^*/y^*)$ in Zou's equation [Zo23, (3.2)] is also 1. Zou's corrected root-sum formula therefore reduces to

$$\Theta_{\widetilde{\pi}}(x^*) = \frac{n}{d_F(d_r)|n|_F^{r/2}}\frac{\Delta(x)}{\Delta(x^n)}\Theta_\pi(x) = b_D\frac{\Delta(x)}{\Delta(x^n)}\Theta_\pi(x), \tag{6.19}$$

where $d_F(m) = m|m|_F^{-1/2}$ and $d_r = \gcd(n, r-1+2rc_v) = n$. Indeed,

$$\frac{n}{d_F(n)|n|_F^{r/2}} = \frac{n}{n|n|_F^{-1/2}|n|_F^{r/2}} = |n|_F^{-(r-1)/2} = b_D.$$

The equality of numerical factors is valid even at wild places; it is *not* a wild-place character-integrality theorem. Here $p \nmid n$ implies $b_D = 1$, and Lemma 6.7.4 gives $\Delta(x) = \Delta(x^n)$. Consequently $\Theta_{\widetilde{\pi}}(x^*) = \Theta_\pi(x)$.

4. Transfer back to the two actual covers. By (6.15), the star values in the last identity coincide with those obtained from the covered division-algebra torus. Multiplying by $\widetilde{\theta}(z)$ and combining (6.17), (6.18) and (6.19) yields (6.16) for every good regular element. The star and central-character identifications in the preceding compatibility audit show that the conclusion is independent of the auxiliary isomorphism to the standard Kazhdan–Patterson cocycle model.

5. Bijection rather than just a character identity. Theorem 6.7.6 is a bijection on genuine irreducibles with fixed central character; classical local Jacquet–Langlands is a bijection onto the corresponding essentially square-integrable linear representations; and Zou's corrected correspondence is a bijection in the specified external range and central-character family. Their composition has an inverse obtained by reversing the three maps. This concludes the claim without inferring irreducibility or integrality from L^2 unitarity alone. □

Remark 6.7.9 (Scope and status of Corollary 6.7.8)**.** The earlier version of this corollary carried status **(O)** because the cocycle, star map and central characters had not been compared explicitly with Zou. Equations (6.13)–(6.15) now perform that comparison; the result is **(P)** *under the written external correspondence hypothesis*. In particular $(n, N_r) = 1$ suffices by the stated Flicker–Kazhdan theorem, as used by Zou. For additional parameter pairs one must verify an applicable extension of that external input; the present proof does not supply it. At places $p \mid n$, integrality of L_D remains open. Global matching of Haar measures, test-function normalizations, and unramified spherical units is also not proved by this local pointwise character identity.

6.8 Places dividing n: non-splitting and constraints

At residue characteristics dividing n, the tame splitting argument fails. The commutator pairing from Chapter 4 gives a precise obstruction, but does not supply an irreducible-character correspondence. The numerical test of Remark 6.5.5 is an additional constraint on any proposed correspondence. The section depends on §§6.2 and 6.5 and on the commutator pairing of §4.3; it introduces no new notation.

Proposition 6.8.1 (Wild-place obstruction to splitting)**.** *Suppose $p \mid n$ and retain the commutator formula and nondegeneracy assertion of Proposition 4.3.5(a),(c). Then $\widetilde{H}$ does not split over H^1, and it has no genuine one-dimensional representations. More precisely, every genuine irreducible representation of $\widetilde{H}$ has dimension divisible by*

$$b_D = |n|_F^{-(r-1)/2} > 1.$$

The existence or integrality of the wild-place transform L_D on irreducible characters is not established by this proposition.

Proof. Put $E^1 = \ker(N_{E/F} : E^\times \to F^\times)$ and $B = E^\times/(F^\times E^{\times n})$. The group B has order $|n|_F^{-(r-1)} > 1$ by Lemma 6.3.3; it is n-torsion. The map $E^1 \to B$ is surjective: for $t \in E^\times$,

$$s_t = t^r/N_{E/F}(t) \in E^1, \qquad [s_t] = [t]^r \in B.$$

Since $\gcd(r,n) = 1$, raising to the rth power is an automorphism of B, so these elements generate B. The commutator pairing on the covered torus, described in Proposition 4.3.5(a), has radical $F^\times E^{\times n}$; it is therefore nondegenerate on B. Restricting to E^1 preserves a nontrivial pairing because $E^1 \to B$ is surjective. Consequently the inverse image of $E^1 \subset H^1$ in $\widetilde{H}$ is nonabelian, so the cover cannot admit a splitting over H^1.

Proposition 4.3.5(c), identifies the dimensions of genuine irreducible representations of the covered torus with $\sqrt{|B|} = b_D$ (for the fixed genuine central character). Restricting a finite-dimensional genuine representation of $\widetilde{H}$ to the covered torus and decomposing it into these constituents proves the divisibility assertion, which excludes genuine one-dimensional representations. □

Remark 6.8.2 (Global use and the remaining difficulty)**.** At a finite place v with $p_v \nmid n$, the local $D_v^\times$-side integrality problem is now resolved under the stated covering assumptions. At places dividing n it remains a distinct and genuinely nonabelian problem. Global division algebras can be chosen to avoid those places subject to the Brauer-invariant sum and local-degree conditions. This removes the *division-side local integrality obstacle* for such a choice, but does not prove global trace-formula comparison, spectral multiplicity statements, or a metaplectic Jacquet–Langlands theorem at the split places.

6.9 Local orbital-integral transfer on $D^\times$

This section depends on §§6.2 and 6.4. It introduces the spaces $\mathscr{C}_{\eta^{-1}}(H)$, the orbital integrals O and $J = \Delta O$, the transforms $L_D^{(\widetilde{\eta})}$, and the transfers $\mathscr{O}_D$ and $\mathscr{B}$. In this section no assumption $p \nmid n$ is imposed. The construction therefore applies at both tame and wild places; it uses the good-element description of Proposition 6.2.6. We keep the quotient probability Haar measures from Section 6.4. The result is a transfer of test functions and geometric distributions; it is not an assertion that the transform of every irreducible character is irreducible.

6.9.1 Test functions and orbital integrals

This subsection defines the test functions with a fixed central character on H and on $\widetilde{H}$, their conjugation averages and their orbital integrals. It also explains why orbital integrals on the cover are taken only at good elements.

For a unitary central character η of Z, let

$$\mathscr{C}_{\eta^{-1}}(H) = \{f : H \to \mathbb{C} : \ f \text{ is locally constant and } f(ah) = \eta(a)^{-1} f(h) \text{ for } a \in Z\}.$$

We use analogous notation for $\mathscr{C}_{\widetilde{\eta}^{-1}}(\widetilde{H})$. Since H/Z and $\widetilde{H}/\widetilde{Z}$ are compact, all these functions are compactly supported modulo the indicated center. These are the test-function central characters *inverse* to the central characters of representations: $\Theta_\rho(ah) = \theta(a)\Theta_\rho(h)$, whereas $f(ah) = \theta(a)^{-1} f(h)$, so that the product is a function on H/Z.

Write dh for Haar probability measure on H/Z, and define the conjugation average

$$(\mathscr{A}_H f)(h) = \int_{H/Z} f(x^{-1}hx)\, dx. \tag{6.20}$$

For the cover define, using any lift $\widetilde{x}$ of x,

$$(\mathscr{A}_{\widetilde{H}} \widetilde{f})(\widetilde{h}) = \int_{H/Z} \widetilde{f}(\widetilde{x}^{-1}\widetilde{h}\widetilde{x})\, dx. \tag{6.21}$$

Centrality of μ_n and $\widetilde{Z}$ makes the latter integrand independent of the chosen lifts and of the representative of $x \in H/Z$. Both averages are smooth, conjugation invariant, and central equivariant; they satisfy $\mathscr{A}_H^2 = \mathscr{A}_H$ and $\mathscr{A}_{\widetilde{H}}^2 = \mathscr{A}_{\widetilde{H}}$.

For $t \in T^{\mathrm{reg}}$, $T = F(t)^\times$, equip H/T with the quotient *probability* measure induced by those on H/Z and T/Z. Define

$$O_H(t, f) = \int_{H/T} f(x^{-1}tx)\, d\dot{x} = (\mathscr{A}_H f)(t). \tag{6.22}$$

For a good regular $\widetilde{t} \in p^{-1}(T)$ the corresponding genuine orbital integral is

$$O_{\widetilde{H}}(\widetilde{t},\widetilde{f}) = \int_{H/T} \widetilde{f}(\widetilde{x}^{-1}\widetilde{t}\widetilde{x})\,d\dot{x} = (\mathscr{A}_{\widetilde{H}}\widetilde{f})(\widetilde{t}). \tag{6.23}$$

The first integral is independent of coset representatives precisely because $\widetilde{t}$ is good. If $\widetilde{t}$ is not good, the full conjugation average in (6.21) vanishes: replacing $\widetilde{x}$ by $\widetilde{u}\widetilde{x}$, for a lift of an element of T with nontrivial commutator against $\widetilde{t}$, multiplies the averaged value by a nontrivial genuine scalar. We use this vanishing rather than attempting to define an H/T integral for a nongood element.

For later use put

$$J_H(t,f) = \Delta(t)O_H(t,f), \qquad J_{\widetilde{H}}(\widetilde{t},\widetilde{f}) = \Delta(p(\widetilde{t}))O_{\widetilde{H}}(\widetilde{t},\widetilde{f}) \tag{6.24}$$

when $\widetilde{t}$ is good. The factor Δ is the *square root* of the ordinary Weyl discriminant as in (6.3).

6.9.2 An explicit transfer theorem

This subsection defines transfer maps between test functions on H and on $\widetilde{H}$ and proves an explicit identity between their orbital integrals.

Let $L_D^{(\widetilde{\eta})}$ denote the map in Definition 6.4.1 for the genuine central character $\widetilde{\eta}$, with its associated ordinary central character $\eta(a) = \widetilde{\eta}(a^*)$. In particular, $L_D^{(\widetilde{\theta}^{-1})}$ acts on invariant test functions having the inverse central character.

Theorem 6.9.1 (Division-side test-function transfer)**.** *Assume the specified cover of Chapter 4 and Proposition 6.2.6. Fix a genuine central character $\widetilde{\theta}$ and put $\theta(a) = \widetilde{\theta}(a^*)$. The following maps are well defined:*

$$\mathscr{O}_{D,\widetilde{\theta}} : \mathscr{C}_{\theta^{-1}}(H) \longrightarrow \mathscr{C}_{\widetilde{\theta}^{-1}}(\widetilde{H}), \qquad \mathscr{O}_{D,\widetilde{\theta}}f = L_D^{(\widetilde{\theta}^{-1})}(\mathscr{A}_H f), \tag{6.25}$$

$$\mathscr{B}_{\widetilde{\theta}} : \mathscr{C}_{\widetilde{\theta}^{-1}}(\widetilde{H}) \longrightarrow \mathscr{C}_{\theta^{-1}}(H), \qquad \mathscr{B}_{\widetilde{\theta}}\widetilde{f} = (L_D^{(\widetilde{\theta}^{-1})})^{-1}(\mathscr{A}_{\widetilde{H}}\widetilde{f}). \tag{6.26}$$

Their images consist of conjugation-invariant test functions; and

$$\mathscr{B}_{\widetilde{\theta}}\mathscr{O}_{D,\widetilde{\theta}} = \mathscr{A}_H, \qquad \mathscr{O}_{D,\widetilde{\theta}}\mathscr{B}_{\widetilde{\theta}} = \mathscr{A}_{\widetilde{H}}.$$

They therefore induce mutually inverse linear bijections on the respective test-function spaces modulo the kernels of their conjugation-averaging (equivalently, normalized orbital-integral) maps.

More explicitly, for every regular $t \in T$ and $z \in \widetilde{Z}$,

$$\boxed{O_{\widetilde{H}}(zt^*, \mathscr{O}_{D,\widetilde{\theta}}f) = b_D\,\widetilde{\theta}(z)^{-1}\frac{\Delta(t)}{\Delta(t^n)}\,O_H(t,f), \qquad b_D = |n|_F^{-(r-1)/2}.} \tag{6.27}$$

Consequently the normalized orbital integrals satisfy

$$\boxed{J_{\widetilde{H}}(zt^*, \mathscr{O}_{D,\widetilde{\theta}}f) = b_D\,\widetilde{\theta}(z)^{-1}J_H(t,f).} \tag{6.28}$$

At nongood regular lifts the genuine orbital-integral distribution vanishes. Conversely every genuine test function $\widetilde{f}$ has the same regular orbital integrals as $\mathscr{O}_{D,\widetilde{\theta}}(\mathscr{B}_{\widetilde{\theta}}\widetilde{f})$.

Proof. Conjugation averaging is a projection onto the invariant locally constant functions, and (6.22)–(6.23) identify the averages with the stated orbital integrals. The local smoothness and invertibility of L_D on invariant test functions follow from Lemma 6.4.3 and the proof of Theorem 6.4.4: on each regular torus quotient the star map is a homeomorphism from T/Z onto the open subgroup $(T/Z)^n$, and near scalar elements it is a local analytic isomorphism. Thus both displayed maps take smooth test functions to smooth test functions. Since the image of $\mathscr{A}_H$ consists of invariant functions, $\mathscr{A}_{\widetilde{H}}\mathscr{O}_{D,\widetilde{\theta}}f = \mathscr{O}_{D,\widetilde{\theta}}f$; applying the inverse L_D gives $\mathscr{B}_{\widetilde{\theta}}\mathscr{O}_{D,\widetilde{\theta}}f = \mathscr{A}_H f$. The other composition is identical. At a good point zt^*, the definition of L_D applied to $\mathscr{A}_H f$ gives (6.27). Multiplying it by $\Delta(t^n)$ gives (6.28). Remark 6.2.5 proves the vanishing on nongood lifts. The inverse composition identity gives the converse statement. No assertion about irreducible-character integrality is used. □

Remark 6.9.2 (Comparison with the root transport of Chapter 5)**.** The orbital-integral operator $\mathscr{O}_{D,\widetilde{\theta}}$ is *not* the local root-section transport denoted $\mathscr{T}$ in Chapter 5. They have distinct domains, definitions and central character conventions. Near the center, the parameters satisfy $t^* = \sigma(t^n)$, with σ the root section in Chapter 5; this identity does not equate the operators on arbitrary test functions. The orbital transfer here also involves conjugation averaging and the factor $b_D\Delta(t)/\Delta(t^n)$.

6.9.3 The duality with character transfer

The inverse central characters in Theorem 6.9.1 are forced by trace pairings, not a cosmetic convention.

Proposition 6.9.3 (Trace-pairing adjunction)**.** *For every smooth invariant class function Φ on H of central character θ, and every $f \in \mathscr{C}_{\theta^{-1}}(H)$,*

$$\boxed{\int_{\widetilde{H}/\widetilde{Z}} (L_D^{(\widetilde{\theta})}\Phi)(\widetilde{h})(\mathscr{O}_{D,\widetilde{\theta}}f)(\widetilde{h})\,d\widetilde{h} = \int_{H/Z} \Phi(h)f(h)\,dh.} \tag{6.29}$$

In particular, for $\rho \in \mathrm{Irr}_\theta(H)$, writing the finite expansion of Corollary 6.5.1 as $L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau$, one has

$$\sum_\tau a_{\rho\tau} \operatorname{tr}\tau(\mathscr{O}_{D,\widetilde{\theta}}f) = \operatorname{tr}\rho(f). \tag{6.30}$$

If $p \nmid n$, Theorem 6.7.6 reduces this equality to $\operatorname{tr}\tau_\rho(\mathscr{O}_{D,\widetilde{\theta}}f) = \operatorname{tr}\rho(f)$ for the corresponding genuine irreducible representation τ_ρ. For $p \mid n$, formula (6.30) is a distributional identity with possibly nonintegral coefficients, not *a representation correspondence.*

Proof. Because Φ is invariant, averaging f does not change the integral on the right. For the left side note that $L_D^{(\widetilde{\theta}^{-1})}(\mathscr{A}_H f)$ is the complex conjugate of $L_D^{(\widetilde{\theta})}(\overline{\mathscr{A}_H f})$: the coefficient $b_D\Delta(t)/\Delta(t^n)$ is positive real, and complex conjugation interchanges the two central characters. The unitarity identity of Theorem 6.4.4, applied to Φ and $\overline{\mathscr{A}_H f}$, proves (6.29). Equivalently, the Weyl integration formula gives the identity directly: on each T/Z, the product of the two transforms contributes $b_D^2\Delta(t)^2/\Delta(t^n)^2$ and the change of variables (6.8) contributes I^{-1}; since $b_D^2 = I$, the Weyl integrals are equal. Both trace identities follow by integrating characters against test functions with inverse central character. In particular (6.30) is a reformulation of unitarity and surjectivity of $\mathscr{O}_{D,\widetilde{\theta}}$, not an independent constraint forcing integrality of the matrix $(a_{\rho\tau})$. □

Corollary 6.9.4 (The identity-element factor)**.** *With the probability measures and central-character conventions of this section, the transferred invariant representative satisfies*

$$\boxed{(\mathscr{O}_{D,\widetilde{\theta}} f)(1) = |n|_F^{-(r^2-1)/2} f(1).} \tag{6.31}$$

The factor equals 1 *at tame places. This is a local pointwise test-function normalization; it is* not *a calculation of the global identity term in a trace formula with different Haar measures and central quotients.*

Proof. The central element 1 is fixed by conjugation, so $(\mathscr{A}_H f)(1) = f(1)$. Evaluate the continuous extension of the transfer at 1, using Lemma 6.3.4 and $b_D = |n|_F^{-(r-1)/2}$. □

Remark 6.9.5 (The running wild-place example)**.** For $r = 2$, $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, $|3|_F = 1/9$ and $b_D = 3$. Thus $J_{\widetilde{H}}(zt^*, \mathscr{O}_{D,\widetilde{\theta}} f) = 3\widetilde{\theta}(z)^{-1} J_H(t,f)$, whereas $(\mathscr{O}_{D,\widetilde{\theta}} f)(1) = 27 f(1)$. This realizes the two normalizations discussed in Remark 6.5.5.

Remark 6.9.6 (Scope of local transfer)**.** The existence of such a transfer is elementary on a group compact modulo its center: conjugation averaging identifies orbital integrals with invariant function values. The content of the theorem is the *specified normalization*, extension across singular central elements, and the trace-pairing relation. The local coefficient b_D does not determine adelic orbital measures. The distinction from Chapter 5's root transport is made precise in Remark 6.9.2.

6.10 Kottwitz's framework for rational orbits and inner forms

This section is expository; it depends on nothing earlier in the chapter, and its terminology (stable and rational conjugacy, inner forms, the torus obstruction, Tamagawa factors) is collected in §3.9. Kottwitz's *Stable trace formula: cuspidal tempered terms* [Kot84] explains how Galois cohomology organizes rational conjugacy classes within stable classes (§§7.7–7.9), how maximal tori are transferred to inner forms (§9.5), and how Tamagawa numbers enter stable trace-formula coefficients (§§5,8). These are statements about *linear* reductive groups; they must not be taken as a metaplectic trace formula. In particular, the stabilization of cuspidal tempered terms sketched in his §12 is conditional, whereas the cohomological calculations used below are independent of that sketch.

Stable and rational classes. For a regular semisimple element $\gamma \in G(K)$ with torus centralizer $T = G_\gamma$, the rational conjugacy classes within its represented stable class are parametrized by

$$\ker\big[H^1(K,T) \longrightarrow H^1(K,G)\big] \tag{6.32}$$

[Kot84, §7.7]. In the present setting, $G = \mathrm{GL}_r$ or $G' = \mathrm{GL}_1(D)$ and $T = \mathrm{Res}_{E/K}\mathbb{G}_m$ for a maximal subfield E/K of degree r. By Shapiro's lemma and Hilbert 90, $H^1(K,T) = H^1(E,\mathbb{G}_m) = 1$. Thus every *represented* regular stable class comprises exactly one rational conjugacy class. This does not decide whether the class has a good lift in a covering group.

The inner-form embedding criterion. A degree-r extension E/K embeds in D as a maximal subfield if and only if

$$[D \otimes_K E] = 0 \quad \text{in } \mathrm{Br}(E). \tag{6.33}$$

By Brauer–Hasse–Noether, this condition can be checked at all completions. Where D_v is division of degree r, the local algebra $E \otimes_K K_v$ must be a field of degree r. Consequently, for these inner forms, local embeddability at every place implies global embeddability. Kottwitz's general §9.5 formulates an obstruction to torus transfer in a class of $H^2(K, T_{\mathrm{sc}})$ (pp. 638–639); the specialized Brauer criterion avoids that additional complication here. For linear orbital-integral transfer see also [Coh18].

Tamagawa factors. Kottwitz §5 identifies the relative Tamagawa number $\tau_1(G) = \tau(G)/\tau(G_{\mathrm{sc}})$ as

$$\tau_1(G) = \frac{|\pi_0(Z(\widehat{G})^{\Gamma_K})|}{|\ker^1(K, Z(\widehat{G}))|}. \tag{6.34}$$

For GL_r and $\mathrm{GL}_1(D)$ the dual center is $\mathbb{C}^\times$ with trivial Galois action, so this expression is 1. Kottwitz §8 uses relative Tamagawa numbers in stable trace-formula coefficients. Our local b_D instead uses *probability* Haar measures on compact quotients; its conversion to a coherent adelic convention remains a separate calculation, not a consequence of (6.34).

6.11 Global power classes and a Hasse principle

The scalar-modulo-nth-power quotient is typically infinite, but its subgroup of classes trivial outside a finite set of places vanishes under the hypotheses below. The proof separates the norm argument from the Kummer–Chebotarev detection step. The section depends on §6.3 and on global class field theory; it introduces the scalar power quotient $\mathscr{Q}(E/K, n)$ and the group $\mathrm{Sha}^1(K, T[n])$. The following large quotient measures the failure of *global surjectivity* of a power map, but its size is not the correct measure of failure of a *local-to-global principle*. Let E/K be a maximal subfield of D, of degree r, and consider the multiplicative quotient

$$\mathscr{Q}(E/K, n) = E^\times / (K^\times E^{\times n}). \tag{6.35}$$

A regular rational class represented by $\gamma \in E^\times$ can lie in the scalar multiple of the rational nth-power image only if its class in (6.35) is trivial. This obstruction must not be confused with the finite local index of Lemma 6.3.3.

Proposition 6.11.1 (The ambient power quotient is infinite)**.** *Let E/K be a number-field extension of degree at least two and let $n > 1$. Then $\mathscr{Q}(E/K, n)$ is infinite. More precisely, for any $m \geq 1$ and any distinct finite places $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ of K splitting completely in E, there is a surjective homomorphism*

$$\mathscr{Q}(E/K, n) \longrightarrow (\mathbb{Z}/n\mathbb{Z})^m.$$

In particular, even when $\mu_n \subset K$ and $\gcd(n, r) = 1$, the map $[t] \mapsto [t^n]$ on $E^\times/K^\times$ cannot be surjective.

Proof. For each i choose two distinct primes $\mathfrak{P}_{i,1}$ and $\mathfrak{P}_{i,2}$ of E above $\mathfrak{p}_i$. Because $\mathfrak{p}_i$ splits completely, for $a \in K^\times$ the normalized discrete valuations satisfy $v_{\mathfrak{P}_{i,1}}(a) = v_{\mathfrak{P}_{i,2}}(a)$. Consequently

$$\partial_i([x]) = v_{\mathfrak{P}_{i,1}}(x) - v_{\mathfrak{P}_{i,2}}(x) \pmod{n}$$

is well defined on $\mathcal{Q}(E/K,n)$: multiplication by $K^\times$ changes neither difference, and multiplication by an nth power changes it by a multiple of n. The weak approximation theorem for E permits independent prescription of the valuations at this finite collection of primes, so the product $(\partial_1,\dots,\partial_m)$ is surjective. Chebotarev's theorem supplies infinitely many completely split primes; thus m is arbitrary and the quotient is infinite. □

Remark 6.11.2 (Effect on the proposed trace formula)**.** The infinitude of $\mathcal{Q}(E/K,n)$ does *not* show that there is an infinite global obstruction among the classes satisfying all local power conditions. Indeed, the classes exhibited by valuation differences are already nontrivial in the local power quotient at the corresponding completely split places. For the split-cover commutator these may also be nongood; that additional implication must be checked in the particular Flicker–Kazhdan cocycle, not inferred merely from the valuation calculation. One must distinguish a large ambient quotient from the kernel of the localization map. That kernel is controlled by the next proposition.

Proposition 6.11.3 (Local-to-global powers modulo scalars)**.** *Let E/K be a separable extension of number fields of degree r, let $n>1$, and assume* $\gcd(n,r)=1$ *and* $\mu_n \subset K$. *For* every finite set of places Σ *of K, the natural map*

$$\mathcal{Q}(E/K,n) \longrightarrow \prod_{v\notin\Sigma} \frac{(E\otimes_K K_v)^\times}{K_v^\times (E\otimes_K K_v)^{\times n}} \tag{6.36}$$

is injective. Thus being a scalar times an nth power at almost all places *already implies being so globally.*

Proof. Let $t \in E^\times$ map to the trivial class outside a finite set Σ. Choose integers u,k satisfying $ru+nk=1$, and set $a = N_{E/K}(t)^u \in K^\times$. For each $v\notin\Sigma$, the assumption supplies $t = a_v y_v^n$ in $(E\otimes_K K_v)^\times$, for $a_v \in K_v^\times$ and $y_v \in (E\otimes_K K_v)^\times$. Taking the norm from this degree-r *étale algebra* produces $N(t) = a_v^r N(y_v)^n$. Consequently

$$\frac{t}{a} = \left(a_v^k \frac{y_v}{N(y_v)^u}\right)^n \quad \text{in } (E\otimes_K K_v)^\times.$$

Thus t/a is an nth power at each completion E_w lying over a place $v\notin\Sigma$. Since $\mu_n \subset E$, a nontrivial Kummer extension $E((t/a)^{1/n})/E$ would be a nontrivial cyclic Galois extension and, by Chebotarev, would possess infinitely many unramified places with nontrivial Frobenius. We may choose one lying over a place $v\notin\Sigma$. At that completion t/a could not be an nth power, a contradiction. Hence $t/a \in E^{\times n}$ and $[t]=1$ in $\mathcal{Q}(E/K,n)$. □

Corollary 6.11.4 (Vanishing of the associated Sha^1)**.** *Under the assumptions of Proposition 6.11.3, put $T = \mathrm{Res}_{E/K}\mathbb{G}_m/\mathbb{G}_m$ and let $T[n]$ be its finite n-torsion group scheme. Then for every finite Σ, the restricted localization kernel*

$$\mathrm{Sha}^1_\Sigma(K,T[n]) := \ker\Big(H^1(K,T[n]) \to \prod_{v\notin\Sigma} H^1(K_v,T[n])\Big)$$

vanishes. In particular, the usual

$$Ш^1(K,T[n]) := \ker\left[H^1(K,T[n]) \longrightarrow \prod_v H^1(K_v,T[n])\right] = 0. \tag{6.37}$$

This is stronger than the general finiteness of $Ш^1$ *for a finite Galois module [Mil06, Chapter I, §4].*

Proof. From $1 \to \mathbb{G}_m \to \mathrm{Res}_{E/K}\mathbb{G}_m \to T \to 1$, Hilbert 90 gives $T(K) = E^\times/K^\times$. The same cohomology sequence gives

$$H^1(K,T) \cong \ker\big(\mathrm{Br}(K) \to \mathrm{Br}(E)\big).$$

Restriction followed by corestriction is multiplication by r, so this group is killed by r. Hence, as $\gcd(n,r) = 1$, $H^1(K,T)[n] = 0$. The Kummer exact sequence for multiplication by n on T therefore gives

$$H^1(K,T[n]) \cong T(K)/nT(K) = \mathcal{Q}(E/K,n).$$

At each K_v the same argument applies to the finite étale algebra $E \otimes_K K_v$ (its total degree is still r), yielding analogous isomorphisms for $H^1(K_v,T[n])$. Under these identifications the localization map is (6.36), which is injective, even after omitting any finite set of places, by the proposition. □

Remark 6.11.5 (Scope of the Hasse principle)**.** The conclusion uses local power conditions at almost all places, not merely at the division places S. Moreover, an nth-power condition in the scalar quotient must not be identified with the good-element condition for the cover of a split group without a separate commutator calculation. The theorem eliminates this particular Kummer obstruction; it does not provide the missing metaplectic geometric comparison.

6.12 The global comparison: remaining problems

The following spectral identity is a target rather than a theorem. The arithmetic in Section 6.11 simplifies its rational-orbit accounting, but leaves the covering-group good-locus and measure comparisons untouched. The section depends on §§6.9–6.11 and introduces the target identity (6.38) and the map Ψ_D. Let K be a number field containing μ_n (for odd $n > 1$ this forces K to be totally imaginary, as in Chapters 4 and 5), and D/K a central division algebra of prime degree r. Denote by S its finite ramification set. We consider the automorphic spectra of $D^\times(\mathbb{A}_K)$ and its chosen metaplectic cover, both modulo their respective centers. The quotient of $D^\times(\mathbb{A}_K)$ by its rational points and adelic center is compact, hence both spectral trace formulas are discrete. *This fact alone does not establish equality of their geometric sides.*

The following proposed identity is the main mathematical target:

$$\sum_{\widetilde{\Pi}_D} m(\widetilde{\Pi}_D)\,\mathrm{tr}\,\widetilde{\Pi}_D(\widetilde{f}) \stackrel{?}{=} \sum_{\Pi_D} m(\Pi_D)\,\mathrm{tr}\,\Pi_D(f), \tag{6.38}$$

for appropriately matched factorizable test functions and compatible global central characters. It is *conditional* on a comparison of the geometric sides with all constants correct. Section 6.9 records a normalized division-side local matching operator on test functions, dual to the character-space

transform. Its existence is automatic from compactness modulo the center, while smoothness and the normalization require the stated local input. Kottwitz's framework in Section 6.10 clarifies the ordinary rational-orbit and measure issues, but neither automatically gives a geometric comparison for the cover nor settles wild-place character integrality.

6.12.1 Four proof obligations

This subsection lists the four steps that a proof of the target identity (6.38) would have to carry out.

(1) **Rational orbit accounting.** Combine the uniqueness of represented regular rational classes from Section 6.10 with the local-to-global power result of Proposition 6.11.3. Then determine which rational elements are locally *good* for the particular cover at split and division places. Vanishing of the Kummer localization kernel alone does not identify the surviving cover orbits.

(2) **Local normalization and compatibility.** At $v \in S$, convert the probability-measure test transfer of Section 6.9 to the fixed global measures and check the local geometric factor against the exact desired global comparison. At split places, cite the exact usable form of Flicker–Kazhdan [FK86], taking into account Zou's correction to their Lemma 27.1 [Zo23]. Check unramified unit elements and archimedean factors separately.

(3) **Identity and volume terms.** With the trace-formula quotient measures fixed, calculate the identity terms, orbital-integral Jacobians, and central character factors. The b_D of Theorem 6.4.4 is a *local probability-measure normalization* and does not dispose of the global normalization problem.

(4) **Spectral separation and integrality.** Derive a finite, or otherwise rigorously summable, character identity. Use classical strong multiplicity one for the linear group [Bad08], but do not infer integrality merely from L^2 unitarity. The proposed Cauchy–Schwarz argument requires an explicit identity, its correct normalization, and control of every other local factor.

Remark 6.12.1 (Restriction on n)**.** The standard hypotheses under which one invokes the original Flicker–Kazhdan trace formula include $(n, N_r) = 1$, where N_r is the least common multiple of the composite integers $\leq r$. Our local Theorem 6.4.4 uses only n odd, $(n, r) = 1$, and the covering result in Proposition 6.2.6; it does not use $(n, N_r) = 1$. The global comparison will provisionally impose the latter condition until the exact later improvements or alternatives are verified.

6.12.2 Conditional representation-theoretic consequences

This subsection records what would follow for representations if the target identity were proved. Everything in it is conditional.

If (6.38) is established in a form that proves character integrality at the remaining *wild* ramified places, while Theorem 6.7.6 handles the tame ones, the resulting map Ψ_D from irreducible genuine representations of the cover to $\mathrm{Irr}(D_v^\times)$ can be composed, subject to central-character and normalization checks, with

$$\mathrm{Irr}(D_v^\times) \xrightarrow{\mathrm{JL}} \mathrm{Irr}_{\mathrm{ds}}(\mathrm{GL}_r(F)) \xleftarrow{\mathrm{MC}} \mathrm{Irr}_{\mathrm{gen,ds}}(\widetilde{\mathrm{GL}}_r(F)),$$

where MC is Zou's corrected form of the Flicker–Kazhdan correspondence, with the required restriction that the linear central character be trivial on μ_n. At tame places this composition gives the proved conclusion of Corollary 6.7.8 in the stated range of the external Flicker–Kazhdan theorem; outside that range it remains conditional on an applicable extension of the external input. This would establish a local metaplectic Jacquet–Langlands correspondence. The fact that local supercuspidals are easier to isolate spectrally does not eliminate the wild-place integrality obstruction for all representations.

Similarly, removing the non-cancellation hypothesis (Hyp'_D) of Definition 5.6.5, would require showing that a prescribed packet has only finitely many S-types, or some other condition contradicting the cancellation criterion from Chapter 5. The original planning memo's argument through strong multiplicity one is a plausible *conditional* route, but its spectral separation and finiteness arguments have not yet been established.

Removal of the extra cuspidal place w_0, or of the converse hypothesis (Hyp') (Corollary 5.6.10), remains a different problem. None of these goals is asserted as a theorem in the present draft.

Chapter 7

Local power maps, toral positivity, and conditional global comparison

Overview. We conclude the fourth part of the metaplectic Jacquet–Langlands series with a precise separation of proved local and arithmetic results from two still-conditional goals. We audit the ambient cocycle and scalar-central parameter, recall the all-torus good-element criterion, compute kernels and Haar moduli of nth-power maps on finite etale tori modulo scalars, and classify globally good regular rational classes. Building on Chapter 6's unitary division-side character transform, we prove that wild irreducible-character integrality is equivalent to positive definiteness of every transferred irreducible character. A new toral permutation-character theorem establishes positive integral Fourier coefficients of the discriminant ratio for *every quadratic field extension* over a nonarchimedean field of odd residue characteristic, for all odd n including $p \mid n$. The complete numerical $r = 2$, $n = 3$ example and the nonabelian gluing problem are developed separately in a standalone companion chapter (Chapter 8). We compare our normalizations with Zou's correction of Flicker–Kazhdan, state explicit conditions for split-place matching, and prove a conditional geometric and trace-distribution comparison with normalization factor Λ. No full wild-place irreducible correspondence, construction of exceptional local matching functions, or unconditional ordinary-to-covered global comparison is claimed.

7.1 Introduction and mathematical status

The problem concerns three different local representation theories. Write $H_v = D_v^\times$, and let $\overline{G_{D_v}}$ denote its specified n-fold central cover; the cover $\overline{G}_v$ of $\mathrm{GL}_r(K_v)$ is defined inside the same ambient cover. An additional *ordinary* group H_v enters through the transform $L_{D,v}$ of Chapter 6. These are not interchangeable. In particular, a comparison between $\overline{G_D}$ and $\overline{G}$ has already been obtained in Chapters 4–5 under their stated test-function and spectral hypotheses. We cite Chapter 6 throughout. The new proposed comparison between H and $\overline{G_D}$ is a different assertion. This concluding version records the separate rank-two wild case in a companion manuscript Chapter 8.

Global assumptions. Let K be a totally imaginary number field with $\mu_n \subset K$, $n \geq 1$ odd. Let $r \geq 2$ be prime, $(r,n) = 1$, let L/K be cyclic of degree r, and let D be a central division K-algebra

of degree r containing L. Put $S = \{v : D_v \text{ is division}\}$. Fix integers r', c such that

$$rr' \equiv 1 \pmod{n}, \qquad r - 1 + 2c \equiv 0 \pmod{n}, \qquad c_v \equiv r'c \pmod{n}. \tag{7.1}$$

Thus $r - 1 + 2rc_v \equiv 0 \pmod{n}$. In fact c_v is independent of v with these choices; the subscript recalls Chapter 4's notation. Locally $F = K_v$ is a nonarchimedean field containing μ_n; p denotes its residue characteristic. We call v *wild* if $p \mid n$.

Status convention. **(P)** denotes a proof included here (possibly invoking **(E)** inputs established in cited publications or earlier parts); **(O)** denotes a proof outlined but not fully established, and **(C)** denotes a conjectural goal requiring new mathematical input. When a proved proposition has explicit unresolved hypotheses, we mark its formal implication (P) and mark the outstanding hypotheses (C) separately, in agreement with the status usage of Chapter 6.

Assertion	Precise basis	Status
Diagonal and nonsplit good-element criterion, including wild p	[FK86, §3]; §4.3	(E)
Compatibility of the Chapter 4 cocycle with σ_{c_v}	Proposition 4.2.4(b); direct calculation	(P)
Power-map kernel, index, and Haar modulus on $T_E(F)/F^\times$	Section 7.4	(P)
Global good-regular class criterion	Section 7.5; Chapter 6 Hasse principle	(P)
Division-side unitary character transform and tame integral case	Theorems 6.4.4 and 6.7.6	(E)
Quadratic toral permutation-character theorem (all odd n)	Theorem 7.6.2; detailed example in companion chapter (Ch. 8)	(P)
Wild irreducible-character integrality for $\widetilde{D^\times}$	Section 7.6	(C)
Covered D–covered GL global simple trace comparison	Chapters 4–5 under their hypotheses	(E)
Covered D–ordinary D global comparison	Conditional implication in §7.8; full matching open	(P)/(C)
Removal of auxiliary support and packet hypotheses	Section 7.9	(C)

The running example

The series' running example is $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, which is ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$ (Example 1.3.1, and [JC26b]). Its wild place $\mathfrak{p}_3$ has $K_v = \mathbb{Q}_3(\zeta_3)$, and there D is the quaternion division algebra treated in Remark 7.6.3 and in the companion chapter (Chapter 8). Theorem 7.6.2 gives the decompositions on all three quadratic tori at $\mathfrak{p}_3$. Theorem 7.5.2 describes the globally good classes, and Example 7.8.9 is the global normalization ledger for this D, in which $\Lambda = 1$ forces $\lambda_\infty = 1/3$. The smallest open case of wild integrality is whether $L_D\mathbf{1}$ at $\mathfrak{p}_3$ is irreducible of dimension 27.

7.2 The exact cocycle convention and star map

This section fixes the one place where a parameter mismatch could invalidate every subsequent constant. The ambient construction is a restriction of a cocycle over L_v, whereas [FK86] uses the

standard $\mathrm{GL}_r(F)$ Kazhdan–Patterson convention.

7.2.1 Restriction and relabeling

This subsection recalls the cocycle of the GL side before and after the relabeling of μ_n, and checks the resulting commutator on the diagonal torus. The two parameters c and c_v are easy to confuse, and every later constant depends on using the right one.

Let σ_F be the parameter-zero Kazhdan–Patterson cocycle with restriction to the diagonal torus

$$\sigma_F(\mathrm{diag}(a_i),\mathrm{diag}(b_i)) = \prod_{i<j}(a_i,b_j)_{n,F}. \tag{7.2}$$

The GL subgroup of the ambient cover in Chapter 4 has, *as an extension*, cocycle model

$$\sigma_F(g,h)^r\,(\det g,\det h)_{n,F}^{c} \quad \text{before relabeling } \mu_n. \tag{7.3}$$

This assertion is Proposition 4.2.4(b). Applying the kernel relabeling $\zeta \mapsto \zeta^{r'}$ yields the model

$$\sigma_{c_v}(g,h) = \sigma_F(g,h)(\det g,\det h)_{n,F}^{c_v}. \tag{7.4}$$

The two parameters c and c_v must be kept separate: c occurs before relabeling and $c_v = r'c \pmod n$ afterward. The isomorphism in (7.3) is an isomorphism of extensions, not a claim that arbitrary choices of cocycle sections give literally equal functions on all of $\mathrm{GL}_r(F)^2$.

Proposition 7.2.1 (Diagonal commutator audit)**.** *For $t=\mathrm{diag}(t_1,\dots,t_r)$ and $u=\mathrm{diag}(u_1,\dots,u_r)$ in $\mathrm{GL}_r(F)$,*

$$[\widetilde{t},\widetilde{u}] = \prod_{i,j}(t_i,u_j)_{n,F}^{A_{ij}}, \qquad A = -I_r + (1+2c_v)J_r. \tag{7.5}$$

Here J_r is the all-ones matrix. Its row sums are $r-1+2rc_v \equiv 0 \pmod n$, so scalar lifts commute with the lifted split torus. The radical on $(F^\times/F^{\times n})^r$ consists precisely of diagonal classes.

Proof. The commutator is the ratio $\sigma_{c_v}(t,u)/\sigma_{c_v}(u,t)$. The Hilbert symbol satisfies $(a,b)_n = (b,a)_n^{-1}$; this is *antisymmetry*, valid for all n. The determinant twist contributes $2c_v$ to each entry of A, and the parameter-zero factor contributes 1 off the diagonal. This gives (7.5). The Hilbert pairing on $X = F^\times/F^{\times n}$ is perfect. As X is killed by n and $(r,n)=1$, the decomposition $X^r = \Delta X \oplus \{(x_i) : \sum x_i = 0\}$ holds by averaging with r^{-1} modulo n. On these summands A acts, respectively, by $r-1+2rc_v = 0$ and -1. Therefore $\ker A = \Delta X$. □

Remark 7.2.2. The assumption $(r,n)=1$ also follows algebraically from $n \mid r-1+2rc_v$, because any common divisor of r and n must divide 1. We retain it because it is a global hypothesis of the series. The oddness of n is used for the ordinary star-map convention below, not for Hilbert-symbol antisymmetry.

7.2.2 Star and central characters

This subsection recalls the star map $h \mapsto h^*$, its behavior on scalars and on tori, and its relation to the conventions of [FK86] and [Zo23]. It then fixes how genuine and ordinary central characters correspond.

For an element h of the ordinary group, choose any lift $\widetilde{h}$ and define $h^* = \widetilde{h}^n$. It is independent of the lift because μ_n is central. In the present scalar-central cover, for $a \in F^\times$,

$$(ah)^* = a^* h^*, \qquad a^* = \widetilde{a}^n. \tag{7.6}$$

If h_1, h_2 commute in an ordinary torus, their lifted commutator is in μ_n, and the class-two power identity

$$(\widetilde{h}_1\widetilde{h}_2)^n = \widetilde{h}_1^n \widetilde{h}_2^n [\widetilde{h}_2, \widetilde{h}_1]^{n(n-1)/2} = \widetilde{h}_1^n \widetilde{h}_2^n \tag{7.7}$$

uses n odd. Thus star is multiplicative on the lifted torus. This agrees with the convention in [FK86, §4] and [Zo23, §3]: Zou writes $x^* = u(x)s(x)^n$, where $u(x) = \pm 1$ is the correction defined in [FK86, §4] and [Zo23, §3.1]. For odd n the extra sign is absent in the Flicker–Kazhdan convention: the star is the nth power of a lift. The correction $u(x)$ is introduced for the even-n convention; we do not infer that an arbitrary sign automatically lies in μ_n. In the odd-degree setting, therefore, $u(x) = 1$ by definition of the relevant convention (compare [FK86, §4] and [Zo23, §3.1]). For a scalar $\zeta \in \mu_n(F)$, $\zeta^* = 1$ in the standard diagonal cocycle model: the self-symbol $(\zeta, \zeta)_n = 1$ for odd n, so $s(\zeta I_r)^n = s(I_r) = 1$. This last statement is also invariant under the extension isomorphisms in Chapter 4. For representation characters we first fix a genuine unitary character $\widetilde{\theta}$ of the central preimage $\widetilde{Z}$ and *then* put $\theta(a) = \widetilde{\theta}(a^*)$. The reverse implication is not unique: θ alone does not specify $\widetilde{\theta}$. For $\zeta \in \mu_n(F)$ we have $\zeta^* = 1$, hence $\theta(\zeta) = 1$. Globally the rational splitting gives $a^* = \iota(a^n)$ for $a \in K^\times$; if $\widetilde{\theta}$ is automorphic, it is trivial on the rational splitting, so $\theta(a) = 1$ on $K^\times$. Test functions use the inverse central characters θ^{-1} and $\widetilde{\theta}^{-1}$.

7.3 Good elements for every maximal torus

A regular maximal torus in $\mathrm{GL}_r(F)$ may be nonsplit even at a place where $D_v \simeq M_r(F)$. This is the situation not treated by the diagonal matrix alone. The following calculation is not new: it is the proof of [FK86, §3, Proposition] in the notation of this series, and also underlies Proposition 4.3.5.

Theorem 7.3.1 (**(E)** Flicker–Kazhdan good-element theorem)**.** *Let $E = \prod_{i=1}^k E_i$ be a finite* étale *F-algebra of dimension r, embedded as a regular maximal torus $T_E(F) = E^\times$ in $\mathrm{GL}_r(F)$. Under (7.1), for any $t \in T_E(F)$ and any lift $\widetilde{t}$,*

$$[\widetilde{t}, \widetilde{u}] = (\mathrm{N}_{E/F}\, t, \mathrm{N}_{E/F}\, u)_{n,F}^{1+2c_v} \prod_{i=1}^{k} (t_i, u_i)_{n,E_i}^{-1}. \tag{7.8}$$

The radical of this commutator pairing is

$$\{t \in T_E(F) : [\widetilde{t}, \widetilde{u}] = 1 \ \forall u \in T_E(F)\} = F^\times T_E(F)^n. \tag{7.9}$$

For regular t, this is exactly the criterion that $\widetilde{t}$ is good. Both identities are valid when $p \mid n$.

Proof from the cited formula, included for clarity. The commutator formula is [FK86, §3], using [KP84, Prop. 0.1.5]; its restriction to a single field is exactly Proposition 4.3.1 after relabeling Chapter 4. Write $N = \mathrm{N}_{E/F}$. Norm compatibility of local Hilbert symbols gives

$$(Nt, Nu)_{n,F} = \prod_i (Nt, u_i)_{n,E_i}.$$

Thus (7.8) equals

$$\prod_i \left((Nt)^{1+2c_v} t_i^{-1}, u_i \right)_{n,E_i}.$$

The nondegeneracy of the nth Hilbert symbol over each E_i (including wildly ramified E_i) implies that the commutator is trivial for all u exactly when

$$t_i \equiv (Nt)^{1+2c_v} \pmod{E_i^{\times n}} \quad (1 \le i \le k). \tag{7.10}$$

This directly says $t \in F^\times E^{\times n}$. Conversely, if $t = ay^n$ for $a \in F^\times$, then $Nt = a^r (Ny)^n$; therefore $(Nt)^{1+2c_v} \equiv a^{r(1+2c_v)} \equiv a \equiv t_i$ modulo $E_i^{\times n}$ by scalar centrality. This proves the criterion. No tameness assumption has been introduced. □

Example 7.3.2 (The diagonal rank-two check)**.** Take $r = 2$, $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$. In the notation of Example 4.3.7, choose $c = 1$, so $c_v \equiv 2 \pmod 3$. Formula (7.5) becomes

$$A \equiv \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3, \qquad [\widetilde{t}, \widetilde{u}] = (t_1/t_2, u_1/u_2)_{3,F}.$$

Thus $\mathrm{diag}(t_1, t_2)$, when regular, is good exactly when $t_1/t_2 \in F^{\times 3}$. If a quadratic field E/F is embedded as a nonsplit maximal torus, the same theorem says that goodness is equivalent to $t \in F^\times E^{\times 3}$; the two tests are instances of the same formula, not separate conjectures.

Remark 7.3.3 (Why goodness is not enough for a regular-root formula)**.** The power of a regular element in a *split* torus need not be regular. For $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, the element $\mathrm{diag}(1, \zeta_3)$ is regular but not good: $\zeta_3 \notin F^{\times 3}$, since F does not contain ζ_9. Its cube is the identity. A character or orbital-integral formula with $\Delta(t^n)$ in the denominator must restrict to t^n regular (as in Zou's equation [Zo23, (3.2)]), or use a separately proved germ extension. On a degree-r *field* torus, by contrast, if t is regular and $t^n \in F$, then Kummer theory gives $[F(t) : F] \mid n$; since $[F(t) : F] = r$ and $(r, n) = 1$, this is impossible. Because r is prime, $F(t^n) = F(t)$ and t^n is regular.

7.4 Local power-map calculation for étale tori

The key local distinction is between a field centralizer (compact modulo scalars) and a product of fields (not compact modulo scalars). Both occur in the split *group*. The next proposition gives an exact count of the roots and the size of the image at wild and tame places, including the factors that a global trace identity must account for.

Lemma 7.4.1 (Kummer injection for a product algebra)**.** *Let $E = \prod_{i=1}^k E_i$ be finite* étale *of dimension r over F, with $\mu_n \subset F$ and $(r, n) = 1$. Then*

$$F^\times / F^{\times n} \longrightarrow E^\times / E^{\times n} \tag{7.11}$$

is injective.

Proof. Suppose $a \in F^\times$ is an nth power in every E_i. The Kummer extension $F(a^{1/n})/F$, since $\mu_n \subset F$, is cyclic of degree $d \mid n$ and embeds in every E_i. Therefore $d \mid [E_i : F]$ for every i, so also $d \mid \sum_i [E_i : F] = r$. Coprimality gives $d = 1$ and $a \in F^{\times n}$. □

Proposition 7.4.2 (Kernel, index, and Haar modulus)**.** *Under the hypotheses of Lemma 7.4.1, put* $A_E = E^\times/F^\times$ *and* $P_n([t]) = [t^n]$. *Then*

$$\ker(P_n) \simeq \mu_n(E)/\mu_n(F) \simeq \mu_n^k/\mu_n, \qquad |\ker(P_n)| = n^{k-1}, \tag{7.12}$$

$$[A_E : P_n(A_E)] = n^{2(k-1)}|n|_F^{-(r-1)}. \tag{7.13}$$

In particular, $P_n(A_E)$ *is open and closed. With* any *Haar measure* da *on* A_E *and its restriction to* $P_n(A_E)$*, for compactly supported integrable* q *on that image,*

$$\int_{A_E} q(a^n)\,da = n^{k-1}|n|_F^{-(r-1)} \int_{P_n(A_E)} q(b)\,db. \tag{7.14}$$

Proof. If $t^n = a \in F^\times$, then a is an nth power in every E_i; by Lemma 7.4.1, write $a = b^n$ with $b \in F^\times$. Thus each $t_i/b \in \mu_n(E_i) = \mu_n(F)$, giving (7.12), including its canonical description.

For a local field M containing μ_n the Kummer quotient has

$$|M^\times/M^{\times n}| = n^2|n|_M^{-1}. \tag{7.15}$$

Indeed the discrete valuation contributes n; the finite torsion unit factor containing μ_n contributes n; and the torsion-free p-adic unit factor contributes $|n|_M^{-1}$ (multiplication by n on its $\mathbb{Z}_p$-lattice). Alternatively, (7.15) follows from local duality. Since $F^\times/F^{\times n}$ injects into $E^\times/E^{\times n}$, the group A_E/A_E^n is its cokernel:

$$|A_E/A_E^n| = \frac{\prod_i n^2|n|_{E_i}^{-1}}{n^2|n|_F^{-1}} = n^{2(k-1)}|n|_F^{-(r-1)},$$

where $|n|_{E_i} = |n|_F^{[E_i:F]}$ for normalized absolute values on local fields. This proves (7.13).

For the Haar statement, write $K_0 = \ker(P_n)$, of order n^{k-1}. Choose a sufficiently small compact open subgroup $U \subset A_E$ consisting of principal-unit classes and having $U \cap K_0 = \{1\}$. The p-adic logarithm identifies U with a free $\mathbb{Z}_p$-module of rank $(r-1)[F : \mathbb{Q}_p]$: the factors $E_i^\times$ have combined principal-unit rank $r[F : \mathbb{Q}_p]$, and quotienting by the scalar principal units subtracts $[F : \mathbb{Q}_p]$. Multiplication by n maps U injectively into itself, with

$$[U : U^n] = p^{v_p(n)(r-1)[F:\mathbb{Q}_p]} = |n|_F^{-(r-1)}.$$

Since $P_n^{-1}(U^n) = K_0U$ and $K_0 \cap U = \{1\}$, its pushforward of Haar measure gives

$$\frac{\mathrm{vol}(P_n^{-1}(U^n))}{\mathrm{vol}(U^n)} = \frac{|K_0|\,\mathrm{vol}(U)}{\mathrm{vol}(U)/[U : U^n]} = n^{k-1}|n|_F^{-(r-1)}.$$

The pushforward is translation-invariant on the open image $P_n(A_E)$, so this ratio is its constant density on that image, proving (7.14). Notice that this argument requires neither a global quotient-index formula nor a choice of scale for Haar measure. □

Remark 7.4.3 (A measure-theoretic warning)**.** For a general open endomorphism $\varphi : A \to A$ of a locally compact abelian group with finite kernel, its pushforward Haar density need not equal $[A : \varphi A]/|\ker\varphi|$. For example, on $A = \mathbb{Z}$ with counting measure, $a \mapsto na$ has density 1 on $n\mathbb{Z}$, not n. The correct local computation here is $|\ker P_n|[U : U^n]$, where U is the compact open subgroup in the proof. For this specific A_E it happens to simplify to $[A_E : P_n A_E]/|\ker P_n|$; that numerical equality is not a general change-of-variables theorem.

Example 7.4.4 (Three local torus types)**.** If E/F is a field of degree r, then $k = 1$ and P_n is injective, with index and Haar modulus $|n|_F^{-(r-1)}$. If $E = F^r$ then $k = r$ and the kernel has order n^{r-1}. For $r = 2, n = 3$, $F = \mathbb{Q}_3(\zeta_3)$ and $|3|_F = 1/9$,

type	$\lvert\ker P_3\rvert$	$[A_E : A_E^3]$	factor in (7.14)
E/F quadratic	1	9	9
$E = F \times F$	3	81	27

These integers quantify why one cannot use the division-torus substitution formula unchanged on the full split group.

7.5 A global classification of good regular classes

The local theorem now applies to the actual torus at *every* place. Combining it with the Hasse principle of Chapter 6 removes the local-to-global ambiguity for power classes of a degree-r field, though not the independent measure and spectral questions.

Proposition 7.5.1 (**(E)** Almost-everywhere powers modulo scalars)**.** *Let E/K be a field extension of degree r with $(r,n) = 1$ and $\mu_n \subset K$. For every finite set Σ of places,*

$$E^\times/(K^\times E^{\times n}) \longrightarrow \prod_{v\notin\Sigma} \frac{(E\otimes_K K_v)^\times}{K_v^\times (E\otimes_K K_v)^{\times n}} \tag{7.16}$$

is injective. This is Proposition 6.11.3 Chapter 6.

Recalled proof. Let t be locally scalar times an nth power outside Σ. Choose $u, k \in \mathbb{Z}$ with $ru + nk = 1$ and put $a = N_{E/K}(t)^u \in K^\times$. If $t = a_v y_v^n$ in the local étale algebra, taking norms gives the identity

$$t/a = \left(a_v^k y_v / N_{E_v/K_v}(y_v)^u\right)^n.$$

Thus t/a is an nth power in every completion of E over all $v \notin \Sigma$. If its class in $E^\times/E^{\times n}$ were nontrivial, the nontrivial Kummer extension $E((t/a)^{1/n})/E$ would, by Chebotarev, admit infinitely many unramified places at which t/a is not an nth power. This contradicts the hypothesis. Hence $t \in K^\times E^{\times n}$. □

Theorem 7.5.2 (Global good-regular criterion)**.** *Assume the covers and rational splittings of Chapter 4. Let G denote $D^\times$ or GL_r, and let $\gamma \in G(K)$ be regular elliptic, so $E = K[\gamma]$ is a field of degree r. The following are equivalent:*

(a) the rational lift $\iota(\gamma)$ is good at every place;

(b) it is good at every place outside some finite set Σ;

(c) $\gamma \in K^\times E^{\times n}$.

The map $[s] \mapsto [s^n]$ is a bijection from the noncentral classes in $E^\times/K^\times$ onto the globally good regular classes belonging to E. Furthermore

$$W_{s^n} = W_s, \qquad W_s = \{\sigma \in \mathrm{Aut}(E/K) : \sigma(s) \in K^\times s\}. \tag{7.17}$$

Proof. At each finite v, the ordinary centralizer is $T_v = (E \otimes_K K_v)^\times$. If D_v splits, apply Theorem 7.3.1, transported through Proposition 4.2.7 (the split-place isomorphism of the specified covers). If D_v is division, use Proposition 4.3.5(a), whose commutator is the same by Proposition 4.3.1. At complex places the nth-power map on T_v is surjective. Thus at every place the local goodness condition is precisely $\gamma \in K_v^\times T_v^n$. Implications (c)⇒(a)⇒(b) follow, while (b)⇒(c) is Proposition 7.5.1.

The power map $E^\times/K^\times \to E^\times/K^\times$ is injective: if $s^n \in K^\times$, Kummer theory shows $[K(s):K] \mid n$, while $[K(s):K] \mid [E:K] = r$. Hence $s \in K^\times$. Because r is prime, every noncentral $s \in E$ generates E, and the same is true of s^n. This proves the bijection. Finally $\sigma(s^n)/s^n \in K^\times$ if and only if $(\sigma(s)/s)^n \in K^\times$, which is equivalent by injectivity to $\sigma(s)/s \in K^\times$. Equation (7.17) follows. □

Remark 7.5.3 (Precise consequence for the geometric side)**.** Proposition 4.6.1, uses the rational regular classes which are good everywhere, with coefficient

$$c(\gamma) = \frac{\mathrm{vol}\big(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt\big)}{|W_\gamma|}. \tag{7.18}$$

Theorem 7.5.2 parametrizes its noncentral summation set by global $[s^n]$ and preserves W_γ. It does *not* replace the local root sums in Zou's character formula: the global root class may be unique even when a localized split torus has n^{k_v-1} roots modulo scalars.

7.6 Wild local integrality: exact reductions and a toral theorem

The division-side transform of Chapter 6 is an isometric isomorphism of spaces of invariant class functions. Wild-place irreducible-character integrality does not follow from this analytic statement. We record here the exact character-theoretic criterion and a positive result on quadratic field tori. Detailed ramified and unramified computations for $F = \mathbb{Q}_3(\zeta_3)$ belong to the separate companion chapter, Chapter 8.

7.6.1 Normalization and the unitary character matrix

This subsection recalls the transform L_D of Chapter 6 with its constants. It then states the criterion that links integrality of the character matrix to positivity.

Let F be a nonarchimedean field containing μ_n, let D/F be a division algebra of prime degree r with $(r,n) = 1$, and put $H = D^\times$ and $Z = F^\times$. Write $\widetilde{H}$ for the scalar-central cover specified in Chapter 4. For a regular element t in a maximal field $E \subset D$, put

$$\Delta(t) = |\det(1 - \mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t})|_F^{1/2}, \qquad R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}. \tag{7.19}$$

The character transform of Chapter 6, with compatible central characters $\theta, \widetilde{\theta}$, is

$$(L_D\Phi)(zt^*) = b_D\widetilde{\theta}(z)R_E(t)\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}; \tag{7.20}$$

it vanishes on nongood regular lifts. Its continuously extended identity value is

$$(L_D\Theta_\rho)(1) = c_D \dim\rho, \qquad c_D = |n|_F^{-(r^2-1)/2}. \tag{7.21}$$

Theorem 6.4.4 and Corollary 6.5.1 prove that L_D is a *unitary onto map* between the fixed-central-character invariant L^2 spaces and admit finite expansions

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle. \tag{7.22}$$

We do not assume that $a_{\rho\tau}$ is integral at wild places.

Proposition 7.6.1 (Integrality–positivity criterion)**.** *For the entire unitary matrix $A = (a_{\rho\tau})$, the following conditions are equivalent:*

(i) *L_D bijects the irreducible-character bases with positive signs;*

(ii) *$a_{\rho\tau} \in \mathbb{Z}$ for all ρ, τ;*

(iii) *$a_{\rho\tau} \geq 0$ is real for all ρ, τ;*

(iv) *every $L_D\Theta_\rho$ is a continuous positive-definite function on $\widetilde{H}$.*

Positive definiteness for one *row is not sufficient for integrality of that row.*

Proof. A row with integral entries and squared norm one has exactly one nonzero entry, equal to ± 1. Equation (7.21) rules out the minus sign, and unitarity and surjectivity give the bijection. Conversely, permutation of the character bases gives integrality. For nonnegative entries, orthogonality of distinct rows implies that their supports are disjoint. Every column has squared norm one; it meets at most one row, and its sole nonzero entry therefore equals one. Each row has norm one, so it also contains exactly one entry equal to one. Finally, Peter–Weyl orthogonality for the compact-mod-center group with fixed unitary central character identifies positive-definite invariant functions with those having nonnegative coefficients in the irreducible-character basis. A single nonnegative norm-one row may nevertheless contain several nonintegral entries (for example $1/\sqrt{2}, 1/\sqrt{2}$). □

7.6.2 The toral Fourier reduction

This subsection restricts the question to a maximal field torus, where it becomes a question about the Fourier coefficients of the discriminant ratio R_E. It proves that these coefficients are nonnegative integers when E/F is quadratic.

For a maximal field E/F, $A_E = E^\times/F^\times$ is compact abelian. The continuous discriminant ratio is locally constant and hence has a finite Fourier expansion

$$R_E(t) = \sum_{\chi\in\widehat{A_E}} m_\chi\chi(t). \tag{7.23}$$

A genuine central character on $\widetilde{E^\times}$ compatible with an ordinary character α identifies the coefficient m_χ with the multiplicity of the corresponding genuine Stone–von Neumann torus representation in the *restricted* transfer $\mathscr{L}_E\alpha$ (Proposition 4.3.5, and Chapter 6). This is a statement about the full preimage of the field torus; it is not an assertion that the torus modules extend to $\widetilde{H}$. Consequently,

$$m_\chi \in \mathbb{Z}_{\geq 0} \quad \text{for all } \chi \tag{7.24}$$

is sufficient for positive integral *torus restrictions* of all transferred ordinary characters. It does not imply any $a_{\rho\tau} \in \mathbb{Z}$ for the full nonabelian cover.

Theorem 7.6.2 (Quadratic toral permutation positivity)**.** *Let $F/\mathbb{Q}_p$ be finite with p odd, let n be odd with $\mu_n \subset F$, and let E/F be a quadratic field extension. Then R_E is a permutation character of a finite quotient of $E^\times/F^\times$. In particular the Fourier coefficients in* (7.23) *belong to $\mathbb{Z}_{\geq 0}$, even when $p \mid n$. At the identity, $R_E(1) = |n|_F^{-1}$.*

Proof. Write $n = p^a m$ with $(m,p) = 1$ and set $G = E^1 = \ker N_{E/F}$. Hilbert 90 identifies $E^\times/F^\times$ with G by $t \mapsto u = t/\sigma(t)$. If $a = 0$ the ratio is identically one, so suppose $a > 0$. Set $G_k = \{u \in G : v_E(1-u) \geq k\}$ and $G_0 = G$. A root $\zeta \in \mu_n$ with nontrivial prime-to-p component cannot have $\zeta u \equiv 1$ modulo the maximal ideal: reduction of u has norm one and its intersection with the scalars is $\{\pm 1\}$. For $\zeta \in \mu_{p^a} \setminus \{1\}$, the ultrametric inequality, together with $\sigma(u) = u^{-1}$ and p odd, gives

$$v_E(1-\zeta u) = \min\big(v_E(1-u), v_E(1-\zeta)\big).$$

Indeed, cancellation at equal valuations would give both $u \equiv \zeta^{-1}$ and $u^{-1} \equiv \zeta^{-1}$ beyond the valuation of $1-\zeta$, contradicting $v_E(\zeta - \zeta^{-1}) = v_E(1-\zeta)$. Factoring $1-u^n$ yields

$$R_E(u) = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(v_E(1-u), v_E(1-\zeta))}, \tag{7.25}$$

where q_E is the size of the residue field of E. This formula also defines the continuous central value.

Put $t_\zeta = v_E(1-\zeta)$, $T = \max_{\zeta \neq 1} t_\zeta$ and $M_k = q_E^{\frac{1}{2}\sum_\zeta \min(k, t_\zeta)}$ ($M_0 = 1$). The M_k are integers, since ζ and ζ^{-1} occur in pairs, and $M_k = M_T$ for $k \geq T$. If E/F is unramified and $q = q_F$, then $q_E = q^2$ and

$$[G : G_k] = (q+1)q^{k-1} \quad (k \geq 1).$$

Writing $s_k = \#\{\zeta \neq 1 : t_\zeta \geq k\}$, we have $M_k/M_{k-1} = q^{s_k}$ and s_k is even. The quantities

$$c_k = \frac{M_k - M_{k-1}}{(q+1)q^{k-1}} \quad (1 \leq k \leq T)$$

are nonnegative integers: $q+1 \mid q^{s_k} - 1$, and when $s_k > 0$, the preceding M_{k-1} is divisible by q^{k-1}. Therefore

$$R_E = \mathbf{1}_G + \sum_{k=1}^{T} c_k [G : G_k] \mathbf{1}_{G_k}.$$

Each summand $[G : G_k]\mathbf{1}_{G_k}$ is a permutation character of G/G_k.

If E/F is ramified, then $q_E = q$, the t_ζ are even, and $G_{2j} = G_{2j+1}$, with $[G : G_1] = 2$ and $[G : G_{2j}] = 2q^j$ for $j \geq 1$. Set

$$c_1 = \frac{M_1 - 1}{2}, \qquad c_{2j} = \frac{M_{2j+1} - M_{2j-1}}{2q^j}.$$

The first is a nonnegative integer because M_1 is odd. When the numerator of c_{2j} is nonzero, some pair ζ, ζ^{-1} has $t_\zeta \geq 2j$; hence $q^j \mid M_{2j-1}$. The quotient M_{2j+1}/M_{2j-1} is a power of odd q, so its difference from one is even. Thus every $c_{2j} \in \mathbb{Z}_{\geq 0}$ and

$$R_E = \mathbf{1}_G + c_1[G : G_1]\mathbf{1}_{G_1} + \sum_{1 \leq j \leq T/2} c_{2j}[G : G_{2j}]\mathbf{1}_{G_{2j}}.$$

The norm-one filtration indices follow directly by writing $E = F(\sqrt{d})$, $v_F(d) = 1$, and parameterizing G_1 by $(1 + c\sqrt{d})/(1 - c\sqrt{d})$ with $c \in \mathscr{O}_F$; congruence modulo G_{2j} records c modulo $\mathfrak{p}_F^j$. This completes the two cases. At the center $R_E(1) = |n|_F^{-1}$, by the cyclotomic product. □

Remark 7.6.3 (A quantified rank-two instance)**.** For $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$ and the quaternion D/F, every quadratic maximal subfield E/F satisfies Theorem 7.6.2. The detailed decompositions are

$$R_{E_{\mathrm{ur}}} = 1 + 2\,\mathrm{Reg}(C_4), \qquad R_{E_{\mathrm{ram}}} = 1 + \mathrm{Reg}(C_2) + \mathrm{Reg}(C_6).$$

These give genuine torus restriction multiplicities $(3,2,2,2)$ and $(3,2,1,1,1,1)$, respectively, of degree-three constituents, hence total dimension 27. Both ramified quadratic isomorphism classes yield the same formula. The full derivations and the status of the unproved nonabelian gluing step are recorded in Chapter 8.

7.6.3 What the local reductions prove, and what they do not

Theorem 7.6.2 establishes toral Fourier positivity for *all* quadratic extensions and odd covering degrees, including wild primes, but does not establish positive definiteness on the full cover. The logical implications are

$$\text{positive character on every covered torus} \quad \not\Longrightarrow \quad \text{positive-definite character on } \widetilde{H}.$$

At a fixed finite quotient this is visible even for $C_2 \times C_2$: the function equal to 2 at the identity and -2 at each of the three involutions restricts to twice the sign character on every cyclic subgroup of order two, but equals $-\chi_0 + \chi_1 + \chi_2 + \chi_3$ on the full group, so it is not positive definite. (The function equal to 2 at the identity and 0 elsewhere shows separately that positivity does not force integrality: it restricts to the regular character on each cyclic subgroup, but is half the sum of the four irreducible characters.) The finite-level Brauer/Clifford and induced-type strategies in Chapter 8 address this additional *nonabelian gluing* requirement. For arbitrary prime $r > 2$, even the toral Fourier-positivity step remains open. No theorem here identifies a wild irreducible genuine image or proves the new ordinary-$D^\times$ to covered-$D^\times$ automorphic comparison unconditionally.

7.7 Connection with Zou's corrected character relation

For the Kazhdan–Patterson cover of the *split group*, Zou [Zo23, §3] revisits the Flicker–Kazhdan metaplectic correspondence and records a corrected scalar in the character relation. This is an essential input, but its objects are $\widetilde{\mathrm{GL}_r(F)}$ and ordinary $\mathrm{GL}_r(F)$, not directly $\widetilde{H}$ and H at a division place.

Proposition 7.7.1 ((E) Zou's root-sum constant in our case)**.** *Write* $d_F(m) = m|m|_F^{-1/2}$ *and* $d_r = \gcd(n, 2rc_v + r - 1)$. *Under* (7.1), $d_r = n$. *In Zou's equation [Zo23, (3.2)], the coefficient preceding the root sum in rank r is*

$$\frac{1}{d_F(d_r)|n|_F^{r/2}} = \frac{1}{n|n|_F^{(r-1)/2}} = \frac{b_D}{n}. \tag{7.26}$$

For regular elliptic x with degree-r field centralizer, x^n *is automatically regular by Remark 7.3.3. The equation* $t^n = x^n$ *then has exactly n solutions* $t = \zeta x$, $\zeta \in \mu_n$. *If the linear character* π *has central character* ω_π *trivial on* μ_n, *these summands agree and the effective coefficient is* b_D. *This restriction on* ω_π *is essential: without it, the scalar sum is* $\sum_{\zeta\in\mu_n} \omega_\pi(\zeta) = 0$, *so [Zo23, (3.2)] instead forces the covered character to vanish at* x^*. *For a genuine lift pair the compatibility* $\omega_\pi(a) = \omega_{\widetilde{\pi}}(a^*)$ *and* $\zeta^* = 1$ *imply* $\omega_\pi|_{\mu_n} = 1$ *automatically.*

Proof. The equation $d_r = n$ is immediate from scalar centrality; (7.26) follows by substitution. Regularity of x^n follows from the field case of Remark 7.3.3. Any solution t centralizes x^n, so $t \in F[x]^\times$. In the field $F[x]$, $(t/x)^n = 1$, and $\mu_n(F[x]) = \mu_n(F)$. Thus $t = \zeta x$. The Weyl discriminant is invariant under central scalar multiplication, the ordinary character takes the same value because it is trivial on μ_n, and $(\zeta x)^* = x^*$ by the explicit odd-n scalar star computation in Section 7.2.2. These are precisely the equal terms in Zou's (3.2). □

Remark 7.7.2 (What is and is not composed)**.** At the ramified places one may compare the *desired* $\overline{G_{D_v}} \leftrightarrow \overline{G}_v$ character relation of Chapters 4–5 with (a) the classical linear Jacquet–Langlands elliptic character identity and (b) Zou's formula. The common coefficient b_D and the familiar sign $(-1)^{r-1}$ are consistent with this comparison. Nevertheless *the existence of an irreducible genuine image on the covered division side is the missing conclusion*; combining two known formulas cannot be used to assume that missing representation exists. For non-elliptic split centralizers, retain Zou's entire root sum, including the central-character factors.

7.8 Global geometric comparison: proved input and new obligations

There are two comparisons to keep distinct:

$$\begin{aligned} \overline{G_D}(\mathbb{A}_K) &\longleftrightarrow \overline{G}(\mathbb{A}_K) \quad \text{(Chapters 4–5, under their hypotheses)}, \\ D^\times(\mathbb{A}_K) &\dashrightarrow \overline{G_D}(\mathbb{A}_K) \quad \text{(proposed additional comparison)}. \end{aligned} \tag{7.27}$$

The first has a proved simple trace-formula comparison with regular-support or later matching hypotheses (Theorem 4.6.7; Theorem 5.6.3). The second is not a formal consequence of the first or of the unitarity of L_D.

7.8.1 Geometric coefficients and the product formula

This subsection records how the global torus coefficients and the local discriminant factors enter the geometric side. It explains why the local constants $b_{D,v}$ cannot be cancelled by the product formula alone.

For a regular elliptic rational γ, with $E = K[\gamma]$, Chapter 4 identifies the torus coefficient $c(\gamma)$, recalled in (7.18). The same torus measure dt must be used whenever this term is compared with the corresponding linear inner-form orbital integral. Chapter 4 uses the normalized orbital integral $J_v(\gamma, f) = \Delta_v(\gamma) O_v(\gamma, f)$. For a rational regular element the product formula implies

$$\prod_v \Delta_v(\gamma) = 1. \tag{7.28}$$

This is because each discriminant is the normalized absolute value of a nonzero element of K (or its square root), including the determinant normalization in (7.19). The local *ratio* of discriminants $\Delta_v(t)/\Delta_v(t^n)$ must be kept until local transfer and the global product are both defined.

At $v \in S$ Theorem 6.9.1, gives the exact local normalized orbital-integral identity

$$J_{\widetilde{H}_v}\left(zt^*, \mathscr{O}_{D,v} f_v\right) = b_{D,v}\, \widetilde{\theta}_v(z)^{-1} J_{H_v}(t, f_v), \tag{7.29}$$

using the same quotient measures on the two copies of H_v/Z_v. At $v \notin S$, $H_v \cong \mathrm{GL}_r(K_v)$ and the appropriate root-sum comparison is not (7.29); it must use Flicker–Kazhdan–Zou, and it can involve the kernel $\mu_n^{k_v}/\mu_n$ of Proposition 7.4.2. In particular the local factors $b_{D,v}$ for $v \in S$ *cannot* simply be cancelled using $\prod_v |n|_v = 1$ without the split-side contributions.

7.8.2 Haar ledger

This subsection lists, in one table, the measure normalizations and the local statements that a global geometric comparison needs, with the status of each.

For $v \in S$, Definition 5.2.1, fixes

$$\mathrm{vol}(D_v^\times / F_v^\times; dg_{D,v}) = d(\mathrm{St}; dg_{\mathrm{GL},v})^{-1}, \tag{7.30}$$

where $d(\mathrm{St}; dg_{\mathrm{GL},v})$ is the formal degree relative to the chosen $\mathrm{GL}_r(F_v)/F_v^\times$ measure. The probability measure used in Chapter 6 is $d(\mathrm{St}; dg_{\mathrm{GL},v})\, dg_{D,v}$. When *both* sides of (7.29) are scaled by the same measure, $b_{D,v}$ is unchanged. Different measures on the GL side must still be reconciled with (7.30).

Item	Required statement or calculation	Status
Ambient cocycle and common center	Lemma 4.5.1 and Proposition 4.5.6; restriction (7.3)	(E)
Regular rational good locus	Theorem 7.5.2	(P)
Global rational/stable classes	Hilbert 90 and Kottwitz's framework; local embeddability into D	(E)
Global torus coefficient	Same dt in (7.18), including $\lvert W_\gamma \rvert$	(E)
Local $v \in S$ orbital transfer	Theorem 6.9.1, equation (7.29)	(E)
Split-place $v \notin S$ invariance	Lemma 7.8.3 (necessary); rank-two spherical calculation below	(P)
Split-place pointwise transfer and λ_v	Proposition 7.8.6; unproved existence	(C)

Item	Required statement or calculation	Status
Unramified unit elements	[FK86, §§11–12] under $(n,N_r)=1$; transport to Chapter 4 cover pending	(E)/(O)
Nonregular powers and singular germs	Remark 7.3.3; germ compatibility or support needed	(C)
Central identity class	Suppressed in Prop. 7.8.6; separate germ/measure test below	(E)/(C)
Spectral expansion	Chapter 4–II packet formula with restrictions; new factor matching	(E)/(C)

Here N_r is the least common multiple of the *composite* positive integers at most r (the convention in [FK86]), with $N_r = 1$ when there are none. The classical unramified unit matching in [FK86, §12, Corollary] is established under $(n,N_r)=1$ and $p \nmid n$. Thus $N_2 = N_3 = 1$, and for prime $r = 5$ we have $N_5 = 4$, automatically prime to the odd integer n. Other ranks must retain the published hypothesis.

7.8.3 Split-place invariance and a geometric comparison criterion

At a place $v \notin S$ the algebra D_v is split, but a rational elliptic field $K[s]$ may decompose locally as a product of fields. By Proposition 7.4.2, its local nth-root fibre modulo scalars can contain n^{k_v-1} points. The global class $[s]$ supplied by Theorem 7.5.2 is nevertheless unique. Therefore a local sum over *all* roots cannot be substituted for the orbital integral of this one rational representative without an additional compatibility statement.

Definition 7.8.1 (FK-compatible ordinary functions)**.** Fix the local central characters $\theta_v, \widetilde{\theta}_v$ with $\theta_v(a) = \widetilde{\theta}_v(a^*)$. A normalized ordinary test function f_v is called *FK-compatible on the regular-power locus* if, whenever $x, x' \in H_v$ have x^n, x'^n regular and their star images satisfy $(x')^* = zx^*$ for a central lift $z \in \widetilde{Z}_v$ (after conjugating representatives), one has

$$J_{H_v}(x', f_v) = \widetilde{\theta}_v(z)^{-1} J_{H_v}(x, f_v). \tag{7.31}$$

The case $x' = \xi x$ with *scalar* $\xi \in \mu_n(F)$ is automatic, since $\theta_v(\xi) = 1$. The substantive condition occurs on a product torus $E_v = \prod_{i=1}^{k_v} E_i$ with $k_v \geq 2$: if $\zeta = (\zeta_i) \in \mu_n(E_v) \simeq \mu_n(F)^{k_v}$ is nonscalar and x^n is regular, then

$$J_{H_v}(\zeta x, f_v) = J_{H_v}(x, f_v). \tag{7.32}$$

The odd-n star is multiplicative on each lifted torus, and $\zeta^* = 1$, so $(\zeta x)^* = x^*$. The kernel on the quotient torus is $\mu_n(F)^{k_v}/\mu_n(F)$, of size n^{k_v-1} (Proposition 7.4.2). Equation (7.31) is the invariant-function condition of [FK86, §6], expressed with the present inverse central transformation convention.

Example 7.8.2 (The condition in rank two)**.** Let $r = 2$ and $n = 3$, and let $v \notin S$, so that $H_v \cong \mathrm{GL}_2(F)$ with $F = K_v$. A regular element lies in a torus $E_v^\times$, where E_v is either a quadratic field or $F \times F$. If E_v is a field, then $k_v = 1$ and every cube root of unity in E_v is a scalar, so (7.32) imposes nothing. If

$E_v = F \times F$, then $k_v = 2$ and the torus is conjugate to the diagonal torus. Take $x = \mathrm{diag}(a,b)$ with $a^3 \neq b^3$, so that x^3 is regular, and take $\zeta = (1,\zeta_3)$. Then (7.32) reads

$$J_{H_v}\big(\mathrm{diag}(a,\zeta_3 b), f_v\big) = J_{H_v}\big(\mathrm{diag}(a,b), f_v\big).$$

The quotient $\mu_3(F)^2/\mu_3(F)$ has 3 elements. So the condition says that the normalized orbital integrals of f_v take the same value at $\mathrm{diag}(a,b)$, $\mathrm{diag}(a,\zeta_3 b)$ and $\mathrm{diag}(a,\zeta_3^2 b)$.

Lemma 7.8.3 (Necessary split-place invariance)**.** *Suppose a genuine $\widetilde{f}_v$ and a fixed nonzero constant λ_v satisfy*

$$J_{\widetilde{H}_v}(x^*, \widetilde{f}_v) = \lambda_v J_{H_v}(x, f_v) \tag{7.33}$$

for all x in a conjugation- and scalar-stable region where x^n is regular, with the identity at central translates understood with the prescribed inverse-central-character law. Then f_v obeys (7.31) *on that region.*

Proof. If $(x')^* = zx^*$, conjugation invariance and the inverse central transformation law of the genuine test function imply $J_{\widetilde{H}_v}((x')^*, \widetilde{f}_v) = \widetilde{\theta}_v(z)^{-1} J_{\widetilde{H}_v}(x^*, \widetilde{f}_v)$. Apply (7.33) twice and divide by the fixed nonzero λ_v. In particular, for $x' = \zeta x$ with nonscalar $\zeta \in \mu_n(E_v)$, one has $(x')^* = x^*$ and obtains (7.32). The scalar-root case is already automatic from the central-character convention. Thus FK-compatibility is a *necessary* condition, not an asserted construction of $\widetilde{f}_v$. □

Proposition 7.8.4 (Rank-two tame spherical-unit check)**.** *Let F be nonarchimedean with $\mu_n \subset F$, $p \nmid n$, $r = 2$, and $K_0 = \mathrm{GL}_2(\mathcal{O}_F)$. Take a spherical ordinary unit f^0, supported on $F^\times K_0$, with an unramified central character, and the standard quotient measures. Then f^0 obeys* (7.32) *for all regular split-torus x with x^n regular. For a quadratic field torus $E^\times$, the root-fibre condition modulo scalars is vacuous. Moreover, when f^0 and the genuine spherical unit $\widetilde{f}^0$ are transported from the standard Flicker–Kazhdan cocycle and measure conventions, [FK86, §12, Corollary] establishes their regular-power matching with $\lambda_v = 1$ in that normalization. Whether the same equality holds* literally *for the selected ambient cover and global measures of Chapters 4–7 requires their explicit splitting and Haar identifications.*

Proof. A split regular element is conjugate to $\mathrm{diag}(a,b)$; multiplying by a scalar, first consider $a,b \in \mathcal{O}_F^\times$. The Iwasawa and unipotent-orbit calculation, with $\mathrm{vol}(\mathcal{O}_F) = 1$, yields

$$O_{\mathrm{GL}_2}(\mathrm{diag}(a,b), 1_{K_0}) = |a-b|_F^{-1}, \qquad \Delta(\mathrm{diag}(a,b)) = |a-b|_F,$$

so $J = 1$. Multiplication of either eigenvalue by a root of unity preserves its valuation, hence the normalized orbital integral. If the two eigenvalue valuations differ, neither x nor such a translate is conjugate into $F^\times K_0$; both orbital integrals vanish. Scalar translations contribute only the specified unramified central-character factor. This proves the ordinary split-torus invariance. For a quadratic field E/F, Proposition 7.4.2 gives $\ker(P_n : E^\times/F^\times \to E^\times/F^\times) = 1$. The final statement invokes the *published* unit matching, not the elementary split-torus calculation. Its passage to the specified ambient cover is a distinct normalization obligation. □

Remark 7.8.5 (Scope of the published fundamental lemma)**.** The Flicker–Kazhdan result is a theorem for the matching unramified Hecke units, not an existence result for arbitrarily prescribed functions at finitely many exceptional places. Its original hypotheses include $p \nmid n$ and $(n, N_r) = 1$. In rank two the only maximal-torus types are split and quadratic field tori, so the ordinary invariance calculation above is exhaustive. At higher ranks, mixed tori do not automatically invalidate spherical matching: the Flicker–Kazhdan theorem covers them as well whenever its hypotheses apply.

A regularity and germ obligation. Equation (7.33) is imposed only where x^n is regular. On a split torus, x can be regular while x^n is singular (Remark 7.3.3); the power map is a local analytic isomorphism on the torus, but normalized orbital integrals of a function at images approaching a singular element may have nonconstant Shalika-germ behavior. A full transfer construction must either arrange support away from these root-collision loci or prove the matching germ conditions there. The same issue occurs at the complex places, although the cover there splits: the ordinary power map can still carry regular elements to singular ones. No universal vanishing of the germ coefficients is claimed here.

Proposition 7.8.6 (Conditional geometric comparison with an explicit normalization factor)**.** *Fix the rational splittings and compatible central characters of Chapters 4–6, and fix* corresponding *local Haar measures, global torus measures, and normalizer conventions on the ordinary and covered sides. Choose factorizable ordinary and genuine test functions in a trace-class, regular elliptic support range for which the two simple trace formulas apply. Assume:*

(i) For each $v \notin S$, f_v is FK-compatible as in Definition 7.8.1, and a genuine test function $\widetilde{f}_v$ satisfies (7.33) for every regular rational local component s_v of a globally good class $[s^n]$, with a nonzero constant λ_v independent of s_v and of its local torus type. The prescribed central-translate identity holds, $\lambda_v = 1$ for almost all v, and the hypotheses include the complex places.

(ii) For each $v \in S$, use the Theorem 6.9.1, orbital-integral transfer $\widetilde{f}_v = \mathscr{O}_{D,v} f_v$, with coefficient $b_{D,v} = |n|_v^{-(r-1)/2}$ and matching quotient measures on H_v/Z_v and $\widetilde{H}_v/\widetilde{Z}_v$. The functions obey the reciprocal central transformation laws.

(iii) The common rational splitting and all local lifts satisfy the star compatibility $\iota(s^n) = \iota(s)^n$; the global torus coefficients (including $|W_s|$), central phases, and quotient measures agree under the stipulated identifications. The support condition removes the central identity contribution. The ordinary and covered simple geometric expansions are absolutely convergent in the selected test-function range.

Put

$$\boxed{\Lambda = \Big(\prod_{v \notin S} \lambda_v\Big)\Big(\prod_{v \in S} b_{D,v}\Big).} \tag{7.34}$$

Then the regular elliptic geometric distributions, and hence the corresponding simple spectral trace distributions, satisfy

$$\boxed{I_{\widetilde{H}}(\widetilde{f}) = \Lambda I_H(f).} \tag{7.35}$$

In particular the distributions agree if $\Lambda = 1$.

Proof. Since the rational splitting ι is a homomorphism, $\iota(s^n) = \iota(s)^n$; at every completion, $\iota(s)_v$ is a lift of s_v, so this equals its star lift s_v^*. By Theorem 7.5.2, globally good noncentral rational classes are parametrized by the unique global root class $[s]$ of $[s^n]$, preserving normalizer orders. Every ordinary regular rational class $[s]$ is paired with its good covered star class $[s^n]$; there is no additional hypothesis that the ordinary class $[s]$ itself be good. The cover has zero regular orbital integrals at nongood lifts, which contribute nothing to the covered geometric expansion. Local compatibility

handles the multiple *local* roots without introducing additional rational summands. Hypothesis (i) supplies $\lambda_v J_{H_v}(s_v, f_v)$ off S, and (ii) supplies $b_{D,v} J_{H_v}(s_v, f_v)$ on S. By (iii) the remaining torus coefficients and central phases agree, and the product of Weyl discriminants is 1 by (7.28). Every regular rational term on the covered side is thus Λ times the corresponding ordinary term. The central identity has been suppressed, and the stipulated trace-class and convergence conditions permit summation and the two trace formulas. This gives (7.35). The proof supplies *no* existence theorem for the matching functions or their exceptional local constants. It only establishes the implication under the explicit hypotheses. □

Remark 7.8.7 (Normalization versus canonical matching)**.** The condition $\Lambda = 1$ is a normalization once nonzero matching functions are known: rescaling one $\widetilde{f}_v$ rescales its λ_v and hence Λ. It is *not* a theorem about canonically normalized units; rescaling a unit changes its canonical value. The meaningful unresolved question is to compute the constants for the prescribed local transfers, especially the finitely many exceptional split places, and to verify the global product in those fixed conventions. Likewise, compactness of $D^\times(K)Z(\mathbb{A}_K)\backslash D^\times(\mathbb{A}_K)$ eliminates continuous spectrum but does not by itself justify every unrestricted trace sum; the support and trace-class conditions above are retained for this reason.

7.8.4 The identity class and an explicit normalization test

Because r is prime and D is division, every noncentral rational element generates a degree-r field and is regular elliptic. Thus, after quotienting by the rational center, the central identity class is the only remaining singular rational class. Proposition 7.8.6 suppresses it, but it can be used as an additional test *if* the test range is enlarged and the necessary local germ identities are supplied.

Remark 7.8.8 (Central discriminants versus orbital-integral germs)**.** Let $d = r(r-1)/2$. For any torus, as a regular $x \to 1$ with x^n regular, the *discriminant ratio* satisfies

$$\frac{\Delta(x)}{\Delta(x^n)} \longrightarrow |n|_v^{-d},$$

because $(1-u^n)/(1-u) \to n$ for each root value $u \to 1$. However this identity *does not by itself yield* $\widetilde{f}_v(1) = \lambda_v |n|_v^{-d} f_v(1)$ from (7.33). On the split group, normalized orbital integrals generally have nontrivial Shalika germs near singular central elements; their limiting coefficients need not be the evaluation of the test function at 1. The requisite relation between *central test-function values* is therefore a separate local matching obligation, even when the regular orbital integrals match. Compare the unit-specific calculation of Proposition 7.8.4 with the general germ warning following Lemma 7.8.3.

At $v \in S$, Corollary 6.9.4 gives $\widetilde{f}_v(1) = c_{D,v} f_v(1)$ with $c_{D,v} = b_{D,v} |n|_v^{-r(r-1)/2}$. If, as an *additional unproved local identity-value hypothesis*, every split place (including complex places) satisfies $\widetilde{f}_v(1) = \lambda_v |n|_v^{-d} f_v(1)$ with $d = r(r-1)/2$, and all denominators are nonzero, the ratio of global identity-function values becomes

$$\prod_v \frac{\widetilde{f}_v(1)}{f_v(1)} = \Lambda \prod_v |n|_v^{-r(r-1)/2} = \Lambda \tag{7.36}$$

by the product formula. This does not imply equality of the *identity trace-formula terms* without a separate choice of matching quotient measures: homeomorphic adelic quotients need not have equal volumes with arbitrarily scaled Haar measures. Nor does it justify retaining the identity term in the simple trace formula without checking its other hypotheses.

Example 7.8.9 (A possible global numerical ledger)**.** Let $K = \mathbb{Q}(\zeta_3)$, $r = 2$, $n = 3$, and choose a quaternion division K-algebra ramified at the place over 3 and at one other finite place; for instance the running example $D = (-1, 4-\zeta_3)_K$, ramified at $\mathfrak{p}_3$ and $\mathfrak{p}_7$. At the wild division place, $b_{D,v} = 3$, $c_{D,v} = 27$. If the finite split places have matching $\lambda_v = 1$, the equation $\Lambda = 1$ would require $\lambda_\infty = 1/3$ at the unique complex place. Since the normalized complex absolute value has $|3|_\infty = 9$, the hypothesized central-value formula would then give the complex identity factor $(1/3)/9 = 1/27$, balancing 27. This is a *necessary numerical consistency check* under the stated assumptions, not a proof of matching complex test functions or a global comparison.

7.8.5 Why Li, Kottwitz, and DKV do not close the gap

The linear Jacquet–Langlands theorem of Deligne–Kazhdan–Vignéras [DKV84] controls the ordinary $D^\times$–GL_r relation, including the elliptic character sign, but it does not automatically manufacture a genuine representation of $\overline{G_{D_v}}$. Kottwitz's analysis of stable conjugacy and relative trace coefficients [Kot84] provides the correct arithmetic setting; for the induced tori $\mathrm{Res}_{E/K}\mathbb{G}_m$ one has $H^1(K,T) = 1$ by Shapiro's lemma and Hilbert 90, and embeddings $E \hookrightarrow D$ are controlled by Brauer splitting. Wen–Wei Li proves an invariant trace formula for adelic covers under an invariant trace Paley–Wiener hypothesis for Levi subgroups at *real* archimedean places [Li14]. Here K is totally imaginary, so that real-place hypothesis is vacuous. At the complex places the specified covers split (§4.5.4, item (R4); see also its ambient-cover construction), and the linear invariant Paley–Wiener theorem of Clozel–Delorme [CD84] supplies the relevant archimedean background. Li also discusses known real metaplectic GL cases, but his general result does not assert the particular *equality* of ordinary-$D^\times$ and covered-$D^\times$ traces required in (7.27). The simpler already-proved trace formulas of Chapters 4–5 remain the appropriate initial comparison.

7.9 Spectral separation and a research program

This section states the spectral identity that the conditional geometric comparison would give. It then explains what further argument is needed to pass from that identity to statements about individual local characters. The section describes a program and claims no new result.

7.9.1 The actual spectral distribution identity

This subsection defines the two global spectral distributions and states the conditional identity between them.

Let $H = D^\times$ and $\widetilde{H} = \overline{G_D}$ globally, with compatible central characters $\theta, \widetilde{\theta}$. The quotient $H(K)Z_H(\mathbb{A}_K)\backslash H(\mathbb{A}_K)$ is compact because D is a division algebra over K; its covered analogue is compact modulo the specified central preimage as well. Thus the fixed-central-character right

regular representations have discrete Hilbert decompositions. For factorizable test functions with the inverse central transformation law, define

$$I_H(f) = \sum_{\rho \in \mathscr{A}(H,\theta)} m_H(\rho) \prod_v \operatorname{tr} \rho_v(f_v), \tag{7.37}$$

$$I_{\widetilde{H}}(\widetilde{f}) = \sum_{\widetilde{\rho} \in \mathscr{A}(\widetilde{H},\widetilde{\theta})} m_{\widetilde{H}}(\widetilde{\rho}) \prod_v \operatorname{tr} \widetilde{\rho}_v(\widetilde{f}_v). \tag{7.38}$$

Here $\mathscr{A}$ denotes the relevant automorphic spectrum and m is its actual spectral multiplicity, not an assumed 1. The expressions are interpreted on trace-class test functions satisfying the simple trace-formula hypotheses; a bare unrestricted infinite sum is not justified by notation alone. Under the assumptions of Proposition 7.8.6, the comparison of geometric expansions supplies $I_{\widetilde{H}}(\widetilde{f}) = \Lambda I_H(f)$; ordinary equality holds when the fixed normalizations give $\Lambda = 1$. It does not identify summands. This is a conditional identity of *distributions*, with both the multiplicities and local trace factors retained.

7.9.2 Individual local characters and integrality

An equality of global spectral sums is a statement about *distributions*. To isolate a single local irreducible at a wild place, one needs an argument that the relevant coefficients are integers (or multiplicities which can be shown to isolate a single constituent). Linear independence alone can separate finite lists of characters; it does not supply the required integrality from a unitary Fourier transform.

Remark 7.9.1 (What an integer spectral coefficient would prove)**.** If a justified global-to-local argument supplies $a_{\rho\tau} \in \mathbb{Z}$ for every τ in the row of a fixed ρ, Proposition 7.6.1 (Corollary 6.5.2) forces $L_{D,v}\Theta_\rho$ to be one irreducible genuine character with positive sign. If integrality holds for all rows, unitarity implies a bijection. This is an application of an already proved lattice criterion, not a second conditional theorem; the missing step is precisely the integrality hypothesis at a wild place.

Separating the remaining tasks. First, Theorem 7.6.2 proves toral permutation positivity in rank two for every odd covering degree. The separate rank-two project Chapter 8 addresses the remaining nonabelian gluing problem and must justify the choice of a finite level before any finite character calculation becomes decisive. Second, at split places the root sum in Zou's (3.2) should be written against the kernel and index in Proposition 7.4.2, including the chosen Haar measures and central phases. Third, one needs the explicit ambient-cover identification that transports the known unramified unit matching of [FK86, §12] to the selected convention, together with functions at the finitely many exceptional split places and a complete global coefficient identity as in Proposition 7.8.6. Every ordinary rational regular class is paired with the star of its global representative; the ordinary class itself need not be good. Finally, one must justify globalization and spectral separation for representations of the covered inner form, and remove the auxiliary supercuspidal and non-cancellation hypotheses of Parts I–II. These are mathematical work packages, not restatements of an already-proved theorem.

Chapter 8

The wild place in degree two: toral positivity and the nonabelian gluing problem

Overview. We study irreducible-character integrality for the scalar-central odd-degree metaplectic cover of a quaternion division algebra at a wild place. The general result is a uniform quadratic toral permutation-character theorem: for every quadratic E/F in odd residue characteristic and every odd n with $\mu_n \subset F$, the discriminant ratio $\Delta(t)/\Delta(t^n)$ is a finite permutation character of $E^\times/F^\times$. The main new result concerns the running example $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, D quaternion. We prove that the division-side transform L_D of Chapter 6 carries the trivial character of $D^\times$ to the character of a single genuine irreducible representation τ of the cover, of dimension 27 (Target A), and we describe τ: its restriction to the pro-3 part of the cover is induced from a canonical genuine character of the covered third congruence subgroup. The proof has three layers: a reduction, via Brauer's characterization and the toral theorem, to the pro-3 Sylow subgroup; a structural analysis of the good elements of the congruence filtration which turns the question into a finite combinatorial problem about 72 vectors in $\mathbb{F}_3^5$ that has exactly two mirror-image solutions; and an explicit cocycle model of the cover, in which the one remaining identity (multiplicativity of the canonical star section on the sixth congruence subgroup) is proved by a short computation with cubic Hilbert symbols. The same holds for every irreducible representation of $D^\times$ trivial on the third congruence subgroup and compatible with the cover (central character trivial on μ_3): the transform of its character is the character of a genuine irreducible representation whose restriction to the pro-3 part is a sum of irreducible representations induced from the covered third congruence subgroup, one for each orbit of genuine constituents of a "cube-root transform" of $\rho|_{K^1}$ (part of Target B). For an arbitrary row we prove the reduction to the pro-3 Sylow subgroup and a sufficient criterion in terms of a cube-root transform of Θ_ρ. Finally, the p-th power identity and the multiplicativity of the canonical star section on the $2d$-th congruence subgroup are proved for every odd prime p and every $F \supset \mu_p$ (part of Target C); what is not uniform is identified.

8.1 Purpose, scope, and status

This chapter separates the wild division-side character problem from the conditional global trace comparison in Chapter 7. The chapter proves the general toral theorem (Section 8.5), works out the three quadratic tori of the running example (Section 8.7), proves the positivity–integrality criterion

(Section 8.4), and formulates three targets:

Target A
For $F = \mathbb{Q}_3(\zeta_3)$, D quaternion and $n = 3$, decide whether $L_D\mathbf{1}$ is the character of a single 27-dimensional genuine irreducible representation of $\widetilde{H}$ with the prescribed central character.

Target B
For every ordinary irreducible ρ with a fixed compatible central character, prove that all coefficients $a_{\rho\tau}$ are integral, or exhibit a nonintegral one.

Target C
Extend Target B to all odd n, all F of odd residual characteristic with $\mu_n \subset F$, and quaternion D/F.

The later sections settle Target A affirmatively (Theorem 8.12.1), settle Target B for all representations trivial on K^3 and their norm twists (Theorems 8.13.5, 8.13.8), prove the reduction theorem for every row (Theorem 8.13.1) and a sufficient criterion for a row (Proposition 8.13.4), and prove the two cocycle-level ingredients of Target C uniformly in p (Theorems 8.14.2, 8.14.3). The dependencies of the main theorem are listed in Remark 8.12.2.

Labels. Proved here is the default and carries no visible tag; **(E)** marks a statement established elsewhere and used as a black box; **(O)** marks one outlined but not fully written out or checked; **(C)** marks one that is conjectural. A remark immediately follows every **(E)**, **(O)** or **(C)** statement, saying where it is proved or, if conjectural or outlined, the motivation or what remains. The label **($\mathbf{P^c}$)** marks a statement whose proof uses, besides hand arguments, an exact finite computation (linear algebra over $\mathbb{F}_3$, or exact arithmetic in a number field) that is *exhaustive* over an explicitly described finite set, and is proved here in that sense, so it also carries no further tag beyond **($\mathbf{P^c}$)** itself. Every such computation is described in the text, and the scripts are listed in Appendix A.2. No statement labeled **($\mathbf{P^c}$)** depends on floating-point arithmetic or on sampling. Sampled computations occur in the chapter only as corroboration of statements that are proved by hand, and are identified as such where they are mentioned.

Statement	Source	Status
Unitary local L_D and identity factor	Chapter 6	**(E)**
Heisenberg commutator and degree on maximal tori	Chapter 4	**(E)**
Quadratic toral permutation-character theorem	Theorem 8.5.1	**(P)**
Three exact cubic wild field-torus calculations	Section 8.7	**(P)**
All-row positivity–integrality equivalence	Proposition 8.4.1	**(P)**
Structure of good elements in the congruence filtration	Section 8.8	**(P)**
Reduction of Target A to the pro-3 Sylow subgroup	Theorem 8.9.5	**(P)**
Finite model; the defect function; uniqueness of the design	Section 8.10	**(P^{c})**
Explicit cocycle model (Kubota–Kazhdan–Patterson)	Lemma 8.11.1	**(E)/(P)**
Cube identity; star section on K^6; proof of (H1)	Section 8.11	**(P)**
$L_D\mathbf{1}$ is a genuine irreducible character (Target A)	Theorem 8.12.1	**(P^{c})**
Reduction theorem for every row	Theorem 8.13.1	**(P)**
Cube-root transform; sufficient criterion for a row	Proposition 8.13.4	**(P)**
All rows trivial on K^2, and norm twists (part of Target B)	Theorem 8.13.5	**(P^{c})**
All rows trivial on K^3 (part of Target B)	Theorem 8.13.8	**(P^{c})**
Target B for level ≥ 3	Sections 8.13.5, 8.15	**(O)**
p-th power identity and (H1) for every odd p (part of Target C)	Section 8.14	**(P)**
Target C: Brauer reduction; finite design	Remark 8.14.4	**(O)/(C)**

8.2 Background

This section collects standard facts used throughout, with references; nothing in it is new. Readers familiar with division algebras and with the Heisenberg–Weil machinery can skip to Section 8.3 and return for the heuristic of Section 8.2.3, which explains how the results of Sections 8.10–8.13 should be read.

8.2.1 Division algebras over local and global fields

The general theory recalled in this subsection is developed with more examples, and tied to the running example throughout, in §3.3; what follows specializes it to this chapter's running example and extends it with the congruence filtration and the notion of level on which the rest of the chapter depends.

Let F be a nonarchimedean local field. A central simple algebra over F has dimension r^2 for an integer r (its degree), and its class in the Brauer group is determined by the Hasse invariant $\mathrm{inv} : \mathrm{Br}(F) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ [Ser79, Ch. XIII]; [Rei75, §31]. The algebras of degree r are the $M_m(D)$ with D a division algebra of degree $d \mid r$, $md = r$, and D is determined by its invariant a/d mod 1 with $(a,d) = 1$; in particular there is exactly one quaternion division algebra over F, with invariant $\frac{1}{2}$. A division algebra D of degree r has the following structure [Rei75, §§12–14]; [Wei73, Ch. I]. It carries a discrete valuation v_D extending v_F, with $v_D(F^\times) = r\mathbb{Z}$ when normalized by $v_D(\Pi) = 1$ for a uniformizer Π; its valuation ring $\mathscr{O}_D$ is the unique maximal order, with unique maximal two-sided ideal $\mathfrak{P} = \Pi\mathscr{O}_D$, and residue field $k_D = \mathscr{O}_D/\mathfrak{P}$ the extension of k_F of degree r. The unramified extension E_0/F of degree r embeds in D, $\mathscr{O}_D = \mathscr{O}_{E_0}[\Pi]$, and D is the cyclic algebra $\bigoplus_{k<r} E_0\Pi^k$ with $\Pi x\Pi^{-1} = \sigma^a(x)$ for a generator σ of $\mathrm{Gal}(E_0/F)$ and $\Pi^r = \varpi$ up to a unit; for the quaternion algebra one may take $\Pi^2 = \varpi$ and $\Pi x\Pi^{-1} = \sigma(x)$, which is the description $D = E_0 \oplus E_0 j$ used in

Example 8.7.1. Every field extension of F of degree r embeds in D as a maximal subfield, and when r is prime (the case of this book), every element of $D \setminus F$ generates a maximal subfield; the maximal subfields of the quaternion algebra are the three quadratic extensions of F (p odd), one unramified and two ramified, and the maximal tori of $D^\times$ are their multiplicative groups. The reduced norm $\mathrm{Nrd} : D^\times \to F^\times$ is surjective, $D^\times/F^\times$ is compact, and $D^\times$ has the filtration by the normal subgroups $K^m = 1 + \mathfrak{P}^m$ ($m \geq 1$) with $K^m/K^{m+1} \simeq k_D$ (additive) and $\mathscr{O}_D^\times/K^1 \simeq k_D^\times$; the group $D^\times/F^\times K^1$ is finite, of order $r(q^r-1)/(q-1)$, and K^1 is a pro-p group. For the running example ($F = \mathbb{Q}_3(\zeta_3)$, $q = 3$, $r = 2$) these are the groups $Q = K^1F^\times/F^\times$, Q_m and $D^\times/F^\times K^1 \simeq D_8$ of Section 8.8. Every irreducible smooth representation of $D^\times$ is finite dimensional and trivial on some K^m; its *level* is the least m with $\rho|_{K^{m+1}}$ trivial. (The general definition of level and the congruence filtration, and the running example's own level-≤ 2 representation count, are given once, for reuse book-wide, in the Congruence filtration and level entry of §3.3; the rest of this paragraph specializes that entry to the combinatorics this chapter actually needs.)

Levels. The level organizes everything below, so we record what it means. Since K^{m+1} is normal and $D^\times/F^\times K^{m+1}$ is finite, the representations of level $\leq m$ with a given central character are the representations of a finite group; for the running example $D^\times/F^\times K^{m+1}$ has order $8 \cdot |Q/Q_{m+1}| = 8 \cdot 3^{m+\lceil m/2 \rceil}$ and the representations of level ≤ 2 with trivial central character are the irreducible representations of the group $\mathrm{Heis} \rtimes D_8$ of order 216: five of level zero (dimensions $1,1,1,1,2$, the representations of D_8), four of level one (dimension 4, those nontrivial on $K^1/K^2 \simeq \mathbb{F}_9$ but trivial on K^2) and four of level two (dimension 6, those nontrivial on K^2/K^3). The level is the integer form of the Moy–Prasad depth: for $D^\times$ with its standard filtration a representation of level m has depth m/r, and depth is preserved by the local Jacquet–Langlands correspondence [BH06, Ch. 13]; [DKV84]. For $p \nmid r$, which includes the running example ($r = 2$, $p = 3$), Howe and Corwin [How77b]; [Cor74] parametrize the irreducible representations of $D^\times$ by admissible pairs (E, χ), $E \subset D$ a subfield and χ a character of $E^\times$, with $\rho_{E,\chi}$ induced from a character of $E^\times K^{m'}$ extending χ; the level of $\rho_{E,\chi}$ is governed by the conductor of χ, and this induction through a congruence subgroup is the model for the inducing data in Sections 8.12 and 8.13. Levels are unbounded: every irreducible representation has some finite level, but no finite set of levels exhausts $\mathrm{Irr}(D^\times)$, which is why the results of Section 8.13, which cover levels ≤ 2, do not by themselves give Target B and why Section 8.15 asks for an inductive mechanism.

Over a number field F the Brauer group is described by the Albert–Brauer–Hasse–Noether theorem: the sequence $0 \to \mathrm{Br}(F) \to \bigoplus_v \mathrm{Br}(F_v) \xrightarrow{\Sigma} \mathbb{Q}/\mathbb{Z} \to 0$ is exact, so a central simple algebra is determined by its local invariants, which vanish at all but finitely many places and sum to zero; the period of a class equals its index, so a class of order r is the class of a division algebra of degree r [Rei75, §32]; [Wei73, Ch. XIII]. A quaternion algebra over a number field is ramified at an even number of places (where its invariant is $\frac{1}{2}$) and split elsewhere. The local components $D_v = D \otimes F_v$ at ramified v are the local division algebras above; the group $D^\times(F)\backslash D^\times(\mathbb{A}_F)/F^\times(\mathbb{A}_F)$ is compact, so the automorphic spectrum of $D^\times$ is discrete, and the Jacquet–Langlands correspondence [JL70, Ch. III]; [DKV84]; [Rog83]; [Bad08] identifies the automorphic representations of $D^\times$ with those of GL_r that are discrete series at the ramified places; its local form is a bijection between the irreducible representations of $D_v^\times$ and the discrete series of $\mathrm{GL}_r(F_v)$ characterized by the character relation $\Theta_\pi(g) = (-1)^{r-1}\Theta_\rho(g')$ on matching regular elliptic elements. Chapters 4–7 concern the metaplectic analogue of this picture: an n-fold cover of $D^\times$ and the Kazhdan–Patterson cover of

GL_r, related by a transfer of characters. The present chapter is purely local and concerns one ramified place of the running example.

8.2.2 The cover, the star map and good elements

We use the following from Chapter 4. Let $\mu_n \subset F$. The n-fold scalar-central cover $\widetilde{H} \to H = D^\times$ is the central extension by μ_n obtained by restricting the Kazhdan–Patterson cover of $\mathrm{GL}_r(E_0)$ [KP84] along $D \hookrightarrow M_r(E_0)$; its preimage $\widetilde{Z}$ of the center $F^\times$ is central, and $\widetilde{Z} \simeq \mu_n \times F^\times$ noncanonically (Proposition 4.2.4). The restriction of the cover to a maximal torus $E^\times$ is a Heisenberg group: the commutator of lifts is a bimultiplicative pairing $E^\times \times E^\times \to \mu_n$ with kernel $F^\times E^{\times n}$ on either side (Proposition 4.3.1 and (8.10)), so the covered torus has center C_E of index $|n|_F^{-(r-1)}$ in $\widetilde{E^\times}$, and every genuine irreducible representation of $\widetilde{E^\times}$ has degree $b_D = |n|_F^{-(r-1)/2}$ and is determined by its central character (Stone–von Neumann; see Section 8.2.3). The *star map* $x \mapsto x^* = \tilde{x}^n$ is well defined on H (independent of the lift, since μ_n is central) and is a homomorphism on each covered torus, because $(\tilde{x}\tilde{y})^n = \tilde{x}^n\tilde{y}^n[\tilde{y},\tilde{x}]^{n(n-1)/2}$ with $[\tilde{y},\tilde{x}] \in \mu_n$ and n odd; an element of H is *good* if it lies in $F^\times \cdot \{x^n\}$, and the division-side transform (8.1) is supported on the star lifts $\tilde{z}t^*$ of good elements. In the running example $n = 3$, $b_D = 3$, $c_D = 27$.

8.2.3 Heisenberg groups, the Weil representation and the Howe correspondence

This subsection recalls the Heisenberg group, the Stone–von Neumann theorem, the Weil representation and the Howe correspondence. They serve as a heuristic for the later sections, and Remark 8.16.1 uses them to restate Target B as a duality.

Let $(W, \langle\ ,\ \rangle)$ be a finite-dimensional symplectic space over a local field or a finite field k, and $\mathscr{H}(W) = W \times k$ the Heisenberg group, $(w,s)(w',s') = (w+w', s+s'+\frac{1}{2}\langle w,w'\rangle)$. The Stone–von Neumann theorem says that for each nontrivial character ψ of the center k there is a unique irreducible (smooth, for local k) representation ρ_ψ of $\mathscr{H}(W)$ with central character ψ; its character is supported on the center. It is realized in the *Schrödinger model*: for a Lagrangian decomposition $W = X \oplus Y$, $\rho_\psi = \mathrm{Ind}_{Y\times k}^{\mathscr{H}(W)}(\mathbf{1}\otimes\psi)$, acting on functions on X [Wei64]; [MVW87, Ch. 2]. The symplectic group $\mathrm{Sp}(W)$ acts on $\mathscr{H}(W)$ fixing the center, so by uniqueness it acts projectively on ρ_ψ; the resulting genuine representation of the metaplectic double cover $\mathrm{Mp}(W)$ is the Weil representation ω_ψ. For a *dual pair* (G,G') of mutually centralizing reductive subgroups of $\mathrm{Sp}(W)$, Howe's duality conjecture [How79], proved by Waldspurger for $p \neq 2$ [Wal90] and in general by Gan and Takeda [GT16], says that the restriction of ω_ψ to $\tilde{G}\times\tilde{G}'$ decomposes as $\bigoplus_\pi \pi\otimes\theta(\pi)$ with $\pi \mapsto \theta(\pi)$ injective on the set of π that occur: a bijection between representations of the two members of the pair, with multiplicity one, whose "kernel" is the character of ω_ψ. The first nontrivial instance is Shimura's correspondence: Waldspurger realized the correspondence between genuine representations of $\widetilde{\mathrm{SL}}_2$ and representations of PGL_2, or of $PD^\times$ for the quaternion algebra D, through the dual pairs $(\widetilde{\mathrm{SL}}_2, O(V_3))$ with V_3 a three-dimensional quadratic space, split or anisotropic [Wal80]; [Wal91]. So for $n = 2$ the link between a metaplectic cover and a quaternion algebra is literally a Howe correspondence, with the cover on the split side. For odd n no classical dual pair produces an n-fold cover, and the metaplectic correspondence of Flicker and Kazhdan [FK86]

is established by the trace formula instead; the transform L_D of Chapter 6 is likewise defined analytically, by (8.1), not by a kernel.

Three points of contact with the present chapter. First, the covered tori are Heisenberg groups and the degree-b_D representations of Section 8.6 are their Stone–von Neumann representations. Second, the finite groups of Sections 8.8–8.10 have a Heisenberg flavor: Q_1/Q_3 is the Heisenberg group of order 27, and the character λ_0 of $\widetilde{U}_3$ with $\tau|_{\widetilde{U}_1} = \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}\lambda_0$ is reminiscent of a Schrödinger model, a character of an abelian normal subgroup of index 27 induced up, with the freeness statement of Proposition 8.13.9 playing the role of nondegeneracy (every genuine character of $\widetilde{U}_3$ of the relevant level is regular, so every such induced representation is irreducible). This is an analogy, not an identification: $\widetilde{U}_3/N_6 \simeq \mu_3 \times V$ is abelian by (H3), V has dimension five, and $\widetilde{U}_1/N_6$ is not a Heisenberg group with a nondegenerate symplectic quotient; what is used is ordinary Clifford theory for an abelian normal subgroup. Third, and most to the point, Target B is itself a Howe-duality statement about the pair $(\widetilde{H}, H)$: if $L_D\Theta_\rho = \Theta_{\tau_\rho}$ with τ_ρ irreducible for every ρ, then $\Omega := \bigoplus_\rho \tau_\rho \otimes \rho^\vee$ is a representation of $\widetilde{H} \times H$ in which each ρ is paired with exactly one τ_ρ, with multiplicity one, and its level-by-level truncations have the explicit characters computed in Remark 8.16.1 (the full sum is a formal object; see there). Whether Ω has an intrinsic construction, as ω_ψ has, is not known to us; the induced models of Theorems 8.13.5 and 8.13.8 are what such a construction would have to produce level by level. Finally, the obstruction that distinguishes the wild place is the one that limits Howe's Kirillov theory for compact p-adic groups [How77a]: at a tame place the n-th root transform of characters is a dilation of coadjoint orbits by $1/n$, hence a Galois conjugation of characters (Lemma 6.7.5); at $p \mid n$ the dilation is not available on the Lie lattice, and Sections 8.10–8.13 show how the corresponding prediction nevertheless survives, with the character λ_0 absorbing the nonlinearity of the cube root.

8.3 Division-side character transform and its normalization

This section recalls the transform L_D of Chapter 6 and its normalizing constants, in the notation of this chapter. It then recalls the criterion that the rest of the chapter tests. The transform permutes irreducible characters exactly when its matrix is integral.

Let F be nonarchimedean, D/F a division algebra of prime degree r, $H = D^\times$, and $\widetilde{H}$ its specified cover. Set $Z = F^\times$ and $\widetilde{Z} = p^{-1}(Z)$, equipped with compatible Haar measures on $H/Z \simeq \widetilde{H}/\widetilde{Z}$. The Chapter 6 transform is

$$(L_D\Phi)(zt^*) = b_D\,\widetilde{\theta}(z)\frac{\Delta(t)}{\Delta(t^n)}\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}, \tag{8.1}$$

for regular t, $z \in \widetilde{Z}$; it vanishes at nongood regular lifts. Here $t^* = \tilde{t}^n$ is the star of any lift $\tilde{t}$ of t, and

$$\Delta(t) = \left|\det\big(1 - \mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t}\big)\right|_F^{1/2} = \prod_{i<j}\left|\frac{(\lambda_i - \lambda_j)^2}{\lambda_i\lambda_j}\right|_F^{1/2}. \tag{8.2}$$

This equals Chapter 4's $|D(t)|^{1/2}|\mathrm{Nrd}(t)|_F^{-(r-1)/2}$. Since the power map on a field torus modulo scalars has index $|n|_F^{-(r-1)}$, Weyl integration gives b_D^2 equal to this index. The factor b_D is independent of *common* rescalings of the two identified quotient Haar measures. The ratio of discriminants

extends at central elements to $|n|_F^{-r(r-1)/2}$ (Lemma 6.3.4), hence

$$(L_D\Theta_\rho)(1) = c_D \dim\rho, \qquad c_D = b_D|n|_F^{-r(r-1)/2} = |n|_F^{-(r^2-1)/2}. \tag{8.3}$$

The identity factor is pointwise and does not depend on Haar measures. Theorem 6.9.1, separately gives the adjoint orbital-integral/test-function transfer. Test functions transform by $\widetilde{\theta}^{-1}$, not by $\widetilde{\theta}$.

8.3.1 Integral-matrix criterion

This subsection writes L_D as a matrix in the bases of irreducible characters and recalls the integrality criterion of Chapter 6. The criterion turns a question about characters into a question about integers.

Choose $\widetilde{\theta}$ first and put $\theta(a) = \widetilde{\theta}(a^*)$. Theorem 6.4.4 and Corollary 6.5.1, show that L_D is a unitary *isomorphism* of invariant L^2 spaces and that

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle, \tag{8.4}$$

with each row and column finite. The characters on either side are orthonormal bases in their fixed-central-character spaces.

Proposition 8.3.1 ((**E**) Integral-lattice criterion, Chapter 6)**.** *The transform L_D permutes the irreducible characters with positive signs if and only if every $a_{\rho\tau}$ in (8.4) is an integer.*

Proof. An integral unit vector in ℓ^2 is a signed basis vector; unitarity gives injectivity and surjectivity because the inverse is the adjoint of a unitary map. The value at identity in (8.3) is positive and fixes the sign. This is Corollary 6.5.2, with its converse. □

Remark 8.3.2 (The actual obstruction)**.** For wild v, Chapter 6 does not prove $a_{\rho\tau} \in \mathbb{Z}$. The tame Adams argument works with a finite quotient Q and a central subgroup C, under the hypothesis $\gcd(n, |Q/C|) = 1$ (Lemma 6.7.5). At a wild place that hypothesis need not hold. For the simplest illustration take $Q = C_p$, $C = 1$, $n = p$: then the Adams operation sends every linear character to the trivial character. This does not prove a nonintegral coefficient for L_D; it only shows why that particular proof method fails. The present chapter replaces the Adams argument, in the running example, by Brauer's characterization of characters and an explicit analysis of the pro-3 part of the cover.

8.4 A useful global criterion: positivity of every row

This section proves that integrality of the whole character matrix is equivalent to positive definiteness of every row. It also explains why the positivity of a single row decides nothing about that row.

Proposition 8.4.1 (Positivity versus integrality)**.** *With the unitary character matrix $(a_{\rho\tau})$ of (8.4), the following are equivalent:*

(a) L_D permutes irreducible characters with positive signs;

(b) every $a_{\rho\tau}$ is an integer;

(c) every $a_{\rho\tau}$ is a nonnegative real number;

(d) every $L_D\Theta_\rho$ is positive definite on $\widetilde{H}$.

The implication from positive definiteness to integrality requires every *row; positivity of a single row does not force integrality or irreducibility for that row.*

Proof. By Chapter 6, each row and column of A has ℓ^2 norm one. An integer row is a signed basis vector, and its sign is positive by the identity value (8.3). If all entries are nonnegative, orthogonal rows have disjoint supports. Every column meets exactly one row, and since its norm is one its nonzero entry is one; row norms then imply the permutation property. The equivalence of nonnegative character coefficients and positive definiteness follows from Peter–Weyl/GNS for a compact-mod-center group with fixed unitary central character. One row could instead be $(2^{-1/2}, 2^{-1/2}, 0, \dots)$. □

Remark 8.4.2. For a *single* row the usable criterion is different and is the one we shall apply: a class function of ℓ^2 norm one with positive value at the identity is an irreducible character as soon as it is a *virtual* character (an integral combination of irreducible characters). Brauer's theorem reduces the virtual-character property to restrictions to elementary subgroups; Section 8.9 carries this out.

8.5 A general quadratic theorem

This section proves that the discriminant ratio R_E is the character of a permutation representation, for every quadratic extension E/F and every odd n. This is the toral positivity statement of the chapter. The proof computes R_E on the filtration of the norm-one group, first in general and then separately for unramified and ramified extensions.

Let F be a finite extension of $\mathbb{Q}_p$ with p odd, residue field of cardinality q, and let $n \geq 1$ be odd with $\mu_n \subset F$. Let E/F be any quadratic field extension, with nontrivial Galois involution σ. Put

$$A = E^\times/F^\times, \qquad G = E^1 = \ker(N_{E/F} : E^\times \to F^\times).$$

Hilbert's Theorem 90 identifies $A \xrightarrow{\sim} G$ via $t \mapsto u = t/\sigma(t)$. The normalized absolute value on E is $|x|_E = q_E^{-v_E(x)}$, so $|x|_E = |N_{E/F}x|_F$.

For noncentral $t \in E^\times$ let

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}, \qquad \Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid D/E)\right|_F^{1/2}.$$

The ratio extends continuously at scalar elements, with value $|n|_F^{-1}$.

Theorem 8.5.1 (Quadratic toral permutation positivity)**.** *For any $E/F, n$ as above, R_E is the character of a finite-dimensional permutation representation of a finite quotient of A. In particular,*

$$R_E(t) = \sum_{\chi \in \widehat{A}} m_\chi \chi(t), \qquad m_\chi \in \mathbb{Z}_{\geq 0},$$

where only finitely many m_χ are nonzero. This includes every wild case $p \mid n$. At the identity $R_E(1) = |n|_F^{-1}$.

8.5.1 A radial valuation formula

This subsection shows that $R_E(u)$ depends only on the depth $v_E(1-u)$ of u in the norm-one group, and gives the formula. The two subsections that follow turn the formula into a sum of permutation characters.

Write $n = p^a m$ with $(m,p)=1$, and put

$$t_\zeta = v_E(1-\zeta), \qquad G_k = \{u \in G : v_E(1-u) \geq k\} \quad (k \geq 1), \qquad G_0 = G.$$

If $a = 0$ then $R_E \equiv 1$, so assume $a \geq 1$.

Lemma 8.5.2. *For $u \in G$ and any $\zeta \in \mu_n$ whose prime-to-p component is nontrivial, $|1-\zeta u|_E = 1$. For $\zeta \in \mu_{p^a} \setminus \{1\}$,*

$$v_E(1-\zeta u) = \min\{v_E(1-u), t_\zeta\}.$$

Consequently, with $k = v_E(1-u) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$,

$$R_E(u) = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(k, t_\zeta)}. \tag{8.5}$$

Proof. The reduction of $u \in E^1$ lies in a norm-one subgroup. In the unramified case its intersection with $k_F^\times$ is $\{\pm 1\}$, and in the ramified case the reduction of u is itself in $\{\pm 1\}$. A root of unity with nontrivial prime-to-p component has reduction in $k_F^\times \setminus \{\pm 1\}$, since n is odd, proving the first assertion.

For a nontrivial p-power root ζ, the ultrametric inequality gives the asserted minimum when the two valuations differ. If both equal t_ζ and $v_E(1-\zeta u) > t_\zeta$, then $u \equiv \zeta^{-1} \pmod{\mathfrak{p}_E^{t_\zeta+1}}$. Applying σ and using $\sigma(u) = u^{-1}$ gives $u^{-1} \equiv \zeta^{-1}$ to the same precision, hence $\zeta \equiv \zeta^{-1}$, a contradiction to $v_E(\zeta - \zeta^{-1}) = t_\zeta$ (as p is odd). Thus equality holds.

For $u = t/\sigma(t)$ we have $\Delta(t) = |N_{E/F}(1-u)|_F^{1/2} = |1-u|_E^{1/2}$. Factor $1-u^n = \prod_{\zeta \in \mu_n}(1-\zeta u)$ and use the two valuation statements. The identity follows for noncentral t and extends continuously to $u = 1$. At $u = 1$, the cyclotomic product $\prod_{\zeta \in \mu_{p^a}\setminus\{1\}}(1-\zeta) = p^a$ gives $R_E(1) = |p^a|_E^{-1/2} = |n|_F^{-1}$. □

Define for $k \geq 0$

$$M_k = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(k, t_\zeta)}, \qquad M_0 = 1,$$

and put $T = \max_{\zeta \neq 1} t_\zeta$. All values M_k are positive integers because roots ζ, ζ^{-1} occur in pairs of equal valuation. We have $M_k = M_T$ for $k \geq T$.

8.5.2 The unramified filtration

This subsection treats an unramified extension E/F. It computes the indices of the filtration subgroups and writes R_E as a nonnegative integral combination of the permutation characters of G on the quotients G/G_k.

Suppose E/F is unramified. Then $q_E = q^2$ and reduction of the norm-one group has order $q+1$; at every deeper level the trace-zero subgroup of the residue-field additive group has order q. Thus the filtration satisfies

$$[G : G_1] = q+1, \qquad [G : G_k] = (q+1)q^{k-1} \quad (k \geq 1).$$

Write $s_k = \#\{\zeta \in \mu_{p^a} \setminus \{1\} : t_\zeta \geq k\}$. Each s_k is even, and $M_k/M_{k-1} = q^{s_k}$. Set

$$c_k = \frac{M_k - M_{k-1}}{(q+1)q^{k-1}} \quad (1 \leq k \leq T).$$

For $k = 1$, $M_0 = 1$ and $q+1$ divides $q^{s_1} - 1$ because s_1 is even. For $k > 1$ with $s_k > 0$, M_{k-1} is divisible by q^{k-1}, since at least two roots contribute at least $k-1$ to its exponent. Again $q+1$ divides $q^{s_k} - 1$. Hence $c_k \in \mathbb{Z}_{\geq 0}$ (and $c_k = 0$ if $s_k = 0$).

By telescoping (8.5),

$$\boxed{R_E = \mathbf{1}_G + \sum_{k=1}^{T} c_k \, [G : G_k] \, \mathbf{1}_{G_k}.} \tag{8.6}$$

Here $[G : G_k]\,\mathbf{1}_{G_k}$ is precisely the permutation character of G on G/G_k.

8.5.3 The ramified filtration

This subsection treats a ramified extension E/F. The positive depths are now odd, so the indices change, but the conclusion has the same form as in the unramified case.

Suppose E/F is ramified. Then $q_E = q$ and $t_\zeta = 2v_F(1-\zeta)$ is even. The norm-one group has filtration

$$[G : G_1] = 2, \qquad G_{2j} = G_{2j+1}, \qquad [G : G_{2j}] = 2q^j \quad (j \geq 1).$$

For completeness, write $E = F(\sqrt{d})$ with $v_F(d) = 1$. Hilbert 90 gives $u = (a + b\sqrt{d})/(a - b\sqrt{d})$; normalize a, b to be integral and not both divisible by the uniformizer of F. If a is a unit, scale it to 1 and write $c = b/a \in \mathscr{O}_F$: then the depth of $u - 1$ is $1 + 2v_F(c)$. Moreover, two such parameters c, c' define elements congruent modulo $\mathfrak{p}_E^{2j}$ precisely when $c \equiv c' \pmod{\mathfrak{p}_F^j}$, so $[G_1 : G_{2j}] = q^j$. If a is not a unit and b is, then $u \equiv -1$ modulo $\mathfrak{p}_E$. These observations give the displayed indices and the fact that positive depths are odd.

Define

$$c_1 = \frac{M_1 - 1}{2}, \qquad c_{2j} = \frac{M_{2j+1} - M_{2j-1}}{2q^j} \quad (1 \leq j \leq T/2).$$

The first coefficient is an integer because $M_1 = q^{(p^a-1)/2}$ is odd. For $j \geq 1$ write

$$\frac{M_{2j+1}}{M_{2j-1}} = q^{e_j}, \qquad e_j = \frac{1}{2} \sum_{\zeta \neq 1} \big(\min(2j+1, t_\zeta) - \min(2j-1, t_\zeta)\big) \in \mathbb{Z}_{\geq 0}.$$

If $e_j = 0$, then $c_{2j} = 0$. Otherwise some pair $\{\zeta, \zeta^{-1}\}$ has $t_\zeta \geq 2j$, so M_{2j-1} is divisible by q^{2j-1}, hence by q^j. Moreover $q^{e_j} - 1$ is even. Therefore $c_{2j} \in \mathbb{Z}_{\geq 0}$.

As all positive depths are odd, telescoping now gives

$$\boxed{R_E = \mathbf{1}_G + c_1 \, [G : G_1] \mathbf{1}_{G_1} + \sum_{j=1}^{T/2} c_{2j} \, [G : G_{2j}] \, \mathbf{1}_{G_{2j}}.} \tag{8.7}$$

Again each summand is the character of a permutation representation. Equations (8.6) and (8.7) prove Theorem 8.5.1.

8.6 From Fourier coefficients to covered-torus restrictions

This section explains what the Fourier coefficients of R_E mean on the cover. They are the multiplicities with which genuine representations of the covered torus occur in the restriction of a transferred character. The section also states the limit of this test. It concerns restrictions to a torus and does not decide integrality on the whole cover.

Lemma 8.6.1 (Fourier test for restrictions to covered field tori)**.** *Let E/F be any maximal field in D, with r prime and $(r,n)=1$, and put $A_E = E^\times/F^\times$. The function*

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}$$

extends continuously at scalars, is a locally constant function on the compact abelian group A_E, and has a finite Fourier expansion

$$R_E(t) = \sum_{\chi \in \widehat{A_E}} m_\chi \chi(t), \qquad m_\chi = \int_{A_E} R_E(t)\overline{\chi(t)}\,dt. \tag{8.8}$$

Here the Haar measure of A_E has mass 1. Choose a compatible genuine central character $\widetilde{\theta}$ of $\widetilde{Z}$ and set $\theta(a) = \widetilde{\theta}(a^*)$. For any smooth character $\alpha : E^\times \to \mathbb{C}^\times$ with $\alpha|_{F^\times} = \theta$, and for each $\chi \in \widehat{A_E}$, let $\eta_{\alpha\chi}$ be the unique genuine irreducible representation of $\widetilde{E^\times}$ whose central character satisfies

$$\zeta_{\alpha\chi}(zt^*) = \widetilde{\theta}(z)\alpha(t)\chi(t) \qquad (z \in \widetilde{Z},\ t \in E^\times).$$

Define a torus-level class function $\mathscr{L}_E\alpha$ to equal $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on $zt^* \in C_E$ and to vanish outside C_E. Then, as a genuine class function on the covered torus*,*

$$\mathscr{L}_E\alpha = \sum_\chi m_\chi\, \Theta_{\eta_{\alpha\chi}}. \tag{8.9}$$

For an ordinary character Θ_ρ of $D^\times$, its actual L_D transform restricts to the sum of these $\mathscr{L}_E\alpha$ with the multiplicities in $\Theta_\rho|_{E^\times}$. Consequently $m_\chi \in \mathbb{Z}_{\geq 0}$ for every χ is sufficient for the restriction of every transferred ordinary irreducible character to this covered torus to be a genuine representation character. It is not sufficient for the full $\widetilde{D^\times}$ character to be a representation character.

Proof. The extension of R_E at scalars follows from Lemma 6.3.4. Since E/F is a degree-r field extension with $(r,n)=1$, the nth power of a noncentral t stays noncentral. The compactness of A_E and local constancy imply that R_E factors through a finite quotient, giving (8.8).

By Chapter 4 the center of the covered torus is $C_E = \widetilde{Z}\cdot\{t^* : t \in E^\times\}$. The displayed formula for $\zeta_{\alpha\chi}$ is well-defined: the power map on A_E is injective, and on scalar overlaps the compatibility $\alpha(a) = \widetilde{\theta}(a^*)$ is precisely what is required. Such genuine central characters extend over C_E, and Stone–von Neumann gives a unique $\eta_{\alpha\chi}$. Its character vanishes off C_E, while on C_E it equals $b_D\zeta_{\alpha\chi}$ because its degree is b_D. The defining formula (8.1) for L_D has the identical vanishing and equals $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on zt^*. Comparing with (8.8) proves (8.9). Finally, restricting a finite-dimensional ordinary representation to $E^\times$ decomposes it into one-dimensional characters, all restricting to θ on $F^\times$; summing the displayed formula proves the last claim. □

8.7 The explicit cubic wild example

The following calculation verifies a *necessary representation-ring integrality test on every maximal torus*, including the two ramified quadratic field types. Sections 8.8–8.12 then settle the full-group statement.

Example 8.7.1 (The field, algebra, and cover)**.** Take

$$F = \mathbb{Q}_3(\zeta_3), \qquad \varpi = \zeta_3 - 1, \qquad 3 = -\zeta_3^{-1}\varpi^2, \qquad q_F = 3.$$

Thus $v_F(3) = 2$ and $|3|_F = 3^{-2} = 1/9$. The compatible covering parameters in [JC26a] and Chapter 4 are respectively $c = 1$ and $c_v = 2 \pmod 3$. Let D/F be the unique quaternion division algebra. With $E_0 = F(i)$, $i^2 = -1$, its unramified quadratic maximal subfield, we may write

$$D = E_0 \oplus E_0 j, \qquad j^2 = \varpi, \qquad jx = \sigma(x)j \quad (x \in E_0),$$

where σ is the nontrivial automorphism of E_0/F. Proposition 4.3.1, gives, for every quadratic maximal subfield E/F, the covered-torus commutator

$$[x,y]_{\widetilde{E^\times}} = (N_{E/F}x, N_{E/F}y)^2_{F,3}(x,y)^{-1}_{E,3}. \tag{8.10}$$

The center C_E of $\widetilde{E^\times}$ is the preimage of $F^\times E^{\times 3}$ (Proposition 4.3.5), and

$$[E^\times : F^\times E^{\times 3}] = 9, \qquad b_D = 3, \qquad c_D = 27.$$

By the Heisenberg-type Stone–von Neumann theorem ([JC26a, Proposition 2.4]; Proposition 4.3.5), every genuine irreducible representation of $\widetilde{E^\times}$ has degree 3. This is *not* a degree calculation for representations of $\widetilde{D^\times}$.

8.7.1 The unramified quadratic torus

This subsection computes R_{E_0} on the unramified quadratic torus of the example. It then decomposes R_{E_0} into characters of a finite quotient.

Put $E_0 = F(i)$ as above. Since $X^2 + 1$ is irreducible over $\mathbb{F}_3$, E_0/F is unramified and its residue field is $\mathbb{F}_9$. For $t = a + bi$ with $a, b \in F$ the reduced norm and trace give the elementary identity

$$R_{E_0}(a+bi) = \frac{|a^2+b^2|_F}{|3a^2-b^2|_F}. \tag{8.11}$$

Multiplying t by an element of $F^\times$, suppose $a, b \in \mathscr{O}_F$ and at least one is a unit. Then $a^2 + b^2$ is always a unit. If b is a unit, so is $3a^2 - b^2$. If $b = \varpi u$, then

$$3a^2 - b^2 = 3(a^2 + \zeta_3 u^2),$$

and a is a unit; the reduction of $a^2 + \zeta_3 u^2$ modulo ϖ is either 1 or 2, never 0. Hence

$$R_{E_0}(a+bi) = \begin{cases} 9, & b \in \varpi\mathscr{O}_F, \\ 1, & b \in \mathscr{O}_F^\times. \end{cases} \tag{8.12}$$

The finite residue quotient is

$$B_0 = k_{E_0}^\times / k_F^\times \simeq C_4.$$

The condition $b \in \varpi \mathscr{O}_F$ says exactly that the image of t in B_0 is the identity. Denote its four characters by $\chi_0 = 1, \chi_1, \chi_2, \chi_3$. The regular character of C_4 equals 4 at the identity and 0 elsewhere. Thus

$$\boxed{R_{E_0} = 1 + 2\,\mathrm{reg}_{C_4} = 3\chi_0 + 2\chi_1 + 2\chi_2 + 2\chi_3.} \tag{8.13}$$

This is a positive integral Fourier decomposition of degree $3+2+2+2 = 9$; it is stronger than merely checking that R_{E_0} takes integer values.

8.7.2 Both ramified quadratic tori

This subsection does the same computation for the two ramified quadratic extensions of F. Proposition 8.7.2 then collects the three cases.

Every ramified quadratic extension of F can be written $E_d = F(j_d)$, $j_d^2 = d = u\varpi$ with $u \in \mathscr{O}_F^\times$. There are two isomorphism classes, distinguished by the square class of u. For $t = a + b j_d$,

$$R_{E_d}(a + b j_d) = \frac{|a^2 - db^2|_F}{|3a^2 + db^2|_F}. \tag{8.14}$$

Again normalize a, b to be integral and not both divisible by ϖ. The terms $3a^2$ and db^2 have different valuations whenever both a and b are units. A direct valuation calculation therefore gives

$$R_{E_d}(a + b j_d) = \begin{cases} 9, & a \in \mathscr{O}_F^\times, \quad b \in \varpi \mathscr{O}_F, \\ 3, & a, b \in \mathscr{O}_F^\times, \\ 1, & a \in \varpi \mathscr{O}_F, \quad b \in \mathscr{O}_F^\times. \end{cases} \tag{8.15}$$

For example, in the original cyclic algebra $d = \varpi$,

t	j	$1+j$	$1+\varpi j$
$R(t)$	1	3	9
$(L_D 1)(t^*)$	3	9	27

for the central-character normalization of Example 8.7.1. The three values illustrate why the unramified torus alone is not an adequate test.

There is a natural finite quotient

$$B_d = E_d^\times / (F^\times U_1) \simeq C_2 \times C_3,$$

where $U_1 \subset \mathscr{O}_{E_d}^\times$ consists of $a + b j_d$ with $a \in \mathscr{O}_F^\times$ and $b \in \varpi \mathscr{O}_F$; equivalently $U_1 = \mathscr{O}_F^\times (1 + \mathfrak{p}_{E_d}^2)$. The C_2 coordinate records the parity of the E_d-valuation, and on even elements the C_3 coordinate is $\bar{c} \in \mathbb{F}_3$ for $1 + c j_d$ with $c \in \mathscr{O}_F$ read modulo ϖ. The multiplication law

$$(1 + c j_d)(1 + c' j_d) = (1 + dcc') \left(1 + \frac{c + c'}{1 + dcc'} j_d\right)$$

shows that $\bar{c}$ is additive, and $j_d^2 = d \in F^\times$ provides the order-two factor.

Write ε for the nontrivial character of C_2 and ξ for a character of order 3 of C_3. The identity (8.15) says that R_{E_d} has values 9 at the identity of B_d, 3 at its two nonidentity elements in C_3, and 1 on its three odd elements. Consequently

$$\boxed{\begin{aligned} R_{E_d} &= 1 + \mathrm{reg}_{C_2} + \mathrm{reg}_{C_2 \times C_3} \\ &= 3 \cdot 1 + 2\varepsilon + \xi + \xi^2 + \varepsilon\xi + \varepsilon\xi^2. \end{aligned}} \tag{8.16}$$

Here the regular representations are inflated from the indicated quotients. All coefficients are nonnegative integers and sum to $3+2+1+1+1+1=9$. This applies to *both* ramified quadratic isomorphism classes, independently of u.

Proposition 8.7.2 (All quadratic torus restrictions pass an integral-character test)**.** *In the wild example $F = \mathbb{Q}_3(\zeta_3)$, $r = 2$, $n = 3$, let D/F be the quaternion division algebra. The function R_E is the character of a nine-dimensional permutation representation of a finite quotient of $E^\times/F^\times$, for every quadratic maximal subfield $E \subset D$. More precisely, for unramified E it is $1 \oplus 2\,\mathbb{C}[C_4]$, and for either ramified E it is $1 \oplus \mathbb{C}[C_2] \oplus \mathbb{C}[C_2 \times C_3]$. For a suitable genuine central character inducing the trivial ordinary central character, the restriction of $L_D\mathbf{1}$ to $\widetilde{E^\times}$ is therefore an* actual genuine representation character *of total dimension 27, namely*

E/F	*multiplicities of three-dimensional genuine torus irreducibles*
unramified	$(3,2,2,2)$
ramified, either type	$(3,2,1,1,1,1)$.

These multiplicities are invariant under the nontrivial F-automorphism of each E.

Proof. A p-adic field of odd residue characteristic has four square classes; hence F has exactly three quadratic field extensions, one unramified and two ramified. Every quadratic field extension embeds in the quaternion division algebra. Equations (8.13) and (8.16) prove the first assertion. Lemma 8.6.1, together with the degree-three torus Stone–von Neumann theorem, converts those Fourier coefficients into the displayed multiplicities; their weighted dimensions are $3 \cdot 9 = 27$. For the trivial ordinary character one can choose $\widetilde{\theta}$ with $\widetilde{\theta}(a^*) = 1$: the odd-$n$ scalar star kills $\mu_3 \subset F^\times$ (Section 4.3), and a faithful character of μ_3 extends across the abelian scalar central group modulo the star image. The nontrivial Galois automorphism acts by inversion on the respective finite quotients' character parameters (with the C_2 factor fixed), so the displayed multiplicities are invariant. □

Remark 8.7.3 (Standing conventions for the rest of the chapter)**.** From here on $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, D is the quaternion division algebra, $E_0 = F(i)$, and j, ϖ are as in Example 8.7.1. We write $\mathcal{O}_D$ for the maximal order, $\mathfrak{P} = \mathfrak{P}_D = j\mathcal{O}_D$ for its maximal ideal, v_D for the valuation with $v_D(j) = 1$ (so $v_D(\varpi) = 2$ and $v_D(3) = 4$), and

$$K^m = 1 + \mathfrak{P}^m, \qquad U_m = F^\times K^m, \qquad \widetilde{U}_m = p^{-1}(U_m) \subset \widetilde{H} \qquad (m \geq 1).$$

Chapter 6's log records that the standard scalar cocycle is trivial; accordingly $\widetilde{Z} \simeq \mu_3 \times F^\times$ and a genuine central character has the form $\widetilde{\theta}(\zeta, a) = \zeta\,\theta_0(a)$. For $\rho = \mathbf{1}$ the compatibility $\widetilde{\theta}(a^*) = 1$ forces $\theta_0^3 = 1$; we fix one such θ_0 (there are 81; by Remark 8.9.6 the choice is immaterial) and note that $\widetilde{\theta}(\widetilde{Z}) = \mu_3$. The central character θ_ρ of any ρ to which L_D applies is trivial on μ_3, since $\theta_\rho(\zeta) = \widetilde{\theta}(\zeta^*) = \widetilde{\theta}(1) = 1$.

8.8 The pro-3 structure of K^1 and the cube map

This section describes the pro-3 group on which the rest of the argument takes place. It determines the good elements of that group by studying the cube map, and it computes the transferred trivial character there.

The whole wild phenomenon lives on the pro-3 group

$$Q := K^1F^\times/F^\times \;\simeq\; K^1/U_F^{(1)}, \qquad Q_m := \text{image of } K^m \quad (m \geq 1),$$

where $U_F^{(k)} = 1+\mathfrak{p}_F^k$ and $K^1 \cap F^\times = U_F^{(1)}$. Since $U_F^{(k)} = F \cap K^{2k}$, the successive quotients are

$$Q_m/Q_{m+1} \simeq \begin{cases} \mathbb{F}_9, & m \text{ odd}, \\ \mathbb{F}_9/\mathbb{F}_3, & m \text{ even}, \end{cases} \qquad [Q : Q_6] = 3^8, \quad [Q : Q_3] = 27, \quad [Q_3 : Q_6] = 3^5. \tag{8.17}$$

Every element of $D^\times \setminus F^\times$ is regular and lies in a unique maximal subfield; in particular, Q is torsion-free, because $E^\times/F^\times$ has no 3-torsion for quadratic E (Lemma 4.3.4) and no 2-torsion inside a pro-3 group. The group $D^\times/F^\times K^1$ has order 8; the subgroup

$$P_0 := \langle 1+i,\ j\rangle F^\times/F^\times \simeq D_8 \tag{8.18}$$

is a complement to Q in $D^\times/F^\times$ (indeed $(1+i)^4 = -4$, $j^2 = \varpi$ and $j(1+i)j^{-1} = 2(1+i)^{-1}$ in $D^\times$), and every finite 2-subgroup of $D^\times/F^\times$ is conjugate into P_0 (Schur–Zassenhaus).

8.8.1 The cube map

This subsection collects the properties of the cube map on Q in one lemma. Its main output is the description of the good elements. They have depth exactly 3 or depth at least 6.

Lemma 8.8.1. *(a) For $k \geq 3$ the cube map $K^k \to K^{k+4}$ is bijective.*

(b) The cube map $Q \to Q$ is injective.

(c) Let $t \in Q_1 \setminus Q_2$ have leading coefficient $\bar{w} \in \mathbb{F}_9^\times$ (that is, $t = 1 + w\pi + \cdots$ with $\pi = j$). Then $t^3 \in Q_3 \setminus Q_4$ with leading coefficient $\bar{w}^5$; in particular every nonzero leading coefficient occurs.

(d) $\mathrm{Cubes}(Q_2) = Q_6$.

(e) The set $\mathrm{Good}(Q)$ *of images of good elements is the set of cubes, and*

$$\mathrm{Good}(Q) = G_3 \sqcup Q_6, \qquad G_3 := \mathrm{Cubes}(Q_1 \setminus Q_2) \subset Q_3 \setminus Q_4.$$

Thus a good element of Q has depth exactly 3 or depth at least 6.

(f) G_3 is stable under multiplication by Q_6.

(g) $V := Q_3/Q_6$ is an elementary abelian group of order 3^5, and the image $\bar{G}_3$ of G_3 in V has exactly 72 elements, with $\bar{G}_3 = -\bar{G}_3$; it spans V.

Proof. (a) For $u \in \mathfrak{P}^k$, $(1+u)^3 = 1+3u+3u^2+u^3$ with $3u \in \mathfrak{P}^{k+4}$, $3u^2 \in \mathfrak{P}^{2k+4}$, $u^3 \in \mathfrak{P}^{3k}$; for $k \geq 3$ both error terms lie in $\mathfrak{P}^{k+5}$, and $u \mapsto 3u$ is a bijection $\mathfrak{P}^k/\mathfrak{P}^{k+1} \to \mathfrak{P}^{k+4}/\mathfrak{P}^{k+5}$. Successive approximation gives surjectivity; injectivity follows from (b) and $\mu_3 \cap K^3 = 1$.

(b) If $t^3 = t'^3$ in Q with $t \neq 1$, both t, t' lie in the torus of the regular element t^3 and $(t'/t)^3 \in F^\times$, hence $t'/t \in F^\times$ by Lemma 4.3.4.

(c) With $t = 1+u$, $u = w\pi + \cdots$: $3u \in \mathfrak{P}^5$, $3u^2 \in \mathfrak{P}^6$ and $u^3 \equiv w\sigma(w)w\pi^3 = \mathrm{N}(\bar{w})\bar{w}\,\pi^3 \pmod{\mathfrak{P}^4}$, since $\pi a = \sigma(a)\pi$ for $a \in E_0$; and $\mathrm{N}(\bar{w})\bar{w} = \bar{w}^5$, with 5 prime to 8.

(d) Cubes of K^2 lie in K^6. Conversely every element of Q_6 is regular or trivial and lies in some torus $A = E^\times \cap F^\times K^1/F^\times$. For $E = E_0$ (so $E_0 \cap K^6 = U_{E_0}^{(3)}$): $U_{E_0}^{(4)}$ consists of cubes of $U_{E_0}^{(2)}$ (cubing $U_{E_0}^{(k)} \to U_{E_0}^{(k+2)}$ is bijective for $k \geq 2$, as $e(E_0/\mathbb{Q}_3) = 2$), and $U_{E_0}^{(3)}/U_{E_0}^{(4)} \simeq \mathbb{F}_9$ is covered by $U_F^{(3)}$ (the $\mathbb{F}_3$-part) together with the cubes $(1+\varpi w)^3 \equiv 1+\varpi^3(w^3-w)$, whose leading coefficients fill the Artin–Schreier image $\mathbb{F}_3 i$. For ramified E (so $E \cap K^6 = U_E^{(6)}$, $e(E/\mathbb{Q}_3) = 4$): $U_E^{(7)}$ consists of cubes of $U_E^{(3)}$, and $U_E^{(6)}/U_E^{(7)} \simeq \mathbb{F}_3$ is covered by $U_F^{(3)}$. In both cases the cube root lies in $E \cap K^2$.

(e) The good elements of H are $F^\times \cdot \{\text{cubes}\}$ (Proposition 4.3.5(a), applied in each torus), so $\mathrm{Good}(Q) = \mathrm{Cubes}(Q)$. By (c), cubes of depth-one elements have depth exactly 3; by (d), cubes of Q_2 exhaust Q_6.

(f) Fix $t \in Q_1 \setminus Q_2$ with leading coefficient a and put $\tau = t-1$. For $u \in \mathfrak{P}^k$ one has $(t(1+u))^3 = t^3\big(1+L(u)+O(\mathfrak{P}^{2k})\big)$ with $L = 1+\mathrm{Ad}(t^{-1})+\mathrm{Ad}(t^{-2})$, and $L(u) \equiv (\mathrm{Ad}(t^{-1})-1)^2 u \equiv [[u,\tau],\tau]$ $(\mathrm{mod}\ \mathfrak{P}^{k+3})$ (ring commutators), because $3u \in \mathfrak{P}^{k+4}$. For $u = b\pi^3$ the double commutator is $-aD\pi^5$ with $D = b\sigma(a) - a\sigma(b)$, which vanishes iff $b \in \mathbb{F}_3 a$. Hence cubing maps $W := t\cdot(1+\mathbb{F}_3 a\pi^3+\mathfrak{P}^4)$ into t^3Q_6. Cubing is a local homeomorphism near regular elements (its differential L is invertible, its determinant on $\mathfrak{d}/\mathfrak{f}$ having absolute value $|3|_F\, R(t)^{-2} = 3^{-4}$ for depth-one t), so the image of W is open in t^3Q_6, and its measure is $\mu(W)\cdot 3^{-4} = 3^{-4}\cdot 3^{-4} = 3^{-8} = \mu(Q_6)$. Therefore cubing maps W onto t^3Q_6, i.e. $t^3Q_6 \subset G_3$. (The finite verification in Appendix A.2, script 2, confirms this statement in Q/Q_7.)

(g) $[K^3,K^3] \subset K^6$ and $(K^3)^3 \subset K^7$, so V is elementary abelian of order $[Q_3 : Q_6] = 3^5$. By (f) and the Jacobian computed in (f), $\mu(G_3) = \frac{8}{9}\cdot 3^{-4}$, which is one third of $\mu(Q_3 \setminus Q_4)$; hence $|\bar{G}_3| = \frac{1}{3}\cdot 216 = 72$. The explicit description is in Proposition 8.8.2; spanning and the symmetry $\bar{G}_3 = -\bar{G}_3$ are read off from it. □

8.8.2 Coordinates on V and the set $\bar{G}_3$

This subsection introduces coordinates on the finite group $V = Q_3/Q_6$ and describes the good classes of depth three explicitly in them. The later finite computations are carried out in these coordinates.

Write $\mathfrak{P}^3 = \varpi\mathscr{O}_{E_0}j \oplus \varpi^2\mathscr{O}_{E_0}$ and $\mathfrak{P}^6 = \varpi^3\mathscr{O}_D$. Since $(1+X)(1+Y) \equiv 1+X+Y$ modulo $\mathfrak{P}^6$ for $X, Y \in \mathfrak{P}^3$, the map $1+X \mapsto X$ identifies K^3/K^6 with the additive group $\mathfrak{P}^3/\mathfrak{P}^6$, and writing

$$X \equiv \varpi(A+B\varpi)j + \varpi^2 C, \qquad A,B,C \in \mathbb{Z}_9 = W(\mathbb{F}_9),$$

gives coordinates $(\bar{A},\bar{B},\bar{C}) \in \mathbb{F}_9^3$ (levels $3,5,4$ respectively). The scalars $U_F^{(2)}$ contribute exactly the $\mathbb{F}_3$-line in the C-coordinate. Hence

$$V = Q_3/Q_6 \simeq \mathbb{F}_9 \oplus (\mathbb{F}_9/\mathbb{F}_3) \oplus \mathbb{F}_9 \ni (\bar{A},\ \bar{C} \bmod \mathbb{F}_3,\ \bar{B}), \tag{8.19}$$

and we use the $\mathbb{F}_3$-basis $(A_{\mathrm{re}}, A_{\mathrm{im}}, C_{\mathrm{im}}, B_{\mathrm{re}}, B_{\mathrm{im}})$ in the tables.

Proposition 8.8.2 ($(\mathbf{P}^{\mathrm{c}})$ The good depth-three classes). *With* $\mathrm{N}(a) = a\bar{a}$ *and* $\mathrm{Tr}(a) = a + \bar{a}$ *on* $\mathbb{F}_9$,

$$\bar{G}_3 = \Big\{ \big(a,\ x_4,\ y + \mathrm{N}(a)\big(\mathrm{Tr}(a\bar{y}) + x_4^2 - 1\big)a\big) \ :\ a \in \mathbb{F}_9^\times,\ x_4 \in \mathbb{F}_3 i,\ y \in \mathbb{F}_9 \Big\}, \tag{8.20}$$

a set of 72 *elements forming* 36 *lines through* 0 *(each line is* $\bar{G}_3 \cap \bar{A}_E$ *for the ramified tori* E *whose image in* V *is a fixed plane). Explicitly, for* $a \in \mathbb{F}_9^\times$ *and* $z \in \mathbb{F}_3\, i\bar{a}^{-1}$,

$$\bar{G}_3 = \{(a, 0, z - \mathrm{N}(a)a)\} \cup \{(a, \pm i, z + \mathrm{N}(a)a)\}.$$

Proof. For a ramified torus $E = F(j')$ with $j' = c\pi + c_2\pi^2 + c_3\pi^3 + \cdots$ pure (so $\mathrm{Tr}(c_2) = 0$, $j'^2 = u'\varpi$ with $\bar{u}' = \mathrm{N}(c)$), the cubes of depth-one elements $1 + j'w$, $w \in \mathscr{O}_E^\times$, are $\equiv 1 + (d'w_0^3 - \varpi^2 w_0)\, j'$ modulo $\mathfrak{P}^6$ and scalars, where $w \equiv w_0 \in \mathscr{O}_F^\times$. Reading off coordinates gives the vectors $\pm(\mathrm{N}(c)c, \mathrm{N}(c)c_2, \mathrm{N}(c)c_3 + (\bar{u}_1' - 1)c)$ with $\bar{u}_1' = \mathrm{Tr}(c\bar{c}_3) + c_2^2$, which after the substitution $a = \mathrm{N}(c)c$, $y = \mathrm{N}(c)c_3$ is (8.20). The map $y \mapsto y + \mathrm{N}(a)\,\mathrm{Tr}(a\bar{y})a$ has kernel $\mathbb{F}_3 a$ and image $\mathbb{F}_3\, i\bar{a}^{-1}$, giving the second description and the count $8 \cdot 3 \cdot 3 = 72$. The formula was verified by exact enumeration of the 648 depth-one classes modulo $K^4 U_F^{(1)}$ (Appendix A.2, script 1). □

8.8.3 The transferred trivial character on $\widetilde{U}_1$

This subsection computes the values of $L_D\mathbf{1}$ on the preimage $\widetilde{U}_1$ of U_1. In particular it shows that the restriction is supported on $\widetilde{U}_3$.

Lemma 8.8.3. *Put* $\Phi := L_D\mathbf{1}|_{\widetilde{U}_1}$ *and, for a good* $g \in U_1$, *write* $g = zt^3$ *with* $z \in F^\times$ *and* $t \in K^1$ *($t = 1$ if g is central); its* star lifts *are the three elements* $\zeta\tilde{z}t^*$, $\zeta \in \mu_3$. *Then*

$$\Phi(\zeta\tilde{z}t^*) = \begin{cases} 27\,\zeta\,\widetilde{\theta}(\tilde{z}), & t \in F^\times K^2 \ (\textit{i.e. } g \in U_6), \\ 9\,\zeta\,\widetilde{\theta}(\tilde{z}), & t \in F^\times K^1 \setminus F^\times K^2 \ (\textit{i.e. } g \in F^\times G_3), \end{cases} \tag{8.21}$$

and Φ *vanishes at all lifts of non-good elements. In particular* Φ *is supported on* $\widetilde{U}_3$.

Proof. By (8.1), $\Phi(\tilde{z}t^*) = 3\widetilde{\theta}(\tilde{z})R_E(t)$ with $E = F(t)$. If $t \in F^\times K^2$ and $E = E_0$, then $t \in F^\times U_{E_0}^{(1)}$ and $R = 9$ by (8.12); if E is ramified, $t \in F^\times U_E^{(2)}$, which is the case $a \in \mathscr{O}_F^\times$, $b \in \varpi\mathscr{O}_F$ of (8.15), so $R = 9$. If $t \in F^\times K^1 \setminus F^\times K^2$, then E is ramified ($E_0 \cap K^1 \subset K^2$) and $t \in F^\times(U_E^{(1)} \setminus U_E^{(2)})$, the case a, b units, so $R = 3$. Good elements of U_1 are $F^\times \cdot \mathrm{Cubes}(K^1)$, whose images in Q are $G_3 \sqcup Q_6$ by Lemma 8.8.1(e). □

8.9 Reduction to the pro-3 Sylow subgroup

This section reduces Target A to a statement about the group $\widetilde{U}_1$. It passes to a finite quotient of $\widetilde{H}$, treats the elements of 2-power order directly, and computes the norm of the restricted function. Theorem 8.9.5 combines these steps.

8.9.1 A finite quotient

This subsection replaces $\widetilde{H}$ by a finite quotient Γ on which $L_D\mathbf{1}$ is defined, so that the character theory of finite groups applies. It also shows that the toral restrictions of Section 8.7 descend to Γ.

Let $\phi := L_D\mathbf{1}$. By Chapter 6 the row of ϕ in (8.4) is finite: $\phi = \sum_{\tau\in S} a_\tau \Theta_\tau$ with S finite. Each $\tau \in S$ is a finite-dimensional smooth representation with central character $\widetilde{\theta}$, so $\ker\tau$ is an open normal subgroup of $\widetilde{H}$ with $\ker\tau \cap \widetilde{Z} = \ker\widetilde{\theta}$. Put

$$N := \widetilde{U}_1 \cap \bigcap_{\tau\in S} \ker\tau \cap N_6, \tag{8.22}$$

where $N_6 \subset \widetilde{U}_6$ is the open normal subgroup introduced in Definition 8.10.1 below (for the present section one may take $N_6 = \widetilde{U}_m$ for any large m). Then N is open and normal in $\widetilde{H}$, $N \cap \widetilde{Z} = \ker\widetilde{\theta}$, and

$$\Gamma := \widetilde{H}/N \quad \text{is finite}, \qquad T := \widetilde{Z}N/N \simeq \widetilde{Z}/\ker\widetilde{\theta} = \mu_3 \text{ is central}, \qquad \Gamma_0 := \widetilde{U}_1/N.$$

Γ_0 is a normal 3-subgroup of index 8 with $\Gamma/\Gamma_0 \simeq D_8$, hence the unique Sylow 3-subgroup; ϕ and $\Phi = \phi|_{\widetilde{U}_1}$ are class functions on Γ and Γ_0. For a maximal subfield E write $T_E := \widetilde{E^\times}N/N \subset \Gamma$.

Lemma 8.9.1 (Toral restrictions descend)**.** *For every maximal subfield E, $\phi|_{T_E}$ is the character of a representation of T_E.*

Proof. By Proposition 8.7.2, $\phi|_{\widetilde{E^\times}}$ is the character of a (finite-dimensional, genuine) representation W. Since $\phi(gm) = \phi(g)$ for $m \in N \cap \widetilde{E^\times}$, in particular $\Theta_W(m) = \Theta_W(1) = \dim W$, so m acts trivially on W. □

8.9.2 Two-groups

This subsection treats the elements of 2-power order in Γ. It computes the restriction of $L_D\mathbf{1}$ to the relevant 2-subgroup and describes the centralizers of 2-elements. Both facts are used in the proof of the reduction theorem.

Lemma 8.9.2 (The star section on 2-groups)**.** *Let $\hat{P}_0 \subset \widetilde{H}$ be the preimage of $P_0F^\times$ and $\Gamma_{P_0} := \hat{P}_0N/N$. Then $\Gamma_{P_0} \simeq \mu_3 \times D_8$, the elements of 2-power order form the unique complement $\tilde{P}_0 \simeq D_8$ to μ_3, and the map $u \mapsto \sigma(u) := (u^{1/3})^*$ ($u^{1/3}$ the cube root in the group P_0 of order 8) is the inverse of the projection $\tilde{P}_0 \to P_0$. Consequently*

$$\phi|_{\Gamma_{P_0}} = 3\,\psi_P + 3\sum_\pi (\dim\pi)\,\Theta_\pi, \tag{8.23}$$

where ψ_P is the genuine linear character trivial on $\tilde{P}_0$ and π runs over the genuine irreducible representations of Γ_{P_0}; in particular $\phi|_{\Gamma_{P_0}}$ is a character.

Proof. $\hat{P}_0 \cap \widetilde{U}_1 = \widetilde{Z}$ because $P_0 \cap Q = 1$, so $\hat{P}_0 \cap N = \ker\widetilde{\theta}$ and Γ_{P_0} is a central extension of D_8 by $T = \mu_3$. Since $|D_8|$ is prime to 3, $H^2(D_8,\mu_3) = H^1(D_8,\mu_3) = 0$: the extension splits and the complement is unique; it consists of the elements of 2-power order. On each cyclic subgroup of P_0 the star map is a homomorphism (it is a homomorphism on every covered torus, as $(\tilde{x}\tilde{y})^3 = \tilde{x}^3\tilde{y}^3[\tilde{y},\tilde{x}]^3$

with $[\tilde{y},\tilde{x}] \in \mu_3$), so $\sigma(u)^8 = 1$ and $\sigma(u) \in \tilde{P}_0$ is the unique element of the complement over u. Every nontrivial $u \in P_0$ is of order 2 or 4 in a quadratic torus and has $R_E(u) = 1$ by (8.12)–(8.15) (the cases b a unit, resp. $a \in \varpi\mathscr{O}_F$), so by (8.1) $\phi(\zeta\tilde{z}\sigma(u)) = 3\zeta\widetilde{\theta}(\tilde{z})$ for $u \neq 1$ and $= 27\zeta\widetilde{\theta}(\tilde{z})$ for $u = 1$. That is $3\psi_P + 24\delta$, with δ the genuine delta function at μ_3, and $8\delta = \sum_\pi (\dim\pi)\Theta_\pi$. □

Lemma 8.9.3 (Centralizers of 2-elements)**.** *Let $y \in \Gamma$ be an element of 2-power order whose image in $\Gamma/\Gamma_0 \simeq D_8$ is nontrivial, $\eta \in D^\times$ a representative of that image in $D^\times/F^\times$ and $E_y = F(\eta)$. (For $\rho = \mathbf{1}$ every nontrivial 2-element qualifies, T being μ_3; in Theorem 8.13.1 the central $\mu_2 \subset T$ is excluded, as its centralizer is all of Γ_0.)*

(a) $C_{\Gamma_0}(y) \subset T_{E_y}$.

(b) If $B_2 \subset \Gamma$ is a 2-subgroup whose image in $\Gamma/\Gamma_0 \simeq D_8$ is noncyclic, then $C_{\Gamma_0}(B_2) = T$.

Proof. (a) Replacing y by y^2 if necessary (which does not change E_y and enlarges the centralizer) we may assume η has order 2 modulo $F^\times$, so $s := \mathrm{Ad}(\eta)$ is an involutive automorphism of the pro-3 group Q. For $x \in Q$ put $b := (x^{-1}s(x))^{-1/2}$ (square roots are unique in a pro-3 group) and $a := xb^{-1}$; one checks $s(a) = a$, $s(b) = b^{-1}$ and $x^{-1}s(x) = b^{-2}$. If $[y,x] \in N$ then $b^{-2} \in \bar{N}$ (the image of N in Q, an s-stable closed normal subgroup), hence $b \in \bar{N}$ and $x \in Q^s\bar{N}$. Finally $Q^s = \{x : \eta x\eta^{-1} \in \pm xF^\times\}$; the sign $-$ would give a nontrivial 2-torsion element of Q, so $Q^s = (E_y^\times \cap F^\times K^1)/F^\times$, and x lifts into $\widetilde{E_y^\times}N\widetilde{Z}$.

(b) The (noncyclic) image of B_2 in D_8 contains the central involution $\bar{a}^2$ (the class of i) and a reflection. By (a) applied to the class of i, an element $b \in C_{\Gamma_0}(B_2)$ is the image of $\tilde{z}u$ with u a lift of $u_0 \in U_{E_0}^{(1)}$. A reflection η_2 acts on E_0 by σ and commutes with b in Γ, so $\sigma(u_0)u_0^{-1} \in \bar{N}$, i.e. $u_0^{-2} \in \bar{N}F^\times$, hence $u_0 \in \bar{N}$ and $b \in T$. □

8.9.3 The norm of Φ

This subsection computes the norm of $L_D\mathbf{1}$ and of its restrictions to the subgroups $\widetilde{U}_m$. These norms are used in the reduction theorem and in the later sections.

Lemma 8.9.4. *With respect to the probability Haar measures, $\langle\phi,\phi\rangle_{\widetilde{H}} = 1$ and, for $m \geq 1$,*

$$\langle\Phi|_{\widetilde{U}_m},\Phi|_{\widetilde{U}_m}\rangle_{\widetilde{U}_m} = [Q_{m'} : Q_m], \qquad m' = \begin{cases} 1, & m \leq 3, \\ 2, & 4 \leq m \leq 6, \\ m-4, & m \geq 7. \end{cases}$$

In particular the norms for $m = 1,3,4,6$ are $1, 27, 27, 729$.

Proof. Weyl integration on the compact group H/Z reads $\int_{H/Z} f\,d\mu = \sum_E \frac{c_E}{2}\int_{E^\times/F^\times}\Delta(t)^2 f(t)\,dt$ for class functions f, with constants c_E fixed by $f \equiv 1$. On each torus the cube map $s \mapsto s^3$ is an injective endomorphism of $E^\times/F^\times$ with image the good elements, of index 9; it pushes Haar measure forward to $\frac{1}{9}$ of Haar measure on the image. Substituting $t = s^3$ and using $|\phi|^2(s^3) = 9R(s)^2$ and $R(s) = \Delta(s)/\Delta(s^3)$ gives $\int_{E^\times/F^\times}\Delta(t)^2|\phi|^2(t)\mathbf{1}_U(t)\,dt = \int\Delta(s)^2\mathbf{1}_U(s^3)\,ds$. For $U = H$ this is unitarity of the row. For $U = U_m$, note that $s^3 \in U_m$ iff $s \in U_{m'}$ by Lemma 8.8.1(c)–(e) and (a), so $\int_{U_m/Z}|\phi|^2 d\mu = \mu(U_{m'}/Z)$, and the normalized inner product on $\widetilde{U}_m$ is $\mu(U_{m'})/\mu(U_m) = [Q_{m'} : Q_m]$. □

8.9.4 The reduction theorem

This subsection states and proves the reduction. Target A holds if and only if the restriction of $L_D\mathbf{1}$ to $\widetilde{U}_1$ is a virtual character.

Theorem 8.9.5 (Reduction to the Sylow 3-subgroup)**.** *The following are equivalent.*

(i) (Target A) $L_D\mathbf{1}$ *is the character of a genuine irreducible representation of* $\widetilde{H}$ *with central character* $\widetilde{\theta}$.

(ii) $L_D\mathbf{1}$ *is a virtual character of* $\widetilde{H}$.

(iii) $\Phi = L_D\mathbf{1}|_{\widetilde{U}_1}$ *is a virtual character of* $\widetilde{U}_1$.

(iv) Φ *is the character of an irreducible representation of* $\Gamma_0 = \widetilde{U}_1/N$, *necessarily of dimension* 27.

Moreover, in that case the representation in (i) has dimension 27, *its restriction to* $\widetilde{U}_1$ *is irreducible, and it is unique.*

Proof. (i)⇒(ii)⇒(iii) are clear, and (iii)⇔(iv) by Lemma 8.9.4 ($\langle\Phi,\Phi\rangle = 1$, $\Phi(1) = 27 > 0$). Assume (iii). By Brauer's characterization of characters it suffices to show that $\phi|_B$ is a virtual character for every elementary subgroup $B = B_2 \times B_3 \subset \Gamma$ (B_ℓ an ℓ-group, one of them cyclic). If $B_2 = 1$, then $B \subset \Gamma_0$ and $\phi|_B = \Phi|_B$. If $B_2 = \langle y\rangle$ is cyclic and nontrivial, then $y \in T_{E_y}$ and $B_3 \subset C_{\Gamma_0}(y) \subset T_{E_y}$ by Lemma 8.9.3(a), so $B \subset T_{E_y}$ and Lemma 8.9.1 applies. If B_2 is noncyclic, then $B_3 \subset T$ by Lemma 8.9.3(b), so after conjugation $B \subset \Gamma_{P_0}$ and Lemma 8.9.2 applies. Hence ϕ is a virtual character of Γ; by unitarity (Lemma 8.9.4) and $\phi(1) = 27 > 0$ it is an irreducible character, which is (i). The restriction of that representation to $\widetilde{U}_1$ has character Φ, of norm one, hence is irreducible; a character determines the representation. □

Remark 8.9.6 (Independence of the choice of $\widetilde{\theta}$)**.** Two compatible choices $\widetilde{\theta}, \widetilde{\theta}'$ for $\rho = \mathbf{1}$ differ by a character θ_0 of $F^\times/F^{\times 3}$. Twisting a representation of $\widetilde{H}$ by $(\theta_0 \circ \mathrm{Nrd} \circ p)^{-1}$ changes its central character by $\theta_0(z)^{-2} = \theta_0(z)$ and multiplies its character at $\tilde{z}t^*$ by $\theta_0(\mathrm{Nrd}(zt^3))^{-1} = \theta_0(z)$; this is exactly the relation between $L_D\mathbf{1}$ computed with $\widetilde{\theta}$ and with $\widetilde{\theta}\theta_0$. So Target A does not depend on the choice.

8.10 The finite model and the defect function

This section studies the restriction of $L_D\mathbf{1}$ to $\widetilde{U}_3$ in a finite model. It introduces Hypothesis (H1) and a defect function, and it shows that Target A is equivalent to (H1).

Throughout this section we assume the following hypothesis, which is necessary for Target A (Lemma 8.10.3) and will be proved in Section 8.11.

Definition 8.10.1 (Hypothesis (H1))**.** Let $\psi_6 : \widetilde{U}_6 \to \mu_3$ be the function $\psi_6(\zeta\tilde{z}t^*) := \zeta\widetilde{\theta}(\tilde{z})$ on star lifts (all elements of $\widetilde{U}_6$ are star lifts, Lemma 8.8.1(d)). **Hypothesis (H1)**: ψ_6 is a (continuous) character of $\widetilde{U}_6$. Under (H1) put $N_6 := \ker\psi_6$, an open subgroup of $\widetilde{U}_6$, normal in $\widetilde{H}$ because the star section is conjugation equivariant and $\widetilde{\theta}$ is central.

Example 8.10.2 (What (H1) says over one maximal subfield)**.** Let $E \subset D$ be a maximal subfield, and consider the elements of $\widetilde{U}_6$ that lie over $E^\times$. The star map is a homomorphism on each covered torus (§8.2.2), and $\widetilde{Z}$ is central. So for two such elements $\zeta\tilde{z}t^*$ and $\zeta'\tilde{z}'t'^*$ with $t,t' \in E^\times$, the product is $\zeta\zeta'\tilde{z}\tilde{z}'(tt')^*$, and

$$\psi_6\big(\zeta\tilde{z}t^* \cdot \zeta'\tilde{z}'t'^*\big) = \zeta\zeta'\,\widetilde{\theta}(\tilde{z}\tilde{z}') = \psi_6(\zeta\tilde{z}t^*)\,\psi_6(\zeta'\tilde{z}'t'^*).$$

Thus ψ_6 is multiplicative on the part of $\widetilde{U}_6$ over any single maximal subfield, and this needs no hypothesis. The content of (H1) is the same identity for star lifts t^* and t'^* with t and t' in different maximal subfields.

Lemma 8.10.3. *If $L_D\mathbf{1} = \Theta_\tau$ with τ irreducible, then $\tau|_{\widetilde{U}_6} = 27\,\psi_6$; in particular (H1) holds.*

Proof. By Lemma 8.8.3, $|\Theta_\tau| = 27 = \dim\tau$ on $\widetilde{U}_6$, so $\widetilde{U}_6$ acts by scalars, by ψ_6. □

Under (H1) define

$$\bar{\Gamma} := \widetilde{U}_1/N_6 \text{ (order } 3^9), \qquad \widetilde{V} := \widetilde{U}_3/N_6, \qquad \mu_3 \simeq \widetilde{U}_6/N_6 \subset \widetilde{V} \to V = Q_3/Q_6 \to 0.$$

$\widetilde{V}$ is a central extension of the elementary abelian group V by μ_3; let $\omega : V \times V \to \mu_3$ be its commutator form. By Lemma 8.8.3, Φ factors through $\bar{\Gamma}$, with $\bar{\Gamma}_6 := \widetilde{U}_6/N_6 = \mu_3$ carrying $\Phi = 27\,\mathrm{id}$. The elements $\bar{\sigma}(g)$ ($g \in G_3$) are the images of the star lifts; by Lemma 8.8.1(f) and the next lemma they depend only on the class of g in $\bar{G}_3 \subset V$.

Lemma 8.10.4 (Three consequences of (H1))**.** *Assume (H1).*

(a) For $a \in H$ and $b \in U_6$, $\psi_6([\tilde{a},\tilde{b}]) = 1$. Hence for $i+j \geq 6$ the map $\kappa_{i,j}(a,b) := \psi_6([\tilde{a},\tilde{b}])$ on $K^i \times K^j$ is bimultiplicative, alternating, conjugation invariant, and trivial if a or b lies in U_6; $\kappa_{1,5}$ factors through $(Q_1/Q_2)\times(Q_5/Q_6) \simeq \mathbb{F}_9 \times \mathbb{F}_9$.

(b) $\kappa_{1,5}(a,b) = \zeta^{\beta\,\mathrm{Tr}(a\bar{b})}$ for a fixed sign $\beta \in \{\pm 1\}$.

(c) (H3) $\omega = 0$: $\widetilde{V}$ is abelian, hence $\widetilde{V} \simeq \mu_3 \times V$.

(d) (H2) For $g \in G_3$ and $u \in Q_6$, $\bar{\sigma}(gu) = \bar{\sigma}(g)$.

Proof. (a) $\ker\psi_6$ is normal in $\widetilde{H}$ and $\widetilde{U}_6/\ker\psi_6$ is central in $\widetilde{H}/\ker\psi_6$ (as ψ_6 is conjugation invariant), so $[\tilde{a},\tilde{b}] \in \ker\psi_6$ for $b \in U_6$. Bimultiplicativity follows from $[xy,z] = x[y,z]x^{-1}[x,z]$. A character of Q trivial on cubes and commutators factors through Q/Q_2, since $[Q,Q]Q^3 \supseteq Q_2$ (level by level: at even levels the commutators $[1+a\pi, 1+b\pi^{k-1}]$ produce the trace-zero line and U_F the rest; at odd levels they produce all of $\mathbb{F}_9$).

(b) Write $\kappa_{1,5}(a,b) = \zeta^{\mathrm{Tr}(\alpha ab)+\mathrm{Tr}(\beta a\bar{b})}$ with $\alpha,\beta \in \mathbb{F}_9$ (these span the bilinear forms). Conjugation by $\xi \in \mathscr{O}_{E_0}^\times$ multiplies both a and b by $\lambda = \xi/\bar{\xi} \in \mu_4$, forcing $\alpha = 0$; conjugation by j applies Frobenius to both, forcing $\beta \in \mathbb{F}_3$. It remains to see $\beta \neq 0$, and for this we use the ramified torus $E = F(j)$, $j^2 = \varpi$, explicitly. Its unit filtration is $U_E^{(k)}$ with $U_E^{(2l)}/U_E^{(2l+1)}$ represented by $1+a\varpi^l$ ($a \in \mathbb{F}_3$, scalars) and $U_E^{(2l+1)}/U_E^{(2l+2)}$ by $1+a\varpi^l j$ ($a \in \mathbb{F}_3$), of D-depth $2l+1$. For $a \in \mathscr{O}_F^\times$, $(1+aj)^3 = 1 + a^3\varpi j + 3a^2\varpi + 3aj$, and $v_E(3) = 4$, so the cube has leading term $a^3\varpi j$ at E-depth

3 and its next j-component $3aj$ at E-depth 5 is determined by a; cubes of $U_E^{(3)}$ lie in $U_E^{(7)}$. Hence modulo $F^\times E^{\times 3}$ the classes of $U_E^{(3)}$ and $U_E^{(k)}$, $k \geq 6$, are trivial, and $E^\times/F^\times E^{\times 3}$, which has order 9 (Proposition 4.3.5 and (8.10)), is the $\mathbb{F}_3$-vector space with basis the classes of $1+j$ (D-depth one, leading coefficient 1) and $1+\varpi^2 j$ (D-depth five, leading coefficient 1). The commutator pairing (8.10) is alternating and nondegenerate on this two-dimensional space (Proposition 4.3.5), so $[\widetilde{1+j}, \widetilde{1+\varpi^2 j}] \neq 1$; by (a) this commutator is $\kappa_{1,5}(1,1) = \zeta^{\beta\,\mathrm{Tr}(1)} = \zeta^{2\beta}$, whence $\beta \neq 0$. (Consistently, $N_{E/F}(1+aj) = 1 - a^2\varpi \in U_F^{(1)}$ and $N_{E/F}(1+a\varpi^2 j) \in U_F^{(5)}$ pair trivially in F, so the commutator is the Hilbert pairing $(1+j, 1+\varpi^2 j)_E^{-1}$ of E at the top level $1+5 = pe_E/(p-1) = 6$, which is nondegenerate [FV02, Ch. IV].)

(c) Let t have depth one and $y \in K^4$, $c := [\tilde{t}, \tilde{y}] \in \tilde{K}^5$, $d := [c^{-1}, \tilde{t}] \in \tilde{K}^6$, $e := [d^{-1}, \tilde{t}]$. Then $[\tilde{t}^3, \tilde{y}] = c^{t^2} c^t c$ and, expanding with $c^t = cd$, $c^{t^2} = cd^2 e$, one gets $[\tilde{t}^3, \tilde{y}] = c^3 \cdot Y$ with Y a product of d^3, e and commutators each involving d or e. As c^3 is a star lift, $d, e \in \widetilde{U}_6$ and (a) holds, $\psi_6([\tilde{t}^3, \tilde{y}]) = 1$. Thus $\omega(\bar{G}_3, V_4) = 0$ with $V_4 = Q_4/Q_6$, hence $\omega(V, V_4) = 0$ since $\bar{G}_3$ spans V (Lemma 8.8.1(g)). So ω is induced from an alternating form on $V/V_4 = Q_3/Q_4 \simeq \mathbb{F}_9$, a two-dimensional $\mathbb{F}_3$-space, on which conjugation by j acts by Frobenius, of determinant -1; invariance forces $\omega = 0$. So $\widetilde{V}$ is abelian. It also has exponent 3: for $v \in K^3$ the cube of any lift is the star lift of $v^3 \in K^9 \subset U_6$, on which $\psi_6 = 1$, and for $z \in F^\times$ the cube of $\tilde{z}$ is the star lift of z^3, on which $\psi_6 = \theta_0(z)^3 = 1$; so every element of $\widetilde{V} = \widetilde{U}_3/N_6$ cubes into N_6. An abelian group of exponent 3 is an $\mathbb{F}_3$-vector space, so the subgroup μ_3 has a complement. (Abelianness alone would not suffice: $C_9 \to C_3$ is a nonsplit abelian extension with genuine characters.)

(d) Let $g = t^3$, t of depth one with leading coefficient a, and $gu = t'^3$ with $t' = t(1+w)$, $w \in \mathbb{F}_3 a\pi^3 + \mathfrak{P}^4$ (proof of Lemma 8.8.1(f)). In $\bar{\Gamma}$ put $x = \tilde{t}$, $y = \widetilde{1+w}$, $c = [y,x]_L = yxy^{-1}x^{-1}$. The ring computation in Lemma 8.8.1(f) shows $[1+w, t] \in K^5$ with level-five coefficient in $\mathbb{F}_3\, ia$, so $c \in \bar{\Gamma}_5$. Since $\bar{\Gamma}_2$ has exponent 3 (cubes of $\widetilde{U}_2$ are star lifts), $c^3 = 1$; $[c,y] \in \bar{\Gamma}_8 = \mu_3$ equals ω-type pairing $\kappa_{5,3}$, trivial by (c); and $d := c^{-1}xcx^{-1} \in \bar{\Gamma}_6 = \mu_3$ is central. A direct expansion then gives $(xy)^3 = d^{-1}x^3y^3$, and $y^3 = 1$ in $\bar{\Gamma}$ because $(1+w)^3 \in Q_9$. Hence $\bar{\sigma}(gu)\bar{\sigma}(g)^{-1} = d^{-1} = \kappa_{1,5}(a, sia)^{\pm 1}$ for some $s \in \mathbb{F}_3$, and $\mathrm{Tr}(a\overline{sia}) = -s\,\mathrm{N}(a)\,\mathrm{Tr}(i) = 0$. □

8.10.1 The 27 characters and the defect function

This subsection describes the genuine characters of the finite group $\widetilde{V}$ and defines the defect function e. It then records what Target A forces on e.

Assume (H1). Fix a complement V_0 to μ_3 in $\widetilde{V}$; genuine characters of $\widetilde{V}$ are then $\mu_h(\zeta, v) = \zeta \cdot \zeta^{h(v)}$, $h \in V^\vee$. The star section gives a canonical lift $\bar{\sigma}(v) \in \widetilde{V}$ for each $v \in \bar{G}_3$, and we define the *defect function* $e : \bar{G}_3 \to \mathbb{F}_3$ by $\bar{\sigma}(v) = (\zeta^{e(v)}, v)$; it satisfies $e(-v) = -e(v)$ (star is a homomorphism on each torus line). The group $\mathrm{Heis} := Q_1/Q_3$ (extraspecial of order 27 and exponent 3) acts on $\widetilde{V}$ by conjugation, fixing μ_3; relative to V_0 the action is $(\zeta, v) \mapsto (\zeta\zeta^{\beta_s(v)}, sv)$ with $\beta_s \in V^\vee$, and equivariance of the star section gives

$$e(sv) - e(v) = \beta_s(v) \qquad (v \in \bar{G}_3,\ s \in \mathrm{Heis}). \tag{8.24}$$

Proposition 8.10.5 (What Target A forces)**.** *Suppose $L_D\mathbf{1} = \Theta_\tau$. Then, with $\Lambda \subset V^\vee$ the set of h with μ_h a constituent of $\tau|_{\widetilde{V}}$:*

(a) $\tau|_{\widetilde{V}}$ is multiplicity free, $|\Lambda| = 27$, and Heis *acts simply transitively on Λ (by $h \mapsto (h - \beta_s) \circ s^{-1}$);*

(b) for $v \in V \setminus (\bar{G}_3 \cup \{0\})$ the values $h(v)$, $h \in \Lambda$, are equidistributed in $\mathbb{F}_3$;

(c) for $v \in \bar{G}_3$ the value $-e(v)$ is taken 15 times and the other two values 6 times each;

(d) equivalently, for every $h \in V^\vee$ the number $N_e(h) := \#\{\text{lines } \ell \subset \bar{G}_3 \cup \{0\} : \ h|_\ell = -e|_\ell\}$ is 20 if $h \in \Lambda$ and 11 otherwise.

Conversely, if (H1) holds and e satisfies (d) for some 27-set Λ on which Heis *acts simply transitively, then $\tau_1 := \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda$, $\lambda = \mu_h$ for any $h \in \Lambda$, is irreducible with character Φ, and Target A holds by Theorem 8.9.5.*

Proof. By Lemma 8.9.4, $\langle \Phi|_{\widetilde{U}_3}, \Phi|_{\widetilde{U}_3} \rangle = 27 = \Phi(1)$, which forces multiplicity one with 27 constituents; Clifford theory for the normal subgroup $\widetilde{U}_3$ of $\widetilde{U}_1$, with $[\widetilde{U}_1 : \widetilde{U}_3] = 27$, gives (a). Evaluating $\sum_{h \in \Lambda} \mu_h = \Phi|_{\widetilde{V}}$ at $(1, v)$ and at $\bar{\sigma}(v)$ gives (b) and (c), using (8.21) and (H2); (d) is the Fourier dual statement (each line $\{v, -v\}$ contributes 2 or -1 to $\sum_{v \in \bar{G}_3} \zeta^{h(v)+e(v)} = 3N_e(h) - 36 \in \{24, -3\}$). The converse is Mackey irreducibility and the Frobenius character formula, whose Fourier transform is (b)–(c), together with the vanishing of Φ and of the induced character off $\widetilde{U}_3$. □

8.10.2 Symmetry pins down the defect function

This subsection uses the conjugation action of $D^\times$ to cut the possible defect functions down to a two-parameter family, and then determines the defect function. The conclusion is Theorem 8.10.7.

The group $D^\times$ acts on everything by conjugation through $G_{\mathrm{sym}} := D^\times / F^\times K^3$, of order 216, an extension of $D_8 = \langle \mathrm{Ad}(1+i), \mathrm{Ad}(j) \rangle$ by Heis. Its action on V is recorded in Appendix A.1; one checks $V^{D_8} = 0$ and $(V^\vee)^{D_8} = 0$. Since $|D_8|$ is prime to 3, there is a D_8-stable complement V_0, and it is unique. With respect to it, e is D_8-invariant. The D_8-orbits on $\bar{G}_3$ have sizes $4, 4, 8, 8, 8, 8, 8, 8, 8, 8$; six of them contain $-v$ together with v, forcing $e = 0$ there, and the other four form two pairs $\{O, -O\}$. Hence e is determined by two parameters in $\mathbb{F}_3$.

Proposition 8.10.6 (**(P^{c})** Uniqueness of the design)**.** *Among the 3^{10} D_8-invariant functions e on $\bar{G}_3$, exactly two are odd and satisfy condition (d) of Proposition 8.10.5, namely e_* and $-e_*$, where e_* is tabulated in Appendix A.1; it vanishes on 20 of the 36 lines and takes the values 1 and 2 on 8 lines each. Among the nine odd D_8-invariant e, exactly three admit linear β_s as in (8.24) for the generators $s = \mathrm{Ad}(1+j), \mathrm{Ad}(1+ij)$ of* Heis*: $e = 0$ and $e = \pm e_*$. For $e = \pm e_*$ the resulting* Heis*-action on $V^\vee$ is simply transitive on $\Lambda_\pm := \{h : N_{\pm e_*}(h) = 20\}$, $0 \in \Lambda_\pm$, and (b)–(c) hold. For $e = 0$ one has $\beta_s = 0$ for all s.*

Proof. Exact enumeration over $\mathbb{F}_3$ (Appendix A.2, scripts 3, 4, 6): the D_8-orbits, the counts $N_e(h)$ for all 243 covectors and all candidates, the linear extension problem (8.24), the orbit of 0 under the affine action, and the Fourier identities. The distribution of $N_0(h)$ is $\{9^{80}, 12^{90}, 15^{72}, 36^1\}$, that of $N_{\pm e_*}(h)$ is $\{11^{216}, 20^{27}\}$. □

Theorem 8.10.7 (**(P^{c})** Target A is equivalent to (H1))**.** *$L_D\mathbf{1}$ is a genuine irreducible character if and only if (H1) holds. In that case the defect function is $\pm e_*$, and $\tau|_{\widetilde{U}_1} \simeq \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda_0$ where λ_0 is the genuine character of $\widetilde{U}_3$ trivial on N_6 and on the unique D_8-stable complement V_0; explicitly*

$$\lambda_0(y) = \Big(\prod_{d \in D_8} \tilde{d}\, y\, \tilde{d}^{-1} \Big)^{-1} \in \mu_3 \qquad (y \in \widetilde{U}_3), \tag{8.25}$$

the product taken in the abelian group $\widetilde{V}$ (it lies in $\widetilde{V}^{D_8} = \mu_3$).

Proof. Necessity is Lemma 8.10.3. Assume (H1). By Lemma 8.10.4, (H2) and (H3) hold, so the defect function e is defined, is odd and D_8-invariant for the D_8-stable complement, and satisfies (8.24); by Proposition 8.10.6, $e \in \{0, e_*, -e_*\}$. If $e = 0$, then $\beta_s = 0$ for $s = \mathrm{Ad}(1+j)$; but for the depth-five vector $v = e_{B_{\mathrm{re}}}$ of the torus $F(j)$, fixed by s, $\zeta^{\beta_s(v)}$ is the commutator of lifts, i.e. $\kappa_{1,5}(1,1) = \zeta^{2\beta} \neq 1$ by Lemma 8.10.4(b). Hence $e = \pm e_*$, and the converse part of Proposition 8.10.5 applies with $\Lambda = \Lambda_\pm \ni 0$; the character μ_0 is λ_0, and (8.25) is the formula for the projection $\widetilde{V} = \mu_3 \times V_0 \to \mu_3$, since $\prod_d d \cdot x = \lambda_0(x)^8 = \lambda_0(x)^{-1}$ for $x \in \widetilde{V}$. Uniqueness of V_0: complements form a torsor under $V^\vee$, and $(V^\vee)^{D_8} = 0$. □

Remark 8.10.8. The two signs correspond to the two identifications of μ_3 with $\mathbb{Z}/3$ (equivalently, to replacing the cover by its opposite); the statement of Theorem 8.10.7 is symmetric. The Heis-cocycle β attached to e_* restricted to $V_5 = Q_5/Q_6$ reproduces $\kappa_{1,5}$ with $\beta = -1$, an independent consistency check between the design computation and the torus data.

8.11 The cover in the Kazhdan–Patterson model

It remains to prove (H1). We do this in an explicit model of the cover. By Lemma 4.2.3 and Proposition 4.2.4(a), the cover of $D^\times$ is the restriction, along

$$\iota : D^\times \hookrightarrow \mathrm{GL}_2(E_0), \qquad x + yj \longmapsto \begin{pmatrix} x & \varpi y \\ \sigma(y) & \sigma(x) \end{pmatrix}, \tag{8.26}$$

of the parameter-c_v Kazhdan–Patterson cover of $\mathrm{GL}_2(E_0)$, whose cocycle is $\sigma_{E_0}(g,h)\,(\det g, \det h)_{E_0}^{c_v}$ with σ_{E_0} the parameter-0 cocycle. We use the Kubota-type representative

$$\sigma_{E_0}(g,h) = \Big(\frac{x(gh)}{x(g)}, \frac{x(gh)}{x(h)\det g}\Big)_{E_0}, \qquad x\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} c, & c \neq 0, \\ d, & c = 0, \end{cases} \tag{8.27}$$

and write $\beta(g,h) \in \mathbb{Z}/3$ for the exponent of $\sigma_{E_0}(g,h)(\det g, \det h)_{E_0}^2$ (before the relabeling $\zeta \mapsto \zeta^{r'}$, which multiplies all exponents by 2 and is irrelevant for everything below). Elements of the cover are pairs (g,e) with $(g,e)(h,f) = (gh, e+f+\beta(g,h))$.

Lemma 8.11.1 ((**E**) The model is the cover)**.** *(a) (8.27) is a 2-cocycle on $\mathrm{GL}_2(E_0)$ whose restriction to the diagonal torus is $(a_1,b_2)_{E_0}$.*

(b) On scalars the twisted cocycle is trivial, so $z \mapsto (z,0)$ is a homomorphic section over $F^\times$; and for $z \in F^\times$, $g \in \mathrm{GL}_2(E_0)$, $\beta(z,g) = (z, x(g)\det g)_{E_0}$.

(c) For $x, y \in E_0^\times$, the commutator of $(\iota x, 0)$ and $(\iota y, 0)$ has exponent $(\mathrm{N}x, \mathrm{N}y)_F + (x,y)_{E_0}$, in accordance with Proposition 4.3.1 (before relabeling, $c = 1$).

(d) The extension so defined is isomorphic, as a central extension of $D^\times$ by μ_3, to the cover of Chapter 4; any two such isomorphisms differ by $\chi \circ \mathrm{Nrd}$ for a character χ of $F^\times$ with values in μ_3, which does not affect star lifts, good elements, or $L_D\mathbf{1}$.

Proof. (a) is Kubota's cocycle for GL_2: for SL_2 it is [Kub67, Thm. 2] and [Kub69, Ch. 1], and the extension to GL_2 by the factor $\det g$ in the second entry is the case $r=2$, $c=0$ of the Kazhdan–Patterson cocycle [KP84, §0.1], with the n-th power Hilbert symbol for the n-fold cover (the same formula serves in Section 8.14); it is the one external input of this section and is labeled **(E)**. (The twist $(\det g, \det h)^c$ is bimultiplicative, hence a cocycle, so the twisted formula is a cocycle as well.) As a safeguard against transcription errors, the cocycle identity was also checked on 250 random triples and the diagonal restriction on 100 random pairs by exact computation (script 7); this check is corroboration, not part of the proof. (b) With $x(zI)=z$, $x(zg)=zx(g)$, $\det(zI)=z^2$ one gets $\sigma(zI,g)=(x(g),z^{-1})=(z,x(g))$ and the twist contributes $(z^2,\det g)^2=(z,\det g)$; for $g=wI$ the two terms cancel, since $(z,w)_{E_0}=(z,w)_F^2$. (c) is a direct evaluation; it was also checked on 200 random pairs. (d) Steps 1–3 of the proof of Proposition 4.2.4(b), apply verbatim to two cocycles on $\mathrm{GL}_2(E_0)$ with the same diagonal restriction; the last assertion is Remark 4.8.1 (the star map $x^*=\tilde{x}^3$ absorbs $\chi\circ\det$). □

The Hilbert symbols. The cubic Hilbert symbols of F and of E_0 enter only through their classes. By Moore's theorem (cited in Proposition 4.2.4), a continuous Steinberg symbol $F^\times\times F^\times\to\mu_3$ is a power of the Hilbert symbol. We therefore computed the alternating bilinear forms on $F^\times/F^{\times 3}\simeq\mathbb{F}_3^4$ and $E_0^\times/E_0^{\times 3}\simeq\mathbb{F}_3^6$ annihilated by all Steinberg relations $(a,1-a)=1$ (exact linear algebra over $\mathbb{F}_3$ on several thousand relations, script 7); in both cases the solution space is one-dimensional and nondegenerate, so each symbol is determined up to the common sign, and the relative sign is fixed by norm compatibility $(a,b)_{E_0}=(a,\mathrm{N}b)_F$, which the computed forms satisfy. Classes modulo cubes are computed from the filtrations $U^{(k)}$ by explicit discrete logarithms. The hand arguments below use only the following standard facts about the cubic symbol of E_0, for which $e(E_0/\mathbb{Q}_3)=2$ and $pe/(p-1)=3$: bimultiplicativity and antisymmetry; the Steinberg relation $(u,1-u)=1$; Galois invariance $(\sigma u,\sigma v)=(u,v)$; triviality on cubes, in particular on -1, on μ_8 and on $U_{E_0}^{(4)}\subset E_0^{\times 3}$; and the depth bound $(U_{E_0}^{(i)},U_{E_0}^{(j)})=1$ for $i+j\geq 4$ (see [FV02, Ch. IV] or [Ser79, Ch. XIV]). A consequence used repeatedly: for $m\in U_{E_0}^{(3)}$ and any $u\in E_0^\times$,

$$(u,m)_{E_0}=(\varpi,m)_{E_0}^{v_{E_0}(u)}, \tag{8.28}$$

because $u=\varpi^{v(u)}\xi u_1$ with $\xi\in\mu_8$ and $u_1\in U_{E_0}^{(1)}$. The computed tables are consistent with these facts (for instance $(1+\varpi,1+\varpi^3)=(1+\varpi^2,1+\varpi^2 i)=1$ while $(1+\varpi,1+\varpi^2)\neq 1$). With ϖ, $1+\varpi$, $1+\varpi^2$, $1+\varpi^3$ as basis of $F^\times/F^{\times 3}$ the form is $(\varpi,1+\varpi^3)=(1+\varpi,1+\varpi^2)=\zeta$ and all other basis pairings trivial; the 6×6 matrix for E_0 is in Appendix A.1.

8.11.1 The cube identity and the star section on K^6

This subsection proves (H1). It first shows that the standard section of the model is compatible with cubes. It then identifies the star section on K^6 and verifies the identity that makes ψ_6 a character.

Proposition 8.11.2 (Cubes of the standard section)**.** *Let $w=x+yj\in D^\times$ with $y\neq 0$, $T:=\mathrm{Tr}_{E_0/F}(x)\neq 0$ and $A:=x^2+x\sigma(x)+\sigma(x)^2+\varpi\,\mathrm{N}(y)\neq 0$. Then $(w,0)^3=(w^3,0)$ in the cover. The hypotheses hold for every $w\in K^1\setminus E_0$; for $w\in E_0^\times$ the conclusion holds as well.*

Proof. Put $n = \mathrm{Nrd}(w) = \mathrm{N}(x) - \varpi\,\mathrm{N}(y)$. From $w^2 = (x^2 + \varpi\,\mathrm{N}\,y) + yT\,j$ and $w^3 = \cdots + yA\,j$ one reads off $x(\imath w) = \sigma(y)$, $x(\imath w^2) = \sigma(y)T$, $x(\imath w^3) = \sigma(y)A$, with $T, A \in F$. Writing h_E, h_F for the exponents of the symbols and using $h_E(a,b) = 2h_F(a,b)$ for $a,b \in F$:

$$\beta(w,w) = h_E(T, T/n) = h_F(T,n), \qquad \beta(w^2,w) = h_E(A/T, A/n^2) = h_F(nT,A) + h_F(T,n),$$

the twists vanishing because $h_E(n,n) = h_E(n^2,n) = 0$. Hence the exponent of $(w,0)^3$ is $h_F(n,T) + h_F(nT,A)$. Now $A = T^2 - n$; with $u := n/T^2$,

$$h_F(nT, T^2 - n) = h_F(nT, T^2) + h_F(nT, 1-u) = 2h_F(n,T) + h_F(uT^3, 1-u) = 2h_F(n,T),$$

by the Steinberg relation $h_F(u, 1-u) = 0$. So the exponent is $3h_F(n,T) = 0$. For $w \in K^1$, $x \equiv 1 \bmod \varpi$ gives $T \equiv 2$ and $v_F(x^2 + x\sigma x + \sigma x^2) = 2$ exactly, so $A \neq 0$. For $w = x \in E_0^\times$, $x(\imath w^k) = \sigma(x)^k$ and $\det = \mathrm{N}(x)^k$, and the same bookkeeping gives $\sigma(w,w) + \sigma(w^2,w) = -h_E(\sigma x, x) + h_E(\sigma x, x) = 0$. □

Corollary 8.11.3 (The star section on K^6)**.** *(a) For every cube $g = t^3$ with $t \in K^1$, the star lift is $\sigma(g) = (g,0)$.*

(b) Every $X \in K^6$ can be written $X = zv^3$ with $z \in U_F^{(1)}$ and $v \in K^2$ (Lemma 8.8.1(d)); z is determined modulo $F^{\times 3}$, and $z \equiv \mathrm{Nrd}(X)^{-1} \pmod{F^{\times 3}}$.

(c) The star lift of $X = x_X + y_X\,j \in K^6$, that is, the element $\ell(X) := \tilde{z}\tilde{v}^3$ with $\tilde{z} = (z,0)$, is

$$\ell(X) = (X, \kappa(X)), \qquad \kappa(X) = \begin{cases} h_E(y_X, \mathrm{Nrd}\,X), & y_X \neq 0, \\ 0, & y_X = 0. \end{cases} \tag{8.29}$$

Equivalently, $\kappa(X) = c(X)\,v_{E_0}(y_X)$, where $c(X) \in \mathbb{F}_3$ is the real part of the level-six digit of X (the digit of $(x_X - 1)/\varpi^3$).

(d) (H1) is equivalent to the identity

$$\kappa(XX') = \kappa(X) + \kappa(X') + \beta(X,X') \qquad (X, X' \in K^6), \tag{8.30}$$

and when it holds, ψ_6 is given on $\tilde{K}^6$ by

$$\psi_6(X,e) = \zeta^{\,e - \kappa(X)}\,\theta_0(\mathrm{Nrd}\,X)^{-1}, \tag{8.31}$$

where $\widetilde{\theta}(\zeta,z) = \zeta\,\theta_0(z)$ on $\widetilde{Z} = \mu_3 \times F^\times$, the splitting being $z \mapsto (z,0)$ of Lemma 8.11.1(b), and θ_0 is any character with $\theta_0^3 = 1$.

Proof. (a) is Proposition 8.11.2. (b) Existence is Lemma 8.8.1(d) ($\mathrm{Cubes}(Q_2) = Q_6$). Taking reduced norms, $\mathrm{Nrd}(X) = z^2\,\mathrm{Nrd}(v)^3$, so $z^2 \equiv \mathrm{Nrd}\,X$ modulo $F^{\times 3}$; if also $X = z'v'^3$ then $(z/z')^2 \in F^{\times 3}$, hence $z/z' \in F^{\times 3}$ because $F^\times/F^{\times 3}$ has exponent 3. Finally $z \equiv z^4 \equiv \mathrm{Nrd}(X)^2 \equiv \mathrm{Nrd}(X)^{-1}$ modulo cubes. (c) $\ell(X) = (z,0)(v,0)^3 = (z,0)(v^3,0) = (X, \beta(z,v^3))$ by (a), and $\beta(z,v^3) = h_E(z, x(\imath v^3)\det \imath v^3)$ by Lemma 8.11.1(b). If $y_X \neq 0$ then $x(\imath v^3) = \sigma(y_X/z)$ and $\det \imath v^3 = \mathrm{Nrd}(v)^3$, so $\beta(z,v^3) = h_E(z, \sigma(y_X)) - h_E(z,z) = h_E(z, y_X)$ by Galois invariance and $h_E(z,z) = h_E(z,-1) = 0$; by (b) this is $-h_E(\mathrm{Nrd}\,X, y_X) = h_E(y_X, \mathrm{Nrd}\,X)$, which does not depend

on the choice of z because $h_E(F^{\times 3},\cdot)=0$. If $y_X=0$ then $x(\iota v^3)=\sigma(x_X/z)$, $\beta(z,v^3)=h_E(z,x_X)=h_E(x_X,\mathrm{Nrd}X)=h_E(x_X,x_X\sigma x_X)=h_E(x_X,\sigma x_X)=0$, since $x_X\in U_{E_0}^{(3)}$ and $3+3\geq 4$. The digit formula follows from (8.28), from $h_E(\varpi,\mathrm{Nrd}X)=h_E(\varpi,\mathrm{N}x_X)=2h_E(\varpi,x_X)=-h_E(\varpi,x_X)$ (as $\mathrm{Nrd}X\equiv\mathrm{N}x_X$ modulo $U_F^{(7)}\subset F^{\times 3}$ and $h_E(\varpi,\sigma x_X)=h_E(\varpi,x_X)$), and from $h_E(1+\varpi^3,\varpi)=1$, $h_E(1+\varpi^3 i,\varpi)=0$ (Appendix A.1); it is recorded because the tables and scripts use it. (d) $\ell(X)\ell(X')=(XX',\kappa(X)+\kappa(X')+\beta(X,X'))$, so (8.30) says exactly that ℓ is a homomorphism on K^6. If it is, then ℓ is a homomorphism on $U_6=F^\times K^6$, because $\ell(zX)=(z,0)\ell(X)$ with $(z,0)$ central and $z\mapsto(z,0)$ a homomorphism (Lemma 8.11.1(b)); and then $\widetilde{U}_6=\mu_3\times\ell(U_6)$ and ψ_6 is the character which is $\zeta\mapsto\zeta$ on μ_3 and $zv^3\mapsto\theta_0(z)$ on $\ell(U_6)$. The latter is well defined and multiplicative on U_6: if $zv^3\cdot z'v'^3=z''v''^3$ then $z''\in zz'F^{\times 3}$ by the norm argument of (b), and $\theta_0^3=1$. Since $(X,e)=\zeta^{e-\kappa(X)}\ell(X)$ and $\theta_0(z)=\theta_0(\mathrm{Nrd}X)^{-1}$ by (b), this is (8.31); the factor $\theta_0(\mathrm{Nrd}X)^{-1}$ is itself a character of K^6, so the choice of θ_0 affects neither (H1) nor ℓ. Conversely (H1) implies (8.30), since ψ_6 is then a character whose kernel meets each fiber of $\widetilde{U}_6\to U_6$ in exactly one point, namely $\ell(u)$. □

Proposition 8.11.4 ((H1) holds)**.** *The identity* (8.30) *holds for all $X,X'\in K^6$. Consequently ψ_6 is a character of $\widetilde{U}_6$, given on $\tilde{K}^6$ by* (8.31).

Proof. Write $X=a+bj$, $X'=a'+b'j$ with $a,a'\in U_{E_0}^{(3)}$, $b,b'\in\varpi^3\mathscr{O}_{E_0}$, and $n=\mathrm{Nrd}X$, $n'=\mathrm{Nrd}X'$, so that $n,n'\in U_F^{(3)}$ and $XX'=(aa'+\varpi b\sigma(b'))+(ab'+b\sigma(a'))j$. Three preliminary remarks. First, $n=\mathrm{N}(a)-\varpi\,\mathrm{N}(b)\equiv\mathrm{N}(a)$ modulo $U_F^{(7)}\subset E_0^{\times 3}$, and likewise $n'\equiv\mathrm{N}(a')$, so in every symbol n and n' may be replaced by $\mathrm{N}a$ and $\mathrm{N}a'$. Second, by (8.28) and Galois invariance, for every $u\in E_0^\times$,

$$h_E(u,\mathrm{N}a)=v(u)\,\lambda(a),\qquad \lambda(a):=h_E(\varpi,\mathrm{N}a)=2h_E(\varpi,a)=h_E(a,\varpi),\tag{8.32}$$

and similarly for a'. Third, all symbols between two of $a,\sigma a,a',\sigma a',n,n'$ vanish ($3+3\geq 4$), and the twist $2h_E(n,n')$ vanishes. We write v for v_{E_0}.

Main case: $b\neq 0$, $b'\neq 0$, $R\neq 0$, where

$$\tau:=\sigma(b'/b),\qquad R:=a'+\sigma(a)\tau,$$

so that $x(\iota X)=\sigma(b)$, $x(\iota X')=\sigma(b')$ and $x(\iota XX')=\sigma(ab'+b\sigma(a'))=\sigma(b)R$. Inserting into (8.27) and using $h_E(R,R)=h_E(R,-1)=0$,

$$\beta(X,X')=h_E\Big(R,\frac{R}{\tau n}\Big)+2h_E(n,n')=-h_E(R,\tau)-h_E(R,n).\tag{8.33}$$

On the other side, $y_{XX'}=ab'+b\sigma(a')=b\,\sigma(R)$, so by (8.29), Galois invariance and the first remark,

$$\kappa(XX')=h_E(b\sigma(R),nn')=h_E(b,n)+h_E(b,n')+h_E(R,n)+h_E(R,n').$$

Hence (8.30) is equivalent to $h_E(b,n')+h_E(R,n)+h_E(R,n')=h_E(b',n')-h_E(R,\tau)-h_E(R,n)$, i.e. (using $h_E(b'/b,n')=h_E(\sigma\tau,n')=h_E(\tau,n')$) to

$$h_E(R,\tau)=h_E(\tau,n')-h_E(R,n')+h_E(R,n)=v(\tau)\lambda(a')-v(R)\big(\lambda(a')-\lambda(a)\big),\tag{8.34}$$

the second equality by (8.32). To prove (8.34), factor $R = a'(1+\tau_2)$ with $\tau_2 := \sigma(a)\tau/a'$; then $\tau_2 \neq 0, -1$ and $\tau = \tau_2 \cdot a'/\sigma(a)$, so

$$h_E(R,\tau) = h_E(a',\tau) + h_E(1+\tau_2,\tau_2) + h_E\big(1+\tau_2, a'/\sigma(a)\big).$$

The first term is $v(\tau)\lambda(a')$ by (8.28) and (8.32). The second vanishes by the Steinberg relation ($h_E(1+\tau_2,\tau_2) = h_E(1-(-\tau_2),-\tau_2) + h_E(1+\tau_2,-1) = 0$). For the third, $a'/\sigma(a) \in U_{E_0}^{(3)}$ and $v(1+\tau_2) = v(R)$, so by (8.28) it equals $v(R)\big(h_E(\varpi,a') - h_E(\varpi,\sigma a)\big) = v(R)\big(h_E(\varpi,a') - h_E(\varpi,a)\big) = -v(R)\big(\lambda(a') - \lambda(a)\big)$. This is (8.34). Note that no case distinction on $v(\tau)$ or on $v(1+\tau)$ was needed.

Case $b = 0$, $b' \neq 0$. Then $x(\imath X) = \sigma(a)$, $x(\imath XX') = \sigma(a)\sigma(b')$, and (8.27) gives $\beta = h_E(\sigma b', \sigma a/n) = h_E(\sigma b', 1/a) = -h_E(b', \sigma a)$. Also $\kappa(X) = 0$, $\kappa(X') = h_E(b',n')$ and $\kappa(XX') = h_E(ab', nn') = h_E(a,n) + h_E(a,n') + h_E(b',n) + h_E(b',n')$, with $h_E(a,n) = h_E(a,\sigma a) = 0$ and $h_E(a,n') = 0$ by the third remark. So (8.30) reduces to $h_E(b',n) + h_E(b',\sigma a) = 0$; by the first remark $n\sigma(a) \equiv a\sigma(a)^2$, so the left side is $h_E(b', a/\sigma a) + 3h_E(b',\sigma a) = v(b')\big(h_E(\varpi,a) - h_E(\varpi,\sigma a)\big) = 0$.

Case $b \neq 0$, $b' = 0$. Then $x(\imath X') = \sigma(a')$, $x(\imath XX') = \sigma(b)a'$, and

$$\beta = h_E\big(a', \sigma(b)a'/(\sigma(a')n)\big) = h_E(a', \sigma b) = v(b)\lambda(a'),$$

the other three symbols vanishing by the third remark. Also $\kappa(X') = 0$ and

$$\kappa(XX') = h_E(b\sigma(a'), nn') = h_E(b,n) + h_E(b,n') = \kappa(X) + v(b)\lambda(a').$$

Both sides of (8.30) equal $\kappa(X) + v(b)\lambda(a')$.

Case $b = b' = 0$. All three κ vanish and $\beta = h_E(a, \sigma a') = 0$.

Case $b, b' \neq 0$, $R = 0$. Then $XX' \in E_0$, so $\kappa(XX') = 0$, and $\tau = -a'/\sigma(a)$, $b' = \sigma(\tau)b = -b\,\sigma(a')/a$. Now $x(\imath XX') = \sigma(aa' + \varpi b\sigma b')$ and $aa' + \varpi b\sigma(b') = aa' + \varpi\,\mathrm{N}(b)\tau = (a'/\sigma a)n$, so $x(\imath XX') = \sigma(a')n/a$ and

$$\beta = h_E\Big(\frac{\sigma(a')n}{a\,\sigma(b)},\ \frac{\sigma(a')}{a\,\sigma(b')}\Big) + 2h_E(n,n').$$

Expanding bimultiplicatively, discarding the symbols that vanish by the third remark, and using Galois invariance, (8.28) and $v(b') = v(b)$:

$$\begin{aligned}\beta &= -h_E(\sigma a', \sigma b') - h_E(n, \sigma b') + h_E(a, \sigma b') - h_E(\sigma b, \sigma a') + h_E(\sigma b, a) + h_E(\sigma b, \sigma b') \\ &= v(b)\big[h_E(\varpi,a') + 2h_E(\varpi,a) - h_E(\varpi,a) - h_E(\varpi,a') + h_E(\varpi,a)\big] + h_E(b,b'),\end{aligned}$$

and $h_E(b,b') = h_E(b, -b\sigma(a')/a) = h_E(b,\sigma a') - h_E(b,a) = v(b)\big[h_E(\varpi,a') - h_E(\varpi,a)\big]$. Altogether $\beta = v(b)\big[h_E(\varpi,a) + h_E(\varpi,a')\big]$, while $\kappa(X) + \kappa(X') = h_E(b,n) + h_E(b',n') = v(b)\big[2h_E(\varpi,a) + 2h_E(\varpi,a')\big]$. The sum $\kappa(X) + \kappa(X') + \beta = 3v(b)[\cdots] = 0 = \kappa(XX')$.

This exhausts all cases, since $x(\imath XX')$ is $\sigma(b)R$ when $b, b' \neq 0$ and the entries used in each case are the ones prescribed by (8.27). The last assertion is Corollary 8.11.3(d). □

Remark 8.11.5 (Checks and a second route)**.** (a) Script 12 evaluates (8.34) and (8.30) in the explicit model on 12,000 structured instances covering all five cases of the proof (including $v(1+\tau) \in$

$\{1,\dots,5\}$ and $X' \in K^8,\dots,K^{12}$); all pass, and $\beta(X,X') \neq 0$ in about 44% of them, so the identity is not vacuous. Scripts 8 and 9 are earlier checks of the same identity through cube roots computed by Hensel's lemma and on a stratified sample. None of these is part of the proof. (b) A second, independent argument reduces (H1) to a finite problem. The subgroup K^4 is a uniform pro-3 group of dimension 8, so central μ_3-extensions of K^4 are classified by alternating forms on K^4/K^8 [Laz65, Ch. V]; [SW00, §5]; the form of the cover is $\mathrm{Ad}(D^\times)$-invariant and vanishes on pairs of commuting elements by Proposition 4.3.1 and the depth bounds for the symbols of E_0 and of the ramified quadratic extensions of F; script 5a finds that the space of such forms is zero, so the cover splits over K^4 and the star section is a homomorphism on K^8. Script 5b then shows that every odd function on Q_6/Q_8 whose non-additivity is $D^\times$-equivariant is linear. Together these reduce (H1) to multiplicativity of ℓ on $K^6 \times K^8$, which is a special case of Proposition 8.11.4. We record this route because the Lazard-form computation is of independent interest for Target B.

8.12 The main theorem

This section assembles the earlier sections into the main theorem, Target A, and lists what its proof depends on. A corollary extends the result to the representations trivial on K^1 and their twists.

Theorem 8.12.1 (**($\mathbf{P}^c$)** Target A)**.** *Let $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, D the quaternion division algebra over F, $\widetilde{H}$ its scalar-central 3-fold cover, and $\widetilde{\theta}$ a genuine central character compatible with the trivial character of $F^\times$. Then $L_D\mathbf{1} = \Theta_\tau$ for a unique genuine irreducible representation τ of $\widetilde{H}$, of dimension 27 and central character $\widetilde{\theta}$. Moreover:*

(a) $\tau|_{\widetilde{U}_1}$ is irreducible and equals $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}\lambda_0$, with λ_0 as in (8.25); $\tau|_{\widetilde{U}_6} = 27\,\psi_6$, and $\tau|_{\widetilde{U}_3}$ is the sum of the 27 distinct genuine characters of $\widetilde{U}_3/N_6$ whose restriction to $\widetilde{U}_4/N_6$ exhausts the genuine characters of that group.

(b) On the covered quadratic tori, τ restricts as in Proposition 8.7.2: $(3,2,2,2)$ on the unramified torus and $(3,2,1,1,1,1)$ on each ramified torus.

(c) Θ_τ is supported on the star lifts of good elements and of the center; its values are $27\zeta\widetilde{\theta}(\tilde{z})$ at $\zeta\tilde{z}t^*$ with $t \in F^\times K^2$, $9\zeta\widetilde{\theta}(\tilde{z})$ for $t \in F^\times K^1 \setminus F^\times K^2$, and $3\zeta\widetilde{\theta}(\tilde{z})$ for $t \notin F^\times K^1$.

(d) τ is the unique extension to $\widetilde{H}$ of $\tau|_{\widetilde{U}_1}$ with character values $3\zeta\widetilde{\theta}(\tilde{z})$ at the star lifts of the elements of 2-power order modulo the center.

Proof. (H1) holds by Proposition 8.11.4; Theorem 8.10.7 gives Target A together with (a), and Theorem 8.9.5 gives uniqueness and irreducibility of the restriction. The dependencies of this chain are itemized in Remark 8.12.2. (b) is Proposition 8.7.2; (c) collects (8.21) and the proof of Lemma 8.9.2; (d) follows from (c) since the four extensions of an $\widetilde{H}$-stable irreducible of $\widetilde{U}_1$ differ by the linear characters of D_8, which are distinguished by their values on P_0. □

Remark 8.12.2 (Dependencies of Theorem 8.12.1)**.** The proof uses the following inputs and nothing else.

(D1) **(E)** From Chapter 4: the cover of $D^\times$ is the restriction along ι of the parameter-c_v Kazhdan–Patterson cover of $\mathrm{GL}_2(E_0)$ (Lemma 4.2.3, Proposition 4.2.4); the commutator formula on covered tori (Proposition 4.3.1); good elements are $F^\times \cdot \{\text{cubes}\}$ in each torus (Proposition 4.3.5(a)); $E^\times/F^\times$ has no 3-torsion (Lemma 4.3.4); the star map (Theorem 4.3.10, Remark 4.8.1). From Chapter 6: the normalization (8.1) of L_D with $b_D = 3$, $c_D = 27$, and the value $R = 1$ at elements of 2-power order.

(D2) **(E)** Kubota's cocycle on $\mathrm{GL}_2(E_0)$ (Lemma 8.11.1(a)), and the standard properties of the cubic Hilbert symbols of F and E_0 listed after Lemma 8.11.1.

(D3) **($\mathbf{P}^{\mathbf{c}}$)** The explicit values of the two symbols (Appendix A.1), obtained as the unique Steinberg symbols by exact linear algebra (script 7). They enter the hand arguments only through $h_E(1+\varpi^3,\varpi) \neq 0$ (used to exclude $e = 0$ in Proposition 8.10.6 via $\kappa_{1,5}(1,1) \neq 1$) and through the digit form of κ used by the scripts; the proof of Proposition 8.11.4 does not use them.

(D4) **($\mathbf{P}^{\mathbf{c}}$)** Four exhaustive finite computations: the 72 elements of $\bar{G}_3$ from the 648 depth-one classes (script 1, also proved by hand in Proposition 8.8.2); the cube map on Q/Q_7 (script 2, corroborating Lemma 8.8.1); the actions on V and their orbits (script 3); and the design search over all odd D_8-invariant functions on the 36 lines together with the Heisenberg cocycle extension and the Fourier identities (scripts 4 and 6). Each is an enumeration of an explicitly described finite set.

(D5) Everything else is proved by hand: the toral theorem and the three torus calculations, the positivity criterion, the depth lemma, the Brauer reduction (Theorem 8.9.5), the consequences of (H1) and the uniqueness of the design given the finite data (Section 8.10), the cube identity, and (H1) itself.

No sampled computation is load bearing. Scripts 5, 8–12 are corroboration (Remarks 8.11.5 and the next remark). Theorem 8.13.8 depends on no further computation; its Steps 2–3 and Proposition 8.13.9 are proved by hand from the items above.

Remark 8.12.3 (End-to-end verification)**.** Independently of the chain of reductions, the character of $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}\lambda_0$ was computed in the explicit model of Section 8.11 from formula (8.25) (products of the eight D_8-conjugates, 27 coset representatives of $\widetilde{U}_1/\widetilde{U}_3$) and compared with (8.21) at star lifts of cubes of depth-one elements, at elements of $\widetilde{U}_6$ with all three values of ψ_6, at products $\sigma(g)\ell(k)\tilde{z}\zeta^m$, and at non-good elements of depths $3,4,5$; all values agree, including phases (scripts 11, 11b). Script 10 checks, again in the explicit model, that all 124 sampled coplanar triples of good lines have trivial defect and that all 186 sampled four-term relations $\sum v_i = 0$ in $\bar{G}_3$ have defect $\sum e_*(v_i)$; that $\omega = 0$ on 300 pairs; and that $\kappa_{1,5}(a,b) = \zeta^{-\mathrm{Tr}(a\bar{b})}$ on 300 pairs.

Corollary 8.12.4 (**($\mathbf{P}^{\mathbf{c}}$)** Level-zero rows and their norm twists)**.** *Let ρ be one of the five irreducible representations of $D^\times$ trivial on K^1 (inflated from $D^\times/F^\times K^1 \simeq D_8$), and let χ be a character of $F^\times$ trivial on μ_3. Then $L_D\Theta_{\rho\otimes(\chi\circ\mathrm{Nrd})}$, computed with any compatible genuine central character, is the character of a genuine irreducible representation of $\widetilde{H}$, of dimension $27\dim\rho$.*

Proof. First let $\chi = 1$. On $\widetilde{U}_1$, $\Theta_\rho \circ p = \dim\rho$, so $L_D\Theta_\rho|_{\widetilde{U}_1} = (\dim\rho)\Phi$ is a character by Theorem 8.12.1; on Γ_{P_0} it is $3\,\psi_P \otimes \rho + 3\dim\rho\sum_\pi(\dim\pi)\Theta_\pi$; and its restrictions to covered tori are characters by Lemma 8.6.1. The proof of Theorem 8.9.5 applies verbatim (Lemma 8.9.4 holds for every row by Chapter 6), giving an irreducible character. For general χ, write $\chi = \chi_1^3$: this is possible because χ is trivial on μ_3, so that its restriction to $U_F^{(1)} \simeq \mu_3 \times \mathbb{Z}_3^2$ factors through $\mathbb{Z}_3^2$, whose character group is 3-divisible, while on $\varpi^{\mathbb{Z}} \times \mu_2$ cube roots exist trivially. Twisting the representation τ_ρ just obtained by $\chi_1 \circ \mathrm{Nrd} \circ p$ multiplies its character at $\tilde{z}t^*$ by $\chi_1(z)^2\chi(\mathrm{Nrd}\,t)$ and changes its central character to $\widetilde{\theta}' := \widetilde{\theta}\cdot(\chi_1^2 \circ p)$, which is compatible with $\theta_\rho\chi^2 = \theta_{\rho\otimes\chi\circ\mathrm{Nrd}}$ because $\widetilde{\theta}'(a^*) = \theta_\rho(a)\chi_1(a^3)^2$. The result is $L_D\Theta_{\rho\otimes\chi\circ\mathrm{Nrd}}$ computed with $\widetilde{\theta}'$; by Remark 8.9.6 any other compatible choice is obtained by a further twist. □

8.13 Target B: other rows

This section turns to the other rows of the character matrix, that is, to $L_D\Theta_\rho$ for a general irreducible ρ. It extends the reduction theorem to every row and compares each row with the row of the trivial character by a product formula. It then proves Target B for the representations trivial on K^3.

Throughout this section ρ is an irreducible smooth representation of $D^\times$ (finite dimensional, as $D^\times/F^\times$ is compact) with central character θ_ρ trivial on μ_3, θ_0 is a character of $F^\times$ with $\theta_0^3 = \theta_\rho$, and $\widetilde{\theta}(\zeta, z) = \zeta\theta_0(z)$ is the corresponding compatible genuine central character, in the coordinates of Lemma 8.11.1(b). We write $\phi_\rho := L_D\Theta_\rho$ and $\Phi_\rho := \phi_\rho|_{\widetilde{U}_1}$. For $\rho = \mathbf{1}$ we take, once and for all, $\theta_1 := \mathbf{1}$, i.e. $\widetilde{\theta}_1(\zeta, z) = \zeta$ (a compatible choice, by Remark 8.9.6; it is the normalization of the scripts), and write ϕ_1, $\widetilde{\theta}_1$, λ_0 for the objects of Sections 8.10–8.12 in this normalization. Two reductions are free.

Twisting. For a character χ of $F^\times$ trivial on μ_3, $\rho \otimes (\chi \circ \mathrm{Nrd})$ is irreducible with central character $\theta_\rho\chi^2$, and by the argument of Corollary 8.12.4, $L_D\Theta_{\rho\otimes\chi\circ\mathrm{Nrd}}$ is an irreducible character if and only if $L_D\Theta_\rho$ is. Squaring is surjective onto the characters of $F^\times = \varpi^{\mathbb{Z}} \times \mu_2 \times U_F^{(1)}$ that are trivial at -1, so we may and do assume

$$\theta_\rho \in \{\mathbf{1},\ \mathrm{sgn}\}, \qquad \mathrm{sgn}(\varpi^k\varepsilon u) := \varepsilon \quad (\varepsilon = \pm 1,\ u \in U_F^{(1)}), \tag{8.35}$$

according to whether $\rho(-1) = +1$ or -1, and accordingly $\theta_0 = \theta_1 = \mathbf{1}$ or $\theta_0 = \mathrm{sgn}$.

Unitarity. By Chapter 6, $\langle\phi_\rho, \phi_\rho\rangle_{\widetilde{H}} = 1$ and $\phi_\rho(1) = 27\dim\rho > 0$, so ϕ_ρ is an irreducible character as soon as it is a virtual character.

8.13.1 The reduction theorem for every row

This subsection extends the reduction theorem of Section 8.9 from the trivial character to every ρ.

Theorem 8.13.1 (Reduction, all rows)**.** *$L_D\Theta_\rho$ is the character of a genuine irreducible representation of $\widetilde{H}$ if and only if Φ_ρ is a virtual character of $\widetilde{U}_1$.*

Proof. Only "if" needs proof. Assume (8.35). The finite quotient $\Gamma = \widetilde{H}/N$ of Section 8.9 is built from the finite row of ϕ_ρ exactly as before, with $N \cap \widetilde{Z} = \ker\widetilde{\theta} \cap \ker\widetilde{\theta}_1$ (so that both ϕ_ρ and ϕ_1 are functions on Γ); then $T = \widetilde{Z}N/N$ is μ_3 if $\theta_\rho = \mathbf{1}$ and $\mu_6 = \mu_3 \times \mu_2$ if $\theta_\rho = \mathrm{sgn}$; $\Gamma_0 = \widetilde{U}_1/N$ is the Sylow 3-subgroup in the first case and is $\Gamma_0^{(3)} \times \mu_2$ with $\Gamma_0^{(3)}$ the Sylow 3-subgroup in the second; in both cases Γ/Γ_0 is D_8. We check the three inputs of the proof of Theorem 8.9.5.

Tori. $\phi_\rho|_{\widetilde{E^\times}}$ is a genuine representation character for every maximal subfield E (Lemma 8.6.1: $\Theta_\rho|_{E^\times}$ is a sum of characters α with $\alpha|_{F^\times} = \theta_\rho$), so Lemma 8.9.1 holds verbatim.

Centralizers. Lemma 8.9.3 concerns Γ_0 and the images of tori and is unchanged; the μ_2 in T is central.

Two-groups. Let $\hat{P}_0$ be the preimage of $P_0F^\times$ and $\Gamma_{P_0} = \hat{P}_0N/N$, a central extension of D_8 by T; it contains a Sylow 2-subgroup of Γ. Every $\eta \in \hat{P}_0 \subset D^\times$ is good: $\eta^4 \in F^\times$, so $\eta = \eta^4 \cdot (\eta^{-1})^3$. Put $x_\eta := \widetilde{\eta^4}\,(\tilde{\eta}^{-1})^3$, a lift of η with $x_{\eta c} = x_\eta \tilde{c}$ for $c \in F^\times$. By (8.1) and $R = 1$ off $F^\times K^1$,

$$\phi_\rho(x_\eta) = 3\,\theta_0(\eta^4)\,\Theta_\rho(\eta^{-1}) = 3\,\theta_0(\eta^4)\,\Theta_{\rho^\vee}(\eta) \qquad (\eta \notin F^\times),$$

and $\phi_\rho = 27\dim\rho \cdot \widetilde{\theta}$ on $\widetilde{Z}$. Let ψ_P be the genuine character of order 3 of Γ_{P_0} trivial on a Sylow 2-subgroup (unique, by Schur–Zassenhaus for the μ_3-part of T); on the image of $\widetilde{Z}$ it is $\widetilde{\theta}_1$, i.e. $\psi_P(z,0) = \theta_1(z)$, and therefore $\psi_P(x_\eta) = \theta_1(\eta^4)\psi_P(\tilde{\eta}^{-1})^3 = \theta_1(\eta^4)$. (We do not twist $\rho^\vee$ by $\theta_1^2 \circ \mathrm{Nrd}$: on scalars that twist is $\theta_1(z^2)^2 = \theta_1(z)$, not trivial.)

If $\theta_\rho = \mathbf{1}$, then $\theta_0 = \theta_1$, so $\phi_\rho(x_\eta) = 3\psi_P(x_\eta)\Theta_{\rho^\vee}(\eta)$, and $\rho^\vee$ is trivial on $F^\times$, so $\Theta_{\rho^\vee}$ is the character of a representation of $D_8 = \hat{P}_0/F^\times$. With $\Gamma_{P_0} = \mu_3 \times \tilde{P}_0$ as in Lemma 8.9.2,

$$\phi_\rho|_{\Gamma_{P_0}} = \psi_P \otimes \Big(3\,\rho^\vee|_{D_8} + 3\dim\rho \sum_{\pi \in \mathrm{Irr}(D_8)} (\dim\pi)\,\Theta_\pi\Big),$$

a character (the second term is $24\dim\rho$ times the delta function at $1 \in D_8$).

If $\theta_\rho = \mathrm{sgn}$, then $\theta_0 = \mathrm{sgn} \cdot \theta_1$ and $\phi_\rho(x_\eta) = 3\psi_P(x_\eta)\,\mathrm{sgn}(\eta^4)\,\Theta_{\rho^\vee}(\eta)$, where now $\rho^\vee$ has central character sgn. Here $\Gamma_{P_0} = \mu_3 \times \Gamma''$ with $\Gamma'' := \Gamma_{P_0}/\mu_3$, the quotient of $\hat{P}_0$ by μ_3 and the lifts of $\ker\mathrm{sgn} \subset F^\times$, a central extension of D_8 by μ_2; $\rho^\vee|_{\hat{P}_0}$ factors through Γ''. With a, b the images of $1+i$ and j one has $a^8 = 1$ (as $(1+i)^4 = -4$ and $\mathrm{sgn}(-4) = -1$), $b^2 = 1$ and $bab^{-1} = a^3$ (as $j(1+i)j^{-1} = 2(1+i)^{-1}$ and $\mathrm{sgn}(2) = -1$), so $\Gamma'' \simeq SD_{16}$, the semidihedral group, with a^4 the image of -1. The function $f(\eta) := \mathrm{sgn}(\eta^4)$ is -1 on the classes $\{a, a^3\}, \{a^5, a^7\}$ and $+1$ elsewhere. Having central character sgn, $\rho^\vee|_{\Gamma''}$ is a sum of copies of the two faithful two-dimensional characters χ_1, χ_2 of SD_{16} ($a \mapsto \pm i\sqrt{2}$, $a^2 \mapsto 0$, $a^4 \mapsto -2$, 0 off $\langle a\rangle$); a direct check on the seven classes gives $f\chi_1 = \chi_2$ and $f\chi_2 = \chi_1$. Hence $f \cdot \Theta_{\rho^\vee}$ is a character of Γ'', and with δ^{sgn} the function ± 1 at a^0, a^4 and 0 elsewhere, $8\delta^{\mathrm{sgn}} = 2\chi_1 + 2\chi_2$, so

$$\phi_\rho|_{\Gamma_{P_0}} = \psi_P \otimes \big(3f\,\Theta_{\rho^\vee|_{\Gamma''}} + 3\dim\rho\,(2\chi_1 + 2\chi_2)\big)$$

is a character.

Brauer's theorem. Let $B = B_2 \times B_3 \subset \Gamma$ be elementary, and let $\bar{B}_2$ be the image of B_2 in $\Gamma/\Gamma_0 = D_8$. (When $T \supset \mu_2$, B_2 may be noncyclic while $\bar{B}_2$ is cyclic, e.g. $B_2 = \langle a^4, b\rangle$; the case distinction must therefore be made on $\bar{B}_2$.) If $\bar{B}_2 = 1$, then $B \subset \Gamma_0$ and $\phi_\rho|_B = \Phi_\rho|_B$ is a virtual character by hypothesis. If $\bar{B}_2 = \langle\bar{y}\rangle$ is cyclic and nontrivial, pick $y \in B_2$ over $\bar{y}$; then $B_2 \subset \langle y\rangle \cdot (B_2 \cap T)$, $B_3 \subset C_{\Gamma_0}(y) \subset T_{E_y}$ by Lemma 8.9.3(a), $y \in T_{E_y}$ and $T \subset T_{E_y}$, so $B \subset T_{E_y}$ and Lemma 8.9.1 applies.

If $\bar{B}_2$ is noncyclic, then $B_3 \subset C_{\Gamma_0}(B_2) \subset T$ by Lemma 8.9.3(b) (whose proof uses only the images of the elements of B_2 in D_8), so $B \subset B_2 T$, which is conjugate into Γ_{P_0} because Γ_{P_0} contains a Sylow 2-subgroup; the two-group computation applies.

Brauer's theorem now gives that ϕ_ρ is a virtual character of Γ, and unitarity finishes the proof. □

8.13.2 The cube-root transform

This subsection writes $L_D\Theta_\rho$ as the product of $L_D\mathbf{1}$ and a function built from Θ_ρ and the cube root. It then gives a sufficient criterion for a row in terms of this product.

Lemma 8.13.2 (Product formula)**.** *Assume* (8.35)*. There is a unique function F_ρ on the set of star lifts of good elements of H such that*

$$\phi_\rho = \phi_1 \cdot F_\rho \quad \text{on that set}, \qquad F_\rho(\zeta \tilde{z} t^*) = \theta_0 \theta_1^{-1}(z)\, \Theta_\rho(t), \tag{8.36}$$

and both sides of (8.36) *vanish elsewhere. F_ρ is invariant under conjugation by $\widetilde{H}$ and under multiplication by μ_3. On the star lifts of good elements of U_3,*

$$F_\rho(\tilde{z} t^*) = \varepsilon_\rho(z)\, \Theta_\rho(t) \qquad (zt^3 \in U_3,\ t \in K^1), \tag{8.37}$$

where $\varepsilon_\rho = \mathbf{1}$ if $\theta_\rho = \mathbf{1}$ and $\varepsilon_\rho = \mathrm{sgn}$ if $\theta_\rho = \mathrm{sgn}$; here $t \in K^1$ is determined by zt^3 up to $U_F^{(1)}$, on which Θ_ρ is invariant, and z up to $(U_F^{(1)})^3 \subset \ker \varepsilon_\rho$. In particular $g \mapsto \varepsilon_\rho(z)$ is the character of $U_3 = F^\times K^3$ trivial on K^3 and equal to ε_ρ on $F^\times$ (well defined because $F^\times \cap K^3 = U_F^{(2)}$), and it is $\mathrm{Ad}(U_1)$-invariant.

Proof. Both ϕ_ρ and ϕ_1 vanish off star lifts of good elements, and there $\phi_\rho(\zeta\tilde{z}t^*) = 3\zeta\theta_0(z)R(t)\Theta_\rho(t)$, $\phi_1(\zeta\tilde{z}t^*) = 3\zeta\theta_1(z)R(t)$ with $R(t) \neq 0$. If $\tilde{z}t^* = \tilde{z}'t'^*$ then $t' = ct$ with $c \in F^\times$ and $z' = zc^{-3}$ (as in Lemma 8.8.3), and $\theta_0\theta_1^{-1}(c^{-3})\Theta_\rho(ct) = \theta_\rho(c)^{-1}\theta_\rho(c)\Theta_\rho(t)$, so F_ρ is well defined. Invariance is clear. For $zt^3 \in U_3$ with $t \in K^1$: $\theta_0\theta_1^{-1} = \varepsilon_\rho$ on $F^\times$ by the normalization, and $c \in F^\times \cap K^1 = U_F^{(1)}$, on which $\theta_\rho = 1$. □

Remark 8.13.3 (Adams operations)**.** F_ρ restricted to $\widetilde{U}_1$ is the Adams-type transform $\psi^{1/3}\Theta_\rho$, "Θ_ρ at the cube root," corrected by ε_ρ. At a tame place $\psi^{1/3}$ is a Galois conjugation of characters (Lemma 6.7.5), which is why Target B is trivial there; at the wild place it is not even a virtual character in general, and the content of the next criterion is that it need not be: it must only be compatible with the character λ_0 on the good set.

Proposition 8.13.4 (**(P^{c})** A sufficient criterion)**.** *Assume* (8.35) *and Theorem 8.12.1 (the proposition uses λ_0 and $\Theta_{\mathrm{Ind}\,\lambda_0} = \Phi_1$, so it inherits the label of Target A). Suppose there is a virtual character Ψ of U_3 (a $\mathbb{Z}$-combination of characters of finite-dimensional smooth representations) which is $\mathrm{Ad}(U_1)$-invariant and satisfies $\Psi(zt^3) = F_\rho(\tilde{z}t^*)$ for all good $zt^3 \in U_3$. Then $L_D\Theta_\rho$ is the character of a genuine irreducible representation τ_ρ of $\widetilde{H}$ of dimension $27\dim\rho$, and*

$$\Theta_{\tau_\rho}|_{\widetilde{U}_1} = \sum_i n_i\, \Theta_{\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0 \otimes \sigma_i \circ p)}, \qquad \Psi = \sum_i n_i \Theta_{\sigma_i},\ n_i \in \mathbb{Z}. \tag{8.38}$$

Proof. For a finite-dimensional representation σ of U_3 with $\mathrm{Ad}(U_1)$-invariant character, the induced character satisfies, for $x \in \widetilde{U}_3$ and g running over $\widetilde{U}_1/\widetilde{U}_3$,

$$\Theta_{\mathrm{Ind}(\lambda_0\otimes\sigma\circ p)}(x) = \sum_g \lambda_0(gxg^{-1})\Theta_\sigma(p(gxg^{-1})) = \Theta_\sigma(p(x))\sum_g \lambda_0(gxg^{-1}) = \Theta_\sigma(p(x))\,\Theta_{\mathrm{Ind}\,\lambda_0}(x),$$

and both sides vanish off $\widetilde{U}_3$. By Theorem 8.12.1(a), $\Theta_{\mathrm{Ind}\,\lambda_0} = \Phi_1$, which is supported on star lifts of good elements of U_3 (Lemma 8.8.3). Summing with the n_i and using (8.36), $\sum_i n_i\Theta_{\mathrm{Ind}(\lambda_0\otimes\sigma_i\circ p)} = \Phi_1\cdot(\Psi\circ p) = \Phi_1 F_\rho = \Phi_\rho$ on $\widetilde{U}_1$. So Φ_ρ is a virtual character of $\widetilde{U}_1$, and Theorem 8.13.1 applies. □

8.13.3 All representations trivial on K^2

This subsection proves Target B for the representations of $D^\times$ that are trivial on K^2.

Let $\pi_3 : U_3 \to U_3/U_4 \simeq Q_3/Q_4 \simeq \mathbb{F}_9$ be the depth-three digit (the A-coordinate of (8.19)), a homomorphism which is $\mathrm{Ad}(U_1)$-invariant because $[U_1,U_3] \subset U_4$, and identify $K^1/K^2 \simeq \mathbb{F}_9$ by $1+wj \mapsto w$.

Theorem 8.13.5 (**(P^c)** Target B for level at most one)**.** *Assume Theorem 8.12.1. Let ρ be an irreducible representation of $D^\times$ trivial on K^2, with $\theta_\rho|_{\mu_3} = 1$ (automatic, since $\mu_3 \subset U_F^{(1)} \subset K^2$). Then $L_D\Theta_\rho$, computed with any compatible genuine central character, is the character of a genuine irreducible representation τ_ρ of $\widetilde{H}$ of dimension $27\dim\rho$. After the normalization (8.35),*

$$\tau_\rho|_{\widetilde{U}_1} = \bigoplus_\psi m_\psi\,\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}\big(\lambda_0\otimes(\varepsilon_\rho\cdot\psi\circ\pi_3)\circ p\big), \qquad \rho|_{K^1} = \bigoplus_\psi m_\psi\,\psi,$$

the sum over the characters ψ of $K^1/K^2 \simeq \mathbb{F}_9$; the induced representations occurring are irreducible and pairwise distinct. The same conclusion holds for $\rho\otimes(\chi\circ\mathrm{Nrd})$ for every character χ of $F^\times$ trivial on μ_3; these are all the irreducible representations of $D^\times$ whose restriction to K^2 is a multiple of a character of the form $\chi\circ\mathrm{Nrd}$.

Proof. By twisting we may assume (8.35). Let $g = zt^3$ be a good element of U_3 with $t \in K^1$. If $t \in F^\times K^2$, then $g \in U_6$, $\pi_3(g) = 0$ and $\Theta_\rho(t) = \dim\rho = \Theta_{\rho|_{K^1}}(0)$. Otherwise $t \equiv 1+wj$ modulo K^2 with $w \neq 0$, and by Lemma 8.8.1(c) the depth-three digit of g is w^5; since $w^4 = \mathrm{N}_{\mathbb{F}_9/\mathbb{F}_3}(w) = \pm1$, $w^5 = \pm w$, so $w = \pm\pi_3(g)$. Because Θ_ρ is a class function and $i(1+wj)i^{-1} = 1-wj$, we get $\Theta_\rho(t) = \Theta_{\rho|_{K^1}}(\pi_3(g))$ in both cases. By (8.37), $F_\rho(\tilde{z}t^*) = \varepsilon_\rho(z)\Theta_{\rho|_{K^1}}(\pi_3(g))$, which is the value at g of the character of the representation $\Psi := \varepsilon_\rho\otimes(\rho|_{K^1})\circ\pi_3$ of U_3 (with $\rho|_{K^1}$ regarded as a representation of K^1/K^2). Its character is $\mathrm{Ad}(U_1)$-invariant, so Proposition 8.13.4 applies with all $n_i = m_\psi \geq 0$, giving the displayed decomposition. The restriction $\tau_\rho|_{\widetilde{U}_1}$ has norm $\int_{U_1/Z}|\Theta_\rho|^2\,d\mu$ by the Weyl-integration computation of Lemma 8.9.4 (substituting $t = s^3$, $|\phi_\rho|^2(s^3) = 9R(s)^2|\Theta_\rho(s)|^2$), and this is $\langle\Theta_\rho|_{K^1},\Theta_\rho|_{K^1}\rangle = \sum_\psi m_\psi^2$. On the other hand $\langle\sum m_\psi\chi_\psi,\sum m_\psi\chi_\psi\rangle \geq \sum m_\psi^2\langle\chi_\psi,\chi_\psi\rangle \geq \sum m_\psi^2$ for the characters χ_ψ of the induced representations, with equality only if each χ_ψ with $m_\psi > 0$ is irreducible and they are pairwise orthogonal. The last sentence: if $\rho|_{K^2}$ is a multiple of $\chi\circ\mathrm{Nrd}$, then $\rho\otimes(\chi^{-1}\circ\mathrm{Nrd})$ is trivial on K^2; and χ may be modified by any character trivial on $\mathrm{Nrd}(K^2) = U_F^{(1)}$ without changing this. □

Remark 8.13.6. (a) The theorem contains Corollary 8.12.4 and extends it to the level-zero representations with $\rho(-1) = -1$ (the two-dimensional representations of $D^\times/K^1$ induced from the characters of μ_8 of odd order), which are not twists of representations of D_8. (b) For the four four-dimensional irreducible representations of $\mathbb{F}_9 \rtimes D_8$ the theorem gives $\dim \tau_\rho = 108$ and $\tau_\rho|_{\widetilde{U}_1}$ a sum of four irreducibles of dimension 27, permuted transitively by D_8. (c) Nothing in the proof used irreducibility of ρ beyond the last sentence: for every finite-dimensional representation ρ of $D^\times$ trivial on K^2, $L_D\Theta_\rho$ is the character of a genuine representation of $\widetilde{H}$.

8.13.4 Level two

This subsection proves Target B for the representations that are trivial on K^3 but not on K^2. It first describes these representations and the finite group through which the argument runs.

Let now ρ be irreducible and trivial on K^3 but not on K^2. Normalize $\theta_\rho \in \{\mathbf{1}, \mathrm{sgn}\}$ as in (8.35); then $\rho|_{K^1}$ factors through $\mathrm{Heis} := Q_1/Q_3 = K^1/K^3U_F^{(1)}$, the Heisenberg group of order 27, and is nontrivial on its center Q_2/Q_3. The group $D_8 = D^\times/F^\times K^1$ permutes the two three-dimensional irreducible characters χ_3, χ_3' of Heis, so $\rho|_{K^1} = e(\chi_3 + \chi_3')$, and $e = 1$ for irreducible ρ: the stabilizer of χ_3 in D_8 is the cyclic group of order 4 generated by the class of $1+i$, which acts trivially on the center Q_2/Q_3 of Heis (while j inverts it), and a character stable under a cyclic group extends. So $\dim\rho = 6$; there are four such ρ for each normalized central character. For these ρ the criterion of Proposition 8.13.4 *fails*: the restriction of F_ρ to the good classes of Q_3/Q_7 is a rational but not an integral combination of the restrictions of the $\mathrm{Ad}(Q_1)$-orbit sums of $\mathrm{Irr}(Q_3/Q_7)$ (script 13). Nevertheless:

Lemma 8.13.7 (The group $H = Q_3/Q_7$)**.** *Let $H = Q_3/Q_7$, of order 3^6. Then $H' = [H,H] = Q_6/Q_7$, of order 3; $Z(H) = Q_4/Q_7$, of index 9; H has 243 linear characters and 54 nonlinear irreducible characters, all of degree 3, namely the irreducible characters nontrivial on H', 27 above each of the two nontrivial characters ω of H'; and*

$$3\sum_{\chi'(1)=3}\chi' = \mathrm{Reg}_H - \mathrm{Inf}_{H/H'}^H \mathrm{Reg}_{H/H'}, \qquad \sum_{\chi'(1)=3,\ \chi'|_{H'}=3\omega}\chi' = 81\,\omega\mathbf{1}_{H'}. \tag{8.39}$$

Proof. $[K^3,K^3] \subset K^6$ and $[K^3,K^4] \subset K^7$, so $H' \subset Q_6/Q_7$ and $Q_4/Q_7 \subset Z(H)$. For $x = 1+\varpi a j+\cdots$, $y = 1+\varpi b j+\cdots$ of depth three, $[x,y] \equiv 1+\varpi^3(a\bar b - b\bar a)$ modulo K^7 (odd-odd bracket), and $a\bar b - b\bar a = 2i\,\mathrm{Im}(a\bar b)$ is nonzero modulo $\mathbb{F}_3$ for suitable b whenever $a \neq 0$; so $H' = Q_6/Q_7 \simeq \mathbb{F}_9/\mathbb{F}_3$, of order 3, and no element of depth exactly three is central, so $Z(H) = Q_4/Q_7$ and $[H : Z(H)] = |Q_3/Q_4| = 9$. Hence $|H/H'| = 243$ linear characters; every nonlinear irreducible character has degree $\le \sqrt{[H : Z(H)]} = 3$, hence 3; their number is $(729-243)/9 = 54$; they are exactly the characters nontrivial on H', and by Schur's lemma each restricts to H' as 3ω. The first identity in (8.39) is the decomposition of the regular character, since the linear characters of H are the inflations of those of H/H'. For the second, the ω-isotypic part of $\mathbb{C}[H]$ under the central subgroup H' is the H-module $\mathbb{C}[H]e_\omega$ with character $243\,\omega\mathbf{1}_{H'}$ (as $|H/H'| = 243$), and it contains each irreducible character above ω with multiplicity equal to its degree 3; dividing by 3 gives the identity, and comparing degrees ($81\cdot 3 = 243 = 27\cdot 9$) gives 27 characters above each ω. □

Theorem 8.13.8 (($\mathbf{P}^c$) Target B for level two)**.** *Assume Theorem 8.12.1. Let ρ be an irreducible representation of $D^\times$ trivial on K^3 with $\theta_\rho|_{\mu_3} = 1$. Then $L_D\Theta_\rho$, computed with any compatible*

genuine central character, is the character of a genuine irreducible representation τ_ρ of $\widetilde{H}$ of dimension $27\dim\rho$. If $\rho|_{K^2}$ is not scalar, then after normalizing $\theta_\rho \in \{\mathbf{1}, \mathrm{sgn}\}$,

$$\tau_\rho|_{\widetilde{U}_1} = (\varepsilon_\rho \circ p) \otimes \big(\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0 \otimes \chi_1' \circ p) \ \oplus\ \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0 \otimes \chi_2' \circ p)\big),$$

where χ_1', χ_2' are three-dimensional irreducible characters of Q_3/Q_7 (one from each of the two orbits described in the proof); both induced representations are irreducible of dimension 81, and they are distinct.

Proof. By Theorem 8.13.5 we may assume $\rho|_{K^2}$ nonscalar, and by twisting, (8.35). By Theorem 8.13.1 it suffices to show that $\Phi_\rho = \Phi_1 F_\rho$ is a virtual character of $\widetilde{U}_1$. The proof has three steps, all by hand; the finite computations of scripts 13–15 corroborate Steps 2 and 3 and are not used.

Step 1: a finite quotient. For $g = zt^3$ good in U_3 with $t \in K^1$, the class of t in Heis depends only on the class of g in Q_3/Q_7: for $g \in Q_6$ this is the linearity of the inverse Artin–Schreier map (Lemma 8.8.1(d)), and for $g \in G_3$ it follows from the proof of Lemma 8.8.1(f) ($(t(1+u))^3 \in t^3 Q_6$ for $u \in \mathfrak{P}^3$); script 13 verifies it on the 219 good classes of Q_3/Q_7 (216 of depth three, 3 in Q_6/Q_7), and verifies that the cube root is $\mathrm{Ad}(Q_1)$-equivariant. Hence $F_\rho(\tilde{z}t^*) = \varepsilon_\rho(z)\Theta_\rho(t)$ (Lemma 8.13.2) is a function of the class of g in Q_3/Q_7 (times the character ε_ρ of U_1). Moreover Φ_ρ is invariant under right multiplication by $\ell(K^7)$: for $n \in K^7$ and g good in U_3, $\ell(gn) = \ell(g)\ell(n)$ (for $g \in U_6$ by (H1); for $g \in G_3$ by Lemma 8.10.4(d), since $\ell(n) \in N_6$ as $\theta_0(\mathrm{Nrd}\, n) = 1$), and gn is good (Lemma 8.8.1(f)). So Φ_ρ is a class function on the finite group $\hat{\Gamma} := \widetilde{U}_1/\ell(K^7)\ker\widetilde{\theta}$, of order 3^{10}, supported on the normal subgroup $\hat{H} := \widetilde{U}_3/\ell(K^7)\ker\widetilde{\theta}$ of index 27, which is a central extension of $H := Q_3/Q_7$ (order 3^6) by μ_3. (In the normalization $\theta_1 = \mathbf{1}$ fixed above, $\ker\widetilde{\theta} = \ker\widetilde{\theta}_1 = \{(1,z)\}$ when $\theta_\rho = \mathbf{1}$; when $\theta_\rho = \mathrm{sgn}$ we use $\ker\widetilde{\theta}_1$ in place of $\ker\widetilde{\theta}$ and carry the character $\varepsilon_\rho \circ p$ of $\widetilde{U}_1$ separately, as in Remark (b) below.)

Step 2: the restriction to $\widetilde{U}_3$. By Theorem 8.12.1(a), $\Phi_1|_{\widetilde{U}_3} = \sum_{t \in \widetilde{U}_1/\widetilde{U}_3} \lambda_0^t = \lambda_0 \cdot (D \circ p)$ with $D(v) = \sum_{h \in \Lambda} \zeta^{h(v)}$, a function on $V = Q_3/Q_6$ which is 27 at $v = 0$, $9\zeta^{-e_*(v)}$ on $\bar{G}_3$ and 0 elsewhere (Proposition 8.10.5). Hence

$$\Phi_\rho|_{\widetilde{U}_3} = (\varepsilon_\rho \circ p)\, \lambda_0 \cdot \big((\Theta_\rho \circ r)\, D\big) \circ p,$$

where $(\Theta_\rho \circ r)D$ is a class function on H supported on the good classes. We compute it by hand, using Lemma 8.13.7, whose second identity reads, for $h \in H$,

$$\sum_{\chi'(1)=3,\ \chi'|_{H'}=3\omega} \chi'(h) = 81\,\omega(h)\,\mathbf{1}_{H'}(h), \qquad \sum_{\chi'(1)=3} \chi'(h) = \begin{cases} 162, & h = 1, \\ -81, & h \in H' \setminus 1, \\ 0, & h \notin H'. \end{cases} \tag{8.40}$$

On the other side, let χ_3 be one of the two three-dimensional characters of Heis, with central character ω_3 on Q_2/Q_3; it vanishes off the center. For a good class g of depth exactly three, $r(g) \in$ Heis has depth one, so $(\chi_3 \circ r)(g) = 0$. For $g \in Q_6/Q_7 = H'$ one has $D(g) = 27$, and r restricted to H' is the inverse of the Artin–Schreier bijection $Q_2/Q_3 \to Q_6/Q_7$ of Lemma 8.8.1(d), an isomorphism of groups of order 3; so $(\chi_3 \circ r)(g) = 3\omega_3(r(g))$ and $(\chi_3 \circ r)D = 81\,(\omega_3 \circ r)\mathbf{1}_{H'}$. Comparing with (8.40),

$$(\chi_3 \circ r) \cdot D = \sum_{\chi'(1)=3,\ \chi'|_{H'}=3\,\omega_3 \circ r} \chi', \qquad \big((\chi_3 + \chi_3') \circ r\big) \cdot D = \sum_{\chi'(1)=3} \chi', \tag{8.41}$$

each of the 54 characters occurring exactly once. (Script 14 confirms this from the character table of H; it also returns, for the orbit sums of linear characters of Heis, the sums of 27, resp. 108, linear characters of H, in accordance with Theorem 8.13.5.) Thus $\Phi_\rho|_{\widetilde{U}_3} = (\varepsilon_\rho \circ p)\sum_{\mu\in S}\mu$, where S is the set of the 54 genuine irreducible characters $\mu = \lambda_0 \cdot \chi' \circ p$ of $\hat{H}$ of degree 3 with central character ζ on μ_3: a multiplicity-free character of dimension 162.

Step 3: Clifford theory. A class function on $\hat{\Gamma}$ supported on the normal subgroup $\hat{H}$ is determined by its restriction to $\hat{H}$, and for $\mu \in \mathrm{Irr}(\hat{H})$ with $\hat{\Gamma}$-orbit O and stabilizer $\hat{\Gamma}_\mu$, the character of $\mathrm{Ind}_{\hat{H}}^{\hat{\Gamma}}\mu$ is supported on $\hat{H}$ with restriction $[\hat{\Gamma}_\mu : \hat{H}]\sum_{\mu'\in O}\mu'$. By Proposition 8.13.9 below, $\mathrm{Ad}(Q_1)$ acts freely on the genuine irreducible characters of $\hat{H}$, so S is the union of two orbits O_1, O_2 of size 27 (one for each ω, by (8.41) and the invariance of the central character), and

$$\Phi_\rho = (\varepsilon_\rho \circ p)\big(\Theta_{\mathrm{Ind}\,\mu_1} + \Theta_{\mathrm{Ind}\,\mu_2}\big), \qquad \mu_i \in O_i,$$

with $\mathrm{Ind}_{\hat{H}}^{\hat{\Gamma}}\mu_i$ irreducible of dimension 81 and distinct. (Had some orbit $O \subset S$ had size $|O| < 27$, then for an irreducible constituent χ of $\mathrm{Ind}\,\mu_O$ with $\chi|_{\hat{H}} = e\sum_O \mu'$ one would have $\langle \Phi_\rho, \chi\rangle = e|O|/27 \notin \mathbb{Z}$, since $e^2 \le [\hat{\Gamma}_\mu : \hat{H}] = 27/|O|$.) This proves the displayed decomposition, and $\dim \tau_\rho = 162 = 27\cdot 6$. □

Proposition 8.13.9 ($\mathbf{P}^{\mathbf{c}}$) Freeness). *$\mathrm{Ad}(\widetilde{U}_1)$ acts freely on the set of genuine irreducible characters of $\hat{H} = \widetilde{U}_3/\ell(K^7)\ker\widetilde{\theta}_1$: for every genuine irreducible μ and every $t \in U_1 \setminus U_3$, $\mu \circ \mathrm{Ad}(t) \neq \mu$. Equivalently, $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}\mu$ is irreducible for every genuine irreducible representation μ of $\widetilde{U}_3$ trivial on $\ell(K^7)\ker\widetilde{\theta}_1$.*

Proof. Let $A := Q_4/Q_7$, central in H, and $\hat{A}$ its preimage in $\hat{H}$. $\hat{A}$ is central in $\hat{H}$: for $x \in K^3$, $y \in K^4$ one has $[x,y] \in K^7$, so $[\tilde{x},\tilde{y}] = \zeta^a \ell([x,y])$ with $\zeta^a = \psi_6([\tilde{x},\tilde{y}]) = \omega(x,y) = 1$ by Lemma 8.10.4(c), and $\ell([x,y]) \in \ell(K^7)$. Hence $\mu|_{\hat{A}} = \mu(1)\,\hat{\nu}$ for a genuine character $\hat{\nu}$ of $\hat{A}$, and $\mu \circ \mathrm{Ad}(t) = \mu$ implies $\hat{\nu} \circ \mathrm{Ad}(t) = \hat{\nu}$. Write $\hat{\nu} = \lambda_0|_{\hat{A}} \cdot (\alpha \circ p)$ with α a character of A, and let $\omega_\mu := \alpha|_{H'}$ (possibly trivial). Since $\lambda_0(\ell(u)) = 1$ for $u \in U_6$ ($\ell(u) \in N_6$, as $\psi_6(\ell(u)) = \theta_1(z_u) = 1$), the restriction of $\hat{\nu}$ to $\ell(Q_6/Q_7)$ is ω_μ. For $y \in A$,

$$\frac{\hat{\nu}\circ\mathrm{Ad}(t)}{\hat{\nu}}(\tilde{y}) = \hat{\nu}([\tilde{t},\tilde{y}]) = \frac{\lambda_0^t}{\lambda_0}(\tilde{y})\cdot\omega_\mu([t,y]) = c_t(y)\,\omega_\mu([t,y]),$$

where $c_t := \lambda_0^t/\lambda_0 = \zeta^{h_t}$ is a character of V with $h_t \in \Lambda$, and $t \mapsto h_t$ is a bijection $Q_1/Q_3 \to \Lambda$ with $h_t = 0$ iff $t \in Q_3$ (because $\sum_t \lambda_0^t = \Phi_1|_{\widetilde{U}_3} = \sum_{h\in\Lambda}\lambda_h$, Theorem 8.12.1(a), and the 27 characters λ_h are distinct). Let $t \notin Q_3$.

If $t \in Q_2$, then $[t,y] \in K^7$ for all $y \in K^4$ (the leading brackets $[\varpi a, \varpi^2 c] = 0$ and $[\varpi a, \varpi^2 d\,j] = \varpi^3 d(a-\bar{a})j$ have depth ≥ 7), so the displayed ratio is $c_t|_A$, which is nontrivial: no nonzero element of Λ vanishes on the coordinates $(C_{\mathrm{im}}, B_{\mathrm{re}}, B_{\mathrm{im}})$ of Q_4/Q_6 (Appendix A.1).

If t has depth exactly one with leading coefficient $a \in \mathbb{F}_9^\times$, take $y \in Q_5$ with leading coefficient b (so $y = 1 + \varpi^2 b\,j + \cdots$). Then $c_t(y) = \lambda_0([\tilde{t},\tilde{y}]) = \zeta^{\kappa_{1,5}(t,y)} = \zeta^{\beta\,\mathrm{Tr}(a\bar{b})}$ with $\beta \neq 0$ (Lemma 8.10.4(b)), and $[t,y] \in Q_6/Q_7 \simeq \mathbb{F}_9/\mathbb{F}_3$ is the class of $\varpi^3(a\bar{b} - \bar{a}b) = 2i\,\mathrm{Im}(a\bar{b})\varpi^3$, so $\omega_\mu([t,y]) = \zeta^{2\gamma\mathrm{Im}(a\bar{b})}$ for some $\gamma \in \mathbb{F}_3$ depending on μ. As b runs over $\mathbb{F}_9$ so does $u = a\bar{b}$, and $u \mapsto 2\beta\,\mathrm{Re}(u) + 2\gamma\mathrm{Im}(u)$ is a nonzero $\mathbb{F}_3$-linear functional. So the ratio is nontrivial on Q_5.

In both cases $\hat{\nu}\circ\mathrm{Ad}(t) \neq \hat{\nu}$, hence $\mu\circ\mathrm{Ad}(t) \neq \mu$. □

Remark 8.13.10 (Certificate)**.** Script 15 computes λ_0 on $\hat{H}$ in the cocycle model, the twisted action $\mu^t = \lambda_0 c_t(\chi' \circ \mathrm{Ad}\, t)$ of the 27 coset representatives on the 54 characters of S and on the 243 genuine linear characters, verifies that no nontrivial t fixes any of them, and writes the orbit table to `level2_orbit_certificate.txt`: two orbits of size 27 on S, nine of size 27 on the linear characters (the nine induced representations of Theorem 8.13.5). For comparison, the untwisted action $\chi' \mapsto \chi' \circ \mathrm{Ad}\, t$ on the 54 characters of H has six orbits of size 9: the λ_0-twist is what makes the action free. These computations corroborate Proposition 8.13.9 and are not used in its proof.

Remark 8.13.11. (a) The criterion of Proposition 8.13.4 is sufficient but not necessary: here $\tau_\rho|_{\widetilde{U}_1}$ *is* of the form $\mathrm{Ind}(\lambda_0 \otimes \sigma)$ with $\sigma = \chi_1' + \chi_2'$, but Θ_σ is not $\mathrm{Ad}(U_1)$-invariant, and the product formula $\Theta_{\mathrm{Ind}(\lambda_0\otimes\sigma)} = \Theta_\sigma \Theta_{\mathrm{Ind}\,\lambda_0}$ fails. (b) The ε_ρ bookkeeping: for $\theta_\rho = \mathrm{sgn}$, ε_ρ is a character of U_1 trivial on K^1, so twisting by $\varepsilon_\rho \circ p$ preserves virtual characters of $\widetilde{U}_1$ and commutes with everything above. (c) Label. The theorem rests on Target A through λ_0, Λ and e_* only (Theorem 8.12.1(a), Proposition 8.10.6, and the table of Λ); Steps 2 and 3 are hand arguments. Scripts 13–15 are corroboration.

8.13.5 Beyond level two

This subsection describes the pattern common to the two proofs above and isolates the statements that an extension to higher levels would need. It is an outline, as its label says.

(O) The proofs of Theorems 8.13.5 and 8.13.8 both run as follows. For $m \geq 1$ put $N_m := \max(3m+1, 6)$, $\mathrm{Heis}_m := Q_1/Q_{m+1}$, $H_m := Q_3/Q_{N_m}$, $\hat{H}_m := \widetilde{U}_3/\ell(K^{N_m})\ker\widetilde{\theta}_1$ and $\hat{\Gamma}_m := \widetilde{U}_1/\ell(K^{N_m})\ker\widetilde{\theta}_1$ (for $m=1$, $\hat{H}_1 = \widetilde{U}_3/N_6 = \widetilde{V}$, whose genuine characters are the 243 characters λ_h, $h \in V^\vee$; for $m=2$ these are the groups of Theorem 8.13.8). The cube root induces a well-defined *surjection* r from the good classes of H_m onto Heis_m, many-to-one (219 good classes of H_2 onto the 27 elements of Heis_2), whose fibers are the sets $\{t^3 Q_{N_m}\}$ for t in a fixed class of Q_1/Q_{m+1}. For ρ trivial on K^{m+1}, $\Phi_\rho|_{\widetilde{U}_3} = \lambda_0 \cdot ((\Theta_\rho \circ r)D)$, and the two statements

(B1) $(\Theta_\rho \circ r)D$ is a character of H_m (for $m \leq 2$: for the D_8-invariant character $\Theta_\rho|_{K^1} = \sum m_\psi \psi$ of Heis_1 it is $\sum_\psi m_\psi \sum_{h\in\Lambda} \zeta^h \cdot \psi \circ \pi_3$, and for $\chi_3 + \chi_3'$ it is (8.41));

(B2) every genuine irreducible constituent μ of $\Phi_\rho|_{\hat{H}_m}$ has trivial inertia in $\hat{\Gamma}_m/\hat{H}_m \simeq Q_1/Q_3$, or more generally the inertia indices and multiplicities satisfy the divisibility conditions of Step 3 (for $m \leq 2$: freeness, Proposition 8.13.9 and the free orbits of Proposition 8.10.6),

give $\tau_\rho|_{\widetilde{U}_1} = (\varepsilon_\rho \circ p) \otimes \bigoplus_i \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \mu_i$, the μ_i running over one representative of each Q_1-orbit of genuine constituents. For $m=1$ the 243 genuine characters of $\hat{H}_1$ fall into nine free orbits of size 27, of which Λ is one; the nine induced representations are those of Theorem 8.13.5.

A warning about the shape of (B1): it holds for D_8-invariant characters of Heis_m, which are the only ones Target B needs, but not constituent-wise. For a single additive character χ of $\mathbb{F}_9$ the function $\chi \circ r$ on the depth-three good classes is $\chi(\pm\pi_3(g))$ with a sign depending on g ($w^5 = \mathrm{N}(w)w$), which is not $\chi' \circ \pi_3$ for any character χ', and $\Phi_1 \cdot (\chi \circ r)$ is not a virtual character of $\widetilde{U}_1$ for $\chi \neq \mathbf{1}$: with $\chi(a+bi) = \zeta^a$ and the character ζ^h of V with $h = (0,0,0,1,1)$, the table of e_* gives $\langle D \cdot (\chi \circ r), \zeta^h \rangle_V = \frac{1}{3}$ (the 72 good vectors contribute exponents with frequencies $28, 22, 22$).

At level two the constituent-wise identity (8.41) holds for χ_3 because χ_3 vanishes off the center of Heis_2 and $1+i$ acts trivially on that center.

For $m \geq 3$ both (B1) and (B2) are open; Section 8.15 describes what a proof would need and why the mechanism of level two does not continue unchanged.

8.14 Target C: uniform statements

This section records which parts of the argument are uniform in the odd prime p and the field F. Throughout: p is an odd prime, $F/\mathbb{Q}_p$ is finite with $\mu_p \subset F$ (so $(p-1) \mid e := e(F/\mathbb{Q}_p)$), $q = |k_F|$, ϖ is a uniformizer, $n = p$, E_0/F is the unramified quadratic extension, $D = E_0 \oplus E_0 j$ with $j^2 = \varpi$ and $jx = \sigma(x)j$, $K^m = 1+\mathfrak{P}^m$, and

$$d := \frac{pe}{p-1}, \qquad k_0 := \frac{2e}{p-1} = \frac{2d}{p}, \tag{8.42}$$

so that $U_{E_0}^{(k)} \subset E_0^{\times p}$ for $k > d$ and $(U_{E_0}^{(i)}, U_{E_0}^{(j)})_p = 1$ for $i+j > d$ [FV02, Ch. IV]; in the running example $p = 3$, $e = 2$, $d = 3$, $k_0 = 2$. The cover is the restriction along ι of the Kazhdan–Patterson cover of $\mathrm{GL}_2(E_0)$ with cocycle $\sigma_{E_0}(g,h)(\det g, \det h)_{E_0}^{c}$, σ_{E_0} as in (8.27) with the p-th power symbol, and $c \equiv -1/4 \pmod p$, which is the condition for scalars to be central (the computation of Lemma 8.11.1(b) gives $\beta(z,g) - \beta(g,z) = (1+4c)(z, \det g)$); for $p = 3$ this is $c = 2$. We write h_E for exponents of the p-th power symbol of E_0 and $(a,b)_E = h_E(a,b)$ additively.

Lemma 8.14.1 (Depth facts)**.** *(a) For $k > k_0$ the p-th power map $K^k \to K^{k+2e}$ is bijective.*

(b) If $w \in D^\times$ and $w^p \in F^\times$, then $w \in F^\times$.

(c) Every element of K^{2d} is of the form zv^p with $z \in U_F^{(1)}$ and $v \in K^{k_0}$; z is unique modulo $F^{\times p}$ and $z \equiv \mathrm{Nrd}(X)^{(p+1)/2} \pmod{F^{\times p}}$.

Proof. (a) As in Lemma 8.8.1(a): $v_D(p) = 2e$, and for $u \in \mathfrak{P}^k$ the terms $\binom{p}{i}u^i$ $(2 \leq i \leq p-1)$ and u^p lie in $\mathfrak{P}^{k+2e+1}$ when $k > k_0$. (b) If $w \notin F$ then $F(w)$ is a quadratic field in which $X^p - c$ $(c = w^p)$ has a root; since $\mu_p \subset F$, $X^p - c$ is either split or irreducible over F, and $p > 2$ excludes the latter, so $w \in \mu_p w_0 \subset F$ for a root $w_0 \in F$. (c) Every $X \in K^{2d} \setminus F^\times$ lies in a unique maximal subfield E (and $X \in F^\times \cap K^{2d} = U_F^{(d)}$ is trivially of the required form). Recall the standard fact that for a local field $L \supset \mu_p$ with $e_L = e(L/\mathbb{Q}_p)$, the p-th power map $U_L^{(k)} \to U_L^{(k+e_L)}$ is bijective for $k > e_L/(p-1)$ [FV02, Ch. I].

E ramified. Then $e_E = 2e$, $k_E = k_F$, $E \cap K^{2d} = U_E^{(2d)}$ (as $v_D = v_E$), and $U_F^{(d)} \subset U_E^{(2d)}$ induces an isomorphism $U_F^{(d)}/U_F^{(d+1)} \to U_E^{(2d)}/U_E^{(2d+1)}$ of residue groups ($\varpi_E^2 = \varpi \cdot$unit). Choose $z \in U_F^{(d)}$ with $z^{-1}X \in U_E^{(2d+1)}$. Since $k_0 + 1 > 2e/(p-1) = e_E/(p-1)$ and $k_0 + 1 + 2e = 2d+1$, the p-th power map $U_E^{(k_0+1)} \to U_E^{(2d+1)}$ is bijective, so $z^{-1}X = v^p$ with $v \in U_E^{(k_0+1)} \subset K^{k_0+1}$.

E unramified. Then $e_E = e$, $k_E = \mathbb{F}_{q^2}$, $E \cap K^{2d} = U_E^{(d)}$ (as $v_D = 2v_E$), and p-th powers of $U_E^{(k)}$ fill $U_E^{(k+e)}$ for $k > e/(p-1)$; the boundary level $k = e/(p-1) = k_0/2$ needs an argument. Write $\zeta_p = 1 + \varpi^{e/(p-1)}a_0 + \cdots$ with $a_0 \in \mathscr{O}_F^\times$ (possible since $v_F(\zeta_p - 1) = e/(p-1)$), so that $p/\varpi^e \equiv -\bar{a}_0^{p-1}$ in k_F (from $\zeta_p^p = 1$). For $u = 1 + \varpi^{e/(p-1)}a$, $u^p \equiv 1 + \varpi^d(a^p + (p/\varpi^e)a)$ modulo

$U_E^{(d+1)}$, so the induced map $k_E \to k_E$ on the graded pieces $U_E^{(e/(p-1))}/U_E^{(e/(p-1)+1)} \to U_E^{(d)}/U_E^{(d+1)}$ is $\bar a \mapsto \bar a^p - \bar a_0^{p-1}\bar a = \bar a_0^p\,\wp(\bar a/\bar a_0)$ with $\wp(x) = x^p - x$. Its image is $\bar a_0^p \ker(\mathrm{Tr}_{k_E/\mathbb{F}_p})$, of index p in k_E. The image of $U_F^{(d)}$ in $U_E^{(d)}/U_E^{(d+1)}$ is k_F, and $k_F \not\subset \bar a_0^p \ker(\mathrm{Tr}_{k_E/\mathbb{F}_p})$: for $u \in k_F$, $\mathrm{Tr}_{k_E/\mathbb{F}_p}(u/\bar a_0^p) = 2\,\mathrm{Tr}_{k_F/\mathbb{F}_p}(u/\bar a_0^p)$, which is nonzero for suitable u because p is odd and the trace $k_F \to \mathbb{F}_p$ is onto. Hence k_F and the image together span k_E: choose $z \in U_F^{(d)}$ and $v_0 \in U_E^{(e/(p-1))}$ with $z^{-1}v_0^{-p}X \in U_E^{(d+1)} = (U_E^{(e/(p-1)+1)})^p$, say $= v_1^p$, and put $v = v_0 v_1 \in U_E^{(e/(p-1))} \subset K^{k_0}$ (as $v_D = 2v_E$).

In both cases $z \in U_F^{(d)} \subset U_F^{(1)}$ and $v \in E \cap K^{k_0}$. Uniqueness and the formula for z: $\mathrm{Nrd}X = z^2\,\mathrm{Nrd}(v)^p$, and squaring is bijective on $F^\times/F^{\times p}$, with inverse $y \mapsto y^{(p+1)/2}$. □

Theorem 8.14.2 (The p-th power identity)**.** *Let $w = x + yj \in K^1$. Then $(w,0)^p = (w^p,0)$ in the cover.*

Proof. If $y = 0$ the diagonal formula gives exponent $\sum_{k=1}^{p-1}\big(h_E(x^k,\sigma x) + 2ck\,h_E(\mathrm{N}x,\mathrm{N}x)\big) = \frac{p(p-1)}{2}h_E(x,\sigma x) = 0$. Let $y \neq 0$, $T = \mathrm{Trd}\,w$, $n = \mathrm{Nrd}\,w$, and define $S_k \in F$ by $S_0 = 0$, $S_1 = 1$, $S_{k+1} = TS_k - nS_{k-1}$, so that $w^k = S_k w - nS_{k-1}$ by Cayley–Hamilton, i.e. $w^k = x_k + y_k j$ with $y_k = S_k y$ and $x_k = S_k x - nS_{k-1}$. By Lemma 8.14.1(b), $S_k \neq 0$ for $1 \le k \le p$ ($S_k = 0$ would put $w^k \in E_0$ hence, as $w \notin E_0$, $w^k \in F$; for $k < p$ this forces $w \in F$ because $K^1F^\times/F^\times$ is pro-p, and for $k = p$ by (b)). In the cocycle $\beta(w^k,w)$ one has $x(\iota w^k) = \sigma(y)S_k$, $x(\iota w) = \sigma(y)$, $x(\iota w^{k+1}) = \sigma(y)S_{k+1}$, $\det \iota w^k = n^k$, so with $\tau = 1/S_k$ and $R = S_{k+1}/S_k$ (all in F), by the computation leading to (8.33),

$$\beta(w^k,w) = -(R,\tau)_E - (R,n^k)_E + c\,(n^k,n)_E = (S_{k+1},S_k)_E - k(S_{k+1},n)_E + k(S_k,n)_E,$$

using $(S_k,S_k)_E = (n,n)_E = 0$. Put $a_k := (S_{k+1},S_k)_E$ and $b_k := (S_k,n)_E$ (so $a_1 = b_1 = 0$). Summing over $k = 1,\dots,p-1$ and collecting the b's,

$$\sum_{k=1}^{p-1}\beta(w^k,w) = \sum_{k=1}^{p-1}a_k + \sum_{k=2}^{p}b_k \pmod p. \tag{8.43}$$

The Cassini identity $S_{k+1}S_{k-1} - S_k^2 = -n^{k-1}$ (induction on k) and the Steinberg relation for $u = S_{k+1}S_{k-1}/S_k^2$, $1-u = n^{k-1}/S_k^2$ (both nonzero for $2 \le k \le p-1$) give, after expanding bi-multiplicatively,

$$2(a_k - a_{k-1}) = (k-1)\,(b_{k+1} - 2b_k + b_{k-1}) \qquad (2 \le k \le p-1).$$

Hence $a_k = \frac12\sum_{j=2}^{k}(j-1)\Delta^2 b_j$, where $\Delta^2 b_j := b_{j+1} - 2b_j + b_{j-1}$, and interchanging the sums,

$$\sum_{k=1}^{p-1}a_k = \sum_{j=2}^{p-1}c_j\Delta^2 b_j, \qquad c_j := \tfrac12(j-1)(p-j) \quad \text{(computations in } \mathbb{Z}[\tfrac12]/p).$$

Extend c_j to all integers j by the same quadratic formula; then $c_{j-1} - 2c_j + c_{j+1} = -1$ for every j, $c_1 = c_p = 0$ and $c_0 = c_{p+1} = -p/2 \equiv 0$. Expanding $\Delta^2 b_j$ and reindexing,

$$\sum_{j=2}^{p-1}c_j\Delta^2 b_j = \sum_{j=1}^{p}b_j\,(c_{j-1} - 2c_j + c_{j+1}) - \big(c_0b_1 + c_1b_2 + c_pb_{p-1} + c_{p+1}b_p\big) \equiv -\sum_{j=1}^{p}b_j,$$

the four correction terms being those of the full double sum $\sum_{j=1}^{p}b_j(c_{j-1} - 2c_j + c_{j+1})$ that do not occur in $\sum_{j=2}^{p-1}c_j\Delta^2 b_j$, each of which vanishes modulo p. Since $b_1 = 0$, (8.43) vanishes. For $p = 3$ this is the computation of Proposition 8.11.2. □

Theorem 8.14.3 ((H1) for every odd p)**.** *Define the star section on $U_{2d} = F^\times K^{2d}$ by $\ell(zv^p) := (z,0)(v,0)^p$ with $v \in K^{k_0}$. It is well defined,* $\mathrm{Ad}(D^\times)$*-equivariant, and a group homomorphism; explicitly $\ell(X) = (X, \kappa(X))$ with*

$$\kappa(X) = -\tfrac{p+1}{2}\,(y_X, \mathrm{Nrd}X)_E \quad (y_X \neq 0), \qquad \kappa(X) = 0 \quad (y_X = 0).$$

Consequently, for θ_0 with $\theta_0^p = 1$ and $\widetilde{\theta}(\zeta, z) = \zeta\,\theta_0(z)$, the analogue ψ_{2d} of ψ_6 ($\zeta\,\widetilde{\theta}(\tilde{z})$ at $\zeta\tilde{z}v^*$) is a character of $\widetilde{U}_{2d}$, given on $\tilde{K}^{2d}$ by

$$(X, e) \longmapsto \zeta^{\,e-\kappa(X)}\,\theta_0(\mathrm{Nrd}X)^{(p+1)/2},$$

since the value at the star lift $\ell(X)$ is $\theta_0(z)$ with $z \equiv \mathrm{Nrd}(X)^{(p+1)/2}$ (Lemma 8.14.1(c)). For $p = 3$ the exponent $2 \equiv -1$ recovers (8.31).

Proof. Theorem 8.14.2 gives $\ell(zv^p) = (zv^p, \beta(z, v^p))$, and $\beta(z,g) = (z, x(g))_E + 2c(z, \det g)_E$ with $\det g = \mathrm{Nrd}(v)^p$, so $\kappa(X) = (z, y_X/z)_E = (z, y_X)_E = \frac{p+1}{2}(\mathrm{Nrd}X, y_X)_E$ by Lemma 8.14.1(c) (and 0 if $y_X = 0$, as $(z, x_X)_E = \frac{p+1}{2}(x_X \sigma x_X, x_X)_E = 0$ for $x_X \in U^{(d)}$). Well-definedness is the uniqueness in Lemma 8.14.1(c) together with $h_E(F^{\times p}, \cdot) = 0$. For multiplicativity we repeat the proof of Proposition 8.11.4 with $m := -\frac{p+1}{2}$, so $2m \equiv -1$, for $X = a + bj$, $X' = a' + b'j \in K^{2d}$: now $a, a' \in U^{(d)}$, $b, b' \in \varpi^d \mathcal{O}$, $n, n' \in U_F^{(d)}$, all mutual symbols among $a, \sigma a, a', \sigma a', n, n'$ vanish ($2d > d$), the twist $c(n, n')_E$ vanishes, $n \equiv \mathrm{N}a$ modulo $U^{(2d+1)} \subset E_0^{\times p}$, and (8.28)–(8.32) hold with $U^{(3)}$ replaced by $U^{(d)}$ (as $1 + d > d$ and $\mu_{q^2-1} \subset E_0^{\times p}$), with $\lambda(a) = (\varpi, \mathrm{N}a)_E = 2(\varpi, a)_E$. In the main case (8.30) becomes $(R, \tau)_E = m(\tau, n')_E - (m+1)(R, n)_E - m(R, n')_E = -v(\tau)(\varpi, a')_E - v(R)(\varpi, a)_E + v(R)(\varpi, a')_E$, which is exactly the value of $(R, \tau)_E$ computed from $R = a'(1 + \tau_2)$; in the cases $b = 0$, $b' = 0$, $b = b' = 0$ and $R = 0$ the required identities are $(2m+1)v(b')(\varpi, a)_E = 0$, $2mv(b)(\varpi, a')_E = -v(b)(\varpi, a')_E$, $0 = 0$ and $(2m+1)v(b)[(\varpi, a)_E + (\varpi, a')_E] = 0$ respectively, all true since $2m + 1 \equiv 0$. The passage from K^{2d} to U_{2d} and the formula for ψ_{2d} are as in Corollary 8.11.3(d). □

Remark 8.14.4 (What is and is not uniform)**.** (i) Uniform in p and F (for $n = p$): the toral theorem (Section 8.5); the positivity criterion; the cocycle model and the p-th power identity; the depth facts; hypothesis (H1) in the form of Theorem 8.14.3; the product formula and the criterion of Section 8.13 (with 3 replaced by p, K^3 by K^p and λ_0 by the character produced by the analogue of Theorem 8.10.7, once available).

(ii) **(O)** Expected to be uniform but not written: the Brauer reduction. $D^\times / F^\times K^1$ is now the group of order $2(q+1)$ generated by $k_{E_0}^\times / k_F^\times$ and j; the centralizer lemma uses only that p is odd; the ℓ-elementary subgroups for $\ell \neq p$ live in tori or in the preimage of a 2-subgroup, and the 2-group lemma needs the same analysis as in Theorem 8.13.1, with SD_{16} replaced by the appropriate extension.

(iii) **(C)** Not uniform: the finite design. The good elements of depth exactly p (the analogue of G_3, with leading coefficients $w^{1+(p-1)(q+1)/2}$, which need not exhaust $k_{E_0}^\times$ when $\gcd(1 + (p-1)(q+1)/2, q^2 - 1) > 1$) and the quotient $V = Q_p / Q_{2d}$ depend on (p, e, q), and the uniqueness of the defect function was a computation. A conceptual replacement for Section 8.10 is the main open problem of Target C; the next test cases are $F = \mathbb{Q}_3(\zeta_3, \sqrt{3})$ ($e = 4$, $d = 6$) and $F = \mathbb{Q}_5(\zeta_5)$ ($e = 4$, $d = 5$).

(iv) For composite odd n and for $p \nmid n$ nothing new is needed at tame places (Chapter 6); for $n = p^a m$ with $a \geq 2$ the depth bookkeeping changes (n-th powers of K^k land in K^{k+2ae}) and we have not examined it.

8.14.1 Proof checkpoints

This subsection lists what a positive solution for a general row must supply, and notes which items are now available.

A positive solution for a row must still supply (i) an explicit cocycle/central-character normalization (now available, uniformly in p), (ii) a finite-level factorization or an induced construction (now: Ind from $\widetilde{U}_3$, Theorems 8.13.5, 8.13.8), (iii) exact full-group character coefficients (now: the Fourier identities of Proposition 8.10.5), and (iv) a uniform all-ρ argument (open beyond level two).

8.15 Future directions

We stop at level two. This section records, as precisely as we can, what a proof of Target B for all levels would need, what the first open case looks like, and which parts of Target C are within reach. Nothing here is claimed.

8.15.1 Why level two is a natural stopping point

Levels are unbounded (Section 8.2.1), so no finite list of levels proves Target B; what is needed is an inductive mechanism, and the proofs of Sections 8.13 do not contain one. They rely on features special to $m \leq 2$: the group $H_2 = Q_3/Q_7$ has class two with derived subgroup of order 3, so its nonlinear characters are controlled by the regular character (Lemma 8.13.7); $\Theta_\rho \circ r$ vanishes off a central subgroup, so (B1) is a one-line identity; and the central subgroup $\hat{A} = \widehat{Q_4/Q_7}$ is large enough for central characters to detect every nontrivial $t \in Q_1/Q_3$ (Proposition 8.13.9). At $m = 3$ all three fail at once: $H_3 = Q_3/Q_{10}$ has order 3^{11}, class 3, and its largest obvious abelian normal subgroup is Q_5/Q_{10} (note that Q_4/Q_{10} is *not* abelian: $[1+\varpi^2 i, 1+\varpi^2 j] \equiv 1 + 2\varpi^4 ij$ modulo K^{10}, of depth 9); $\Theta_\rho \circ r$ for ρ of level three is supported on the preimages of the noncentral elements of $\mathrm{Heis}_3 = Q_1/Q_4$ as well; and the natural central subgroup $\hat{A}_3 = \widehat{Q_7/Q_{10}}$ lies inside $\widetilde{U}_6$, where λ_0 is trivial and only the brackets $[t,\cdot]$ remain to separate characters. The level-two theorem is therefore a complete result about a finite family of rows, and an honest statement of the present state is: Target B is proved for every row of level ≤ 2, and open beyond.

8.15.2 The two statements to prove at each level

For ρ of level m, with the notation of Section 8.13.5, Target B for ρ follows from (B1) and (B2); conversely, if Φ_ρ is a virtual character of $\widetilde{U}_1$ then its restriction to $\hat{H}_m$ is one, so (B1) is necessary. The two should be tested separately: a failure of (B1) (a non-integral multiplicity in $(\Theta_\rho \circ r)D$) would be a genuine obstruction to Target B, exhibiting a row with a non-integral coefficient $a_{\rho\tau}$; a failure of universal freeness in (B2) would not, since the divisibility conditions of Step 3 of Theorem 8.13.8 ($e|O| \in 27\mathbb{Z}$ for each constituent) can hold with nontrivial inertia.

8.15.3 Level three

The first open case is $m = 3$, ρ trivial on K^4. A finite verification of (B1) and (B2) is feasible but needs a different method from scripts 13–15: H_3 is too large for Burnside's algorithm, and one should instead describe $\mathrm{Irr}(H_3)$ by Clifford theory relative to the abelian normal subgroup Q_5/Q_{10} (abelian because $[K^5, K^5] \subset K^{10}$), determine the action of H_3 on its dual, and compute the inner products of $(\Theta_\rho \circ r)D$ with the resulting characters, where now Θ_ρ is a D_8-invariant character of Heis_3 of order 3^5 and the good classes of H_3 number $216 \cdot 3^4 + \cdots$. For (B2) one needs the twisted action of Q_1/Q_3 on the genuine constituents, i.e. λ_0 on $\hat{H}_3$ and the brackets; the certificate format of script 15 would carry over. Whatever the outcome, level three decides whether the mechanism of level two (multiplicity-free restriction, free action, induced irreducibles of dimension $27 \dim \mu$) is the general shape of $\tau_\rho|_{\widetilde{U}_1}$ or a low-level accident.

8.15.4 Toward an inductive mechanism

Two ideas seem to us the most promising. First, an orbit-theoretic description. Howe's Kirillov theory [How77a] describes the characters of the congruence subgroups K^m (for m large relative to p) by coadjoint orbits on a Lie lattice; the genuine constituents of $\Phi_\rho|_{\widetilde{U}_3}$ should correspond to orbits shifted by the fixed "metaplectic" datum that λ_0 represents, and the twisted action of Q_1/Q_3 on them to an affine coadjoint action. The question is whether the nonvanishing of $\kappa_{1,5}$, which forced regularity at level two, continues to force it at all depths or whether deeper stabilizers can absorb it. Second, an inductive form of freeness: for $m \geq 3$ the subgroup $\hat{A}_m = \widehat{Q_{3m-2}/Q_{3m+1}}$ is central in $\hat{H}_m$ with no hypothesis (for $x \in K^3$, $y \in K^{3m-2} \subset U_6$, $[\tilde{x}, \tilde{y}] = \ell([x,y]) \in \ell(K^{3m+1})$ because ℓ is an Ad-equivariant homomorphism on U_6), so a genuine irreducible μ of $\hat{H}_m$ has a central character $\lambda_0 \alpha_\mu$ on $\hat{A}_m$ and $\mu \circ \mathrm{Ad}(t) = \mu$ forces $\alpha_\mu([t, A_m]) = 1$; freeness for the constituents of exact level m would follow from nondegeneracy of the brackets $Q_1/Q_2 \times Q_{3m-2}/Q_{3m-1} \to Q_{3m-1}/Q_{3m}$ and $Q_1/Q_2 \times Q_{3m-1}/Q_{3m} \to Q_{3m}/Q_{3m+1}$ on the relevant characters, and for the others from level $m-1$; but the characters α_μ that actually occur must first be identified, which is (B1) again.

8.15.5 Target C

Of the uniform statements, the Brauer reduction (Theorem 8.13.1) should go through for every odd p and F with the same proof, the 2-group step being replaced by an analysis of the preimage of the group $D^\times/F^\times K^1$ of order $2(q+1)$; the finite design of Section 8.10 is the part with no uniform argument, and the two test cases named in Remark 8.14.4 ($e = 4$ with $p = 3$, and $p = 5$) would show whether the uniqueness of the defect function is special to $(p, e, q) = (3, 2, 3)$.

8.15.6 The duality

Remark 8.16.1 reformulates Target B as the existence of a representation Ω_θ of $\widetilde{H} \times H$ with the multiplicity-one property of a Howe duality, whose level-m truncations have explicit characters. A construction of Ω_θ that does not go through L_D, for instance on a space of functions on which $\widetilde{H}$ and H act through the cube-root map and the star section, would prove Target B and explain it; the induced models of Theorems 8.13.5 and 8.13.8 are what such a construction must produce at levels ≤ 2.

8.16 Interface with the larger Jacquet–Langlands program

The companion problem is local. Theorem 8.12.1 and Corollary 8.12.4 remove, for the rows they cover, the "L^2-unitarity-to-integrality" gap recorded in Chapter 6, at the wild place $\mathfrak{p}_3$ of the running example; in particular the character relation (D2) of Chapter 4 now has, at $\mathfrak{p}_3$, a proved instance for $\rho = \mathbf{1}$ and the level-zero rows. Even full success on Target B would not on its own prove the additional global comparison of ordinary $D^\times$ and the covered group: that comparison remains conditional on matching functions at exceptional split places, complex-place normalization, a global orbital-coefficient identity, and the support/germ hypotheses stated in Chapter 7. Conversely, the global comparison could be established in a restricted test range without first settling every local irreducible-character row. These projects should be developed in parallel but not conflated.

Remark 8.16.1 (Target B as a Howe duality)**.** Fix a character θ of $F^\times$ trivial on μ_3 and a compatible $\widetilde{\theta}$, and for $m \geq 0$ let $\mathrm{Irr}_\theta^{\leq m}$ be the finite set of irreducible representations of H with central character θ and level $\leq m$, i.e. the irreducible representations of the finite group H/K^{m+1} with that central character. By Lemma 8.13.2 (whose proof does not use (8.35)), $L_D\Theta_\rho = \Theta_{\tau_1} \cdot F_\rho$ with $F_\rho(\zeta\tilde{z}t^*) = \widetilde{\theta}\widetilde{\theta}_1^{-1}(\tilde{z})\Theta_\rho(t)$ on star lifts of good elements, and $L_D\Theta_\rho = 0$ elsewhere. Hence the finite sum

$$K_\theta^{(m)}(x,g) := \sum_{\rho \in \mathrm{Irr}_\theta^{\leq m}} L_D\Theta_\rho(x)\,\overline{\Theta_\rho(g)} = \Theta_{\tau_1}(x)\,\widetilde{\theta}\widetilde{\theta}_1^{-1}(\tilde{z})\,\Delta_\theta^{(m)}(t,g), \qquad x = \zeta\tilde{z}t^*,\ g \in H,$$

where $\Delta_\theta^{(m)}(t,g) := \sum_{\rho \in \mathrm{Irr}_\theta^{\leq m}} \Theta_\rho(t)\overline{\Theta_\rho(g)}$ is the θ-isotypic class delta of the finite group H/K^{m+1} (it vanishes unless $g \equiv zt$ modulo K^{m+1} for some $z \in F^\times$ up to conjugacy, and is then $\theta(z)$ times the order of the centralizer in $H/F^\times K^{m+1}$), is a well-defined class function on $\widetilde{H} \times H/K^{m+1}$, supported where g is, up to the center and modulo K^{m+1}, the cube root of $p(x)$, and explicit because the trivial row is. (The sum over all ρ does not converge pointwise, $K_\theta(1,1) = 27\sum_\rho(\dim\rho)^2$; only the truncations, or the pairing of the full sum against test functions in g, make sense.) Since the Θ_ρ are orthonormal, the H-Fourier coefficients of $K_\theta^{(m)}$ are the $L_D\Theta_\rho$, $\rho \in \mathrm{Irr}_\theta^{\leq m}$; so $K_\theta^{(m)}$ is the character of a finite-dimensional representation $\Omega_\theta^{(m)}$ of $\widetilde{H} \times H/K^{m+1}$ if and only if every $L_D\Theta_\rho$ with ρ of level $\leq m$ is a character, and in that case $\Omega_\theta^{(m)} = \bigoplus_\rho \tau_\rho \otimes \rho^\vee$ with each τ_ρ irreducible and the τ_ρ pairwise distinct (Proposition 8.4.1 and unitarity). Target B for the central character θ is therefore equivalent to: for every m, $K_\theta^{(m)}$ is a character. When it holds, the formal direct sum $\Omega_\theta = \varinjlim \Omega_\theta^{(m)}$ has the multiplicity-one Howe-duality property of Section 8.2.3, with the $K_\theta^{(m)}$ in the role of the character of the Weil representation restricted to the pair. Theorems 8.13.5 and 8.13.8 say that $K_\theta^{(2)}$ is a character and describe $\Omega_\theta^{(2)}$ on $\widetilde{U}_1 \times H/K^3$ as a sum of induced representations. This is a reformulation, not a result: we know no construction of $\Omega_\theta^{(m)}$ that does not go through L_D.

Chapter 9

Summary of results

This chapter lists the results of Chapters 4–8 with their status labels (§3.1) and pointers, one section per chapter.

9.1 Local matching and a global transfer under elliptic-support conditions (Chapter 4)

The results of Chapter 4, with status labels (§3.1) and pointers; notation as in Assumption 4.1.1 and §3, with $v \in S$ and $F = K_v$.

1. **One ambient cover** (Lemma 4.2.3, Proposition 4.2.4; **(P)** using **(E)**). The cover $\overline{G}_D(K_v)$ of [JC26a] and the Kazhdan–Patterson cover $\overline{G}_v$ of $\mathrm{GL}(r,K_v)$ are subgroups of one cover $\overline{\mathscr{G}}_v$ of $\mathrm{GL}(r,L_v)$, with the same central subgroup $\overline{Z}_v$. As an extension, $\overline{G}_v$ is the parameter-c_v cover after relabeling μ_n, and the parameter is forced (Remark 4.3.3). At $v \notin S$ they are identified canonically by conjugation (Proposition 4.2.7; **(P)**).

2. **Theorem A: covered elliptic tori** (Proposition 4.3.1, Proposition 4.3.5, Theorem 4.3.10, Corollaries 4.3.13 and 4.3.14; **(P)**). For every maximal subfield $E \subset D_v$ the two covered tori have the commutator pairing $((Nx)^{1+2c_v}x^{-1},y)_E$ and the good elements $p^{-1}(F^\times E^{\times n})$, of index $|n|_F^{-(r-1)}$. Every isomorphism between them over $\iota_{\mathrm{GL}} \circ j \circ \iota_D^{-1}$ that is the identity on $\overline{Z}_v$ has the same restriction $\jmath_{E,j}$ to the good elements. This gives a canonical bijection $\jmath$ of good regular elliptic classes and of genuine representations of covered tori.

3. **Theorem B: matching on the regular set** (Theorem 4.4.7, §4.4.6; **(P)**). Matching (Definition 4.4.4) carries the sign $(-1)^{r-1}$. Every genuine $\tilde{f}_D$ vanishing near $\overline{Z}_v$ has a matching $\tilde{f}$ supported on the good regular elliptic set, and conversely every regular-supported $\tilde{f}$ with vanishing non-elliptic orbital integrals has a matching $\tilde{f}_D$. Near the center the method fails (Proposition 4.4.8; **(P)**).

4. **Rational elements and trace formulas** (Theorem 4.5.7, Corollary 4.5.8, Propositions 4.6.1 and 4.6.2, Theorem 4.6.7; **(P)** using the **(E)** inputs of Remark 4.6.8). The elliptic terms of the two trace formulas agree term by term; on the GL side we use a variant of [FK86, §18] with the supercusp form at a place $w_0 \notin S$.

5. **Packet identity** (Proposition 4.7.6; **(P)**, unconditional). The D-side and cuspidal GL-side near-equivalence packets with the same components outside $S \cup \{w_0\}$ have equal traces at S against matching regular-supported test functions; both sums are locally finite.

6. **Theorem C: global transfer** (Theorem 4.7.12; **(P)** under (Hyp_D)). A genuine automorphic $\overline{\pi}_D$ with a supercuspidal component at some finite $w_0 \notin S$ transfers to a genuine cuspidal $\overline{\pi}$ with the same components outside S, if its packet satisfies (Hyp_D); this holds when the packet is finite (Proposition 4.7.11; **(P)**). The converse holds under (Hyp) (Theorem 4.7.13), and Lemma 4.7.8 (**(P)**) shows exactly how (Hyp_D) could fail.

7. **What is left** (§4.8; **(C)**). (D1) matching of arbitrary functions, (D2) local character relations, (D3) finiteness of packets, (D4) composition with [JC26a]. Chapter 5 proves (D1), and Chapter 6 proves (D2) at tame places in the range of the corrected Flicker–Kazhdan correspondence (Remark 4.8.1); the rest is open.

Running example (Example 4.3.7): the covered elliptic tori are non-abelian at $\mathfrak{p}_3$ (index 9, dimension 3) and abelian at $\mathfrak{p}_7$; the sign in Theorem B is -1.

9.2 Matching of arbitrary functions and an improved global transfer (Chapter 5)

The results of Chapter 5, with status labels (§3.1) and pointers; notation as in §5.2 and §3. "Relative to Chapter 4" means that the cited results of Chapter 4, restated in §5.2.3, are used.

1. **Compatible measures** (Definition 5.2.1, Remark 5.2.3; **(P)**). At $v \in S$, $\mathrm{vol}(D_v^\times / F^\times) = d(\mathrm{St})^{-1}$; every result of Chapter 4 holds unchanged with these measures.

2. **Theorem D: the section σ and transport** (Lemma 5.3.1, Propositions 5.3.4 and 5.3.7; **(P)**). At every finite place, including those above n, and on both covers: a canonical n-th root on a conjugation-invariant neighborhood V_M of the unipotent set gives a section $\sigma(g) = \widetilde{g^{1/n}}^{\,n}$ that is equivariant, good at regular points, compatible with the splittings and with $\jmath$. The map $\mathscr{T}$ identifies genuine functions supported on $p^{-1}(ZV_M)$ with functions on G/Z supported on ZV_M, preserving orbital integrals.

3. **Germs at central good elements** (Corollary 5.3.8; **(P)** using **(E)**). They are the Shalika germs of $\mathrm{GL}(r,F)$; on the D side only the trivial germ occurs.

4. **Orbital integrals of arbitrary genuine functions** (Lemmas 5.4.1 and 5.4.2; **(P)**). Away from $\overline{Z}$ they are admissible in the sense of §4.4.3 (r prime).

5. **Linear inputs and the Steinberg-type pair** (§5.5.1, Proposition 5.5.5). (L1) is **(E)**; (L2) is **(P)** using **(E)** (Lemma 5.5.3); $f_{\mathrm{St}}(1) = d(\mathrm{St})$ (Lemma 5.5.2; **(P)**). The genuine pair $(e_D, \tilde{f}^{\mathrm{St}})$ matches (**(P)**), but $\tilde{f}^{\mathrm{St}}$ is not a pseudo-coefficient of a single genuine representation (Remark 5.5.6).

6. **Theorem E: matching of arbitrary functions** (Theorem 5.5.8; **(P)** using (L1), (L2)). Every genuine $\tilde{f}_D$ has a matching $\tilde{f} = \tilde{f}_D(1)\tilde{f}^{\mathrm{St}} + \tilde{f}_{\mathrm{reg}}$, and every genuine $\tilde{f}$ with vanishing non-elliptic orbital integrals has a matching $\tilde{f}_D$.

7. **The required germ** (Theorem 5.5.14; **(P)**; Lemma 5.5.10 and Corollary 5.5.11, **(P)** using **(E)**). Near $\overline{Z}_v$ a function matches $\tilde{f}_D$ exactly when its non-trivial unipotent orbital integrals vanish and $\tilde{f}(1) = \tilde{f}_D(1)$.

8. **Theorem C′** (Theorem 5.6.3, Proposition 5.6.4; **(P)** relative to Chapter 4; Theorem 5.6.7, **(P)** under (Hyp'_D)). The packet identity holds with arbitrary matching functions at all places of S but one, and $\overline{\pi}_D$ transfers under (Hyp'_D), which is implied by (Hyp_D). The converse holds under (Hyp') (Theorem 5.6.9), in particular for a single-member packet with square-integrable components at S (Corollary 5.6.10; **(P)** using **(E)**).

9. **When (Hyp'_D) fails** (Proposition 5.6.11, Corollary 5.6.13; **(P)**). Exactly when the S-multiplicities are proportional to the Plancherel weights; it holds if the levels at S are bounded.

10. **What is left** (§5.7; **(C)**). Local character relations (since proved by Chapter 6 at tame places in the range of the corrected Flicker–Kazhdan correspondence), an unconditional global correspondence, removal of (Hyp'_D) and (Hyp'), and completing [JC26a, Remark 10.3].

Running example: Theorem E applies at both $\mathfrak{p}_3$ and $\mathfrak{p}_7$; at the wild place $\mathfrak{p}_3$ matching near the center uses Theorem D.

9.3 The division-algebra side and its local character transform (Chapter 6)

The results of Chapter 6, with status labels (§3.1) and pointers. Throughout, F is a p-adic field containing μ_n, D/F is a central division algebra of prime degree r with $(n,r) = 1$, $H = D^\times$, and $\widetilde{H}$ is the cover of Chapter 4.

1. **Good elements** (Proposition 6.2.6; **(E)** from Chapter 4). For each maximal subfield E, the good elements of $p^{-1}(E^\times)$ are the zt^*, $z \in \widetilde{Z}$, $t \in E^\times$.

2. **Power maps** (Lemmas 6.3.1, 6.3.3, 6.3.4; **(P)**). The n-th power map creates no new singular elements; its degree on elliptic tori and the discriminant ratio $\Delta(t)/\Delta(t^n)$ are computed, including at wild places.

3. **The transform L_D** (Definition 6.4.1, Lemma 6.4.3, Theorem 6.4.4; **(P)**). L_D is a unitary isomorphism between θ-equivariant invariant L^2 functions on H and genuine $\tilde{\theta}$-equivariant ones on $\widetilde{H}$, with coefficient $b_D = |n|_F^{-(r-1)/2}$ forced by unitarity.

4. **The character matrix** (Corollaries 6.5.1, 6.5.2, Lemma 6.5.4; **(P)**). The matrix of L_D in irreducible characters is unitary; if it is integral, then $\dim \tau = c_D \dim \rho$ with $c_D = |n|_F^{-(r^2-1)/2}$; b_D and c_D are positive integers.

5. **Adams operations** (Proposition 6.6.1; **(P)**). The Adams pullback and its weight, the bridge used at tame places.

6. **Tame places** (Lemma 6.7.1, Proposition 6.7.2, Theorem 6.7.6; **(P)**). If $p \nmid n$, the cover splits uniquely over the norm-one group, and L_D maps irreducible characters bijectively to genuine irreducible characters, preserving dimensions.

7. **The character relation (D2) at tame places** (Corollary 6.7.8; **(P)** in the stated range of the corrected Flicker–Kazhdan correspondence, using **(E)**). Composing L_D, the linear Jacquet–Langlands correspondence and Zou's corrected correspondence gives (D2) of Chapter 4 with the sign $(-1)^{r-1}$.

8. **Wild places** (Proposition 6.8.1; **(P)**; Remark 6.5.5). If $p \mid n$, the cover does not split over the norm-one group and genuine irreducibles have dimension divisible by $|n|_F^{-(r-1)/2}$. Integrality of L_D there is **(C)**.

9. **Orbital-integral transfer** (Theorem 6.9.1, Proposition 6.9.3, Corollary 6.9.4; **(P)**). A transfer of invariant test functions at every place of S, adjoint to L_D under the trace pairing.

10. **Global powers** (Propositions 6.11.1, 6.11.3, Corollary 6.11.4; **(P)**). A local-to-global principle for n-th powers modulo scalars outside any finite set of places, and the vanishing of the associated $Ш^1$. Section 6.10 reviews Kottwitz's framework (**(E)**).

11. **The global comparison** (§6.12; **(C)**). The spectral identity (6.38) is a target, with four proof obligations.

Running example: at $\mathfrak{p}_7$, $b_D = c_D = 1$ and L_D is a dimension-preserving bijection; at $\mathfrak{p}_3$, $b_D = 3$, $c_D = 27$, and integrality is open here (Chapter 7 and the companion chapter settle the trivial character and all rows trivial on the third congruence subgroup).

9.4 Local power maps, toral positivity, and conditional global comparison (Chapter 7)

This section summarizes Chapter 7. That chapter has no separate list of results, so its overview is reproduced here.

We conclude the fourth part of the metaplectic Jacquet–Langlands series with a precise separation of proved local and arithmetic results from two still-conditional goals. We audit the ambient cocycle and scalar-central parameter, recall the all-torus good-element criterion, compute kernels and Haar moduli of nth-power maps on finite etale tori modulo scalars, and classify globally good regular rational classes. Building on Chapter 6's unitary division-side character transform, we prove that wild irreducible-character integrality is equivalent to positive definiteness of every transferred irreducible character. A new toral permutation-character theorem establishes positive integral Fourier coefficients of the discriminant ratio for *every quadratic field extension* over a nonarchimedean field of odd residue characteristic, for all odd n including $p \mid n$. The complete numerical $r = 2$, $n = 3$ example and the nonabelian gluing problem are developed separately in a standalone companion chapter (Chapter 8). We compare our normalizations with Zou's correction of Flicker–Kazhdan, state explicit conditions for split-place matching, and prove a conditional geometric and trace-distribution comparison with normalization factor Λ. No full wild-place

irreducible correspondence, construction of exceptional local matching functions, or unconditional ordinary-to-covered global comparison is claimed.

The status of each statement is tabulated in §7.1.

9.5 The wild place in degree two: toral positivity and the non-abelian gluing problem (Chapter 8)

This section lists the results of Chapter 8, with status labels (§3.1) and pointers.

Throughout, $F = \mathbb{Q}_3(\zeta_3)$ unless said otherwise, D is the quaternion division algebra over F, $H = D^\times$, $\widetilde{H}$ is the 3-fold scalar-central cover of Chapter 4, and L_D is the division-side character transform of Chapter 6, normalized in Section 8.3. The labels are those of Section 8.1.

1. **Toral positivity** (Theorem 8.5.1, **(P)**). For every quadratic E/F with p odd and every odd n with $\mu_n \subset F$, the discriminant ratio $\Delta(t)/\Delta(t^n)$ is the character of a finite permutation representation of $E^\times/F^\times$. In the running example the genuine torus multiplicities are $(3,2,2,2)$ for the unramified torus and $(3,2,1,1,1,1)$ for either ramified torus (Proposition 8.7.2), and integrality of the whole character matrix is equivalent to positive definiteness of every row (Proposition 8.4.1).

2. **Target A** (Theorem 8.12.1, **(P^{c})**). $L_D\mathbf{1}$ is the character of a single genuine irreducible representation τ of $\widetilde{H}$ of dimension 27, and $\tau|_{\widetilde{U}_1} = \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda_0$ for the explicit character λ_0 of (8.25). The proof has three layers: a Brauer reduction to the pro-3 Sylow subgroup (Theorem 8.9.5); a finite design problem on $V = Q_3/Q_6 \simeq \mathbb{F}_3^5$ with a unique solution up to sign (Proposition 8.10.6); and hypothesis (H1), that the canonical star section is a homomorphism on K^6, proved by a Hilbert-symbol computation in an explicit Kubota–Kazhdan–Patterson model of the cover (Proposition 8.11.4). The label **(P^{c})** records that the finite design and the symbol tables are exhaustive exact computations (Remark 8.12.2); every other step is a hand proof.

3. **Target B, reduction and criterion** (Theorem 8.13.1, **(P)**; Proposition 8.13.4, **(P^{c})**). For every irreducible ρ with $\theta_\rho|_{\mu_3} = 1$, $L_D\Theta_\rho$ is an irreducible genuine character if and only if its restriction to $\widetilde{U}_1$ is a virtual character, and $L_D\Theta_\rho = \Theta_\tau \cdot F_\rho$ with F_ρ the cube-root transform of Θ_ρ (Lemma 8.13.2).

4. **Target B at levels ≤ 2** (Theorems 8.13.5, 8.13.8, **(P^{c})**). For every irreducible ρ trivial on K^3 (and every norm twist of one), $L_D\Theta_\rho$ is the character of a genuine irreducible representation τ_ρ of dimension $27\dim\rho$, and $\tau_\rho|_{\widetilde{U}_1}$ is an explicit sum of irreducible representations induced from $\widetilde{U}_3$: nine of dimension 27 indexed by the characters of K^1/K^2 at level ≤ 1, and two of dimension 81 for the six-dimensional representations of level two, where the proof rests on Lemma 8.13.7 and on the freeness of the twisted action of Q_1/Q_3 on the genuine characters of $\hat{H}$ (Proposition 8.13.9). Levels ≥ 3 are open (Sections 8.13.5, 8.15).

5. **Target C, uniform ingredients** (Theorems 8.14.2, 8.14.3, **(P)**). For every odd prime p and every $F \supset \mu_p$ with $n = p$: the p-th power identity $(w,0)^p = (w^p,0)$ in the Kubota model, and the multiplicativity of the canonical star section on $F^\times K^{2d}$, $d = pe/(p-1)$. The Brauer reduction is expected to be uniform; the finite design is not (Remark 8.14.4).

6. **Reformulation** (Remark 8.16.1). Target B for a central character θ is equivalent to the statement that, for every level m, the explicit kernel $K_\theta^{(m)} = \Theta_\tau \cdot \widetilde{\theta}\widetilde{\theta}_1^{-1} \cdot \Delta_\theta^{(m)}$ on $\widetilde{H} \times H/K^{m+1}$ is the character of a representation, which then has the multiplicity-one property of a Howe duality.

What remains, and why we stop at level two, is the subject of Section 8.15.

Chapter 10

Glossary

This chapter is a glossary of the technical terms used in every chapter. Each entry gives a short definition and a pointer to the place where the term is defined in full. The symbols for the global and local data are in Chapter 3, and the notation specific to one chapter is listed there as well. The chapter proves nothing and can be consulted at any point.

10.1 Covers and the cocycle normalization

This section lists the terms that describe covering groups and fixes the cocycle normalization used in every chapter. The book compares covers built in different ways, and a comparison of constants is only meaningful once one normalization is fixed.

Cover, p, $\overline{H'}$ An *n-fold cover* of a locally compact group H is a topological central extension $1 \to \mu_n \to \overline{H} \xrightarrow{p} H \to 1$. For a subgroup $H' \subset H$ we write $\overline{H'} := p^{-1}(H')$. Chapters 4 and 5 write $\overline{H}$; Parts III, IV and the companion chapter write $\widetilde{H}$, $\widetilde{Z}$.

Commutators For lifts $\tilde{x}, \tilde{y}$ of commuting elements, $[x,y] := \tilde{x}\tilde{y}\tilde{x}^{-1}\tilde{y}^{-1} \in \mu_n$; it does not depend on the lifts, and it is bimultiplicative on a torus [FK86, §3].

Kazhdan–Patterson cocycle σ_F The cocycle of [KP84, §0.1] and [FK86, §2] on $\mathrm{GL}(r,F)$ with twisting parameter 0. On the diagonal torus,

$$\sigma_F(\mathrm{diag}(a_i), \mathrm{diag}(b_i)) = \prod_{i<j} (a_i, b_j)_F .$$

Parameter-c' cover $\overline{G}^{(c')}$ The cover of $\mathrm{GL}(r,F)$ with cocycle

$$\sigma_F(g,h)\,(\det g, \det h)_F^{c'} .$$

The comparison uses $c' = c_v$; the parameter is forced (Remark 4.3.3). On the diagonal torus the commutator of $\mathrm{diag}(t_i)$ and $\mathrm{diag}(u_j)$ is

$$\prod_{i,j} (t_i, u_j)_F^{A_{ij}}, \qquad A = -I_r + (1 + 2c_v) J_r,$$

where J_r is the all-ones matrix (§7.2.1).

Ambient cover $\overline{\mathscr{G}}_v$ The cover of $\mathscr{G}_v = \mathrm{GL}(r, L_v) = \prod_{w|v} \mathrm{GL}(r, L_w)$ with cocycle

$$\prod_{w|v} \sigma_{L_w}(g_w, h_w)\,(\det g_w, \det h_w)_{L_w}^{c_v},$$

the copies of μ_n being identified (§4.2.2 and equation (4.1)). Both covers compared in the series are subgroups of it, so their common central subgroup $p^{-1}(K_v^\times)$ is literally the same subgroup on the two sides.

Restriction and relabeling Pulled back along $\iota_D : D_v^\times \hookrightarrow \mathscr{G}_v$, the ambient cocycle is the local component of the cocycle $\beta = \sigma_L \cdot (\mathrm{Nrd}, \mathrm{Nrd})^c$ of the corrected 1988 paper [JC26a, equation (4)]. Pulled back along the diagonal embedding of $\mathrm{GL}(r, K_v)$, it is isomorphic, *as an extension*, to the extension with cocycle $\sigma_F^r(\det, \det)_F^c$ (Proposition 4.2.4). The group μ_n is relabeled *once*, at every place and globally, by $\zeta \mapsto \zeta^{r'}$; after relabeling the second extension is $\overline{G}^{(c_v)}$. Formulas state whether they hold before or after relabeling. The isomorphism is one of extensions, not an equality of cocycles, and it is unique up to $\tilde{g} \mapsto \chi(\det p(\tilde{g}))\,\tilde{g}$ with $\chi : F^\times \to \mu_n$ continuous. Central characters are therefore never transported through it (Remark 4.2.6).

Scalar centrality Because $n \mid r - 1 + 2rc_v$, the whole inverse image $\overline{Z} = p^{-1}(F^\times)$ of the scalars is central (Lemma 4.2.3). A cover with this property is called *scalar-central*.

Global normalization Globally one uses the normalized cocycle

$$\tau_w(g,h) = \sigma_w(g,h)\,\kappa_w(g)\kappa_w(h)\kappa_w(gh)^{-1}$$

of [FK86, §2], where κ_w is trivial for $w \mid n\infty$. The compact splitting is $\kappa_w^*(k) = (k,1)_\tau$ on $\mathrm{GL}(r, \mathscr{O}_{L_w})$ for $w \nmid n\infty$, and the rational splitting is $\iota(x) = s_\tau(x)\,\kappa_L(x)^{-1}$; in the unnormalized models the rational splitting is $x \mapsto (x,1)_\sigma$ at every place (Lemma 4.5.1). The global ambient cover $\overline{\mathscr{G}}(\mathbb{A}_L)$ has cocycle $\tau_L \cdot (\det, \det)_{\mathbb{A}_L}^{c_v}$.

Star map For h in the base group and any lift $\tilde{h}$, $h^* := \tilde{h}^n$. It does not depend on the lift, it is conjugation equivariant, $(ah)^* = a^* h^*$ for scalars a, and it is multiplicative on every torus because n is odd (Definition 6.2.1; §7.2.2). For $\zeta \in \mu_n(F)$ one has $\zeta^* = 1$.

10.2 Genuine objects, central characters and good elements

This section lists the terms for objects on a cover: genuine representations and functions, their central characters, and good elements. Harmonic analysis on a cover takes place on the good elements, so these terms are used in almost every statement of the book.

Genuine representation A representation of $\overline{H}$ on which μ_n acts by the tautological character $\zeta \mapsto \zeta$.

$\tilde{\theta}$, θ A genuine character $\tilde{\theta}$ of $\overline{Z}$ (unitary from §4.5.1 on; globally, a character of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^\times)$). The same $\tilde{\theta}$ serves both covers, since $\overline{Z}$ is a common subgroup. The ordinary character attached to it is $\theta(a) := \tilde{\theta}(a^*)$; it is trivial on μ_n, and θ does not determine $\tilde{\theta}$ (§7.2.2). Chapter 6 requires only that $\tilde{\theta}|_{\mu_n}$ be faithful.

Genuine test functions $C_c^\infty(\overline{H},\tilde{\theta})_{\mathrm{gen}}$ is the space of smooth functions with support compact modulo $\overline{Z}$ and $f(z\tilde{h}) = \tilde{\theta}(z)^{-1}f(\tilde{h})$ for $z \in \overline{Z}$; in particular $f(\zeta\tilde{h}) = \zeta^{-1}f(\tilde{h})$. Characters and invariant class functions of genuine representations transform by $\tilde{\theta}$ itself (§6.2).

Regular, elliptic An element of $\mathrm{GL}(r,F)$ is *regular* if its eigenvalues are distinct, and *elliptic regular* if moreover its characteristic polynomial is irreducible. Since r is prime, the regular elements of $D_v^\times$ are the noncentral ones, and they are all elliptic. An element of a cover is regular (elliptic) if its image is.

Good element A regular $\tilde{\gamma}$ is *good* if it commutes with $p^{-1}(T_\gamma)$, where T_γ is the centralizer of $\gamma = p(\tilde{\gamma})$; equivalently, $\tilde{\gamma}$ is central in the covered torus $\overline{T_\gamma}$. Genuine invariant functions vanish at regular elements that are not good. For a maximal subfield E of D_v (or an elliptic torus $E^\times$ of $\mathrm{GL}(r,F)$), the good elements of $\overline{E^\times}$ are $p^{-1}(F^\times E^{\times n}) = \overline{Z}\cdot\{t^*\}$, of index $|n|_F^{-(r-1)}$, and every genuine irreducible representation of $\overline{E^\times}$ has dimension $|n|_F^{-(r-1)/2}$ (Proposition 4.3.5). Covered elliptic tori are abelian exactly at the places not dividing n.

10.3 Orbital integrals and matching

This section lists the terms used to compare test functions on the two groups: the normalized orbital integrals, the identification of tori, matching, and germs. Matching of orbital integrals is the local input to the comparison of trace formulas.

Δ For regular semisimple γ, $\Delta(\gamma) = |D(\gamma)|_F^{1/2}\,|\det\gamma|_F^{-(r-1)/2}$, where $D(\gamma)$ is the discriminant of the (reduced) characteristic polynomial and det = Nrd on the division-algebra side. Thus $\Delta(\gamma)^2$ is the Weyl discriminant $|\det(1-\mathrm{Ad}(\gamma^{-1}))|_{\mathfrak{g}/\mathfrak{t}}|_F$ (Definition 4.4.1 and Lemma 4.4.14).

Φ, $F = \Delta\Phi$ $\Phi(\tilde{\gamma},\tilde{f})$ is the orbital integral over $T_\gamma\backslash G$ of a genuine test function at a good regular $\tilde{\gamma}$, and $F(\tilde{\gamma},\tilde{f}) = \Delta(\gamma)\Phi(\tilde{\gamma},\tilde{f})$ is the normalized orbital integral, set to 0 at regular elements that are not good (Definition 4.4.1). Chapter 6 writes $J(t,f) = \Delta(t)\,O(t,f)$ for the same function, with its own measure conventions.

Canonical identification $\jmath$ At $v \in S$ and for a maximal subfield $E \subset D_v$ with an embedding $j : E \hookrightarrow M_r(F)$, every isomorphism of covered tori that is the identity on $\overline{Z}$ has the same restriction $\jmath_{E,j}$ to the good elements; these restrictions induce a bijection $\jmath$ from the good regular classes of the cover of $D_v^\times$ to the good regular elliptic classes of the cover of $\mathrm{GL}(r,F)$ (Theorem 4.3.10 and Corollary 4.3.13).

Matching Genuine test functions $\tilde{f}_D$ and $\tilde{f}$ *match* if (M1) $F(\tilde{\gamma},\tilde{f}_D) = (-1)^{r-1}F(\jmath(\tilde{\gamma}),\tilde{f})$ at every good regular $\tilde{\gamma}$, and (M2) $F(\tilde{\delta},\tilde{f}) = 0$ at every good regular non-elliptic $\tilde{\delta}$ (Definition 4.4.4). The sign sits in the matching of functions, so that the expected character relation is $\Theta_{\overline{\pi}_D} = (-1)^{r-1}\Theta_{\overline{\pi}}\circ\jmath$. At a place $v \notin S$, matching is simply the identity map in the coordinates fixed by ψ_v, with no sign and nothing to solve; see Example 4.2.8 for this worked out at a named split place of the running example.

Supercusp form A genuine test function on the cover of $\mathrm{GL}(r,F)$ whose integrals along the unipotent radicals of all proper parabolic subgroups vanish, for example a matrix coefficient of a supercuspidal representation [FK86, §13].

Germ The asymptotic behavior of orbital integrals near a singular element, expressed through Shalika germs [FK86, §8]; used near the center in Chapter 5.

10.4 Haar measures

This section records the Haar measures used in each chapter. The chapters normalize measures differently, and constants such as formal degrees depend on the choice, so the conventions are listed side by side.

Chapter 4 At $v \in S$ the Haar measures on $D_v^\times/F^\times$ and $\mathrm{GL}(r,F)/F^\times$ are arbitrary but fixed, and matching is defined with respect to them; torus measures are transported by conjugation and do not affect matching. At $v \notin S$ the measures correspond under the identifications ψ_v (§§4.4.1, 4.5.3).

Chapter 5 At $v \in S$, $\mathrm{vol}(D_v^\times/F^\times) = d(\mathrm{St})^{-1}$, where $d(\mathrm{St})$ is the formal degree of the Steinberg representation for the chosen measure on $\mathrm{GL}(r,F)/F^\times$ (Definition 5.2.1). Example 5.5.13 computes the resulting trivial-germ sign at the running example's tame place.

Chapter 6 For the transform L_D, the quotients H/Z and T/Z carry their Haar *probability* measures (§6.4).

Formal degree For a representation σ that is square integrable modulo the center, the constant $d(\sigma)$ in the Schur orthogonality relations; it depends on the Haar measure.

10.5 The division-side transform

This section lists the terms attached to the transform L_D of class functions on the division-algebra side. The transform is the main object of Chapters 6–8.

$H, \widetilde{H}, L_D$ For a local division algebra D of degree r over F, $H = D^\times$ and $\widetilde{H}$ is its cover from Chapter 4. The transform L_D sends θ-equivariant invariant L^2 functions φ on H to genuine $\tilde{\theta}$-equivariant invariant functions on $\widetilde{H}$ by $(L_D\varphi)(zt^*) = b_D\,\tilde{\theta}(z)\Delta(t)\Delta(t^n)^{-1}\varphi(t)$ at good regular elements, and 0 at the others; it is unitary (Definition 6.4.1 and Theorem 6.4.4).

b_D, c_D $b_D = |n|_F^{-(r-1)/2}$ and $c_D = |n|_F^{-(r^2-1)/2}$ (§§6.4–6.5). Both equal 1 at a tame place; in the running example $b_D = 3$ and $c_D = 27$ at $\mathfrak{p}_3$.

Integrality of L_D The question whether L_D carries irreducible characters to irreducible genuine characters. It holds at tame places (Chapter 6) and is open in general at wild places (Chapter 7; companion chapter).

10.6 Named results and hypotheses

This section lists the theorems and hypotheses that the book refers to by name, with the place where each is stated. It lets a reader who meets a name such as Theorem C or (Hyp_D) find its statement quickly.

Theorems A, B, C Chapter 4: the comparison of covered elliptic tori, the matching of regular-supported test functions, and the global transfer under (Hyp_D) (with its converse under (Hyp)).

Theorems D, E, C′ Chapter 5: the equivariant section near the center and its compatibility with $\jmath$; the matching of arbitrary genuine test functions; and the global transfer under the weaker hypothesis (Hyp'_D) (with its converse under (Hyp')).

(Hyp_D)**, (Hyp),** (Hyp'_D)**,** (Hyp') Non-cancellation hypotheses: the character of a near-equivalence packet at the places of S is not identically zero on the relevant matching test functions (Definition 4.7.9 and Theorem 4.7.13; Definition 5.6.5 and §5.6.3). (Hyp_D) holds when the D-side packet is finite (Proposition 4.7.11); sufficient conditions for (Hyp) are in Remark 4.7.14; and $(\mathrm{Hyp}_D) \Rightarrow (\mathrm{Hyp}'_D)$, $(\mathrm{Hyp}) \Rightarrow (\mathrm{Hyp}')$ (§5.6.3).

(D1)–(D4) The steps left open by Chapter 4: matching of arbitrary functions, local character relations, finiteness of near-equivalence packets, and composition with [JC26a] (§4.8). Their status in the later Parts is recorded in §4.1.3.

Toral positivity; the discriminant ratio For a regular element t in a maximal field $E \subset D_v$, write $\Delta(t) = |\det(1-\mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t})|_F^{1/2}$ for the Weyl-type discriminant factor and $R_E(t) := \Delta(t)/\Delta(t^n)$ for the *discriminant ratio* ((7.19)); this ratio is locally constant on the compact torus $A_E = E^\times/F^\times$ and so has a finite Fourier expansion there. *Toral positivity* is the hypothesis that this expansion has only nonnegative integer coefficients, equivalently that R_E is the character of a finite permutation representation of A_E; by Proposition 7.6.1 this is equivalent to integrality of the entire unitary character-transform matrix $(a_{\rho\tau})$ of L_D (§10.5 below), which is what Chapters 6 and 7 need but cannot prove in general. Chapter 7's global comparison is conditional on toral positivity holding at every place it is needed; Chapter 8 proves it unconditionally for every quadratic E at the running example's wild place (Theorem 8.5.1), with explicit multiplicities $(3,2,2,2)$ on the unramified torus and $(3,2,1,1,1,1)$ on either ramified one.

Targets A, B, C The companion chapter's three local problems at the wild place of the running example and beyond (Chapter 8, §8.1).

Appendix A

Tables and verification scripts for Chapter 8

A.1 Tables

This section collects the tables that Chapter 8 refers to. They are written in the coordinates of (8.19).

Coordinates. $V = Q_3/Q_6 \simeq \mathbb{F}_3^5$ with basis $(A_{\mathrm{re}}, A_{\mathrm{im}}, C_{\mathrm{im}}, B_{\mathrm{re}}, B_{\mathrm{im}})$ as in (8.19); a vector is written as a string of five digits. The generators act by the matrices

$$\mathrm{Ad}(1+j):\begin{pmatrix}1&0&0&0&0\\0&1&0&0&0\\0&1&1&0&0\\0&0&0&1&0\\0&2&1&0&1\end{pmatrix},\quad \mathrm{Ad}(1+ij):\begin{pmatrix}1&0&0&0&0\\0&1&0&0&0\\2&0&1&0&0\\2&0&2&1&0\\0&0&0&0&1\end{pmatrix},$$

$$\mathrm{Ad}(1+i):\begin{pmatrix}0&2&0&0&0\\1&0&0&0&0\\0&0&1&0&0\\0&0&0&0&2\\0&0&0&1&0\end{pmatrix},\quad \mathrm{Ad}(j):\begin{pmatrix}1&0&0&0&0\\0&2&0&0&0\\0&0&2&0&0\\0&0&0&1&0\\0&0&0&0&2\end{pmatrix}.$$

The 36 good lines and the defect function e_*. Each entry is a representative v followed by $e_*(v)$; $e_*(-v) = -e_*(v)$.

```
01002:0  01012:0  01022:0  01101:0  01111:1  01121:2
01201:0  01211:2  01221:1  10020:0  10021:0  10022:0
10110:0  10111:2  10112:1  10210:0  10211:1  10212:2
11002:0  11011:0  11020:0  11101:1  11110:2  11122:0
11201:2  11210:1  11222:0  12001:0  12012:0  12020:0
12102:2  12110:1  12121:0  12202:1  12210:2  12221:0
```

The 27 covectors $\Lambda = \Lambda_+$ (those h with $N_{e_*}(h) = 20$):

```
00000 00100 00200 01001 01121 01211 02002 02112 02222
10010 10111 10212 11022 11110 11201 12021 12102 12210
20020 20122 20221 21012 21101 21220 22011 22120 22202
```

The Heis-cocycle attached to e_* is $\beta_{\mathrm{Ad}(1+j)} = (1,0,0,1,0)$, $\beta_{\mathrm{Ad}(1+ij)} = (0,1,0,0,1)$. The numbers of ordered coplanar triples of distinct good lines is $B = 720$, all of them defect free.

The Hilbert pairings. On the basis $\varpi,\ 1+\varpi,\ 1+\varpi i,\ 1+\varpi^2,\ 1+\varpi^2 i,\ 1+\varpi^3$ of $E_0^\times/E_0^{\times 3}$ (the last five being a basis of $U_{E_0}^{(1)}/U_{E_0}^{(1)3}$; recall $1+\varpi^3 i$ is a cube) the cubic Hilbert pairing of E_0, written as exponents of ζ, is antisymmetric with upper-triangular entries

$$(\varpi, 1+\varpi^3) = 2, \quad (1+\varpi, 1+\varpi i) = 1, \quad (1+\varpi, 1+\varpi^2) = 2, \quad (1+\varpi i, 1+\varpi^2 i) = 1,$$

and all other upper-triangular entries 0. On the basis $\varpi, 1+\varpi, 1+\varpi^2, 1+\varpi^3$ of $F^\times/F^{\times 3}$ the pairing of F has $(\varpi, 1+\varpi^3) = 1$, $(1+\varpi, 1+\varpi^2) = 1$ and all other upper-triangular entries 0. Both are nondegenerate, Galois invariant, and satisfy $(a,b)_{E_0} = (a, \mathrm{N}b)_F$ for $a \in F$; the common sign is the choice of the identification $\mu_3 \simeq \mathbb{Z}/3$.

A.2 Verification scripts

The scripts are in the directory `part-IV/scripts/` of the repository `github.com/wdjoyner/metaplectic-jl`; the version used for this draft is that of commit `67e6439` (November 8, 2026), and the README there lists, for each script, the statement it supports and whether it is load bearing. Script 13 certifies the group law of Q_3/Q_7 deterministically (associativity on all 729^3 triples, identity, inverses) and verifies the character table exactly by checking that each rounded character is a common eigenvector of all class-sum matrices with the correct norm, which characterizes the irreducible characters; script 15 verifies that λ_0 is a character of $\hat{H}$ on all pairs (generator, element), which suffices by induction on word length; and `verify_certificate.py` checks the orbit certificate independently of script 15 and of the cover (that the tables define an action of the Heisenberg group of order 27, whose multiplication table is part of the certificate, that it has no fixed points, and that the orbits are as listed). All scripts are in pure Python with exact rational or modular arithmetic, with two exceptions: script 11 uses floating point only to display complex character values, and scripts 13–14 obtain the character table of Q_3/Q_7 by Burnside's algorithm in floating point and then round the values to $\mathbb{Z}[\zeta_3]$ (maximal rounding error $2\cdot 10^{-13}$) and re-verify all orthogonality relations exactly; this is numerical discovery followed by exact verification, and those two scripts are corroboration only (Theorem 8.13.8 is proved by hand). They are kept with the paper under `part-IV/scripts/`.

1. `step1_G3.py`: exact arithmetic in $\mathcal{O}_D$; the 648 depth-one classes modulo $K^4 U_F^{(1)}$; their cubes in V; the 72 elements of $\bar{G}_3$, the 36 lines, agreement with (8.20).

2. `step2_finite.py`: the finite group Q/Q_7 (3^9 elements); the cube map; depths of cubes (3 or ≥ 6); $\mathrm{Cubes}(Q_2) = Q_6$; Q_6-invariance of G_3.

3. `step3_actions.py`: the actions of $\mathrm{Ad}(1+j), \mathrm{Ad}(1+ij), \mathrm{Ad}(1+i), \mathrm{Ad}(j)$ on V; orders of the groups they generate (27, 8, 216); orbits on $\bar{G}_3$ and on the lines; $B = 720$.

4. `step4_design.py`, `step6_orbit.py`: the design search (the nine, resp. 3^{10}, candidates), the cocycle extension test, the simply transitive orbit, the Fourier identities.

5. `step5_H1.py`: (a) the Lazard form on K^4/K^8: invariant alternating forms vanishing on torus subspaces are 0; (b) odd functions on Q_6/Q_8 with equivariant defects are linear.

6. `cover.py`, `step7_symbols.py`: Steinberg relations and the Hilbert symbols of F and E_0; nondegeneracy, Galois invariance, norm compatibility; the cocycle identity for (8.27) (250 triples); the diagonal formula; triviality on scalars; the torus commutator formula of Chapter 4 (200 pairs).

7. `step8_H1test.py`, `step9_H1_stratified.py`, `step12_H1_proof_check.py`: corroboration of Proposition 8.11.4: (H1) through cube roots (Hensel in $F(X)$), through (8.30) (7600 stratified instances), and through (8.34) and (8.30) on 12,000 instances covering every case of the proof; script 12 also verifies the closed form (8.29) against the digit form and the symbol facts (8.28).

8. `step10_defects.py`: defects of 3- and 4-term relations in $\bar{G}_3$ against e_*; $\omega = 0$; $\kappa_{1,5}$.

9. `step11_endtoend.py`, `step11b.py`: the character of $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda_0$ from (8.25) against (8.21).

10. `step13_level2.py`, `zsolve.py`: the group $H = Q_3/Q_7$ (multiplication table, 297 classes), its character table by Burnside's algorithm with exact re-verification, the $\mathrm{Ad}(Q_1)$-orbits on $\mathrm{Irr}(H)$, the 219 good classes and the cube root; the criterion of Proposition 8.13.4 tested by exact integer linear algebra (Smith normal form) for the four D_8-orbit sums of $\mathrm{Irr}(\mathrm{Heis})$.

11. `step14_level2_restriction.py`: the decomposition of $(\chi \circ r)D$ into irreducible characters of H for the orbit sums and for χ_3, χ_3'.

12. `step15_level2_clifford.py`: λ_0 on $\hat{H}$ in the cocycle model, the twisted action of Q_1/Q_3 on the genuine irreducible characters of $\hat{H}$, an explicit stabilizer check (no nontrivial coset representative fixes any genuine character), and the orbit table written to `level2_orbit_certificate.txt` (two orbits of size 27 on the degree-three characters; nine of size 27 on the linear ones), together with the multiplication table of Q_1/Q_3 and the indexed character values needed to check it. `verify_certificate.py` re-reads the certificate and checks it without using the cover. Running time about ten minutes. Corroborates Proposition 8.13.9.

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This book was written by David Joyner together with two AI systems, Claude (Anthropic) and ChatGPT (OpenAI). Both systems are listed as authors. Their roles were as follows.

Each of Chapters 4–8 began as a separate manuscript. Each was drafted by David Joyner together with Claude or ChatGPT, and refereed by whichever of the two AI systems did not draft it. Chapter 6 grew instead from discussion among all three. The order of authors on each manuscript was as follows.

Chapter	Authors, in the order on the manuscript
4	Joyner, Claude, ChatGPT
5	Joyner, Claude, ChatGPT
6	Joyner, ChatGPT, Claude
7	Joyner, ChatGPT, Claude
8	Joyner, ChatGPT, Claude

Claude assembled the manuscripts into this book and revised Draft 2 to follow the author's style guide. In that revision Claude wrote the abstract, the orienting paragraphs that open sections, and the examples that follow the definitions. The revision log lists each of these additions.

The paragraphs below are reproduced from the manuscripts. They record what is specific to each chapter, such as a referee's correction, an independently supplied proof or an observed consequence.

Chapter 4. ChatGPT's independent proofs and corrections, and the drafting and referee history, are recorded in the revision log.

Chapter 5. ChatGPT contributed referee reports on the planning memo and the drafts, and independent proofs of Proposition 5.6.11 and of the descent argument in Lemma 5.5.10, written before it consulted our sketches. Those proofs agreed with our sketches and supplied the references [How74], [HC78] and [Rog81]; the arguments given here follow them. ChatGPT also observed Corollary 5.6.13.

Chapter 6. The tame splitting and Adams-operation arguments were developed and checked through discussion among the three contributors. The mathematical claims remain subject to independent specialist verification.

Chapter 7. Claude (Anthropic) supplied referee reports on Drafts 1–3, including the correction to the Haar-modulus proof and the split-place matching obligations. The quadratic toral theorem is

included with proof in this concluding draft; the detailed wild rank-two calculations now form the standalone companion chapter, Chapter 8. The remaining research problems are recorded in the status table and Section 7.9.

Chapter 8. The computations stem from the collaborative Chapter 7 project of D. Joyner, ChatGPT (OpenAI), and Claude (Anthropic). Claude's referee report on Chapter 7 suggested the positivity criterion and a uniform rank-two toral analysis; the reduction to the Sylow 3-subgroup, the finite model, the explicit cocycle model and the verification scripts of Drafts 1–3 were produced by Claude; the proof of Proposition 8.11.4 in Draft 2 replaces a case-by-case argument of Draft 1 whose first case was incorrect, as pointed out in ChatGPT's referee report on Draft 1. Sources labeled (E) are imported from Chapters 4–6 or [JC26a]; results labeled (P) are proved here; results labeled (P^c) rely in addition on the exhaustive exact computations listed in Appendix A.2 and itemized in Remark 8.12.2. Independent refereeing of the hand arguments and re-execution of the scripts are advisable before the result is cited.

Limits of AI review. Review by an AI system is not an independent certification of the results. Independent human review is the next step, and the mathematical claims should be assessed independently. Further details on the AI usage are in the revision log of this draft and in the revision logs and referee reports of the manuscripts, which are in the project repository (`github.com/wdjoyner/metaplectic-jl`).

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